

Magnetic Effects of Current & Magnetism

FEW WORDS FOR STUDENTS

First time playing with magnets is always a fun for every child. With age everyone gets aware that magnets have two poles, designated as north and south poles. All of you might have seen a compass used by travellers for the purpose of navigation. In ancient times compass was the only tool available for determining the directions. The needle of such a compass is a small, thin magnet. Magnets and magnetic effects are very important in different industrial applications as well as these are very important in understanding many natural and lab phenomenon related to magnetism. Even many birds use the magnetic field of Earth for navigation along with directions from the location of Sun and stars during long distance migrations. This whole chapter is covering effects and phenomenon of magnetism to strengthen understanding of this topic to a decent level.

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COVER APPLICATION

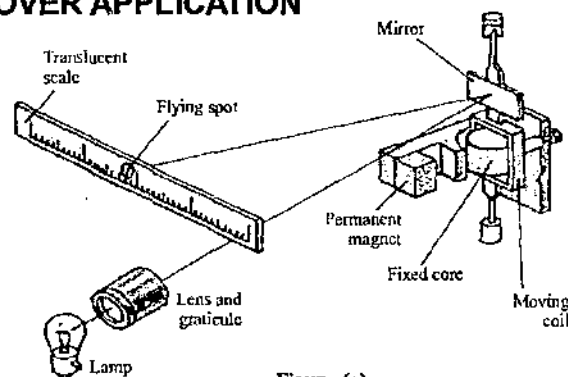


Figure-(a)

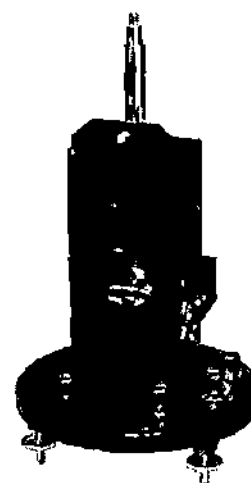


Figure-(b)

A ballistic galvanometer is used to measure charge which passes through the device. When charge is suddenly passed through a coil in magnetic field, it imparts an angular impulse to the coil which deflects the coil once and a small mirror attached to the coil axis also rotates along with the coil due to this. Torsional springs restores the position of coil back to normal but the deflection of a light beam falling on the mirror is measured on a scale and amount of charge can be calculated. Figure-(a) shows the setup of ballistic galvanometer experiment and figure-(b) shows the industrial or lab ballistic galvanometer at the center of which there is a small glass window through which a narrow light beam falls on the mirror and on the path of reflected beam a scale is placed.

Magnetic forces of attraction and repulsion between magnetic poles are similar to the interaction between charges but magnetic poles and charges are not the same thing. However the way electric charges produce electric field in their surrounding is similar to that magnetic poles also have an associated magnetic field in their surrounding. Magnetic lines of forces are also similar to electric lines of forces used to trace the pattern of electric field in surrounding of charges. Tangent to a magnetic line of force in space gives the direction of magnetic field strength vector at that point and density of magnetic field lines in a region is a measure of the magnitude of magnetic field in that region. Similar to charges as we approach closer to a magnetic pole, density of magnetic lines increase.

It is also observed that in a region variation of electric field gives rise to induction of magnetic field and vice versa. A Danish scientist Oersted observes magnetic field in the surrounding of current carrying conductor. Figure-4.1 shows the setup of Oersted's Experiment in which when a current is switched on, deflection in magnetic needle is observed which verified the presence of magnetic field in surrounding of current carrying wires. This is called '*Magnetic Effects of Electric Current*'.

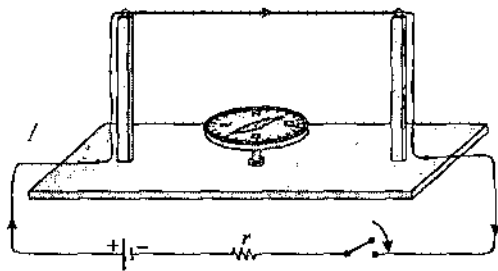


Figure 4.1

In this experiment it is also seen that the direction of deflection in magnetic needle is opposite when the needle is placed below or above the current carrying wire which indicates that magnetic field direction is opposite on the two sides of a current carrying wire.

In previous chapters we've already studied about electric field and electric forces. There are some facts listed below concerning to electric and magnetic field and forces. Students are advised to keep these below points always in mind as these points form the basis of understanding magnetic effects.

(1) A static charge produces only electric field in its surrounding and only electric field can exert a force on static charges.

(2) A moving charge produces both electric and magnetic fields in its surrounding and both electric and magnetic fields can exert force on moving charges.

(3) A current carrying wire produces only magnetic field in its surroundings and only magnetic field can exert a force on current carrying wires.

4.1 Biot Savart's Law

Biot Savart's Law is a basic law concerning electricity and magnetism which describes the magnetic field generated by a current carrying wire in its surrounding. The equation of Biot Savart's Law gives the strength of magnetic field at a specific point in surrounding of a current carrying wire. The magnetic field which is associated with electric field at any point for which we were discussing before this article is often referred as '*Magnetic Induction*' instead of magnetic field in technical terms. This is denoted by ' $\vec{B}$ ' and also referred as magnetic flux density like electric field intensity ' $\vec{E}$ '.

This law analyzes that the magnetic induction produced due to an elemental length of a current carrying wire depends upon four factors. Consider a wire XY carrying a current I . To find the magnetic induction at a point P in its neighbourhood, we consider an elemental segment AB on this wire of length dl as shown.

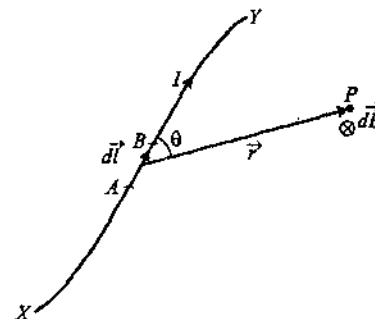


Figure 4.2

According to Biot Savart's Law the magnetic induction dB at point P due to the elemental wire segment AB depends upon four factors which are given as

(i) dB is directly proportional to the current in the element.

$$dB \propto I \quad \dots (4.1)$$

(ii) dB is directly proportional to the length of the element

$$dB \propto dl \quad \dots (4.2)$$

(iii) dB is inversely proportional to the square of the distance r of the point P from the element

$$dB \propto \frac{1}{r^2} \quad \dots (4.3)$$

(iv) dB is directly proportional to the sine of the angle θ between the direction of the current flow and the line joining the element to the point P

$$dB \propto \sin \theta \quad \dots (4.4)$$

Combining above factors, we have

$$dB \propto \frac{Idl \sin \theta}{r^2}$$

$$\Rightarrow dB = K \frac{Idl \sin \theta}{r^2} \quad \dots (4.5)$$

Where K is a proportionality constant and its value depends upon the nature of the medium surrounding the current carrying wire. If the current carrying wire is placed in vacuum, then in SI Units its value is given as

$$K = \frac{\mu_0}{4\pi} = 10^{-7} \text{ T-m/A}$$

Here μ_0 is called permeability of vacuum or free space. It is a physical quantity for a given medium which is a measure of a medium's ability on the extent to which it allows external magnetic field to polarize the material inside the volume of the body. Therefore Biot Savart's Law is written as

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2} \quad \dots (4.6)$$

This constant μ_0 is similar in nature to the constant ϵ_0 we discussed in the topic of electrostatics. For a given medium its magnetic permeability is defined as

$$\mu = \mu_0 \mu_r$$

Where μ_r is called 'Relative Permeability of Medium' which is casually also referred as diamagnetic constant of the medium. Unlike to the case of electrostatics where dielectric constant of any medium is always greater than unity, in case of magnetism, depending upon the types of mediums value of μ_r can be greater or smaller than unity. In the topic of magnetic properties of material in next chapter we will discuss about the magnetic permeability in detail. As of now students can consider that if the current carrying wire is placed in a material medium then equation-(4.6) can be rewritten as

$$dB = \frac{\mu}{4\pi} \frac{Idl \sin \theta}{r^2} = \frac{\mu_0 \mu_r}{4\pi} \frac{Idl \sin \theta}{r^2} \quad \dots (4.7)$$

Above equation-(4.7) gives the magnitude of magnetic induction produced due to a small current element. The direction of this magnetic induction is given by right hand thumb rule stated as "Grasp the conductor in the palm of right hand so that the thumb points in the direction of the flow of

current, then the direction in which the fingers curl, gives the direction of magnetic field lines". This is shown in figure-4.3.

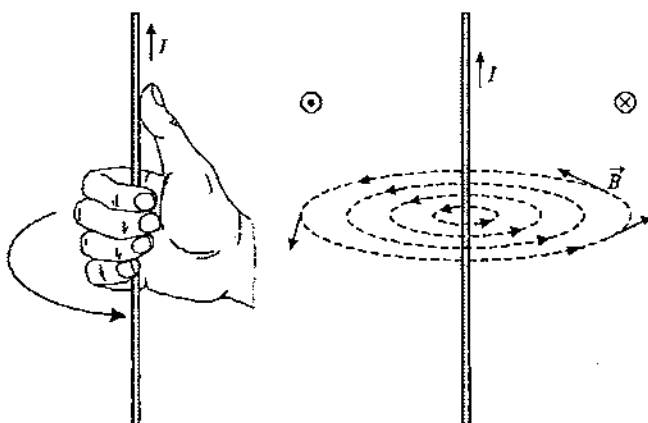


Figure 4.3

In the figure-4.2 using above rule we can see that direction of vector $\vec{dB}$ is into the plane of paper which is represented by $\otimes$ similarly on the points located on the left side of wire direction of $\vec{dB}$ is in outward direction from the plane of paper and it is represented by $\odot$. We can accommodate this direction in equation-(4.6) which can be rewritten as

$$\vec{dB} = \left(\frac{\mu_0}{4\pi} \right) \frac{Id\vec{l} \times \vec{r}}{r^3} \quad \dots (4.8)$$

In above equation you can see that the direction of cross product of the vector of elemental length and position vector of point P with respect to the element is giving the same direction as stated by right hand thumb rule.

At point P in figure-4.2 magnetic induction due to whole wire XY can be obtained by integrating the above expression for the entire length of the wire within proper limits according to the shape of the wire and it is given as

$$\vec{B} = \int d\vec{B} = \int \left(\frac{\mu_0}{4\pi} \right) \frac{Id\vec{l} \times \vec{r}}{r^3} \quad \dots (4.9)$$

In above equations many times the term ' Idl ' which is the product of current and length of element is also referred as a separate physical quantity called 'Current Element' which is measured in units of 'Ampere-meter' or 'A-m'. In later articles of the chapter we will discuss its physical significance.

4.1.1 Magnetic Induction Direction by Right Hand Palm Rule

In previous article we studied that the direction of magnetic induction in surrounding of a current carrying wire can be obtained by right hand thumb rule. Same can also be obtained by 'Right Hand Palm Rule' stated as "Stretch your palm of

4.1.1 Magnetic Induction due to Current

right hand keeping fingers straight and put the thumb along the direction of current and point your fingers toward the point of application. The area vector of your palm face gives you the direction of magnetic induction vector at that point". This is shown in figure-4.4.

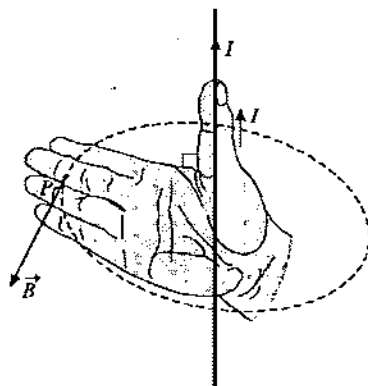


Figure 4.4

4.1.2 Magnetic Induction due to Moving Charges

As already discussed that moving charges produce magnetic field in their surrounding. Biot Savart's law is also defined for moving charges like current carrying wires we studied in article-4.1. To analyze the magnetic induction in surrounding of a moving charge we consider a charge q moving at velocity v as shown in figure-4.5. In this situation we will determine the magnetic induction at a point P located at a position vector r from the instantaneous position of charge as shown in figure.

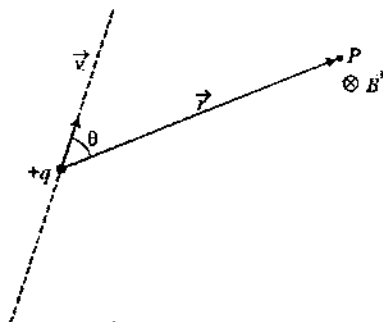


Figure 4.5

The magnetic induction B due to moving charges in surrounding depends upon four factors which are given as

- (i) B is directly proportional to the charge

$$B \propto q \quad \dots (4.10)$$

- (ii) B is directly proportional to the speed of charge

$$B \propto v \quad \dots (4.11)$$

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- (iii) B is inversely proportional to the square of the distance r of the point P from the element

$$B \propto \frac{1}{r^2} \quad \dots (4.12)$$

- (iv) B is directly proportional to the sine of the angle θ between the direction of the motion of positive charge and the line joining the charge to the point P

$$B \propto \sin \theta \quad \dots (4.13)$$

Combining above factors we have

$$B \propto \frac{qv \sin \theta}{r^2}$$

$$\Rightarrow B = K \frac{qv \sin \theta}{r^2} \quad \dots (4.14)$$

For free space we can use $K = \frac{\mu_0}{4\pi}$ so we have

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r^2} \quad \dots (4.15)$$

Vectorially above result can be rewritten including the direction of direction vector of magnetic induction is given as

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \vec{r})}{r^3} \quad \dots (4.16)$$

In above expression of magnetic induction at point P due to the moving charge the cross product of velocity and position vector is written in such a way that the direction of magnetic field obtained by right hand thumb rule is same as that given by this cross product. In figure-4.5 direction of magnetic induction at point P is into the plane of paper (inward) as shown in this figure.

4.1.3 Units used for magnetic induction

In SI units magnetic induction is measured in 'Tesla' denoted as 'T' and magnetic flux in space through any area is measured in units of 'weber' and denoted as 'Wb'. As magnetic induction is also referred as magnetic flux density or flux passing through a unit normal area in a region like electric field was defined. The units tesla(T) and weber(Wb) are related as

$$1 \text{ T} = 1 \text{ Wb/m}^2 \quad \dots (4.17)$$

4.2 Magnetic Induction due to different Current Carrying Conductors Configurations

Equation of Biot Savarts Law gives the magnetic induction due to a current element in its surrounding. For different shapes of current carrying conductors this needs to be integrated

differently to obtain the net magnetic induction due to the conductors in their surrounding at specific points. Next we are going to discuss some standard configurations of the current carrying conductors.

4.2.1 Magnetic Induction due to a Long Straight Current Carrying Conductor

Figure-4.6 shows a long straight wire carrying a current I . To determine the magnetic induction due to the wire at point P situated at a normal distance r from the wire we consider an element of length dy in the wire at a distance y from the point O as shown. By right hand palm rule we can see that the direction of magnetic field at point P is into the plane of paper (inward).

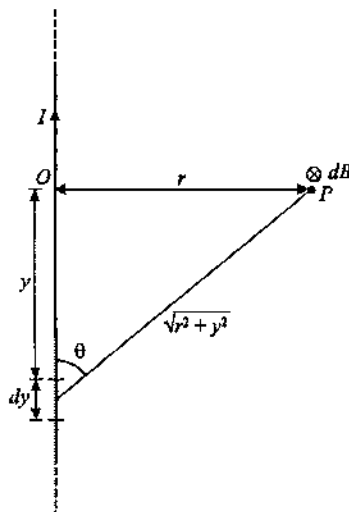


Figure 4.6

The magnetic induction dB at point P is into the plane of paper and given by Biot Savart's Law using equation-(4.6) as

$$dB = \frac{\mu_0}{4\pi} \times \frac{Idy \sin \theta}{(r^2 + y^2)^{3/2}}$$

In above equation we can substitute $\sin \theta = \frac{r}{\sqrt{r^2 + y^2}}$ gives

$$dB = \frac{\mu_0}{4\pi} \times \frac{Idy r}{(r^2 + y^2)^{3/2}} \quad \dots (4.18)$$

As due to all the elements in the wire the magnetic induction is in inward direction at point P , the net magnetic induction at point P is given by integrating above equation-(4.18) for the whole length of wire within limits of y from $-\infty$ to $+\infty$ as

$$B = \int dB = \int_{y=-\infty}^{y=+\infty} \frac{\mu_0}{4\pi} \times \frac{I dy r}{(r^2 + y^2)^{3/2}}$$

$$\Rightarrow B = \frac{\mu_0 I r}{4\pi} \int_{-\infty}^{+\infty} \frac{dy}{(r^2 + y^2)^{3/2}}$$

To integrate the above expression we substitute

$$y = r \tan \theta$$

$$dy = r \sec^2 \theta d\theta$$

With the above substitution limits of integration also changes as

$$\text{at } y = -\infty \rightarrow \theta = -\frac{\pi}{2}$$

$$\text{at } y = +\infty \rightarrow \theta = +\frac{\pi}{2}$$

and

$$\Rightarrow B = \frac{\mu_0 I r}{4\pi} \int_{-\pi/2}^{+\pi/2} \frac{r \sec^2 \theta d\theta}{(r^2 + r^2 \tan^2 \theta)^{3/2}}$$

$$\Rightarrow B = \frac{\mu_0 I r}{4\pi} \int_{-\pi/2}^{+\pi/2} \frac{r \sec^2 \theta d\theta}{r^3 \sec^3 \theta}$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi r} \int_{-\pi/2}^{+\pi/2} \cos \theta d\theta$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi r} [\sin \theta]_{-\pi/2}^{+\pi/2}$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi r} [1 - (-1)]$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi r} \quad \dots (4.19)$$

As already discussed that the direction of magnetic induction due to a long straight wire in its surrounding is along the tangent to the concentric magnetic lines as shown in figure-4.7. This is given by either of right hand thumb rule or right hand palm rule.

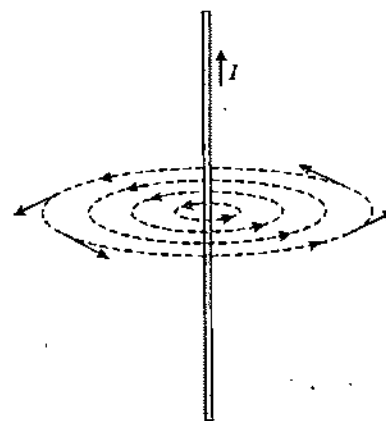


Figure 4.7

4.2.2 Magnetic Induction due to a Finite Length Current Carrying Wire

Figure-4.8 shows a wire AB of length L carrying a current I . In the surrounding of wire consider a point P as shown which is located at a perpendicular distance r from the wire such that the point is subtending angles θ_1 and θ_2 at the end points of the wire as shown in figure.

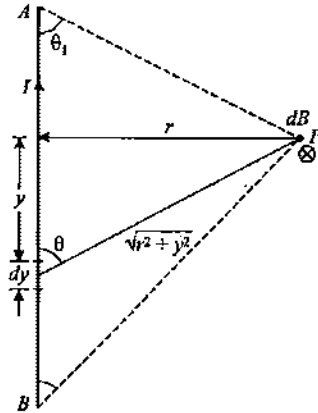


Figure 4.8

To calculate the magnetic induction at point P we consider an element of length dl at a distance y from the point O as shown in figure-4.8. By using Biot Savart's law the magnetic induction dB at point P due to this current element is given as

$$dB = \frac{\mu_0}{4\pi} \cdot \frac{Idl \sin \theta}{(r^2 + y^2)^{3/2}}$$

As due to all the elements in the wire the magnetic induction is in inward direction at point P , the net magnetic induction at point P due to wire AB is given by integrating above equation-(4.18) for the whole length of wire AB within limits of y from $-r \cot \theta_2$ to $+r \cot \theta_1$ as

$$B = \int dB = \int_{y=-r \cot \theta_2}^{y=+r \cot \theta_1} \frac{\mu_0}{4\pi} \times \frac{Idy r}{(r^2 + y^2)^{3/2}}$$

$$\Rightarrow B = \frac{\mu_0 I r}{4\pi} \int_{-r \cot \theta_2}^{+r \cot \theta_1} \frac{dy}{(r^2 + y^2)^{3/2}}$$

To integrate the above expression we substitute

$$y = r \cot \theta$$

$$dy = -r \operatorname{cosec}^2 \theta d\theta$$

With the above substitution limits of integration also changes as

$$\text{at } y = -r \cot \theta_2 \rightarrow \theta = -\theta_2$$

$$\text{and at } y = +r \cot \theta_1 \rightarrow \theta = +\theta_1$$

$$\begin{aligned} \Rightarrow B &= -\frac{\mu_0 I r}{4\pi} \int_{-\theta_2}^{+\theta_1} \frac{r \csc^2 \theta d\theta}{(r^2 + r^2 \cot^2 \theta)^{3/2}} \\ \Rightarrow B &= -\frac{\mu_0 I r}{4\pi} \int_{-\theta_2}^{+\theta_1} \frac{r \csc^2 \theta d\theta}{r^3 \csc^3 \theta} \\ \Rightarrow B &= -\frac{\mu_0 I}{4\pi r} \int_{-\theta_2}^{+\theta_1} \sin \theta d\theta \\ \Rightarrow B &= \frac{\mu_0 I}{4\pi r} [\cos \theta]_{-\theta_2}^{+\theta_1} \\ \Rightarrow B &= \frac{\mu_0 I}{4\pi r} [\cos \theta_1 - (-\cos \theta_2)] \\ \Rightarrow B &= \frac{\mu_0 I}{4\pi r} [\cos \theta_1 + \cos \theta_2] \quad \dots (4.20) \end{aligned}$$

The direction of magnetic induction in surrounding of a finite length current carrying wire can also be given by right hand thumb rule as shown in figure-4.7. Equation-(4.20) can also be used to calculate the magnetic induction due to an infinite current carrying straight wire in which both the side angles θ_1 and θ_2 will tend to zero. In this equation if we substitute both angles zero then it gives the expression given in equation-(4.19).

4.2.3 Magnetic Induction due to a Semi-Infinite Straight Wire

Figure-4.9 shows a very long wire AB carrying a current I and P is a point at a perpendicular distance r from the end A of the wire. Using right hand thumb rule here we can state that the magnetic induction at point P is in outward direction.

The magnitude of the magnetic Induction at P can be calculated by using equation-(4.20). If we look at figure-4.9(b), we can see that at point A point P is making an angle 90° with the length of wire and at point B which is located far away distance, the angle between the line PB and wire can be considered as 0° . Thus magnetic induction at point P due to this semi infinite wire is given as

$$\begin{aligned} B &= \frac{\mu_0 I}{4\pi r} [\cos(90^\circ) + \cos(0^\circ)] \\ \Rightarrow B &= \frac{\mu_0 I}{4\pi r} [0 + 1] \\ \Rightarrow B &= \frac{\mu_0 I}{4\pi r} \quad \dots (4.21) \end{aligned}$$

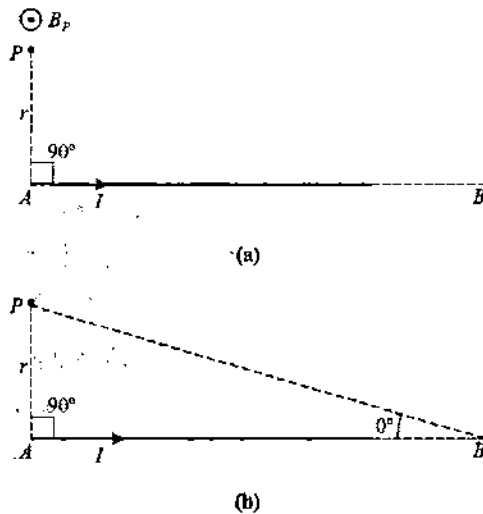


Figure 4.9

The magnetic induction given in above expression given in equation-(4.21) is half of that obtained due to an infinitely long wire as given in equation-(4.19) which can be directly stated as both the halves of an infinite wires are identical with respect to point P in figure-4.9 on the two sides of it and if one half is removed then due to the remaining half the magnetic induction is also reduced to half.

$$B = \frac{\mu_0 I}{4\pi r} [\cos \theta + \cos (0^\circ)]$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi r} [1 + \cos \theta] \quad \dots (4.22)$$

In both of above cases the direction of magnetic induction is given by right hand thumb rule as shown in these figures.

4.2.4 Magnetic Induction at the Center of a Circular Coil

Figure-4.11(a) shows a circular coil of radius R carrying a current I . If we consider an element of length dl along its circumference as shown then due to this element by right hand thumb rule or palm rule we can see that the direction of magnetic induction dB at the center of coil is in outward direction as shown in figure-4.11(b). By Biot Savart's Law the magnetic induction at the center of this coil due to this current element is given as

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin(90^\circ)}{R^2}$$

$$\Rightarrow dB = \frac{\mu_0 Idl}{4\pi R^2} \quad \dots (4.23)$$

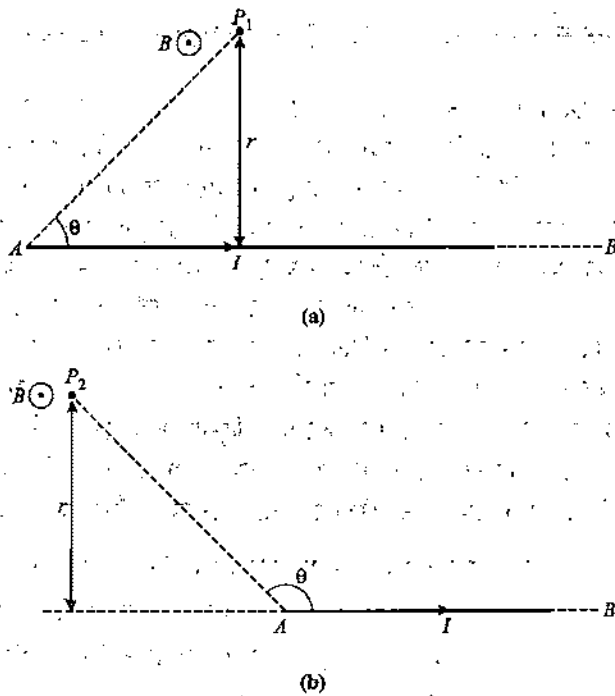
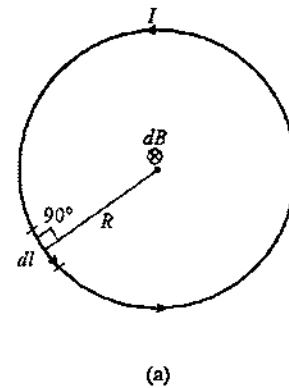
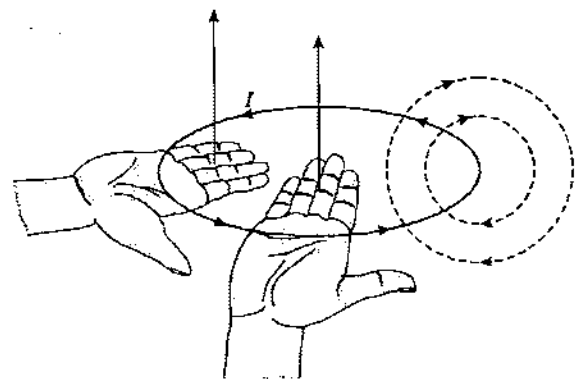


Figure 4.10

Figure-4.10(a) and (b) shows another semi infinite wire but in this case we will calculate the magnetic induction at points P_1 and P_2 which are located at same distance from end point A of wire at same perpendicular distance r from the line of wire. Again using equation-(4.20) we can calculate the magnetic induction at these points which is given as



(a)



(b)

Figure 4.11

As shown in figure-4.11(b) the direction of magnetic induction due to all the elements of the coil are in same outward direction at every interior point of the coil and the magnetic lines are making closed loops from outside of the coil as shown. Thus net magnetic induction at center of coil can be given by integrating equation-(4.20) for the whole length of the coil which is given as

$$B = \int dB = \frac{\mu_0 I}{4\pi R^2} \int_0^{2\pi R} dl$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi R^2} [2\pi R - 0]$$

$$\Rightarrow B = \frac{\mu_0 I}{2R} \quad \dots (4.24)$$

If there are N turns of wire in the coil then total length of wire will be $2\pi RN$ so for N turns in coil, the magnetic induction at the center of coil is given as

$$B = \frac{\mu_0 IN}{2R} \quad \dots (4.25)$$

Like a circular coil for any closed current carrying coil of any shape in a plane the direction of magnetic induction can also be calculated by using right hand thumb rule in a different way. For this we circulate our right hand fingers along the direction of current in the loop as shown in figure-4.12(a) then the direction of our right hand thumb gives the direction of magnetic induction at interior points in the plane of the loop and at all the points outside the loop in its plane the direction of magnetic induction is opposite as shown in figure-4.12(b).

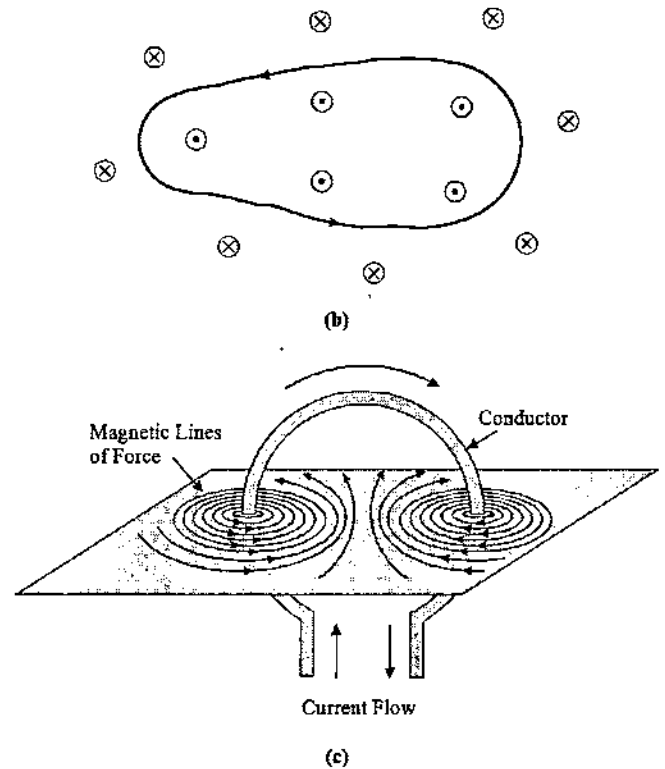
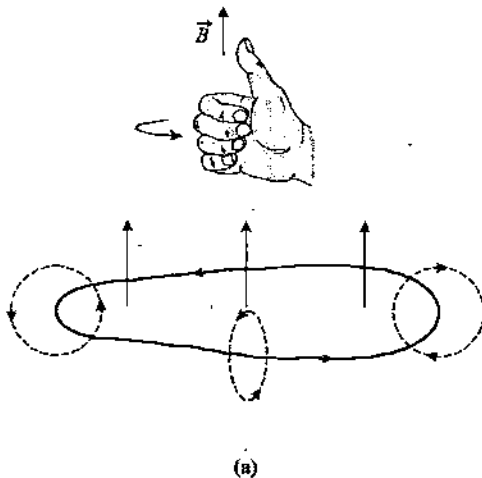


Figure 4.12

Figure-4.12(c) shows the configuration of magnetic lines in a diametrical plane of a circular current carrying coil which is explained in figure-4.11.

4.2.5 Magnetic Induction at Axial Point of a Circular Coil

Figure-4.13 shows a current carrying circular coil of radius R mounted in YZ plane carrying a current I . We will calculate the magnetic induction due to this coil at a point P located on the axis of coil at a distance x from its center as shown. By using right hand thumb rule as described in previous article, we can see that the direction of magnetic induction at point P due to the current in this coil is in rightward direction.

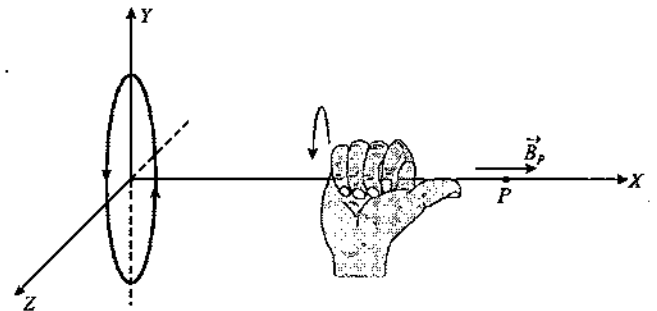


Figure 4.13

To calculate the magnitude of magnetic induction at P due to the coil we consider a small element of length dl at the top of the coil as shown in figure-4.14. Due to the current flowing in

this element if we find the magnetic induction dB at point P then its direction can be given by right hand thumb rule as shown in figure. To understand the direction of dB at point P students can imagine their right hand palm placed in the figure with thumb along the direction of current in dl and fingers pointing toward the point P then you can feel that the direction of dB is perpendicular to the palm of your right hand.

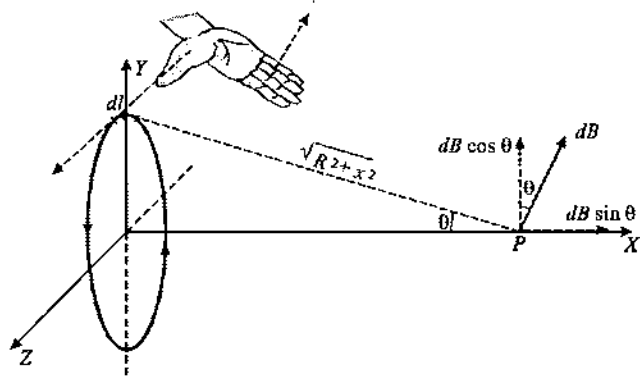


Figure 4.14

If we resolve the magnetic induction dB in mutually perpendicular directions along the axis and perpendicular to the axis as shown in figure then the component $dB \cos \theta$ which is normal to axis gets cancelled out due to the element on coil which is diametrically opposite to the element considered and the other component $dB \sin \theta$ which is along the axis will all get added up as due to all the elements on the coil these components are in same direction and the resultant magnetic induction at point P is in the direction as stated in figure-4.13.

The magnetic induction at point P due to the element considered can be given by Biot Savart's law as

$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin(90^\circ)}{(R^2 + x^2)^{3/2}}$$

$$\Rightarrow dB = \frac{\mu_0}{4\pi} \frac{Idl}{(R^2 + x^2)^{3/2}} \quad \dots (4.26)$$

The magnetic induction at P due to the coil is given by integrating the above expression for the whole length of the circumference of the coil. Thus it is given as

$$\Rightarrow B = \int dB \sin \theta$$

$$\Rightarrow B = \int_0^{2\pi R} \frac{\mu_0}{4\pi} \frac{Idl}{(R^2 + x^2)^{3/2}} \cdot \frac{R}{\sqrt{R^2 + x^2}}$$

$$\Rightarrow B = \frac{\mu_0 IR}{4\pi(R^2 + x^2)^{3/2}} \int_0^{2\pi R} dl$$

$$\Rightarrow B = \frac{\mu_0 IR}{4\pi(R^2 + x^2)^{3/2}} [2\pi R - 0]$$

$$\Rightarrow B = \frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}} \quad \dots (4.27)$$

Above expression given in equation-(4.27) will be used as a standard result for many advance cases of determining magnetic induction. This result can be modified if coil has N turns and it can be given as

$$B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}} \quad \dots (4.28)$$

4.2.6 Helmholtz Coils

A setup of two identical circular coaxial coils separated by a distance equal to their radius when carries current in same direction then in the region between the two coils near their common axis magnetic induction is found to be uniform. Such a system of two coils is called '*Helmholtz Coils*'. Figure-4.15(a) shows the setup of Helmholtz coils and the variation graph of magnetic field with distance of the two coils on their common axis.

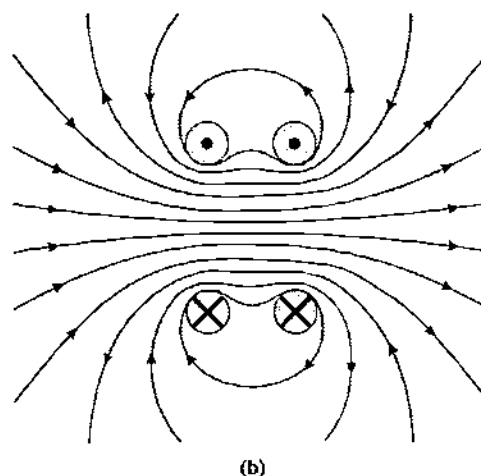
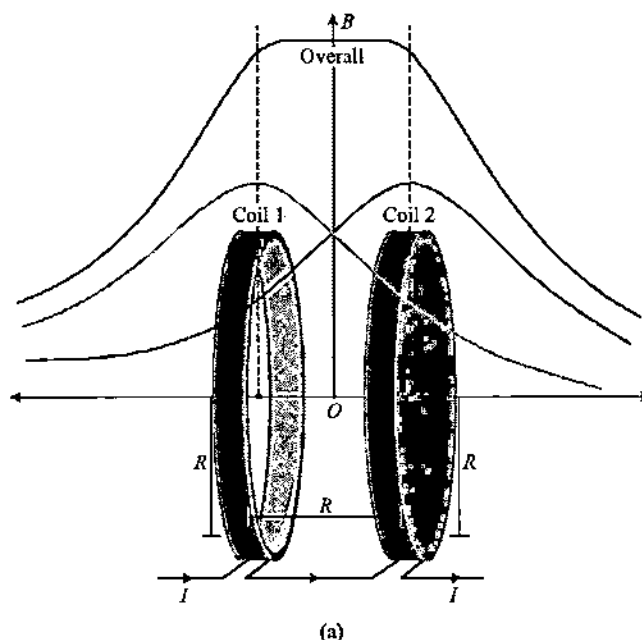


Figure 4.15

In above graphs it is observed that the drop in magnetic

induction due to one coil on either side from $\pm R/2$ position (mid point) is equal to the increase in magnetic induction due to other coil because of which the overall magnetic induction between the two coils remains almost uniform. The magnitude of this magnetic induction can be calculated by using equation-(4.28) at $x = R/2$ due to both the coils.

$$\begin{aligned}
 B_{\text{Helmholtz}} &= 2 B_{x=R/2} \\
 \Rightarrow B_{\text{Helmholtz}} &= 2 \left(\frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}} \right)_{x=R/2} \\
 \Rightarrow B_{\text{Helmholtz}} &= \frac{\mu_0 N I R^2}{(R^2 + (R/2)^2)^{3/2}} \\
 \Rightarrow B_{\text{Helmholtz}} &= \left(\frac{8}{5\sqrt{5}} \right) \frac{\mu_0 N I}{R} \quad \dots (4.29)
 \end{aligned}$$

Above magnitude of magnetic induction as given in equation-(4.29) is approximately constant in the region between the two coils as shown in figure-4.15(b). This setup of Helmholtz coils is used to setup uniform magnetic field in lab frame for various experiments and experimental demonstrations.

4.2.7 Magnetic Induction at Center of a Circular Arc

Figure-4.16 shows a circular arc of radius R carrying a current I . Due to all the current elements on this arc magnetic induction at its center O are in same inward direction which is given by right hand thumb rule as shown. For the angle $\theta = 2\pi$ it will be a circular coil and for angle θ subtended by arc at its center the magnetic induction can be directly given by using the expression of magnetic induction due to a circular coil, given as

$$\begin{aligned}
 B &= \frac{\mu_0 I}{2R} \times \frac{\theta}{2\pi} \\
 \Rightarrow B &= \frac{\mu_0 I \theta}{4\pi R} \quad \dots (4.30)
 \end{aligned}$$

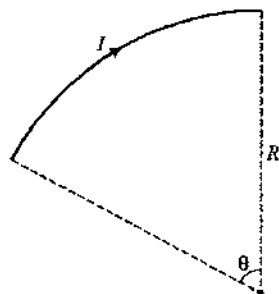


Figure 4.16

Using above expression given in equation-(4.30) we can calculate the magnetic induction due to a semicircular current carrying wire which is given by substituting $\theta = 2\pi$ as

$$B = \frac{\mu_0 I}{4R} \quad \dots (4.31)$$

Above expression in equation-(4.31) is half of the magnetic induction due to a circular coil as given in equation-(4.24) because due to all elements of a circular coil magnetic induction is in same direction so making the coil half reduced the magnetic induction also to half. Similarly we can state that due to a quarter circular arc carrying a current i magnetic induction at its center will be one fourth of that of a circular coil given as

$$B = \frac{\mu_0 I}{8R} \quad \dots (4.32)$$

Illustrative Example 4.1

A current $i = 1\text{A}$ circulates in a round thin wire loop of radius $r = 100\text{mm}$. Find the magnetic induction

- At the centre of the loop
- At a point lying on the axis of the loop at a distance $x = 100\text{mm}$ from its centre.

Solution

- At the center of a circular loop magnetic induction is given as

$$\begin{aligned}
 B &= \frac{\mu_0 i}{2r} \\
 \Rightarrow B &= \frac{4\pi \times 10^{-7} \times 1.0}{2 \times 0.1} \\
 \Rightarrow B &= 6.28 \times 10^{-6} \text{ Wb/m}^2 \approx 6.3 \mu\text{T}
 \end{aligned}$$

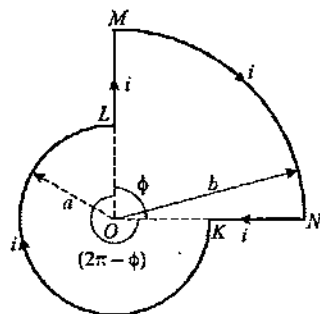
- As studied at a point on the axis of the loop magnetic induction is given as

$$\begin{aligned}
 B &= \frac{\mu_0}{4\pi} \times \frac{ir^2}{(x^2 + r^2)^{3/2}} \\
 \Rightarrow B &= \frac{4\pi \times 10^{-7} \times 1.0 \times (0.1)^2}{2 \times (0.02)^{3/2}} \\
 \Rightarrow B &= 2.22 \times 10^{-6} \text{ T} \\
 \Rightarrow B &= 2.22 \mu\text{T}
 \end{aligned}$$

Illustrative Example 4.2

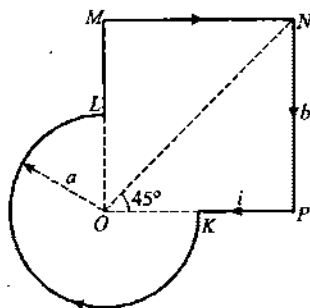
Find the magnetic induction at the point O due to the loop current i in the two cases given below. The shape of the loops are illustrated as

(a) In figure-4.17(a), the radii a and b , as well as the angle ϕ are known.



(a)
Figure 4.17

(b) In figure-4.17(b), the radius a and the side b are known.



(b)
Figure 4.17

Solution

(a) Magnetic field at O due to segment KL which is a circular arc is given as

$$B_1 = \frac{\mu_0 i}{4\pi a} \times (2\pi - \phi)$$

Magnetic field at O due to part LM and KN is zero because the point O is located on the line of current

Magnetic field at O due to arc MN which subtend an angle ϕ is given by

$$B_2 = \frac{\mu_0 i}{4\pi b} \times \phi$$

Total magnetic induction at O is given as

$$B = B_1 + B_2 = \frac{\mu_0 i}{4\pi} \times \left[\frac{2\pi - \phi}{a} + \frac{\phi}{b} \right]$$

(b) Field at O due to arc KL is given as

$$B_1 = \frac{\mu_0 i}{4\pi a} \times \frac{3\pi}{2}$$

Field at O due to parts LM and PK is zero because the point O is located on the line of current.

Field at O due to part MN is given as

$$B_2 = \frac{\mu_0 i}{4\pi b} \times \sin 45^\circ = \frac{\mu_0 i}{4\pi b} \times \frac{1}{\sqrt{2}}$$

Field at O due to part NP is given as

$$B_3 = \frac{\mu_0 i}{4\pi b} \times \sin 45^\circ = \frac{\mu_0 i}{4\pi b} \times \frac{1}{\sqrt{2}}$$

Total magnetic induction at O is given as

$$B = B_1 + B_2 + B_3$$

$$\Rightarrow B = \frac{\mu_0 i}{4\pi} \left[\frac{3\pi}{2a} + \frac{1}{\sqrt{2}b} + \frac{1}{\sqrt{2}b} \right]$$

$$\Rightarrow B = \frac{\mu_0 i}{4\pi} \left[\frac{3\pi}{2a} + \frac{\sqrt{2}}{b} \right]$$

Illustrative Example 4.3

Figure-4.18 shows a current carrying wire bent at an angle α . Find magnetic induction at a point P located on the angle bisector at a distance x from the point O at the bend.

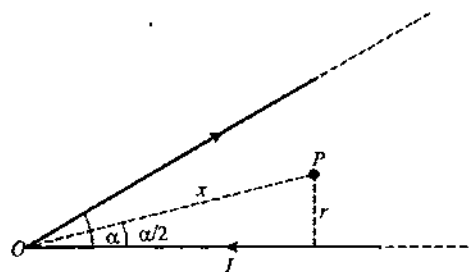


Figure 4.18

Solution

As shown in figure-4.19 below the angle subtended by point P at point O is $\alpha/2$ and at the other point at infinity is 0° so we can use the expression of magnetic induction due to a finite wire by considering these angles to find the magnetic induction at point P due to the two wires which is given as

$$B_P = 2B_{\text{one wire at } P}$$

$$\Rightarrow B_P = \left[\frac{\mu_0 I}{4\pi x \sin \alpha/2} [\cos \alpha/2 + \cos 0] \right]$$

$$\Rightarrow B_P = \frac{\mu_0 I}{4\pi x \sin \alpha/2} [1 + \cos \alpha/2]$$

$$\Rightarrow B_P = \frac{\mu_0 I}{2\pi x \sin \alpha/2} [2 \cos \alpha/2]$$

$$\Rightarrow B_P = \frac{\mu_0 I}{4\pi x \sin \alpha/2 \cos \alpha/4} \times 2 \cos^2 \alpha/4$$

$$\Rightarrow B_P = \frac{\mu_0 I}{2\pi x} \cot \frac{\alpha}{4}$$

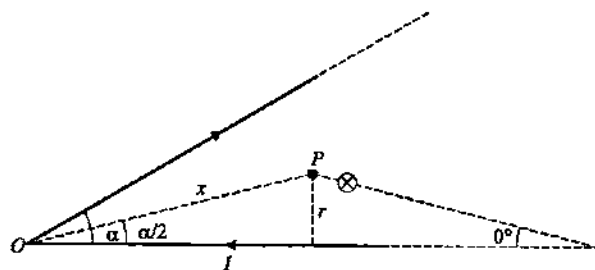


Figure 4.19

Illustrative Example 4.4

A current i flows along a thin wire shaped as a regular polygon with n sides which can be inscribed into a circle of radius R . Find the magnetic induction at the centre of the polygon. Analyse the obtained expression at $n \rightarrow \infty$.

Solution

The situation described in question is shown in figure-4.20.

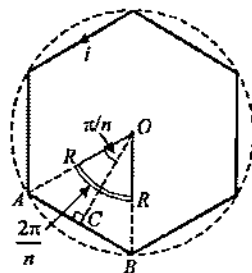


Figure 4.20

In figure, OC is the perpendicular distance of one segment of polygon from the centre. Here $\angle AOB = (2\pi/n)$ as there are n elements like AB thus distance OC is given as

$$OC = R \cos (\pi/n)$$

The magnetic induction at O due to a straight current carrying element AB is given as

$$B_1 = \frac{\mu_0 i}{4\pi} \times \frac{1}{R \cos (\pi/n)} \left[\sin \left(\frac{\pi}{n} \right) + \sin \left(\frac{\pi}{n} \right) \right]$$

$$\Rightarrow B_1 = \frac{\mu_0 i}{4\pi} \times \frac{1}{R \cos (\pi/n)} \times 2 \sin \left(\frac{\pi}{n} \right) \quad \dots (4.33)$$

As there are n sides in the polygon, the total magnetic induction due to polygon is given as

$$B = \frac{\mu_0 n i}{2\pi R} \tan \left(\frac{\pi}{n} \right) \quad \dots (4.34)$$

When $n \rightarrow \infty$, $\tan (\pi/n) \approx \pi/n$ which gives

$$B = \frac{\mu_0 n i}{2\pi R} \left(\frac{\pi}{n} \right) = \frac{\mu_0 i}{2R} \quad \dots (4.35)$$

Above expression in equation-(4.35) is a result of magnetic induction due to a circular coil as a polygon with infinite sides transforms into a circular coil.

Illustrative Example 4.5

A pair of stationary and infinitely long bent wires are placed in the X - Y plane as shown in figure-4.21. The wires carry currents of $i = 10\text{A}$ each as shown in figure. The segments L and M are along the X -axis and the segments P and Q are parallel to the Y -axis such that $OS = OR = 0.2\text{m}$. Find the magnitude and direction of the magnetic induction at the origin O of the coordinate system as shown in figure.

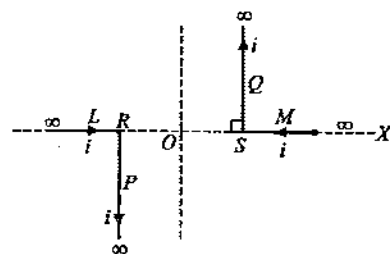


Figure 4.21

Solution

We consider the magnetic induction $\vec{B}$ at O due to each segment separately. Magnetic induction at O due segment L and M is zero because current point O is located on the line of these currents.

Magnetic induction at O due to segments P and Q are in same direction which can be seen by using right hand thumb rule in outward direction and equal in magnitude due to symmetry which is say B_1 can be calculated by using the result of magnetic induction due to a semi infinite wire given as

$$B_1 = \left(\frac{\mu_0 i}{4\pi r} \right)$$

$$\Rightarrow B_1 = \left(\frac{4\pi \times 10^{-4}}{4\pi} \times \frac{10}{10.02} \right) \text{ Wb/m}^2$$

$$\Rightarrow B_1 = 0.5 \times 10^{-4} \text{ Wb/m}^2$$

As magnetic induction due to segments P and Q are equal, net magnetic induction at O is given as

$$B_O = 2B_1$$

$$\Rightarrow B_O = 10^{-4} \text{ Wb/m}^2$$

As discussed that the direction of B_O is perpendicular to X - Y plane and directed outward from the plane of paper.

Illustrative Example 4.6

Figure-4.22 shows two long wires A and B , each carrying a current I , separated by a distance l and oriented in a plane perpendicular to the plane of paper. The directions of currents are shown in figure. Find the magnetic induction at a point P located at a distance l from both wires as shown in figure-4.22.

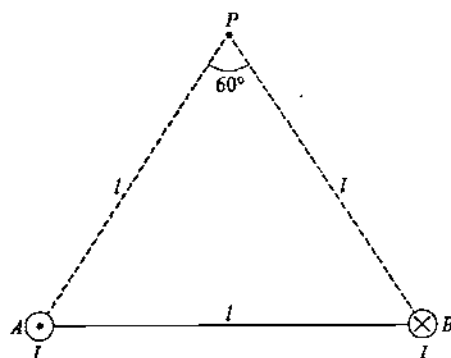


Figure 4.22

Solution

Magnetic induction at point P due to both wires is given as

$$B = \frac{\mu_0 I}{2\pi l}$$

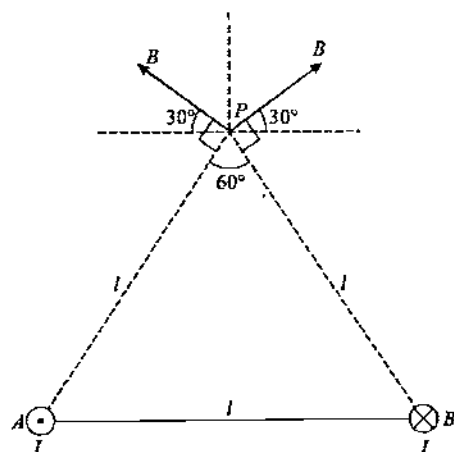


Figure 4.23

The directions of magnetic induction due to individual wires is perpendicular to the line joining the wire to point P . From above figure at point P net magnetic induction is given as

$$B_P = 2B \sin 30^\circ = 2 \times \frac{\mu_0 I}{2\pi l} \times \frac{1}{2}$$

$$\Rightarrow B_P = \frac{\mu_0 I}{2\pi l}$$

Illustrative Example 4.7

In figure-4.24 a current carrying wire configuration is shown with current I . Find the vector of magnetic induction at origin of co-ordinate system O .

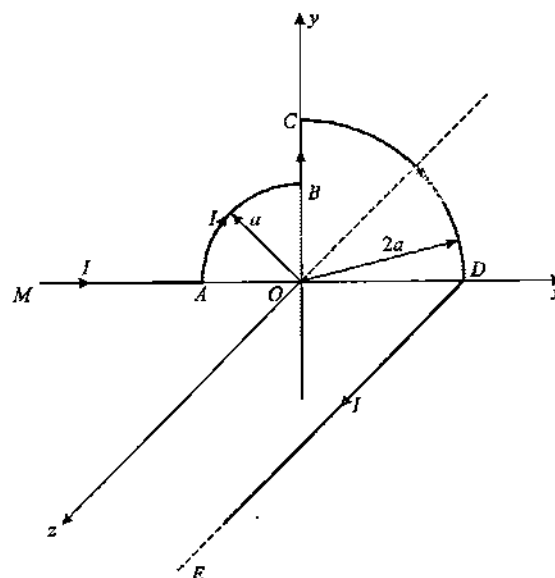


Figure 4.24

Solution

The magnetic induction at origin can be calculated by vector sum of magnetic inductions due to separate segments of this wire configuration which is given as

$$\vec{B}_0 = \vec{B}_{AB} + \vec{B}_{CD} + \vec{B}_{DE}$$

In above vector sum we have not considered the wire segments MA and BC as origin lies along the line of currents so

$$\vec{B}_0 = \frac{\mu_0 I}{8a} (-\hat{k}) + \frac{\mu_0 I}{16a} (-\hat{k}) + \frac{\mu_0 I}{8\pi a} (-\hat{j})$$

$$\Rightarrow \vec{B}_0 = -\frac{\mu_0 I}{8a} \left[\frac{3}{2} \hat{k} + \frac{1}{\pi} \hat{j} \right]$$

Illustrative Example 4.8

Figure shows a circular wire loop in which a current I enters at point A and leaves from point B through straight wires. Find magnetic induction at centre O .

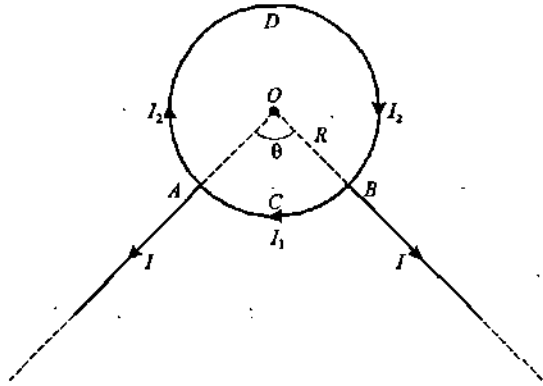


Figure 4.25

Solution

If we consider current is divided as I_1 and I_2 in the two parts of the circular loop which will be divided in inverse ratio of resistance or length of the arcs of the loop and these currents produces magnetic inductions B_1 and B_2 at center O respectively then net magnetic induction at O is given as

$$B_0 = B_1 - B_2 \quad \dots (4.36)$$

$$\Rightarrow B_0 = \frac{\mu_0 I_1 \theta}{4\pi R} - \frac{\mu_0 I_2 (2\pi - \theta)}{4\pi R}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{2\pi - \theta}{\theta}$$

$$\Rightarrow I_1 = I \left(\frac{2\pi - \theta}{\theta} \right)$$

$$\Rightarrow I_2 = I \left(\frac{\theta}{2\pi} \right)$$

Substituting above current values in equation-(4.36) we have

$$B_0 = B_1 - B_2 = 0$$

Illustrative Example 4.9

A current i flows in a long straight wire with cross-section having the form of a thin half-ring of radius R as shown in figure-4.26. Find the induction of magnetic field at the point O .

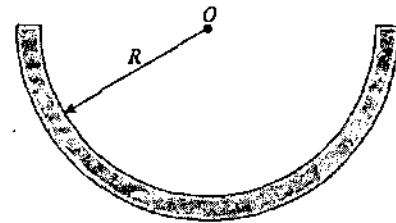


Figure 4.26

Solution

Consider elemental wire of width dx in the cross section as shown in figure-4.27

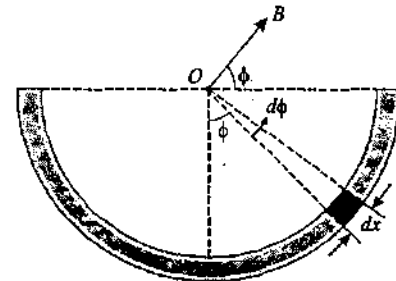


Figure 4.27

Current in the elemental wire is given as

$$di = \frac{idx}{\pi R} = \frac{iRd\phi}{\pi R} = \frac{id\phi}{\pi}$$

The magnetic induction due to a long wire can be used to calculate the magnetic induction due to this elemental wire at point O which is given as

$$dB = \frac{\mu_0 di}{2\pi R} = \frac{\mu_0 id\phi}{2\pi^2 R}$$

All the components $dB \sin \phi$ will be cancelled out due to the corresponding elemental wires on the other half of the cross section as these are in opposite directions and the components $dB \cos \phi$ due to all the elemental wires are in same directions so will be added up. Thus net magnetic induction due to the total current in the complete wire with the cross section shown is given by integrating $dB \cos \phi$ which is given as

$$B_0 = \int dB \cos \phi$$

$$\Rightarrow B_0 = 2 \int_0^{\pi/2} \left(\frac{\mu_0 id\phi}{2\pi^2 R} \right) \cos \phi$$

$$\Rightarrow B_0 = \frac{\mu_0 i}{\pi^2 R} [\sin \phi]_0^{\pi/2}$$

$$\Rightarrow B_0 = \frac{\mu_0 i}{\pi^2 R} [1 - (0)]$$

$$\Rightarrow B_0 = \frac{\mu_0 i}{\pi^2 R}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Magnetic Effects of Current

Module Number - 1 to 20

Practice Exercise 4.1

(i) Find the magnetic induction at the centre of a rectangular wire frame whose diagonal length is equal to $d = 16\text{cm}$ and the angle between the diagonals is equal to $\phi = 30^\circ$, the current flowing in the frame equals $i = 5.0\text{A}$.

[0.1 mT]

(ii) Find the magnetic induction at the point O due to a current i flowing in a current carrying wire which has the shape as shown in figures-4.28 (a, b, c). The radius of the curved part of the wire is R and the linear parts are assumed to be very long.

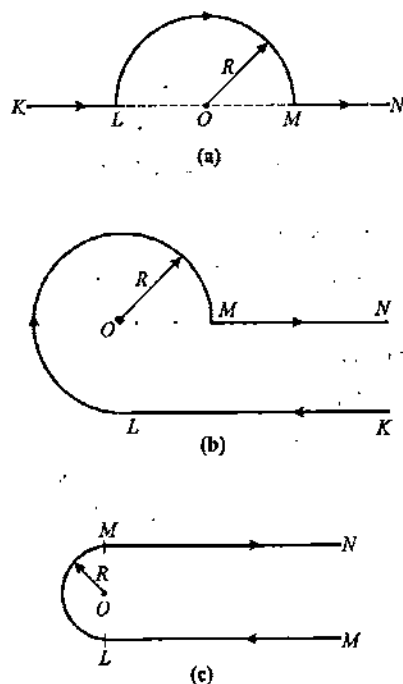


Figure 4.28

$$[(a) \frac{\mu_0 i}{4R}; (b) \frac{\mu_0 i}{4\pi R} \left[1 + \frac{3\pi}{2} \right]; (c) \frac{\mu_0 i}{4\pi R} (2 + \pi)]$$

(iii) Find the magnetic induction at the point O if the wire carrying a current I has a shape as shown in figure-4.29 (a, b, c). The radius of the curved part of the wire is R and consider the linear parts of the wire are very long.

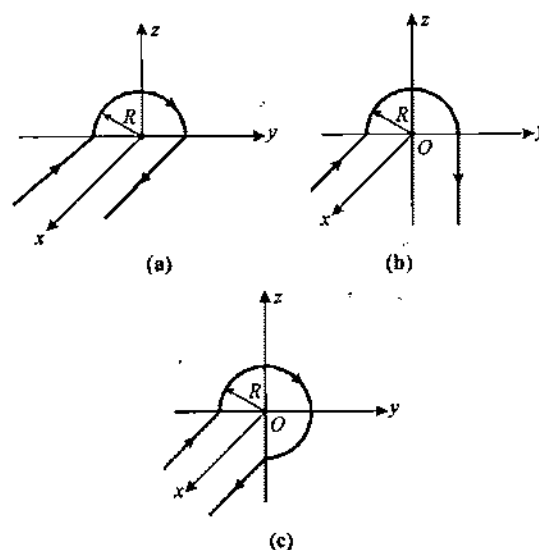


Figure 4.29

$$\left[\frac{\mu_0 I}{4\pi R} \sqrt{\pi^2 + 4}, \frac{\mu_0 I}{4\pi R} \sqrt{1 + (\pi + 1)^2}, \frac{\mu_0 I}{4\pi 2\pi} \sqrt{9\pi^2 + 8} \right]$$

(iv) In the figure-4.30 shown two circular arcs are joined to make a closed loop carrying current I . Find the magnetic induction at the common centre O .

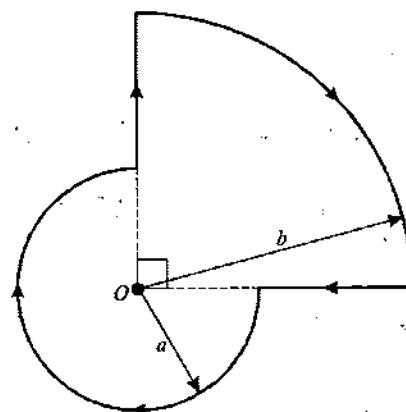


Figure 4.30

$$\left[\frac{\mu_0 I}{8} \left(\frac{3}{a} + \frac{1}{b} \right) \right]$$

(v) Find the magnetic induction at point P due to a current carrying wire AB as shown in figure-4.31

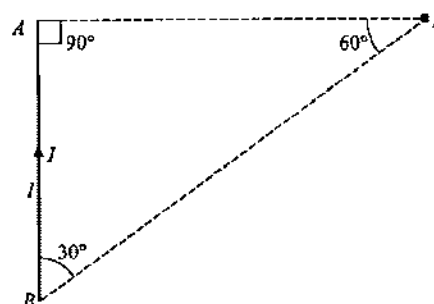


Figure 4.31

$$\left[\frac{\sqrt{3}\mu_0 I}{8\pi l} \right]$$

(vi) Find magnetic induction at point O in the figure-4.32 shown due to the current carrying loop with current I as shown.

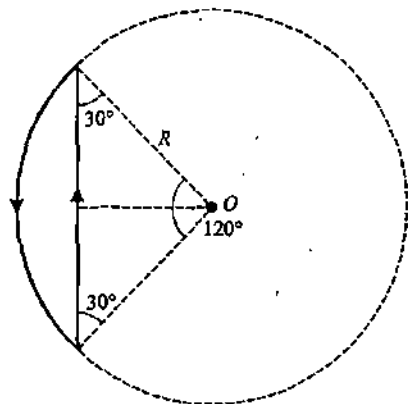


Figure 4.32

$$\left[\frac{\mu_0 I}{2R} \left[\frac{\sqrt{3}}{\pi} - \frac{1}{3} \right] \right]$$

(vii) Find the magnetic induction vector at origin O due to the current carrying wire configuration as shown in figure-4.33.

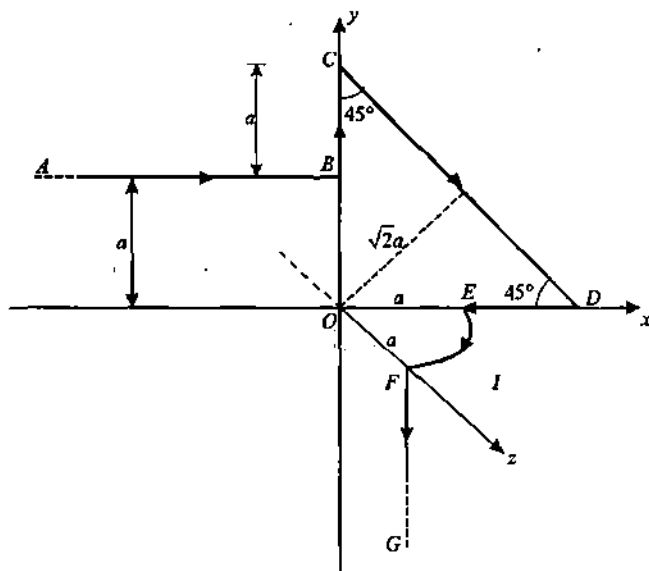


Figure 4.33

$$\left[\frac{\mu_0 I}{4\pi a} \left[\hat{i} - \frac{\pi}{2} \hat{j} - 2\hat{k} \right] \right]$$

(viii) Figure-4.34 shows a spiral of inner radius a and outer radius b with total N turns wound in it. If a current i flows in it, calculate the magnetic induction at the centre of spiral.

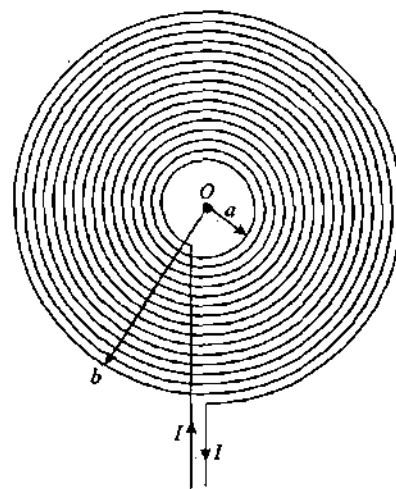


Figure 4.34

$$\left[\frac{\mu_0 IN}{2(b-a)} \ln \left(\frac{b}{a} \right) \right]$$

(ix) Figure shows a long and thin strip of width b which carries a current I . Find magnetic induction due to the current in strip at point P located at a distance r from the strip in the plane of strip as shown in figure-4.35.

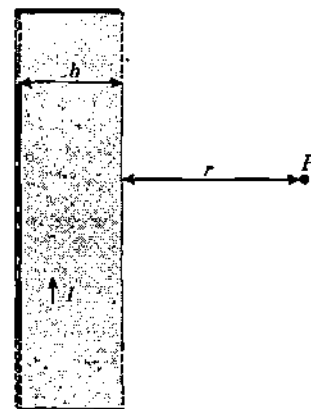


Figure 4.35

$$\left[\frac{\mu_0 I}{2\pi b} \ln \left(\frac{r+b}{r} \right) \right]$$

(x) Two long parallel wires carrying currents 2.5A and 4A in the same direction directed into the plane of the paper are held at points P and Q as shown in figure-4.36 such that they are perpendicular to the plane of paper. The points P and Q are located at a distance of 5m and 2m respectively from a collinear point R as shown.

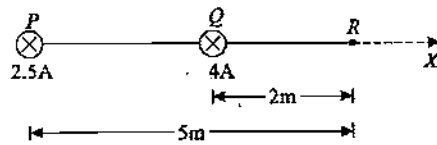


Figure 4.36

Find all the positions at which a third long parallel wire carrying a current of magnitude 2.5A may be placed so that the magnetic induction at R is zero.

[± 1 m from point R along x-axis]

4.3 Magnetic Induction due to Extended Current Configurations

In previous articles we've studied the magnetic induction due to current carrying long and finite straight wires, circular coil and arcs. In this section we will study the magnetic induction due to some current configurations in which the results obtained in previous articles will be used as building blocks. Such current configurations are called extended current configurations.

4.3.1 Magnetic Induction Inside a Long Solenoid

Figure-4.37 shows a long tightly wound solenoid with core radius R which carries a current I . The number of turns per unit length of solenoid are n . We consider tightly wound solenoid because in this case we can consider each turn of wire wound on solenoid core as a circular coil otherwise if it takes a helical shape then we cannot use the expression of magnetic induction due to a circular coil.

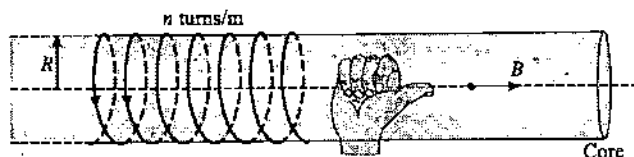


Figure 4.37

To determine the magnetic induction at the axial point of the solenoid we consider an elemental ring of width dx on the solenoid core at a distance x from an axial point O as shown in figure-4.38. Number of turns in this elemental ring will be ndx and due to this elemental ring, magnetic induction at point O is given by using equation-(4.28) as

$$dB = \frac{\mu_0 (ndx) IR^2}{2(R^2 + x^2)^{3/2}} \quad \dots (4.37)$$

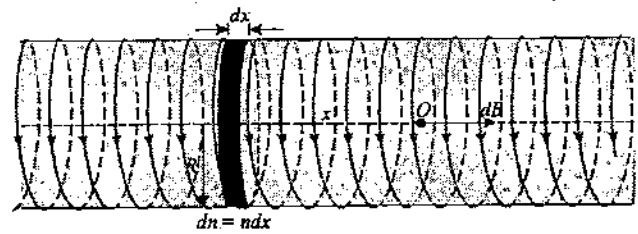


Figure 4.38

Net magnetic induction at point O due to whole solenoid can be calculated by integrating above expression within limits from $-\infty$ to $+\infty$ which is given as

$$B = \int dB = \int_{-\infty}^{+\infty} \frac{\mu_0 (ndx) IR^2}{2(R^2 + x^2)^{3/2}}$$

$\Rightarrow$

$$B = \frac{\mu_0 n I R^2}{2} \int_{-\infty}^{+\infty} \frac{dx}{(R^2 + x^2)^{3/2}}$$

To integrate the above expression we substitute

$$x = R \tan \theta$$

$$dx = R \sec^2 \theta d\theta$$

With the above substitution limits of integration also changes as

$$\text{at } x = -\infty \rightarrow \theta = -\frac{\pi}{2}$$

and

$$\text{at } x = +\infty \rightarrow \theta = +\frac{\pi}{2}$$

$\Rightarrow$

$$B = \frac{\mu_0 n I R^2}{2} \int_{-\pi/2}^{+\pi/2} \frac{R \sec^2 \theta d\theta}{(R^2 + R^2 \tan^2 \theta)^{3/2}}$$

$\Rightarrow$

$$B = \frac{\mu_0 n I R^2}{2} \int_{-\pi/2}^{+\pi/2} \frac{R \sec^2 \theta d\theta}{R^3 \sec^3 \theta}$$

$\Rightarrow$

$$B = \frac{\mu_0 n I}{2} \int_{-\pi/2}^{+\pi/2} \cos \theta d\theta$$

$\Rightarrow$

$$B = \frac{\mu_0 n I}{2} [\sin \theta]_{-\pi/2}^{+\pi/2}$$

$\Rightarrow$

$$B = \frac{1}{2} \mu_0 n I [1 - (-1)]$$

$\Rightarrow$

$$B = \mu_0 n I \quad \dots (4.38)$$

Above expression given in equation-(4.38) is valid only for the case when the solenoid is very long and tightly wound on its core. If there is a gap between turns of solenoid then these turns cannot be considered as circular coils-using which above result is derived.

4.3.2 Direction of Magnetic Induction Inside a Long Solenoid

In upcoming articles we will discuss if in a region at every point magnetic induction is in same direction then it must be uniform throughout any cross section in that region. This will be discussed in details under article-4.4.9 but students must keep this fact in mind for using this in different applications. As at every point on the axis of a long solenoid magnetic induction is given by equation-(4.38) which is a constant thus we can conclude that throughout the volume of solenoid at every cross section magnetic induction will remain uniform as shown in figure-4.39. It is also observed that for closed pack or tightly wound solenoid outside magnetic induction is zero as all magnetic lines form closed loop from negative infinity to positive infinity along the length of solenoid and at every exterior point of solenoid magnetic induction due to different parts on diametrically opposite sides of solenoid cancelled out.

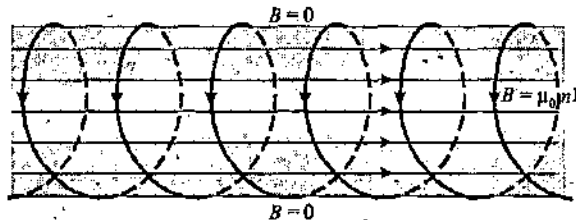


Figure 4.39

Figure-4.40 shows a loosely wound solenoid on a core. As the wires in turns are not closely packed magnetic induction due to each wire turn will be in form of concentric circles close to the wire and at some distance inside and outside the resulting magnetic induction is shown by the magnetic lines in this figure. As wire turns are not closely packed outside magnetic field exist in opposite direction. If turns are very close or almost touching each other the magnetic induction between the wire turns will be zero as due to the two close turns the magnetic field cancels because these are in opposite direction (See point A in figure).

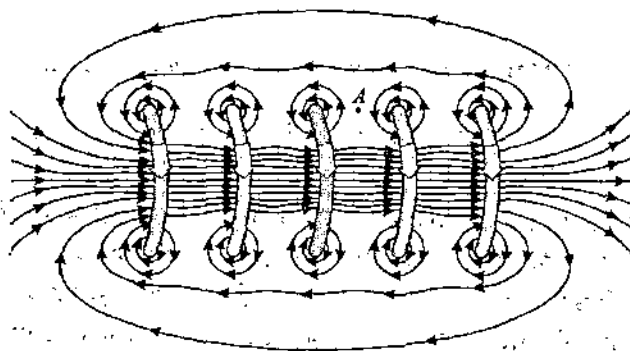


Figure 4.40

4.3.3 Magnetic Induction due to a Semi-Infinite Solenoid

Figure-4.41 shows a semi-infinite solenoid carrying a current I , core radius R and number of turns per unit length n . The magnetic induction at the end of solenoid at its axial point O can be directly given as half of that due to the long solenoid at any of its axial point. This is because if we consider another semi-infinite solenoid adjoining to this one to complete the infinite solenoid then at point O the magnetic induction due to two halves of the long solenoid will be equal which is given by equation-(4.38). Thus the magnetic induction at the point O on the axis at terminal cross section of this semi-infinite solenoid is given as

$$B = \frac{1}{2} \mu_0 n I \quad \dots (4.39)$$

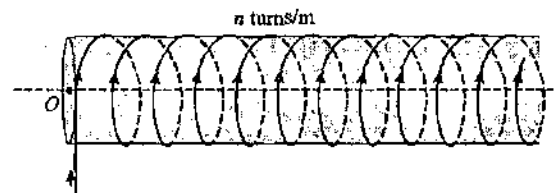


Figure 4.41

Students can also derive the above expression given in equation-(4.39) by integrating the result due to an elemental circular coil as considered in article-4.3.1 within limits from 0 to infinity for the magnetic induction at point O due to this semi-infinite solenoid.

4.3.4 Magnetic Induction due to a finite length Solenoid

Figure-4.42 shows a finite length tightly wound solenoid of core radius R and current I with number of turns per unit length n . In this case we will determine the magnetic induction at a point P at axis of solenoid where the terminal circumference of solenoid cross sections subtend angles θ_1 and θ_2 as shown in figure.

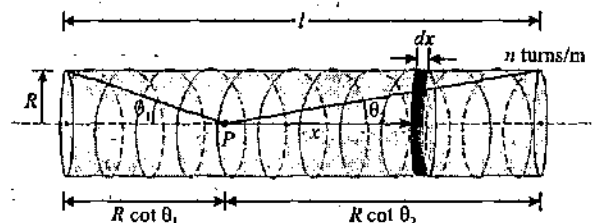


Figure 4.42

To find the magnetic induction at point P we consider an elemental coil of width dx at a distance x from P . Number of turns in this elemental coil are $n dx$. Magnetic induction due to the elemental coil dB at point P is given by using equation-(4.28) as

$$dB = \frac{\mu_0 (n dx) IR^2}{2(R^2 + x^2)^{3/2}} \quad \dots (4.40)$$

Net magnetic induction at point P due to whole solenoid can be calculated by integrating above expression within limits from $-R \cot \theta_1$ to $+R \cot \theta_2$ which is given as

$$B = \int dB = \int_{-R \cot \theta_1}^{+R \cot \theta_2} \frac{\mu_0 (n dx) IR^2}{2(R^2 + x^2)^{3/2}}$$

$$\Rightarrow B = \frac{\mu_0 n I R^2}{2} \int_{-R \cot \theta_1}^{+R \cot \theta_2} \frac{dx}{(R^2 + x^2)^{3/2}}$$

To integrate the above expression we substitute

$$x = R \cot \theta$$

$$dx = -R \csc^2 \theta d\theta$$

With the above substitution, limits of integration also changes as

$$\text{at } x = -R \cot \theta_1 \rightarrow \theta = -\theta_1$$

and

$$\text{at } x = +R \cot \theta_2 \rightarrow \theta = +\theta_2$$

$$\Rightarrow B = \frac{\mu_0 n I R^2}{2} \int_{-\theta_1}^{+\theta_2} \frac{R \csc^2 \theta d\theta}{(R^2 + R^2 \cot^2 \theta)^{3/2}}$$

$$\Rightarrow B = -\frac{\mu_0 n I R^2}{2} \int_{-\theta_1}^{+\theta_2} \frac{R \csc^2 \theta d\theta}{R^3 \csc^3 \theta}$$

$$\Rightarrow B = -\frac{\mu_0 n I}{2} \int_{-\theta_1}^{+\theta_2} \sin \theta d\theta$$

$$\Rightarrow B = \frac{\mu_0 n I}{2} [\cos \theta]_{-\theta_1}^{+\theta_2}$$

$$\Rightarrow B = \frac{1}{2} \mu_0 n I [\cos \theta_2 - (-\cos \theta_1)]$$

$$\Rightarrow B = \frac{1}{2} \mu_0 n I [\cos \theta_1 + \cos \theta_2] \quad \dots (4.41)$$

Using above expression we can also calculate the magnetic induction due to the semi-infinite solenoid by substituting angles $\theta_1 = 90^\circ$ and $\theta_2 = 0^\circ$ which also verifies the expression obtained qualitatively as equation-(4.39) in article-4.3.3.

4.3.5 Magnetic Induction due to a Large Toroid

Figure-4.43 shows a large toroid with its radius R to be very large compared to its cross sectional radius r ($R \gg r$). Due to

large radius turns can be tightly wound on the toroidal core (also called as tore) of toroid. If there are total N turns of wire wound on the toroid then number of turns per unit length can be given as

$$n = \frac{N}{2\pi R} \quad \dots (4.42)$$

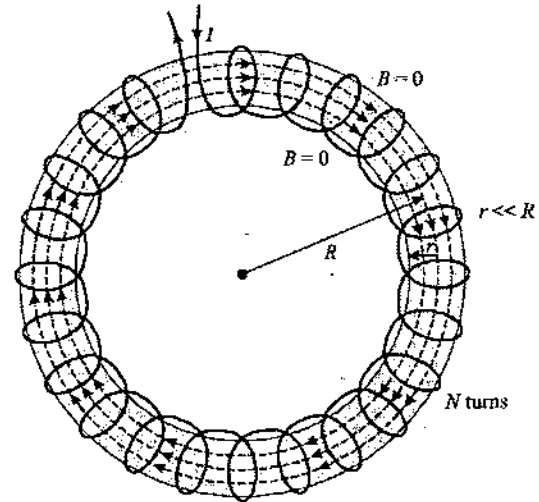


Figure 4.43

This toroid can also be imagined by considering a long solenoid is bent in circular shape so that its both far ends are joined together to close the loop for inside magnetic lines. In case of tightly wound turns with $R \gg r$ the magnetic induction inside the cross sectional region of toroid can be given by equation-(4.38) as

$$B = \mu_0 \left(\frac{N}{2\pi R} \right) I$$

$$\Rightarrow B = \frac{\mu_0 N I}{2\pi R} \quad \dots (4.43)$$

If the cross sectional radius of toroid is not very small compared to the average radius of the toroid then number of turns per unit length at inside and outside periphery of the toroid will be different due to which the magnetic induction inside the toroid cross section will also be different near to inside and outside edge as shown in figure-4.44 which approximately shows the magnetic lines density inside. As near to inner edge (at point 1 shown in figure-4.44) turns are closer so magnetic induction will be higher and at outer edge (at point 2 shown in figure-4.44) turns separation is large so magnetic induction will be lesser. In this case at outside points of toroid also magnetic induction will be non zero because of the same reason as discussed in article-4.3.2.

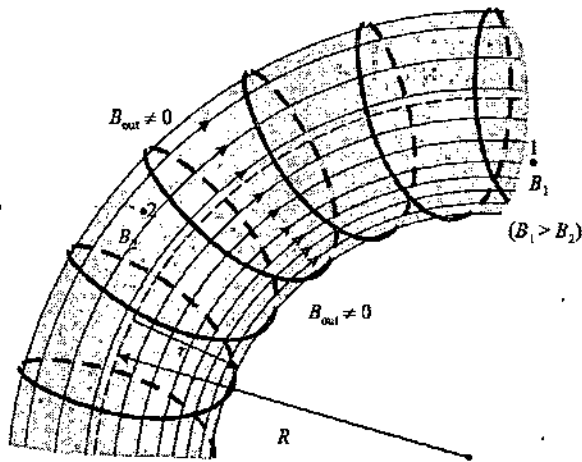


Figure 4.44

4.3.6 Magnetic Induction due to a Large Current Carrying Sheet

Figure-4.45 shows a large metal sheet carrying current throughout its cross section with linear current density i ampere per meter. The direction of magnetic induction due to this sheet above and below can be given by right hand palm rule as shown in figure-4.45(a). We can see that the direction of magnetic induction above and below the sheet are opposite and as the sheet is considered to be very large the direction of magnetic induction is considered uniform in the region. Figure-4.45(b) shows the cross sectional view of the sheet and the direction of magnetic induction due to the current in sheet.

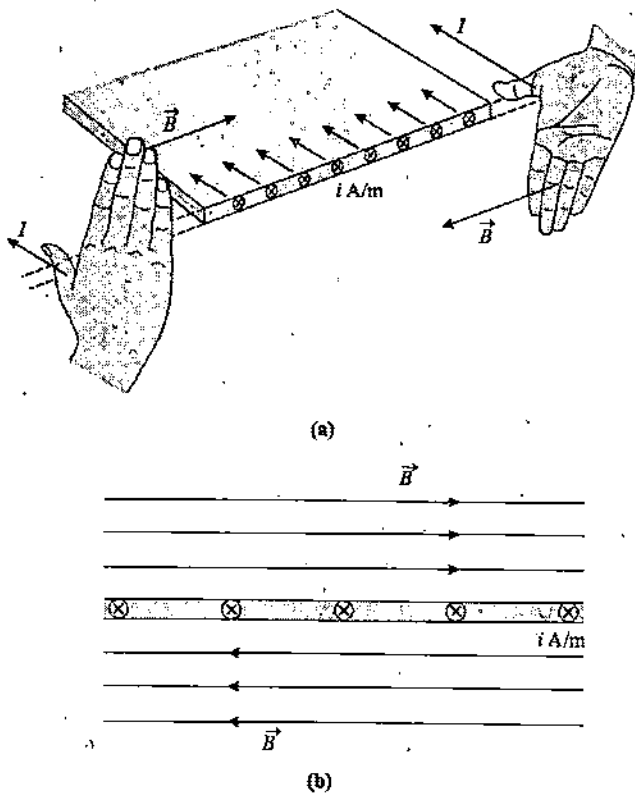


Figure 4.45

To determine the magnitude of magnetic induction at a point P located a distance r above the sheet we consider an elemental wire in the sheet of width dx at a distance x from the sheet as shown in figure-4.46. This figure shows the cross section of the sheet which carries a linear current density i A/m. The current in the elemental wire is given as

$$di = i dx$$

Magnetic induction dB due to the elemental wire on sheet at point P can be calculated by equation-(4.19) as discussed in article-4.2.1 which is given as

$$dB = \frac{\mu_0 di}{2\pi\sqrt{x^2 + r^2}}$$

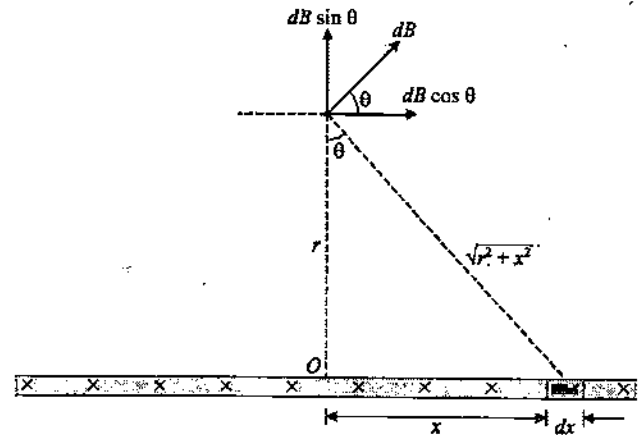


Figure 4.46

The direction of dB is shown in above figure which students can verify by placing their right hand thumb along the elemental wire pointing inward and fingers toward the point P so that right hand palm gives the direction of dB .

In figure-4.46 we can see that the component of dB normal to the sheet $dB \sin \theta$ will get cancelled out by the opposite direction component at P due to another elemental wire on sheet on the other side of point O at the same distance x . Other component of magnetic induction along the surface of sheet $dB \cos \theta$ will be added up due to all the elemental wires on the sheet thus net magnetic induction at point P due to whole sheet will be given by integrating $dB \cos \theta$ for the whole length of sheet from $-\infty$ to $+\infty$ given as

$$B = \int dB \cos \theta$$

$$\Rightarrow B = \int_{-\infty}^{+\infty} \frac{\mu_0 (i dx)}{2\pi\sqrt{x^2 + r^2}} \cdot \frac{r}{\sqrt{x^2 + r^2}}$$

$$\Rightarrow B = \frac{\mu_0 i r}{2\pi} \int_{-\infty}^{+\infty} \frac{dx}{(x^2 + r^2)}$$

$$\Rightarrow B = \frac{\mu_0 i r}{2\pi} \left[\frac{1}{r} \tan^{-1} \left(\frac{x}{r} \right) \right]_{-\infty}^{+\infty}$$

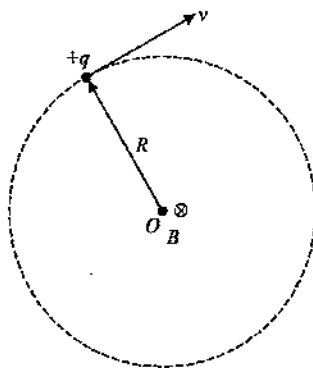
$$\Rightarrow B = \frac{\mu_0 i}{2\pi} \left[\left(\frac{\pi}{2} \right) - \left(-\frac{\pi}{2} \right) \right]$$

$$\Rightarrow B = \frac{1}{2} \mu_0 i \quad \dots (4.44)$$

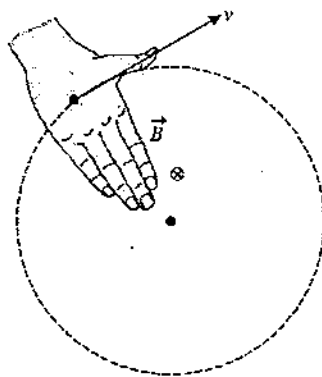
4.3.7 Magnetic Induction due to a Revolving Charge Particle

Figure-4.47(a) shows a point charge q is revolving in a circle of radius R with velocity v . If the charge is revolving at slow speed then the magnetic induction at its center due to motion of charge can be calculated by equation-(4.15) as discussed in article-4.1.2 which is given as

$$B = \frac{\mu_0 q v}{4\pi R^2} \quad \dots (4.45)$$



(a)



(b)

Figure 4.47

The direction of above magnetic induction is given by right hand thumb rule as shown in figure-4.47(b). If the charge is revolving at high speed then it can be considered as a continuously flowing current in coil. For a charge revolving at an angular speed ω then its equivalent current in the circular path is given as

$$\text{Current} = \text{Charge} \times \text{Frequency of revolution}$$

$$\Rightarrow I = q \times \frac{\omega}{2\pi} = \frac{q\omega}{2\pi} \quad \dots (4.46)$$

Thus magnetic induction at the center of this circle can be calculated by using equation-(4.24), given as

$$B = \frac{\mu_0 I}{2R}$$

$$B = \frac{\mu_0}{2R} \left(\frac{q\omega}{2\pi} \right)$$

$$B = \frac{\mu_0 q \omega}{4\pi R}$$

$$B = \frac{\mu_0 q v}{4\pi R^2} \quad [\text{As } \omega = \frac{v}{R}] \quad \dots (4.47)$$

Above equation-(4.47) is same as equation-(4.45). In case of magnetic induction at center of circle both of these results can be considered same but in case of a point on the axis of circle in first case due to revolution of charge particle the instantaneous direction of magnetic induction is shown in figure-4.48 which changes with the location of the particle on circular path in YZ plane as shown and its magnitude is given as

$$B = \frac{\mu_0 q v}{4\pi(R^2 + x^2)} \quad \dots (4.48)$$

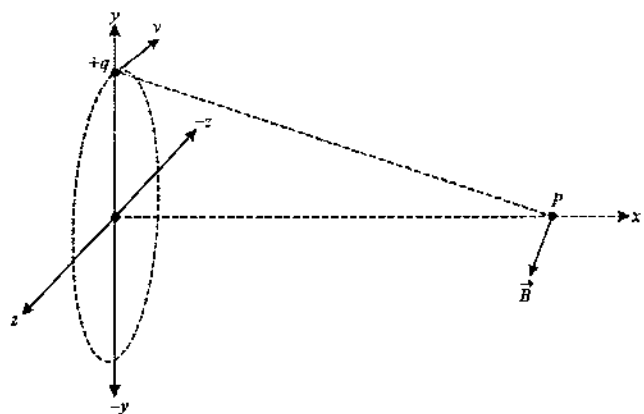


Figure 4.48

If we consider revolving charge as a continuous circulating current of magnitude given by equation-(4.46) then for an axial

point at a distance x from the center of circle the magnetic induction direction will be given by right hand thumb rule as shown in figure-4.49 and its magnitude can be calculated by using equation-(4.28) given as

$$B = \frac{\mu_0 R^2}{2(R^2 + x^2)^{3/2}} \left(\frac{q\omega}{2\pi} \right)$$

$$\Rightarrow B = \frac{\mu_0 q \omega R^2}{4\pi(R^2 + x^2)^{3/2}} \quad \dots (4.49)$$

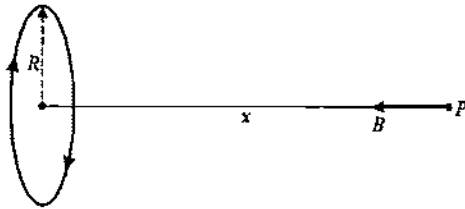


Figure 4.49

4.3.8 Magnetic Induction due to a Surface Charged Rotating Disc

Figure-4.50 shows a circular disc of radius R and charged with a surface charge density $\sigma \text{ C/m}^2$. The disc is rotating at a constant angular velocity ω and due to the rotating charges we will determine the magnetic induction at the center of the disc.

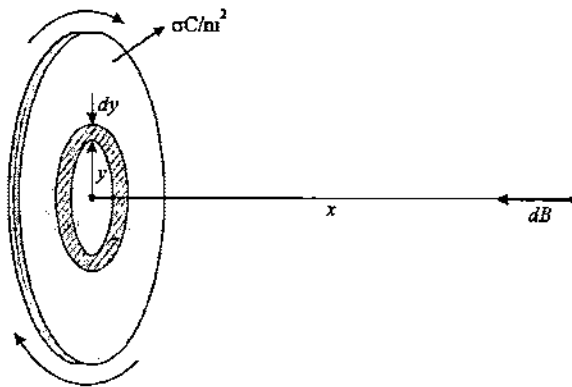


Figure 4.50

To calculate the magnetic induction due to rotating disc we consider an elemental circular ring of radius y and width dy on the disc surface as shown in figure. The charge dq on this elemental ring is given as

$$dq = \sigma \times 2\pi y \cdot dy$$

The current in elemental ring due to its rotation is given as

$$dI = \frac{dq\omega}{2\pi} = \sigma\omega y \cdot dy$$

The magnetic induction dB at the center of disc due to this elemental ring can be calculated by using the result of magnetic induction of a coil which is given as

$$dB = \frac{\mu_0 dI}{2y} = \frac{1}{2} \mu_0 \sigma \omega dy$$

Net magnetic induction due to the whole disc at its center is given by integrating above expression within limits from 0 to R for the complete surface of disc which is given as

$$B = \int dB = \int_0^R \frac{1}{2} \mu_0 \sigma \omega dy$$

$$\Rightarrow B = \frac{1}{2} \mu_0 \sigma \omega R \quad \dots (4.50)$$

Magnetic induction at the axis point of above rotating disc can be calculated by integrating the magnetic induction at axial point due to the above chosen elemental ring. Students can try on their own to get the result and verify it with the result given below.

$$B_{\text{axis}} = \frac{\mu_0 \sigma \omega R^4}{8(R^2 + x^2)^{3/2}} \quad \dots (4.51)$$

Illustrative Example 4.10

A very long straight semi-infinite solenoid has a cross-section radius R and n turns per unit length. A direct current I flows through the solenoid. Suppose that x is the distance from the one end of the solenoid, measured along its axis. Calculate

- The magnetic induction B on the axis of solenoid as a function of distance x .
- Draw an approximate plot of B v/s ratio x/R .
- The distance x_0 to the point on the axis at which the value of B differs by $\eta = 1\%$ from that in the middle section of the solenoid. (Consider $x_0 \gg R$)

Solution

- In terms of angle θ_1 and θ_2 , the magnetic field at a point on the axis is given as

$$B = \frac{\mu_0 n I}{2} [\cos \theta_1 + \cos \theta_2]$$

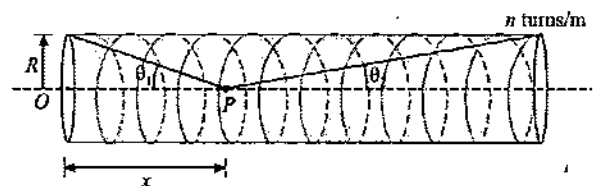


Figure 4.51

For a semi-infinite solenoid we use $\theta_1 = \theta$ and $\theta_2 = 0^\circ$ and from figure we have

$$\cos \theta = \frac{x}{(x^2 + R^2)^{1/2}}$$

Thus magnetic induction as a function of distance x is given as

$$B = \frac{\mu_0 n i}{2} \times \left[\frac{x}{(x^2 + R^2)^{1/2}} + 1 \right] \quad \dots (4.52)$$

(b) Equation-(4.52) is plotted as shown in figure-4.52.

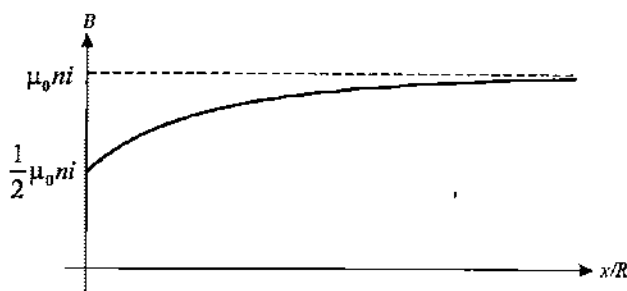


Figure 4.52

(c) At $x = x_0$ magnetic induction is taken as $0.99\mu_0 n i$ so from equation-(4.52) we use

$$\begin{aligned} \frac{\mu_0 n i}{2} \times \left[\frac{x_0}{(x_0^2 + R^2)^{1/2}} + 1 \right] &= \frac{99}{100} \mu_0 n i \\ \Rightarrow 50 \left[\frac{x_0}{(x_0^2 + R^2)^{1/2}} + 1 \right] &= 99 \end{aligned}$$

In above expression $x_0 \gg R$ so we can use binomial approximation as

$$\begin{aligned} 50 \left[\left(1 + \left(\frac{R}{x_0} \right)^2 \right)^{-1/2} + 1 \right] &= 99 \\ \Rightarrow \left[\left(1 - \frac{R^2}{2x_0^2} \right) + 1 \right] &= \frac{99}{50} \\ \Rightarrow 2 - 1.98 &= \frac{R^2}{2x_0^2} \\ \Rightarrow 0.04 x_0^2 &= R^2 \\ \Rightarrow x_0 &= 5R \end{aligned}$$

Illustrative Example 4.11

A straight long solenoid produces magnetic induction B at its centre. If it is cut into two equal parts and same number of turns wound on one part in double layer. Calculate magnetic field produced by new solenoid at its centre.

Solution

Magnetic induction produced by a long solenoid having n turns per unit length on its core and carrying a current I at its center is given as

$$B = \mu_0 n I$$

If same number of turns wound over half length in double layer on the solenoid core then number of turns per unit length on the new solenoid becomes $2n$ so for the same current in the solenoid the magnetic induction will become $2B$.

Illustrative Example 4.12

A very long straight solenoid carries a current I . The cross-sectional area of the solenoid is equal to S , the number of turns per unit length is equal to n . Find the magnetic flux of vector $\vec{B}$ through the end plane of the solenoid.

Solution

Magnetic induction at the end plane of a semi infinite solenoid is given as

$$B = \frac{1}{2} \mu_0 n i$$

Similar to the way we calculate electric flux using electric field strength as flux density, in the same manner magnetic flux can also be calculated using the magnetic induction at a plane. In this case as throughout the cross section magnetic induction is uniform and direction is perpendicular so to the cross sectional surface, here the magnetic flux can be given as

$$\begin{aligned} \phi &= \int \vec{B} \cdot d\vec{S} \\ \Rightarrow \phi &= \int B dS \cos(0^\circ) \\ \Rightarrow \phi &= B \int dS = BS \\ \Rightarrow \phi &= \frac{1}{2} \mu_0 n i \times S = \frac{1}{2} \mu_0 n i S \end{aligned}$$

Illustrative Example 4.13

A non-conducting sphere of radius R charged uniformly with surface density σ rotates with an angular velocity ω about a diametrical axis passing through its centre. Find the magnetic induction due to the rotating charge at the centre of the sphere.

Solution

Figure-4.53 shows the charged sphere on which we consider an elemental ring at an angle θ from the axis of angular width $d\theta$. The surface area of the elemental ring is given as

$$dS = 2\pi R \sin \theta \times R d\theta$$

The charge on the elemental ring surface is given as

$$dq = \sigma dS$$

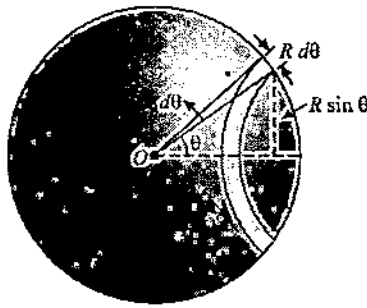


Figure 4.53

Current due to rotating charge on the elemental ring is given as

$$di = \frac{dq\omega}{2\pi}$$

$$\Rightarrow di = \frac{(2\pi\sigma R^2 \sin \theta d\theta)\omega}{2\pi}$$

$$\Rightarrow di = \sigma\omega R^2 \sin \theta d\theta$$

Magnetic induction at the centre O due to the current in elemental ring is given as

$$dB = \frac{\mu_0 di (R \sin \theta)^2}{2(R^2 \cos^2 \theta + R^2 \sin^2 \theta)^{3/2}}$$

$$\Rightarrow dB = \frac{\mu_0 di \sin^2 \theta}{2R}$$

Substituting the value of current di we get

$$dB = \frac{\mu_0}{2} \times \sigma\omega R^2 \sin \theta d\theta \times \frac{\sin^2 \theta}{R}$$

$$\Rightarrow dB = \frac{1}{2} \mu_0 \sigma \omega R \sin^3 \theta d\theta$$

Net magnetic induction at O is given by integrating above expression within limits of θ from 0 to π given as

$$B = \int dB = \frac{1}{2} \mu_0 \sigma \omega R \int_0^\pi \sin^3 \theta d\theta$$

$$\Rightarrow B = \frac{1}{2} \mu_0 \sigma \omega R \left[\frac{1}{3} \cos^3 \theta - \cos \theta \right]_0^\pi$$

$$\Rightarrow B = \frac{1}{2} \mu_0 \sigma \omega R \left[\frac{1}{3} (-1) - (-1) - \frac{1}{3} (1) + (1) \right]_0^\pi$$

$$\Rightarrow B = \frac{2}{3} \mu_0 \sigma \omega R$$

4.4 Ampere's Law

'Ampere's Law' also called 'Ampere's Circuital Law' relates the line integral of magnetic field along a given closed path and the currents which are enclosed by that closed path. The law is stated as

"Line integral of magnetic induction along a closed path is equal to μ_0 times the total current coming out and enclosed by that closed path."

In figure-4.54 M is a closed path in free space where some current carrying wires are present with direction of currents indicated in the figure. Such a path chosen for application of Ampere's law is called 'Ampere's Path'. Along the path at a point we consider an element of length $d\vec{l}$ and at the location of this element if net magnetic induction due to all the wires in the region is $\vec{B}$ then line integral of magnetic induction along the given path M is written as

$$L_i = \oint_M \vec{B} \cdot d\vec{l}$$

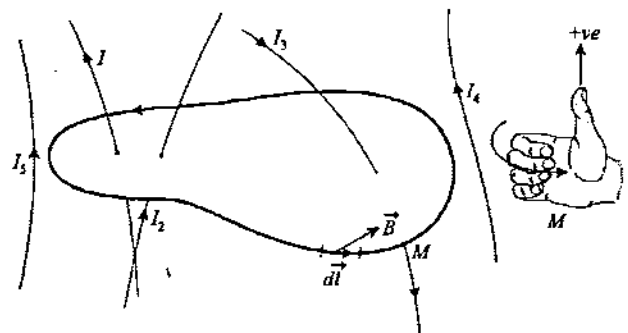


Figure 4.54

According to Ampere's law this line integral is equal to the product of μ_0 and the total current enclosed by the closed path. This gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} \quad \dots (4.53)$$

Expression in equation-(4.53) is called 'Equation of Ampere's law'. For the given situation as shown in figure there are three currents enclosed by the path M so we use

$$I_{\text{enclosed}} = I_1 + I_2 - I_3 \quad \dots (4.54)$$

Substituting the above value of I_{enclosed} in equation-(4.53) gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 (I_1 + I_2 - I_3) \quad \dots (4.55)$$

In equation-(4.54) direction of currents on right hand side of equality is decided by right hand thumb rule as shown in figure-4.54. Here right hand fingers are curled along the direction of line integral on the path M and thumb gives the positive direction or the currents in the direction of thumb are taken positive and opposite ones are taken negative.

It is important to note that in equation-(4.55) on LHS the magnetic induction $\vec{B}$ is due to all the currents in the region but on RHS only enclosed currents are included. This is because due to every individual currents magnetic field is closed loop in surrounding of that current thus line integral of magnetic field for a closed path in surrounding of any current is always zero if the path is not enclosing the current as shown in figure-4.55. If B_1 is the magnetic induction at the location of an element dl chosen on the path Z then we have

$$\oint_Z \vec{B}_1 \cdot d\vec{l} = 0$$

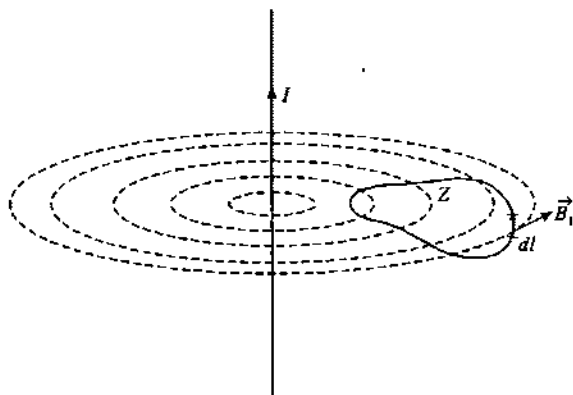


Figure 4.55

Qualitatively we can state that total inward magnetic lines on a path are equal to total outward magnetic lines from the path.

4.4.1 Applications of Ampere's Law

In application of Ampere's law under different situations we need to choose the path in such a way that the line integral of magnetic induction along the path can be carried out easily. If the path is randomly chosen then many times the integral can't be solved so students must be careful about the below mentioned general points for selection of Ampere's path for its application in different situations. These points are

- (i) Path should be chosen in such a way that at every point of path magnetic induction should be either tangential to the path elements or normal to it so that 'dot' product can be easily handled.
- (ii) Path should be chosen in such a way that at every point of path magnetic induction should either be uniform or zero so that calculations become easy.

To have better understanding of applications of Ampere's law we consider some cases of determining magnetic field using Ampere's law which is one of the most common application.

Just like in electrostatics we've studied that in cases of symmetric and uniform charge distribution calculation of electric field becomes easier if we use Gauss's law similarly, in cases of uniform current configurations Ampere's law is easier than Biot Savart's law that is discussed in upcoming articles.

4.4.2 Magnetic Induction due to a Long Straight Wire

Figure-4.56 shows a straight wire carrying a current I and we will determine the magnetic induction due to this wire at a point P located a distance r from wire as shown in figure. This we've already done in article-4.2.1 but here we'll do this by using Ampere's law.

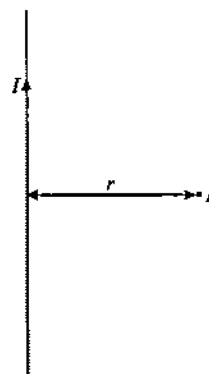


Figure 4.56

To apply Ampere's law we need to choose a path along which we will calculate the line integral of magnetic induction. As

shown in figure-4.57 we consider a circular path M with point P located on this path. We've chosen this path because we know that direction of magnetic induction at every point of this path is tangential to the path or parallel to every element on this path and its magnitude remain constant because of symmetry.

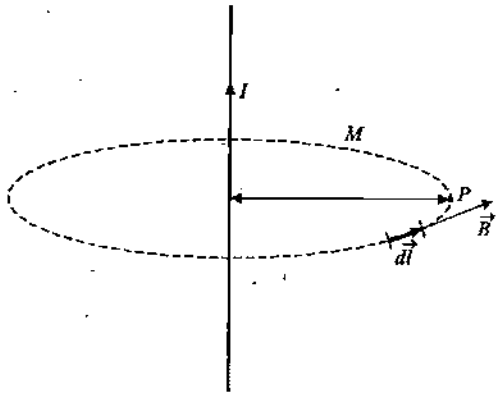


Figure 4.57

Total current enclosed by this path is I so on applying Ampere's law on this path, it gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$\Rightarrow \oint_M B dl \cos(0^\circ) = \mu_0 I$$

$$\Rightarrow B \oint_M dl = \mu_0 I$$

$$\Rightarrow B \cdot 2\pi R = \mu_0 I$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi R} \quad \dots (4.56)$$

Above equation-(4.56) is same as equation-(4.19) of article-4.2.1 which was obtained by Biot Savart's law. Students should note that for application of Ampere's law it is essential to know the direction of magnetic induction in surrounding of current configuration with the use of right hand thumb rule so that we can choose the Ampere's path properly.

4.4.3 Magnetic Induction due to a Long Solenoid

In article-4.3.1 also we've analyzed the magnetic induction due to a long solenoid which now we'll calculate again by using Ampere's law. Figure-4.58 shows a tightly wound long solenoid with core radius R , number of turns per unit length n and carrying a current I .

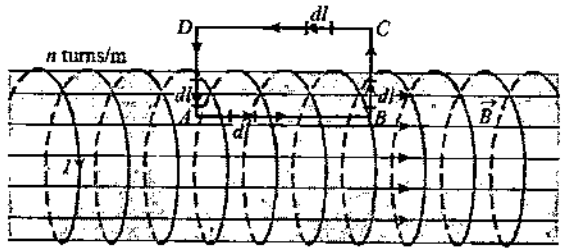


Figure 4.58

To apply Ampere's law we consider a rectangular path $ABCD$ as shown in figure such that the path lies in a radial plane passing through the axis of cylindrical core of solenoid and part CD of the path is lying outside and parallel to the axis of solenoid. The dimensions of the path are taken as $AB = x$ and $CD = y$ so the total number of turns passing through the path $ABCD$ are nx . Now applying Ampere's law on path $ABCD$ gives

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$\Rightarrow \oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 (nxI) \quad \dots (4.57)$$

In above equation to solve the line integral on LHS of equation-(4.57) we split the integral for the different parts of the closed path $ABCD$ as

$$\int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} = \mu_0 (nxI) \quad \dots (4.58)$$

As the magnetic induction inside a tightly wound long solenoid is uniform, for path length AB magnetic induction vector is along the element length so first term in above equation-(4.58) can be given as

$$\int_A^B \vec{B} \cdot d\vec{l} = B \int_A^B dl = Bx \quad \dots (4.59)$$

For the parts BC and DA we can see that the portions which are outside the solenoid, magnetic induction is zero and the portions which are inside the solenoid magnetic induction is perpendicular to the path elements thus for the whole length of these paths second and fourth terms on LHS of equation-(4.58) will be zero which are given as

$$\int_B^C \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.60)$$

$$\text{and} \quad \int_D^A \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.61)$$

For the remaining path CD as it is completely outside the solenoid where magnetic induction is zero so this term will also be zero as at every point of path CD , $B = 0$ given as

$$\oint_C \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.62)$$

Substituting above values in equation-(4.58) gives

$$Bx = \mu_0 (nxI) \\ \Rightarrow B = \mu_0 nI \quad \dots (4.63)$$

Expression in above equation-(4.63) is same as equation-(4.38) of article-4.3.1 which was calculated by integrating the magnetic induction due to ring elements over the whole length of solenoid.

4.4.4 Magnetic Induction due to a Tightly Wound Toroid

In article-4.3.5 also we've analyzed the magnetic induction due to a large toroid which now we'll calculate it again by using Ampere's law. Figure-4.59 shows a tightly wound large toroid with average torus radius R , cross sectional radius r ($r \ll R$), number of turns per unit length n and carrying a current I .

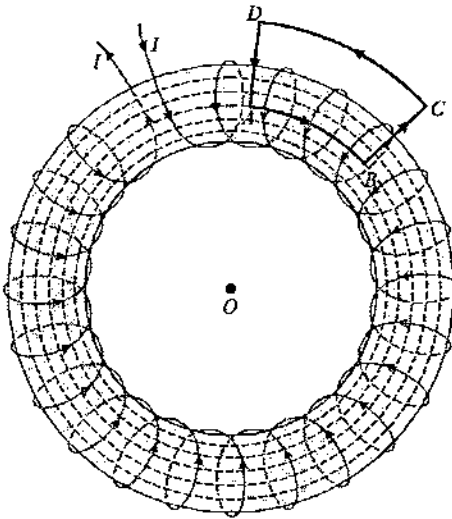


Figure 4.59

To apply Ampere's law we consider an approximately rectangular path $ABCD$ as shown in above figure similar to the case of solenoid such that the path lies in the plane perpendicular to the axis of toroid at point O and part CD of the path is lying outside the toroid. The dimensions of the paths are taken as $AB = x$ and $CD = y$ so the total number of turns passing through the path $ABCD$ are nx . Now applying Ampere's law on path $ABCD$ gives

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} \\ \Rightarrow \oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 (nxI) \quad \dots (4.64)$$

In above equation to solve the line integral on LHS of

equation-(4.64) we split the integral for the different parts of the closed path $ABCD$ similar to the analysis we did for solenoid as

$$\int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} = \mu_0 (nxI) \quad \dots (4.65)$$

As the magnetic induction inside a tightly wound long solenoid is uniform, for path length AB magnetic induction vector is along the element length so first term in above equation-(4.65) can be given as

$$\int_A^B \vec{B} \cdot d\vec{l} = B \int_A^B dl = Bx \quad \dots (4.66)$$

For the parts BC and DA we can see that the portions which are outside the toroid magnetic induction is zero and the portions which are inside the toroid magnetic induction is perpendicular to the path elements thus for the whole length of these paths second and fourth terms on LHS of equation-(4.65) will be zero which are given as

$$\int_B^C \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.67)$$

and

$$\int_D^A \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.68)$$

For the remaining path CD as it is completely outside the solenoid where magnetic induction is zero so this term will also be zero as at every point of path CD , $B = 0$ given as

$$\int_C^D \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.69)$$

Substituting above values in equation-(4.65) gives

$$Bx = \mu_0 (nxI) \\ \Rightarrow B = \mu_0 nI \quad \dots (4.70)$$

If total number of turns in toroid are N then n is given as

$$n = \frac{N}{2\pi R}$$

Thus equation-(4.70) becomes

$$B = \mu_0 \left(\frac{N}{2\pi R} \right) I \\ \Rightarrow B = \frac{\mu_0 NI}{2\pi R} \quad \dots (4.71)$$

Expression in above equation-(4.71) is same as equation-(4.43) of article-4.3.5 which we've now calculated using Ampere's law.

4.4.5 Magnetic Induction due to a Cylindrical Wire

Figure-4.60 shows a long straight cylindrical wire of radius R carrying a current I . To determine the magnetic induction at outside point P located a distance x ($x > R$) from its axis we consider a circular Ampere's path of radius x on which applying Ampere's law gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$\Rightarrow \oint_M B dl \cos(0^\circ) = \mu_0 I$$

$$\Rightarrow B \oint_M dl = \mu_0 I$$

$$\Rightarrow B \cdot 2\pi x = \mu_0 I$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi x} \quad \dots (4.72)$$

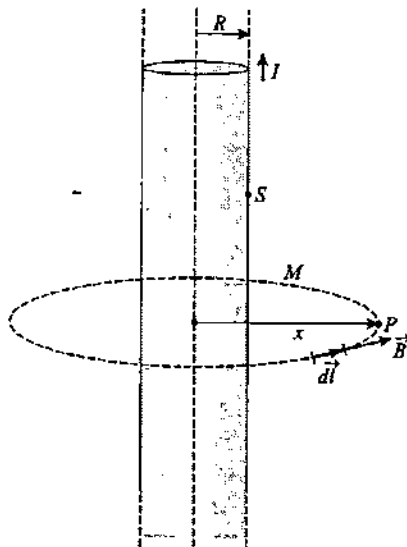


Figure 4.60

If point P is located on the surface of cylindrical wire we can use $x = R$ for which the magnetic induction is given by replacing x with R in equation-(4.72) as

$$B_s = \frac{\mu_0 I}{2\pi R} \quad \dots (4.73)$$

To determine the magnetic induction at an interior point P at a distance x from the axis of wire again we consider a circular path as shown in figure-4.61 and apply Ampere's law on this path which gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} \quad \dots (4.74)$$

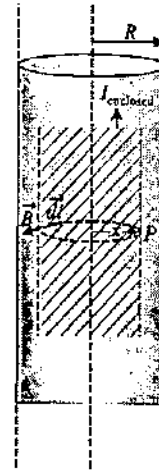


Figure 4.61

As the current I is considered to be uniformly distributed over the cross section of cylindrical wire, the current enclosed in the circular path M of radius x as shown in above figure is given as

$$I_{\text{enclosed}} = \frac{I}{\pi R^2} \times \pi x^2$$

$$\Rightarrow I_{\text{enclosed}} = \frac{Ix^2}{R^2}$$

Thus from equation-(4.74) we have

$$\Rightarrow \oint_M B dl \cos(0^\circ) = \mu_0 \left(\frac{Ix^2}{R^2} \right)$$

$$\Rightarrow B \oint_M dl = \mu_0 \left(\frac{Ix^2}{R^2} \right)$$

$$\Rightarrow B \cdot 2\pi x = \mu_0 \left(\frac{Ix^2}{R^2} \right)$$

$$\Rightarrow B = \frac{\mu_0 Ix}{2\pi R^2} \quad \dots (4.75)$$

For outside and surface points of the cylindrical wire the expressions of magnetic induction as obtained in equations-(4.72) and (4.73) are same as obtained in equation-(4.56) thus for a uniform current carrying cylindrical wire on its outer and surface points we can determine magnetic induction by considering its current concentrated along the axis of wire.

If current density in the cylindrical wire is J then we can rewrite above expression in equation-(4.75) as

$$B = \frac{1}{2} \mu_0 J x \quad \dots (4.76)$$

If position vector of point P is considered as $\vec{x}$ then direction of magnetic induction at point P is perpendicular to $\vec{x}$ thus equation-(4.76) can be vectorially written as

$$\vec{B} = \frac{1}{2} \mu_0 J \vec{x}_\perp \quad \dots (4.77)$$

Above equation-(4.77) can also be written as

$$\vec{B} = \frac{1}{2} \mu_0 (\vec{J} \times \vec{x}) \quad \dots (4.78)$$

Using equation-(4.72), (4.73) and (4.75) we can plot the variation curve of magnetic induction magnitude in surrounding of the cylindrical current carrying wire which is shown in figure-4.62.

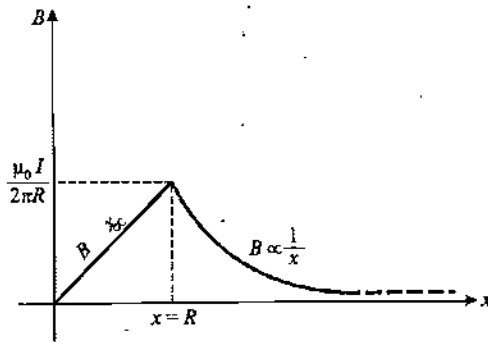


Figure 4.62

4.4.6 Magnetic Induction due to a Hollow Cylindrical Wire

Figure-4.63 shows a hollow thin walled long cylindrical wire carrying a current I . Similar to what we discussed in previous article here also we can say that due to uniform current distribution for outer and surface points we can consider that current is concentrated on the axis of cylinder thus magnetic induction at outer and surface points of the wire is given as

$$\text{For } x > R \quad B_{\text{out}} = \frac{\mu_0 I}{2\pi x} \quad \dots (4.79)$$

$$\text{For } x = R \quad B_s = \frac{\mu_0 I}{2\pi R} \quad \dots (4.80)$$

Above results can also be obtained by considering a circular Ampere's path and applying Ampere's law like the way we did for a solid cylindrical wire.

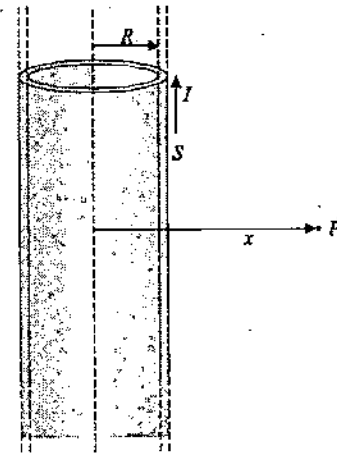


Figure 4.63

For interior points of the hollow cylindrical wire if we consider a point P as shown in figure-4.64 and apply Ampere's law on the circular path M shown then we have

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

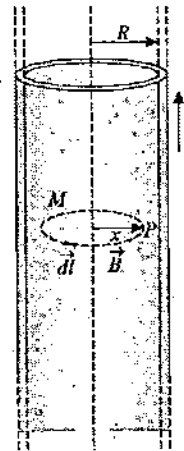


Figure 4.64

In this case for the path shown enclosed current will be zero as current is flowing only on the surface of the hollow cylinder. By symmetry of path we can assume that magnetic induction is uniform along the path M which gives

$$\oint_M \vec{B} \cdot d\vec{l} = 0 \quad \Rightarrow \quad B_{\text{inside}} = 0 \quad \dots (81)$$

Using equation-(4.79), (4.80) and (4.81) we can plot the variation curve of magnetic induction magnitude in surrounding of the cylindrical current carrying wire which is shown in figure-4.65.

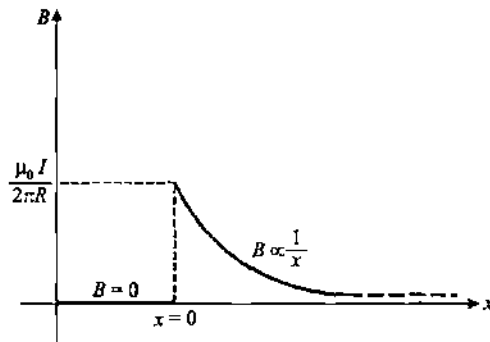


Figure 4.65

4.4.7 Magnetic Induction due to a Large Current Carrying Sheet

Figure-4.66 shows the cross section of a current carrying large sheet with linear current density i A/m flowing in it. In article-4.3.6 we have already discussed about the direction of magnetic induction above and below the sheet as shown in figure-4.45. To determine the magnetic induction at a point P located at a distance r above the sheet using Ampere's law, we consider a rectangular path $ABCD$ as shown in figure-4.66. The dimensions of the path chosen are $AB = x$ and $BC = y$. Along path AB and CD we can see that magnetic induction direction is parallel to the direction of element chosen and for paths BC and DA direction of magnetic induction is perpendicular to the element. Thus by applying Ampere's law on the path $ABCD$ we have

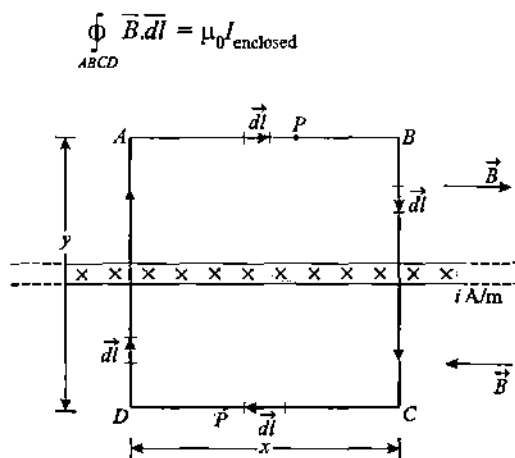


Figure 4.66

As linear current density in the sheet is i A/m, the current in length x of the sheet which is enclosed by the path $ABCD$ is given as ix ampere which gives

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 (ix) \quad \dots (4.82)$$

In above equation to solve the line integral on LHS of

equation-(4.82) we split the integral for the different parts of the closed path $ABCD$ similar to the analysis we did for solenoid as

$$\int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} = \mu_0 (ix) \quad \dots (4.83)$$

Second and fourth term in LHS, the magnetic induction is perpendicular to the element along paths BC and DA thus dot product will result zero and for paths AB and CD magnetic field is uniform and parallel to the element this gives

$$B \int_A^B dl + B \int_C^D dl = \mu_0 (ix)$$

$$\Rightarrow Bx + Bx = \mu_0 (ix)$$

$$\Rightarrow 2Bx = \mu_0 (ix)$$

$$\Rightarrow B = \frac{1}{2} \mu_0 i \quad \dots (4.84)$$

Expression in above equation-(4.84) is same as obtained in equation-(4.44) which we solved by using long integration process.

4.4.8 Magnetic Induction due to a Large and Thick Current Carrying Sheet

Figure-4.67 shows the cross section of a thick large sheet of thickness d . The current flowing in the sheet is having a current density J A/m². As discussed earlier in this case also the direction of magnetic induction above and below the sheet is given by right hand thumb rule as shown in figure-4.45. To determine the magnetic induction magnitude at point P outside the sheet we consider the rectangular path $ABCD$ with point P located on the line AB of the path as shown in figure-4.67. The dimensions of the path are $AB = x$ and $BC = y$. By applying Ampere's law on this path we have

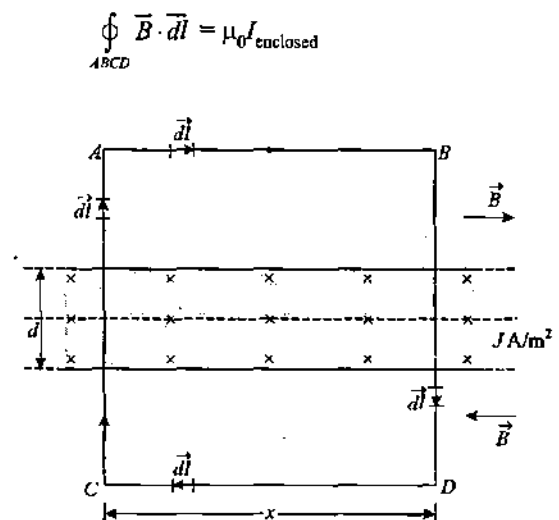


Figure 4.67

As current density in the cross section of sheet is $J \text{ A/m}^2$, the current in length x of the sheet which is enclosed by the path $ABCD$ is given as Jxd ampere which gives

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0(Jxd) \quad \dots (4.85)$$

In above equation to solve the line integral on LHS of equation-(4.85) we split the integral for the different parts of the closed path $ABCD$ similar to the analysis we did earlier as

$$\int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} = \mu_0(Jxd) \quad \dots (4.86)$$

Second and fourth term in LHS, the magnetic induction is perpendicular to the element along paths BC and DA thus dot product will result zero and for paths AB and CD magnetic field is uniform and parallel to the element which gives

$$B \int_A^B dl + B \int_C^D dl = \mu_0(Jxd)$$

$$\Rightarrow Bx + Bx = \mu_0(Jxd)$$

$$\Rightarrow 2Bx = \mu_0(Jxd)$$

$$\Rightarrow B = \frac{1}{2} \mu_0 Jd \quad \dots (4.87)$$

To determine the magnetic induction at interior points of the thick sheet at a point P at a distance x from the central plane of the sheet as shown in figure-4.68, we consider a rectangular path $ABCD$ as shown with dimensions $AB = y$ and $BC = 2x$. On applying Ampere's law on this path, we have

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

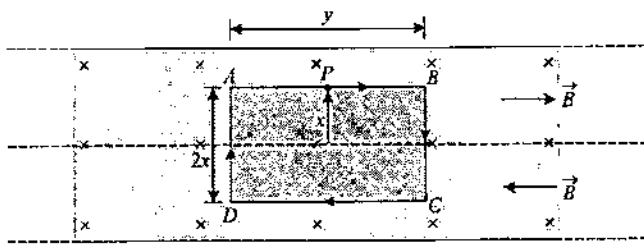


Figure 4.68

As current density in the cross section of sheet is $J \text{ A/m}^2$, the current enclosed by the Ampere's path $ABCD$ is given as $J(2xy)$ which gives

$$\oint_{ABCD} \vec{B} \cdot d\vec{l} = \mu_0(2Jxy) \quad \dots (4.88)$$

In above equation to solve the line integral on LHS of equation-(4.88) we split the integral for the different parts of the closed path $ABCD$ similar to the analysis we did earlier as

$$\int_A^B \vec{B} \cdot d\vec{l} + \int_B^C \vec{B} \cdot d\vec{l} + \int_C^D \vec{B} \cdot d\vec{l} + \int_D^A \vec{B} \cdot d\vec{l} = \mu_0(2Jxy) \quad \dots (4.89)$$

Again like previous analysis second and fourth term in LHS of the expression above, the magnetic induction is perpendicular to the element along paths BC and DA thus dot product will result zero and for paths AB and CD magnetic field is uniform and parallel to the element this gives

$$B \int_A^B dl + B \int_C^D dl = \mu_0(2Jxy)$$

$$\Rightarrow By + By = \mu_0(2Jxy)$$

$$\Rightarrow 2By = \mu_0(2Jxy)$$

$$\Rightarrow B = \mu_0 Jx \quad \dots (4.90)$$

From equation-(4.87) and (4.90) we can plot the variation of magnetic induction with distance from the central plane of a thick sheet which is shown in figure-4.69.

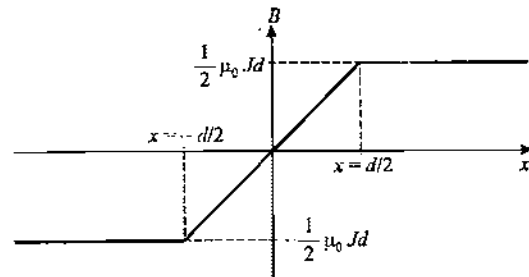


Figure 4.69

4.4.9 Unidirectional Magnetic Field in a Region

Consider the configuration of magnetic lines of forces as shown in figure-4.70 in a region. In this region at every point in space magnetic field is in same direction but the density of magnetic lines is different that means in this region magnetic induction is not uniform everywhere. If in this region we consider a rectangular path $PQRS$ as shown in figure and we apply Ampere's law on this path $PQRS$ then it gives

$$\oint_{PQRS} \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.91)$$

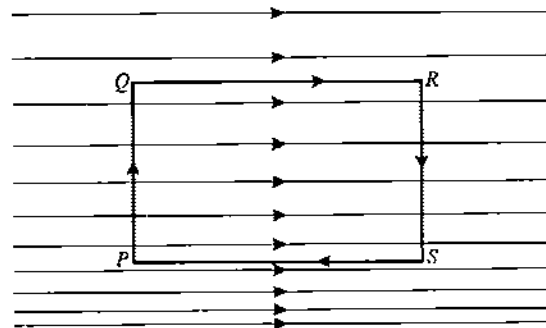


Figure 4.70

Right hand side of the above equation is zero as there is no current enclosed in the closed path. Equation-(4.91) can be expanded for the integrals on the four straight line paths as

$$\int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^R \vec{B} \cdot d\vec{l} + \int_R^S \vec{B} \cdot d\vec{l} + \int_S^P \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.92)$$

For the path segments PQ and RS we can see that magnetic induction vector and $d\vec{l}$ are perpendicular to each other so first and third term of the above equation will vanish and if B_1 and B_2 are the magnetic inductions at path segments QR and SP which are of lengths x then we have

$$0 + B_1 x + 0 - B_2 x = 0$$

$$\Rightarrow B_1 = B_2 \quad \dots (4.93)$$

Thus it is clear that above situation is possible if and only if magnetic induction at path segments QR and SP are equal. This means the configuration of magnetic lines which is shown in figure-4.70 is practically not possible. With the above analysis we can also state "If in a region of space magnetic field is unidirectional then at every point of space it must be uniform in magnitude also."

Illustrative Example 4.14

Figure-4.71 shows a coaxial cable which consist of an inner solid cylindrical conductor of radius a and outer hollow cylindrical shell with inner radius b and outer radius c . A current I flows in both conductors in opposite directions as shown. Find the magnetic induction at a point located at a distance r from central axis with $b < r < c$.

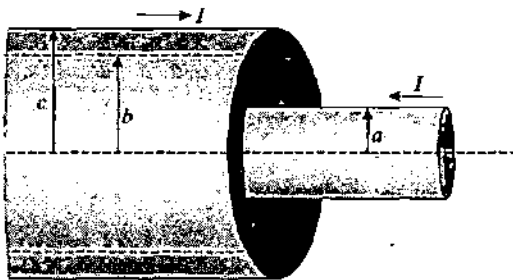


Figure 4.71

Solution

In the figure-4.72 shown below we consider a point P at a radial distance r from the axis and consider a circular path M which passes through point P .

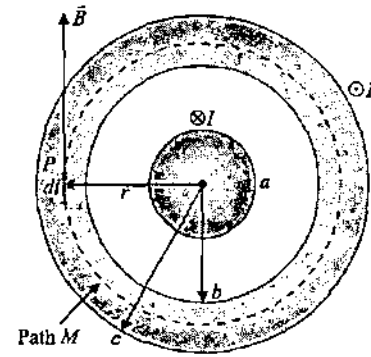


Figure 4.72

The current enclosed by the closed path M can be given as

$$I_{\text{enclosed}} = I - \frac{I}{\pi(c^2 - b^2)} \times \pi(r^2 - b^2)$$

Using Ampere's law on the path M gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 \left[I - \frac{I}{\pi(c^2 - b^2)} \times \pi(r^2 - b^2) \right]$$

$$\Rightarrow B(2\pi r) = \mu_0 I \left[\frac{c^2 - r^2}{c^2 - b^2} \right]$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi r} \left(\frac{c^2 - r^2}{c^2 - b^2} \right)$$

Illustrative Example 4.15

Inside a long straight uniform wire of round cross-section, there is a long round cylindrical cavity whose axis is parallel to the axis of the wire and displaced from the axis by a distance $\vec{l}$. A direct current of density $\vec{j}$ flows along the wire. Find the magnetic induction at a general point inside the cavity. Consider, in particular, the case $\vec{l} = 0$.

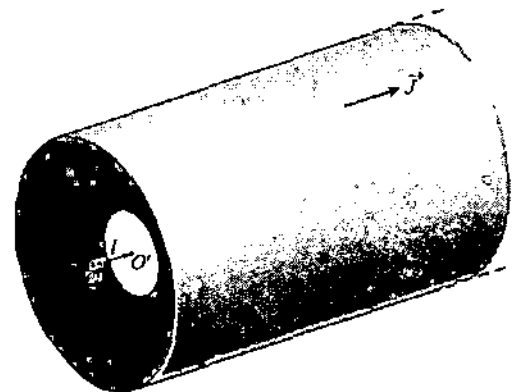


Figure 4.73

Solution

Consider a cross-section of the cylinder perpendicular to its axis as shown in figure-4.73. The cylindrical cavity is shown at a distance l from the axis O of the cylinder. We will calculate the magnetic induction at the point P inside the cavity. The cavity means an absence of current which we consider as an equivalent opposite current of same current density in the cavity. Magnetic induction at point P inside cavity is given by subtracting the magnetic induction due to opposite current considered in cavity region from the total magnetic induction at P due to complete solid cylindrical wire which is given as

$$\vec{B}_P = \vec{B}_{\text{Wire}} - \vec{B}_{\text{Cavity}}$$

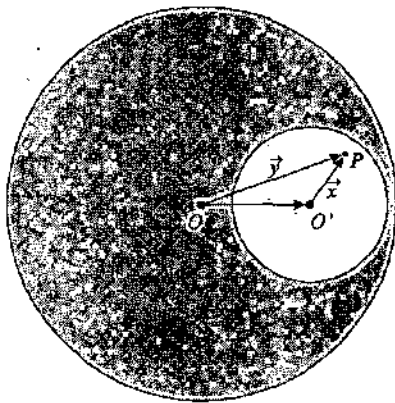


Figure 4.74

Expressions for the magnetic induction due to wire and that due to the opposite current in cavity is given by using equation-(4.77) as

$$\vec{B}_P = \frac{1}{2} \mu_0 J \vec{y}_\perp - \frac{1}{2} \mu_0 J \vec{x}_\perp$$

$$\Rightarrow \vec{B}_P = \frac{1}{2} \mu_0 J (\vec{y}_\perp - \vec{x}_\perp)$$

$$\Rightarrow \vec{B}_P = \frac{1}{2} \mu_0 J (\vec{l}_\perp) \quad \dots (4.94)$$

Above expression of magnetic induction as given in equation-(4.94) is a uniform magnetic induction in the direction normal to the line joining OO' as shown in figure-4.74.

If separation between O and O' becomes zero then the cavity will be coaxial with the axis of cylindrical wire and from above expression given in equation-(4.91) it gives there is no magnetic field in the cavity or inside of a hollow cylindrical current carrying wire.

Illustrative Example 4.16

In a long cylindrical wire of radius R , magnetic induction varies with the distance from axis as $B = cr^\alpha$. Find the function of current density in wire with the distance from axis of wire.

Solution

At a distance r from the axis of wire we consider a closed path M as shown in figure-4.75. We apply Ampere's law on this path gives

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

If $J(x)$ is the current density inside the wire as a function of distance x from the axis of wire, the enclosed current within the closed path M is given by integrating the current in the elemental ring of radius x and width dx considered in the cross section of wire as shown in figure-4.75 given as

$$I_{\text{enclosed}} = \int_0^r J(x) \cdot 2\pi x \cdot dx$$

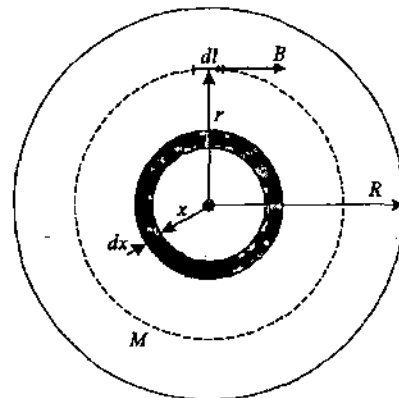


Figure 4.75

$$\Rightarrow B \oint_M dl = \mu_0 \int_0^r J(x) \cdot 2\pi x \cdot dx$$

$$\Rightarrow B(2\pi r) = \mu_0 \int_0^r J(x) \cdot 2\pi x \cdot dx$$

$$\Rightarrow cr^{\alpha+1} = \mu_0 \int_0^r J(x) \cdot x \cdot dx$$

Differentiating the above expression with respect to x gives

$$(\alpha + 1)cr^\alpha = \mu_0 J(r) \cdot r$$

$$\Rightarrow J(r) = \left(\frac{(\alpha + 1)c}{\mu_0} \right) r^{\alpha-1}$$

Illustrative Example 4.17

Figure-4.76 shows a region of space in which a uniform magnetic induction B_0 is present between the planes $z = 0$ and $z = a$. Using Ampere's law prove that such a field cannot exist between two planes as described here.

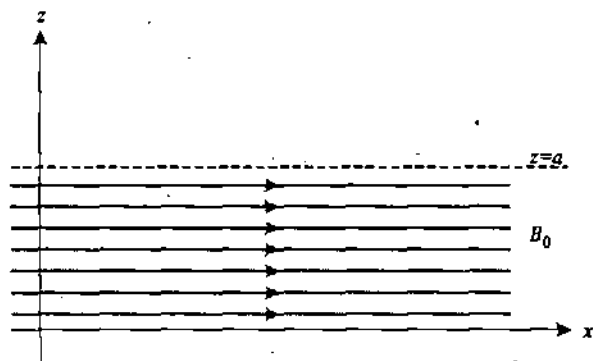


Figure 4.76

Solution

For the given magnetic field we consider a small rectangular path $PQRS$ partly above and partly below the plane $z = a$ as shown in figure-4.77. Now we apply Ampere's law on this path which gives

$$\oint_{PQRS} \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.99)$$

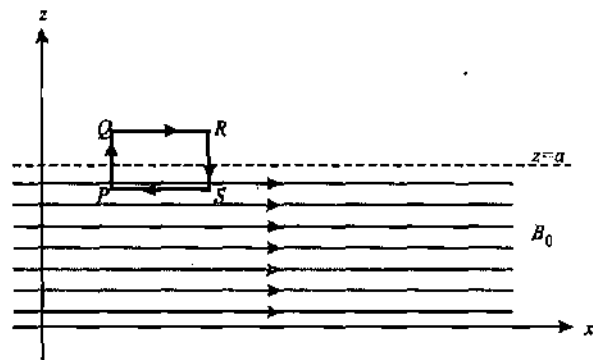


Figure 4.77

Right hand side of the above equation is zero as there is no current enclosed in the closed path. Equation-(4.95) can be expanded for the integrals on the four straight line paths as

$$\int_P^Q \vec{B} \cdot d\vec{l} + \int_Q^R \vec{B} \cdot d\vec{l} + \int_R^S \vec{B} \cdot d\vec{l} + \int_S^P \vec{B} \cdot d\vec{l} = 0 \quad \dots (4.96)$$

For the path segments PQ and RS we can see that magnetic induction vector and $d\vec{l}$ are perpendicular to each other and for the path segment QR it is located outside the magnetic field where no magnetic induction exist so first, second and third term of the above equation will vanish and as B_0 is the magnetic inductions at path segments SP which is considered of lengths x then we have

$$0 + 0 + 0 - B_0 x = 0$$

$$\Rightarrow B_0 = 0$$

Thus with by applying Ampere's law on the path shown in above figure we get $B_0 = 0$ that means such a non-zero magnetic field cannot practically exist.

Illustrative Example 4.18

Figure-4.78 shows a toroidal solenoid whose cross-section is rectangular in shape. Find the magnetic flux through this cross-section if the current through the toroidal winding is I , total number of turns in winding is N , the inside and outside radii of the toroid are a and b respectively and the height of toroid is equal to h .

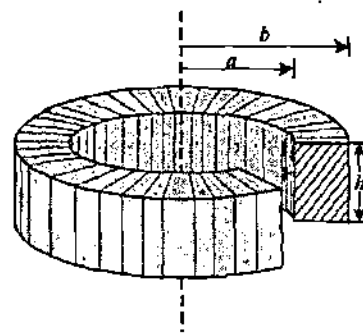


Figure 4.78

Solution

To find the magnetic flux through the cross section shown in figure, we consider an elemental strip of width dx at a distance x from the axis of toroid as shown in figure-4.79.

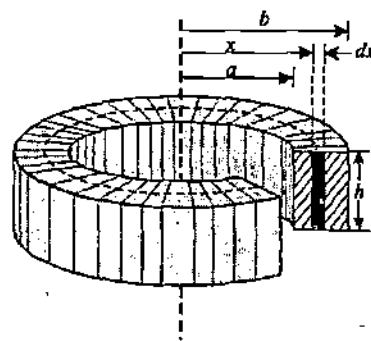


Figure 4.79

The magnetic field at a distance x from the axis of the toroid is given as

$$B = \mu_0 \left(\frac{N}{2\pi x} \right) I = \frac{\mu_0 N I}{2\pi x}$$

Magnetic flux through an elemental strip of radial thickness dx and height h as shown in figure is given as

$$d\phi = \vec{B} \cdot \vec{dS} = B dS$$

$$\Rightarrow d\phi = \frac{\mu_0 NI}{2\pi x} \cdot h dx$$

Total magnetic flux through the cross section of the toroid is given by integrating the above elemental flux within limits from a to b which is given as

$$\phi = \int_a^b \frac{\mu_0 NI h}{2\pi x} dx$$

$$\Rightarrow \phi = \frac{\mu_0 NI h}{2\pi} \int_a^b \frac{1}{x} dx$$

$$\Rightarrow \phi = \frac{\mu_0 NI h}{2\pi} [\ln x]_a^b$$

$$\Rightarrow \phi = \frac{\mu_0 NI h}{2\pi} [\ln b - \ln a]$$

$$\Rightarrow \phi = \frac{\mu_0 NI h}{2\pi} \ln \left(\frac{b}{a} \right)$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Magnetic Effects of Current

Module Number - 21 to 35

Practice Exercise 4.2

(i) A tightly wound long solenoid having n turns per unit length is kept above a large metal sheet carrying a surface current of linear current density i A/m. The solenoid is oriented such that magnetic induction at its center is found to be zero when a current is passed through it.

(a) Find the current in the solenoid

(b) If the solenoid is rotated by 90° then find the magnetic induction at the center of solenoid.

$$[(a) \frac{i}{2n}; (b) \frac{\mu_0 i}{\sqrt{2}}]$$

(ii) A coaxial cable carries the a current i in the inside conductor of radius a as shown in figure-4.80 and the outer conductor of inner radius b and outer radius c carries a current i' in opposite direction. Find the magnetic induction due to the coaxial cable at distance r from the central axis of

the cable for (a) $r < a$; (b) $a < r < b$ and (c) $b < r < c$

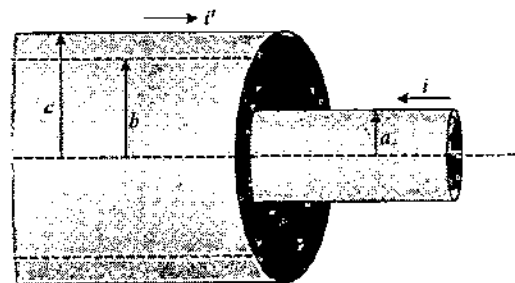


Figure 4.80

$$[(a) \frac{\mu_0 i r}{2\pi a^2}; (b) \frac{\mu_0 i}{2\pi r}; (c) \frac{\mu_0}{2\pi r} \left[i - i' \left(\frac{r^2 - b^2}{c^2 - b^2} \right) \right]]$$

(iii) A thin conducting strip of width h is tightly wound in the shape of a very long cylindrical coil with cross-sectional radius R to make a single layer straight solenoid as shown in figure-4.81. A direct current I flows through the strip. Find the magnetic induction inside and outside the solenoid as a function of the distance r from its axis.

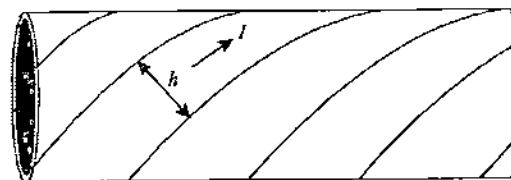


Figure 4.81

$$\left[\frac{\mu_0 I}{h} \sqrt{1 - \left(\frac{h}{2\pi R} \right)^2}, \frac{\mu_0 I}{2\pi r} \right]$$

(iv) A long cylinder of uniform cross-section and radius R is carrying a current i along its length and the current density is uniform. There is a cylindrical cavity of uniform cross-section and radius r in the cylinder parallel to its length. The axis of the cylindrical cavity is separated by a distance d from the axis of the cylinder. Find the magnetic field at the axis of cylinder.

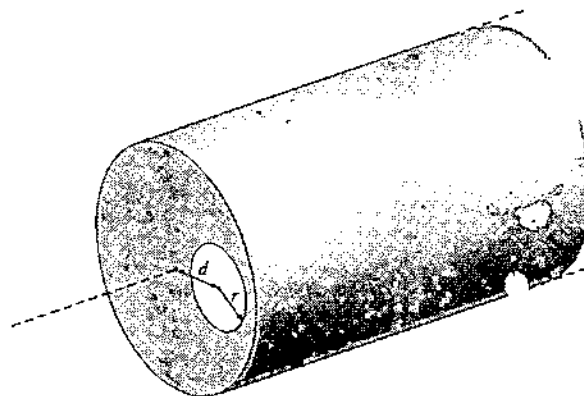


Figure 4.82

$$\left[\frac{\mu_0 i r^2}{2\pi d(R^2 - r^2)} \right]$$

- (v) Two straight infinitely long and thin parallel wires are spaced 0.1m apart and carry a current of 10A each. Calculate the magnetic induction at a point distant 0.1m from both wires in the two cases when the currents are in the (a) same and (b) opposite directions.

[(a) 3.46×10^{-5} T; (b) 2×10^{-5} T]

- (vi) A capacitor of capacitance C is connected to a battery of EMF E for a long time and then disconnected. The charged capacitor is then connected across a long solenoid having n turns per meter in its closely packed winding on its core. After connections it is found that the voltage across the capacitor drops to E/η in a time Δt . In this period estimate the average magnetic induction at the center of solenoid.

$$\left[\frac{\mu_0 n C E}{\Delta t} \left(1 - \frac{1}{\eta} \right) \right]$$

- (vii) A current I flows along a lengthy thin walled tube of radius R with a longitudinal slit of width h . Find the magnetic induction inside the tube under the condition $h \ll R$.

$$\left[\frac{\mu_0 I h}{4\pi^2 R r} \right]$$

- (viii) A direct current I flows along a lengthy straight wire which terminates perpendicularly on an infinitely large conducting plane. From the point O , the current spreads all over the infinite conducting plane. Find the magnetic induction at all points of space above and below the conducting plane at a distance r from the axis of wire.

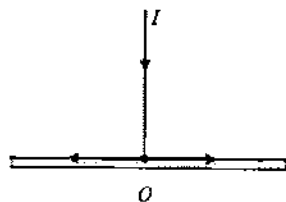


Figure 4.83

$$\left[\frac{\mu_0 I}{2\pi r}, 0 \right]$$

- (ix) A toroid of total 1000 turns is made by using a core of average radius 25cm. What is the magnetic induction inside the core if a current of 2A is passed through it and the relative magnetic permeability of the core material is $\mu_r = 100$.

$[16 \times 10^{-2}$ T]

4.5 Electromagnetic Interactions

We've already discussed in beginning of this chapter about how electric and magnetic fields interact with static and moving charges. Electric field can exert force on both static and moving charges whereas magnetic field can only exert force on moving charges. As motion is always measured in a specific reference frame thus magnetic force is also dependent upon frame. Actually the electric and magnetic forces acting on a charge particle are two types of a single force called 'Electromagnetic Interaction' which is the net force applied by these force fields on a charge. When the situation is observed from different frames of references then for all inertial reference frames the net electromagnetic force on charge particle remain same like what we discussed in topic of statics and dynamics but from different frames the electric and magnetic force vectors may be measured differently. Always the vector sum of electric and magnetic forces experienced by a charge particle remain constant in all inertial frames which gives the total electromagnetic force on the specific charge particle. In upcoming articles we will discuss upon these forces in more details.

4.5.1 Electromagnetic Force on a Moving Charge

When a charge q moving at velocity $\vec{v}$ enters in a region of magnetic field with magnetic induction $\vec{B}$, it experiences a force given as

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad \dots (4.97)$$

If in space electric field $\vec{E}$ also exist then the net electromagnetic force on the charge particle is given as

$$\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B}) \quad \dots (4.98)$$

The expression of force as given in equation-(4.98) is called 'Lorentz Force Equation' which gives net electromagnetic force on a particle. First term in above equation is the electric force on charge and second term is the magnetic force on charge. The two forces may be different as observed from different frames of reference but for all inertial reference frames the total force given by this equation will be a constant. This is because in different frames due to their motion the effective electric and magnetic field changes as variation in one field induces the other and vice versa. This phenomenon of induction of one field due to variation in other will be studied in next chapter of electromagnetic induction.

For some upcoming articles we will restrict our discussion only upto magnetic forces as we've already discussed about electric forces at a decent level in previous chapters.

4.5.2 Direction of Magnetic Force on Moving Charges

As already discussed in previous article that the magnetic force on a moving charge in magnetic field is given by equation-(4.97).

The direction of magnetic force on charge particle can be given by the cross product using right hand thumb rule as shown in figure-4.84.

This figure shows a region with magnetic field in inward direction and a charge q enters the region with a velocity v toward right as shown.

Using cross product in equation-(4.97) we can rotate our right hand fingers from the direction of velocity vector (right) to the direction of magnetic induction vector (inward) and in this state direction of right hand thumb gives the direction of magnetic force shown in figure-4.84.

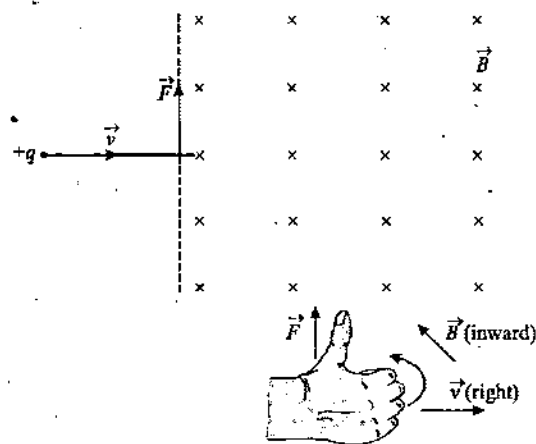


Figure 4.84

Right hand thumb rule is a common way of finding the direction of magnetic force on a moving charge but in many three dimensional cases another method 'Right Hand Palm Rule' is also very helpful in determining the direction of magnetic force which is stated as

"Stretch your right hand palm with fingers straight and thumb in same plane. If you point fingers along the direction of magnetic induction and thumb along the direction of velocity of positive charge motion then area vector of your right hand palm gives the direction of magnetic force."

For the above case shown in figure-4.84 we can also determine the direction of magnetic force using right hand palm rule as shown in figure-4.85.

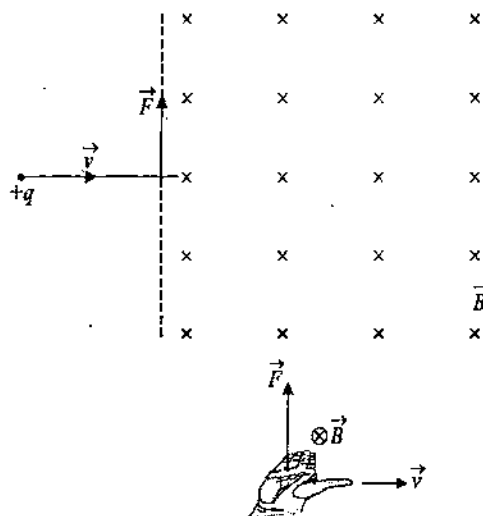


Figure 4.85

In many cases right hand palm rule is advantageous to apply so in upcoming cases we will prefer to use right hand palm rule however students are free to use any of the rule whichever they feel comfortable.

4.5.3 Work Done by Magnetic Force on Moving Charges

From equation-(4.97) it is clear that direction of magnetic force is always perpendicular to instantaneous velocity of the particle thus it can never do work on freely moving charge particles.

Thus in magnetic field we can state that speed of a freely moving charge particle can never be changed due to magnetic forces acting on it.

As magnetic force acts in direction perpendicular to the instantaneous velocity of particle, it can change the direction of motion of the moving charges. In upcoming articles we will discuss different cases of projection of charge particles in magnetic field.

4.5.4 Projection of a Charge Particle in Uniform Magnetic Field in Perpendicular Direction

When a charge particle with charge q is projected with an initial velocity v in magnetic field from point A as shown in figure-4.86 then as velocity is perpendicular to magnetic field, the magnetic force on charge at point A can be calculated by equation-(4.97) which is given as

$$F = qvB \quad \dots (4.99)$$

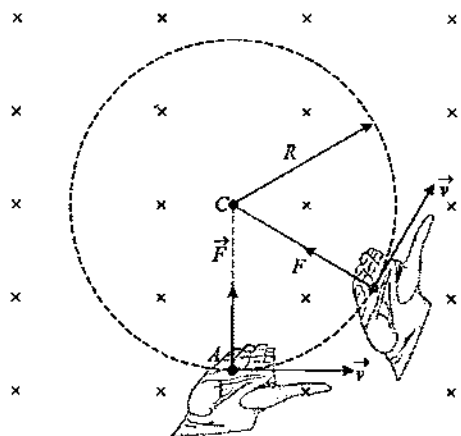


Figure 4.86

By right hand palm rule direction of force on q at point A is in vertically upward direction as shown. Due to this force the direction of motion of charge changes and with the change in direction of velocity of charge, the direction of magnetic force on it also changes with motion. As speed of particle is constant and force on it is always in normal direction so motion of particle will be circular and the magnetic force will provide the necessary centripetal force for this uniform circular motion. If particle is moving in a circle of radius R , we have

$$qvB = \frac{mv^2}{R}$$

$$\Rightarrow R = \frac{mv}{qB} \quad \dots (4.100)$$

The angular speed of particle in circular motion is given as

$$\omega = \frac{v}{R} = \frac{qB}{m} \quad \dots (4.101)$$

Time period of revolution of particle in circular motion is given as

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB} \quad \dots (4.102)$$

From equation-(4.101) and (4.102) we can see that the revolution angular speed and time period does not depend upon the projection speed of particle, it only depends upon the specific charge and magnetic induction magnitude in the region.

4.5.5 Projection of a Charge Particle from Outside into a Uniform Magnetic Field in Perpendicular Direction

Figure-4.87 shows a situation in which a charge particle with charge q , mass m and moving at a velocity v is projected toward

a region of magnetic field such that its velocity is perpendicular to the magnetic field as well as its boundary as shown.

As soon as the particle enters the region of magnetic field, it experiences a magnetic force $F = qvB$ of which the direction is given by right hand palm rule shown in figure. As discussed in previous article, the path of motion will be circular with center lying on the line of action of the force acting on it. The particle will follow a circular arc and come out of the field as shown. The radius of the circular arc is given by equation-(4.100).

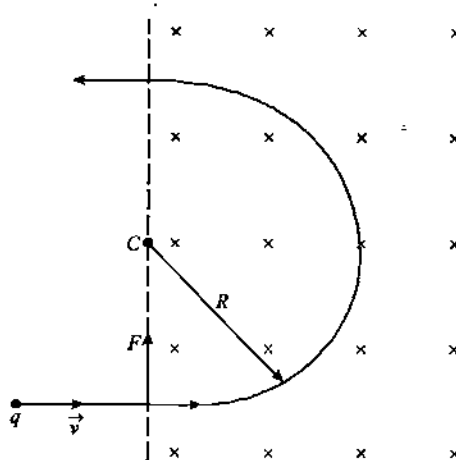
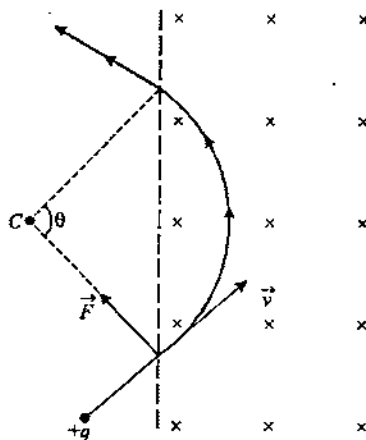


Figure 4.87

There can be many different cases of motion of a charge particle which enters into a region of uniform magnetic field with its velocity perpendicular to the magnetic field direction. In all cases the motion of particle will be a circular arc after it enters into the magnetic field and till it comes out. Some of the cases are shown in figure-4.88.



(a)

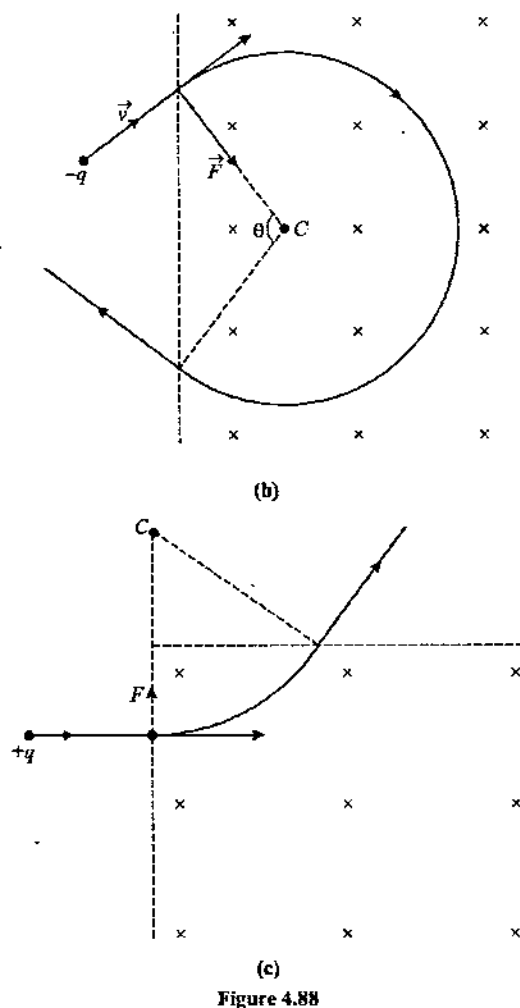


Figure 4.88

In each case when particle enters the magnetic field we can find the direction of magnetic force on particle and somewhere on the line of force we can locate the center of circular trajectory of particle and draw the arc until it comes out of the field.

Always remember that a charge particle which is projected into a magnetic field from outside space will never be able to complete the circle inside the field, it will certainly come out at some point. To complete the circle in magnetic field particle must be projected from a point inside the field as discussed in article-4.5.4 or this is possible if magnetic field vary with time in some cases, discussion of such cases is beyond the scope of this book.

4.5.6 Projection of a Charge Particle in Direction Parallel to the Magnetic Field

When a particle is projected such that its velocity vector is parallel to the direction of magnetic induction then from equation-(4.97) the magnetic force on the particle is zero and hence the trajectory of particle will be a straight line as shown in figure-4.89.

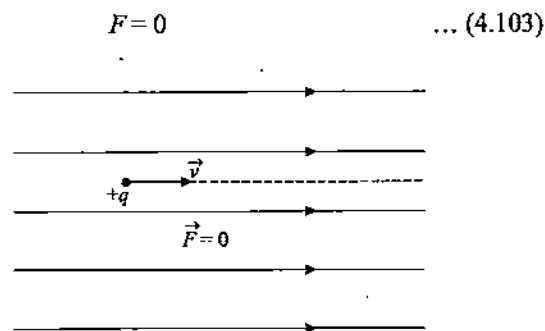


Figure 4.89

4.5.7 Projection of a Charge Particle at Some Angle to the Direction of Magnetic Field

Figure-4.90 shows a particle with charge q and mass m is projected with initial speed v at an angle θ with the direction of magnetic field vector of magnetic induction B . In this case the magnetic force on the charge particle can be calculated by equation-(4.97) which is given as

$$F = qvB \sin\theta \quad \dots (4.104)$$

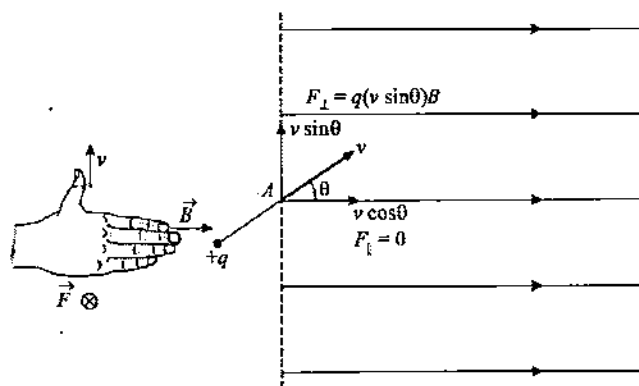


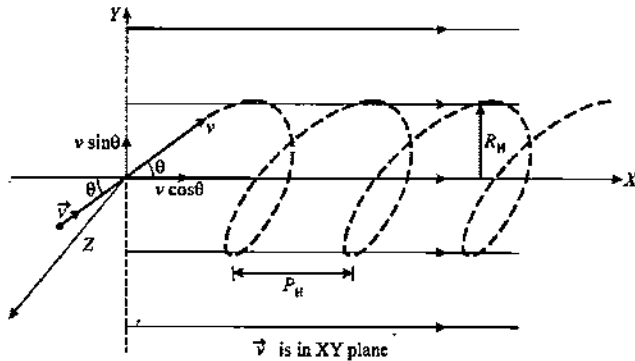
Figure 4.90

At the point A where the particle enters the magnetic field region, we resolve the particle's velocity in two rectangular components along and perpendicular to the magnetic field which are $v \cos\theta$ and $v \sin\theta$.

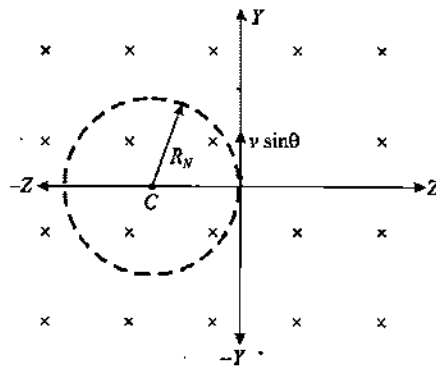
In this case the component $v \cos\theta$ is parallel to the magnetic field direction so it will not experience any magnetic force and due to this component we can consider the motion of charge particle will be straight line.

Other component $v \sin\theta$ is perpendicular to the magnetic field direction on which the magnetic force will be given by equation-(4.104) and the direction of this force is into the plane of paper (inward) which is given by right hand palm rule as

shown in figure-4.90. As the component $v \sin \theta$ is perpendicular to the direction of magnetic induction, the motion due to this component will be circular and effectively if we see the resulting motion of charge particle due to the two velocity components then it will be a helical trajectory as shown in figure-4.91(a).



(a)



(b)

Figure 4.91

Figure-4.91(b) shows the side view of figure-4.91(a) if it is seen from left side. In this view only the component $v \sin \theta$ is seen as other component $v \cos \theta$ is into the plane of paper.

Thus whenever a charge is projected in a magnetic field at some angle, its trajectory is helical as shown above with radius of helical path is given as

$$R_H = \frac{mv \sin \theta}{qB} \quad \dots (4.105)$$

Another characteristic of the helical path is considered as its pitch P_H which is the distance travelled by the particle along the axis of helical path in one revolution as shown in figure-4.91(a) which is given as

$$P_H = v \cos \theta \times T$$

$$\Rightarrow P_H = v \cos \theta \times \frac{2\pi m}{qB}$$

$$\Rightarrow P_H = \frac{2\pi m v \cos \theta}{qB} \quad \dots (4.106)$$

The helical trajectory of the particle is enveloping a cylindrical region of which the cross section is seen in figure-4.91(b). In this view particle is observed in uniform circular motion with speed $v \sin \theta$.

4.5.8 Deflection of a Moving Charge by a Sector of Magnetic Field

Figure-4.92 shows a sector of magnetic field of width d . A charged particle of mass m and charge q is projected toward this sector in perpendicular direction at velocity v . When the charge enters into the magnetic field, it experience a magnetic force $F = qvB$ and follows a circular trajectory of radius given by equation-(4.100) about the center C as shown in figure. If width of the sector is less than its radius then particle will come out of it after getting deflected by an angle of deviation δ as shown in figure.

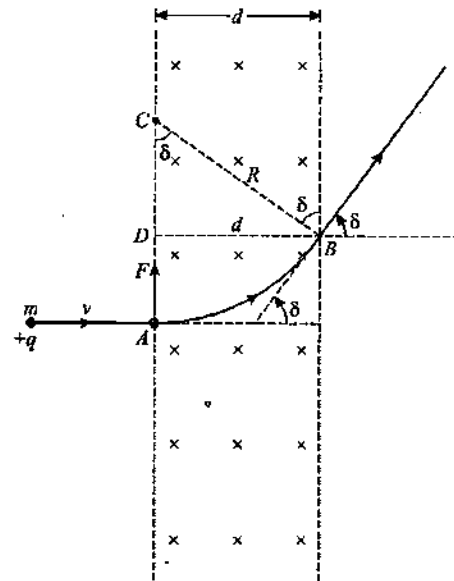


Figure 4.92

From the above figure we can calculate the angle of deviation in direction of motion of particle which is given from $\triangle BCD$ as

$$\sin \delta = \frac{d}{R}$$

$$\Rightarrow \sin \delta = \frac{d}{(mv/qB)}$$

$$\Rightarrow \delta = \sin^{-1} \left(\frac{qBd}{mv} \right) \quad \dots (4.107)$$

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If in above situation the sector width is more than the radius of circular trajectory of the particle then as shown in figure-4.87 we can state that angle of deviation will be 180° and particle will return in the direction opposite to that at which it enters the magnetic field after following a semi-circle in the field.

Illustrative Example 4.19

A charge $4\mu\text{C}$ enters in a region of uniform magnetic field with a velocity $(4\hat{i} - 7\hat{j})$ m/s experiences a force $(5\hat{i} - C\hat{j})$ N. Find the value of C .

Solution

As we know magnetic force is always $\perp$ to velocity of charge particle, given as

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

As $\vec{F}$ is $\perp$ to $\vec{v}$ we use

$$\vec{F} \cdot \vec{v} = 0$$

$$\Rightarrow (5\hat{i} - C\hat{j}) \cdot (4\hat{i} + 7\hat{j}) = 0$$

$$\Rightarrow 20 - 7C = 0$$

$$\Rightarrow C = \frac{20}{7}$$

Illustrative Example 4.20

A charge particle of mass m and charge q is accelerated by a potential difference V volt. It enters in a region of uniform magnetic field B as shown in figure-4.93. Find the time after which it will come out from the magnetic field.

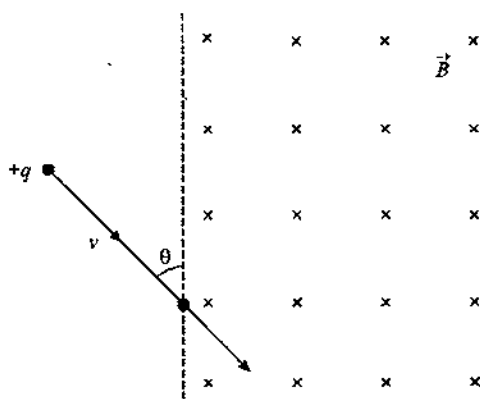


Figure 4.93

Solution

As charge is accelerated at a voltage V , the kinetic energy gained by the particle is written as

$$qV = \frac{1}{2}mv^2$$

As particle's velocity is perpendicular to the applied magnetic field, it follows a circular path with center lying on the line of force which acts on the particle when it enters the field as shown in figure-4.94. The angular speed of particle during its circular motion is given as

$$\omega = \frac{qB}{m}$$

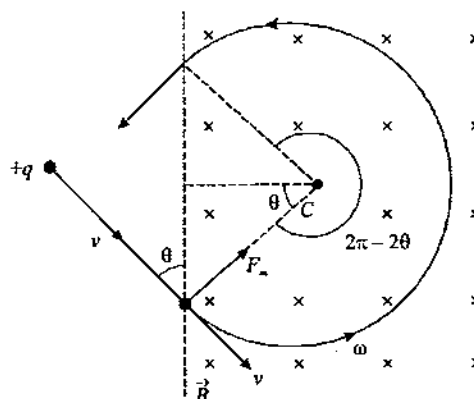


Figure 4.94

From above figure we can see that in the magnetic field the particle follows a circular arc subtending an angle $2\pi - 2\theta$ at C so time spent any particle in magnetic field is given as

$$t = \frac{2\pi - 2\theta}{\omega}$$

$$\Rightarrow t = \frac{2m}{qB}(\pi - \theta)$$

Illustrative Example 4.21

A proton of charge $1.6 \times 10^{-19}\text{C}$ and mass $m = 1.67 \times 10^{-27}\text{kg}$ is shot with a speed $8 \times 10^6\text{ m/s}$ at an angle of 30° with the X -axis. A uniform magnetic field $B = 0.30\text{T}$ exists along the X -axis. Show that path of the proton is a helix. Find the radius and pitch of the helical path followed by the proton.

Solution

The velocity components of the proton are given as

$$v_x = v \cos 30^\circ = (8 \times 10^6)(0.866)$$

$$\Rightarrow v_x = 6.93 \times 10^6\text{ m/s}$$

$$v_y = v \sin 30^\circ = (8 \times 10^6) \times (0.5)$$

$$\Rightarrow v_y = 4.0 \times 10^6 \text{ m/s}$$

The magnetic force due to v_x is given as

$$F = qv_x B \sin 0^\circ = 0$$

The magnetic force due to v_y has no x -component so the motion along x -axis is uniform at speed v_x and due to v_y proton moves in circular path and overall motion will be helical.

The radius of circular transverse motion is given as

$$r = \frac{mv_y}{qB} = \frac{(1.67 \times 10^{-27})(4 \times 10^6)}{(1.6 \times 10^{-19}) \times 0.3} = 0.139 \text{ m}$$

Thus radius of the helix is 0.139 m.

Time taken to complete one circle in transverse motion is

$$T = \frac{2\pi r}{v_y} = \frac{2 \times 3.14 \times 0.139}{4 \times 10^6} = 2.16 \times 10^{-7} \text{ s}$$

The pitch of the helix is the distance travelled by the proton along x -axis in time T which is given as

$$p = v_x \times T$$

$$\Rightarrow p = (6.93 \times 10^6) \times (2.19 \times 10^{-7})$$

$$\Rightarrow p = 1.515 \text{ m}$$

Illustrative Example 4.22

An α -particle is describing a circle of radius 0.45m in a field of magnetic induction of 1.2T. Find its speed, frequency of revolution and kinetic energy. What potential difference will be required which will accelerate the particle so as to give this much of the kinetic energy to it? The mass of α -particle is $6.8 \times 10^{-27} \text{ kg}$ and its charge is twice the charge of electron.

Solution

In case of circular motion in magnetic field we use

$$F = evB = \frac{mv^2}{r}$$

$$\Rightarrow v = \frac{eBr}{m}$$

Substituting the given values, we get

$$v = \frac{3.2 \times 10^{-19} \times 1.2 \times 0.45}{6.8 \times 10^{-27}} = 2.6 \times 10^7 \text{ m/s}$$

The frequency of revolution is given as

$$n = \frac{v}{2\pi r} = \frac{2.6 \times 10^7}{2 \times 3.14 \times 0.45} = 9.2 \times 10^6 \text{ s}^{-1}$$

The kinetic energy of α -particle is given by

$$K = \frac{1}{2}mv^2$$

$$\Rightarrow K = \frac{1}{2} \times 6.8 \times 10^{-27} \times 2.6 \times 10^7 \text{ J}$$

$$\Rightarrow K = 2.3 \times 10^{-12} \text{ J}$$

$$\Rightarrow K = \frac{2.3 \times 10^{-12}}{1.6 \times 10^{-19}} \text{ eV}$$

$$\Rightarrow K = 14 \times 10^6 \text{ eV} = 14 \text{ MeV}$$

An electron will acquire this amount of energy (14MeV) when it is accelerated through a potential difference of $14 \times 10^6 \text{ V}$. Since α -particle carries a charge twice that of an electron it will gain this much amount of kinetic energy when accelerated through half of the potential difference of electron which is $7 \times 10^6 \text{ V}$.

Illustrative Example 4.23

A small ball having mass m and with a charge q is suspended from a rigid support by means of an inextensible string of length l . It is made to revolve on a horizontal circular path in a uniform magnetic field of magnetic induction B which is in vertically upward direction. The time period of revolution of the ball is T . If the thread is always tight, calculate the radius of circular path on which the ball moves.

Solution

The situation described in question is shown in figure-4.95

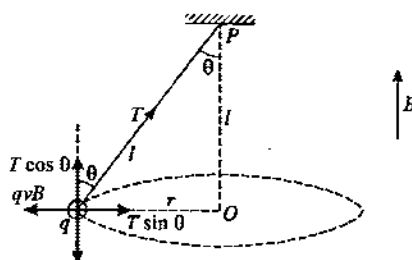


Figure 4.95

For stable circular path we use

$$T \cos \theta = mg \quad \dots (4.108)$$

$$\text{and } T \sin \theta - qvB = \frac{mv^2}{r} \quad \dots (4.109)$$

From equation-(4.108) and (4.109) we have

$$\left(\frac{mg}{\cos \theta}\right) \sin \theta - qBr\omega = mr\omega^2$$

From figure, we have

$$\sin \theta = \frac{r}{l} \text{ and } \cos \theta = \frac{\sqrt{l^2 - r^2}}{l}$$

$$\Rightarrow \frac{mgr}{\sqrt{l^2 - r^2}} - qBr\omega = mr\omega^2$$

$$\Rightarrow \frac{1}{\sqrt{l^2 - r^2}} = \frac{\omega^2}{g} + \frac{qB\omega}{mg}$$

$$\Rightarrow (l^2 - r^2) = \frac{(l/\omega)^2}{[(\omega/g) + (qB/mg)]^2}$$

$$\Rightarrow (l^2 - r^2) = \frac{(T/2\pi)^2}{[(2\pi/gT) + (qB/mg)]^2}$$

$$\Rightarrow r = \left[l^2 - \frac{(T/2\pi)^2}{\{(2\pi/gT) + (qB/mg)\}^2} \right]$$

Illustrative Example 4.24

A particle of mass $m = 1.6 \times 10^{-27}$ kg and charge $q = 1.6 \times 10^{-19}$ C enters a region of uniform magnetic field of strength 1T along the direction as shown in figure-4.96, the speed on the particle is 10^7 m/s. The magnetic field is directed along the inward normal to the plane of the paper. The particle leaves the region of the field at the point F.

(a) Find the distance EF and angle θ .

(b) If the direction of the magnetic field is reversed, find the time spent by the particle in the region of magnetic field after entering at E.

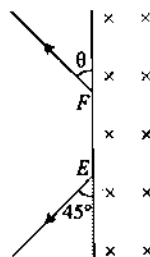


Figure 4.96

Solution

(a) If the direction of magnetic field is directed along the inward normal to the plane of the paper, the path of the particle will be an arc of a circle of radius OE as shown in figure-4.97.

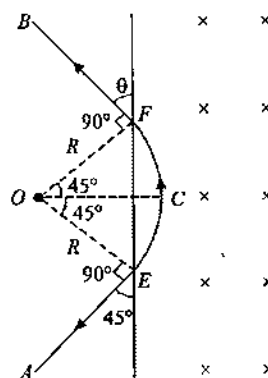


Figure 4.97

From the figure, we can see that $\theta = 45^\circ$ and we have

$$EF = 2R \cos 45^\circ$$

Radius of circular motion is given as

$$R = \frac{mv}{qB} = \frac{1.6 \times 10^{-27} \times 10^7}{1 \times 1.6 \times 10^{-19}} = 0.1 \text{ m}$$

$$\Rightarrow EF = 2 \times 0.1 \times (1/\sqrt{2}) = 0.141 \text{ m}$$

(b) If the direction of the magnetic field is directed along the outward normal to the paper, the path ECH of the particle in the region of magnetic field will be a circular arc in the clockwise direction with radius R as shown in figure-zz and from figure we can see that

$$\angle EO'H = 90^\circ$$

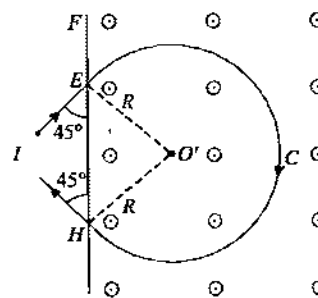


Figure 4.98

Radii OE and O'H are perpendicular at points E and H so the length of arc is given as

$$ECH = 2\pi R - \frac{\pi R}{2} = \frac{3\pi R}{2}$$

Time spent by particle along arc ECH in the magnetic field is given as

$$t = \frac{3\pi R}{2v} = \frac{3 \times 3.14 \times 0.1}{2 \times 10^7} \text{ s}$$

$$\Rightarrow t = 4.71 \times 10^{-8} \text{ s}$$

Illustrative Example 4.25

In a right handed coordinate system XY plane is horizontal and Z -axis is vertically upward. A uniform magnetic field exist in space in vertically upward direction with magnetic induction B . A particle with mass m and charge q is projected from the origin of the coordinate system at $t=0$ with a velocity vector given as

$$\vec{v} = v_1 \hat{i} + v_2 \hat{k}$$

Find the velocity vector of the particle after time t .

Solution

The velocity component v_2 of the particle is along Z -axis and parallel to the magnetic field direction thus it will not experience any force and will remain constant. Due to the other velocity component v_1 which is in positive x -direction, particle will experience a force in negative y -direction and it will follow a circular path with center lying at a point C with position coordinates $(0, -R, 0)$ initially. Due to both velocity components the resulting motion of the particle will be a helical trajectory with axis parallel to Z -axis passing through the point $(0, -R, 0)$. The angular speed of the particle in its helical path is given as

$$\omega = \frac{qB}{m}$$

In a time t the particle will revolve by an angle θ along XY plane which is given as $\theta = \omega t$. Thus at time t after projection of the particle the velocity vector of particle is given as

$$\vec{v} = v_1 \cos \theta \hat{i} - v_1 \sin \theta \hat{j} + v_2 \hat{k}$$

$$\Rightarrow \vec{v} = v_1 \cos \left(\frac{qBt}{m} \right) \hat{i} - v_1 \sin \left(\frac{qBt}{m} \right) \hat{j} + v_2 \hat{k}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Electromagnetic Force

Module Number - 1 to 13

Practice Exercise 4.3

(i) A beam of singly ionized charged particles, having kinetic energy 1000eV contain particles of masses 8×10^{-27} kg and 1.6×10^{-26} kg emerge from one end of an accelerator tube. There is a plate at a distance 0.01m from end of tube placed perpendicular to it. Find minimum magnetic induction in the region so that beam will not strike to plate.

[$\sqrt{2}$ T]

(ii) An electron moving with a velocity 10^8 m/s enters a magnetic field at an angle of 20° to the direction of the field. Calculate

(a) The value of magnetic induction so that the helical path radius will be 2m.

(b) The time required to execute one revolution of the helical path.

(c) The pitch of helical path followed by the electron.

[(a) 9.6×10^{-5} T; (b) 3.69×10^{-7} s; (c) 34.68m]

(iii) A stream of protons and deuterons in a vacuum chamber enters a uniform magnetic field. Both protons and deuterons have been subjected to same accelerating potential, hence the kinetic energies of the particles are the same. If the ion-stream is perpendicular to the magnetic field and the protons move in a circular path of radius 15cm, find the radius of the path traversed by the deuterons. Given that mass of deuteron is twice that of a proton.

[0.212m]

(iv) An electron gun G emits electron of energy 2keV travelling in the positive X -direction. The electrons are required to hit the spot S where $GS = 0.1$ m, and the line GS makes an angle of 60° with the X -axis as shown in figure-4.99 a uniform magnetic field B parallel to GS exists in the region outside the electron gun. Find the minimum value of B needed to make the electrons hit point S .

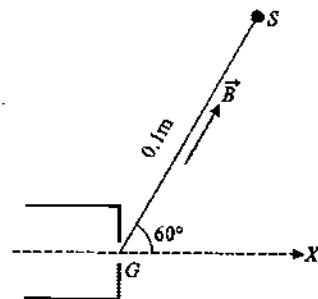


Figure 4.99

[4.73×10^{-3} T]

(v) A 15000V electron is describing a circle in a uniform field of magnetic induction 250G acting at right angle to it. Calculate the radius of the circle.

[1.66cm]

(vi) A slightly divergent beam of charged particles is accelerated by a potential difference V propagates from a point A along the axis of a solenoid. The beam is focussed at a point P at a distance l from A by two successive values of magnetic inductions B_1 and B_2 one after another. Find specific charge of the particles.

$$\left[\frac{8\pi^2 V}{l^2 (B_2 - B_1)^2} \right]$$

(vii) A direct current flowing through the winding of a long cylindrical solenoid of radius R produces a uniform magnetic induction B in it. An electron travelling at velocity v enters into the solenoid along the radial direction between its turn at right angles to the solenoid axis. After a certain time the electron deflected by magnetic field leaves the solenoid. Calculate the time which the electron spends inside the solenoid.

$$\left[\frac{2m}{eB} \tan^{-1} \left(\frac{eBR}{mv} \right) \right]$$

(viii) Two long parallel wires carrying currents 2.5A and I in the same direction directed into the plane of the paper are held at points P and Q as shown in figure-4.100 such that they are perpendicular to the plane of paper. The points P and Q are located at a distance of 5m and 2m respectively from a collinear point R as shown. An electron moving with a velocity of 4×10^5 m/s along the positive x -direction experiences a force of magnitude 3.2×10^{-20} N at the point R . Find the value of I .

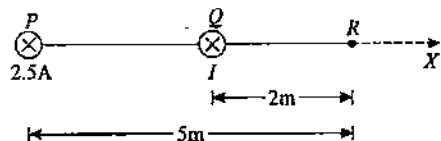


Figure 4.100

[4A]

(ix) In a right handed coordinate system XY plane is horizontal and Z -axis is vertically upward. A uniform magnetic field exist in space in vertically upward direction with magnetic induction B . A particle with mass m and charge q is projected from the origin of the coordinate system at $t = 0$ with a velocity vector given as

$$\vec{v} = v_1 \hat{i} + v_2 \hat{k}$$

Find the position coordinates of the particle after time t .

$$\left[\frac{mv_1}{qB} \sin \left(\frac{qBt}{m} \right), -\frac{mv_1}{qB} \left(1 - \cos \left(\frac{qBt}{m} \right) \right), v_2 t \right]$$

(x) Inside a cylindrical capacitor of inner radius a and outer radius b , an electron is projected from the surface of inner cylindrical shell perpendicular to it with an initial velocity. In the annular region between the two cylindrical shells a uniform magnetic induction B exist in direction parallel to the axis of capacitor. Find the maximum initial velocity with which the electron is to be projected so that it will not hit the outer shell.

$$\left[\frac{eB(b^2 - a^2)}{2mb} \right]$$

4.6 Magnetic Force on Current Carrying Conductors

In previous articles we've studied that a moving charge in magnetic field experiences a force given by equation-(4.97). A current carrying conductor contains free electrons continuously flowing at drift speed. So if a current carrying conductor is placed in a magnetic field then all its free electrons will experience same magnetic force and due to this, overall the conductor will experience a net magnetic force which is the sum of magnetic forces on all its moving free electrons.

There are many cases under which we'll study the force experienced by current carrying conductors. This force may cause translational or rotational motion of the conductors and will affect the overall motion as well. In upcoming articles and illustrations we will extend our knowledge of magnetic force for current carrying conductors.

4.6.1 Force on a Wire in Uniform Magnetic Field

Figure-4.46 shows a current carrying wire XY with current I placed in a region of uniform magnetic induction B . If the free electrons inside the conductor are moving at drift speed v_d then each free electron will experience a magnetic force $F_m = ev_d B$.

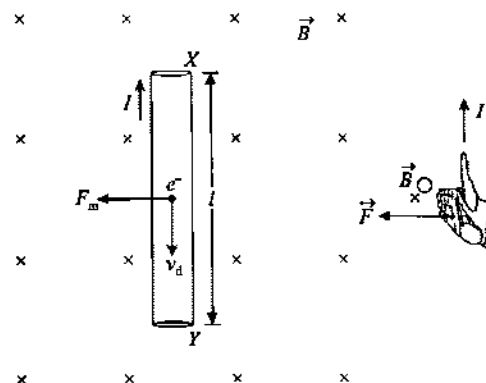


Figure 4.101

The direction of these forces on all free electron is given by right hand palm rule which is toward left as shown in figure. For applying the right hand palm rule in case of current carrying conductors we point thumb in the direction of current flow which is the flow of positive charges in the conductor as shown in figure-4.101.

If n be the free electron density of the conductor, S is the cross sectional area of this conductor then total number of free electrons in this wire XY are nIS . Thus total magnetic force on wire XY is given as

$$F = ev_d B \times nIS \quad \dots (4.110)$$

In above equation-(4.110) nev_d is the current density and product with the cross sectional area S gives the current so it can be rewritten as

$$F = BIl \quad \dots (4.111)$$

If the direction of force is expressed in form of cross product then we use the vectors for length of conductor $\vec{l}$ and that for magnetic field $\vec{B}$ and the magnetic force vector on current carrying conductor can be given as

$$\vec{F} = I(\vec{l} \times \vec{B}) \quad \dots (4.112)$$

Thus if a wire is placed in a uniform magnetic field making an angle θ with the direction of magnetic field as shown in figure-4.102 then the magnetic force on this wire is given by equation-(4.112) as

$$F = BIl \sin \theta \quad \dots (4.113)$$

The direction of this force can be given by cross product of equation-(4.112) which is into the plane of paper as shown.

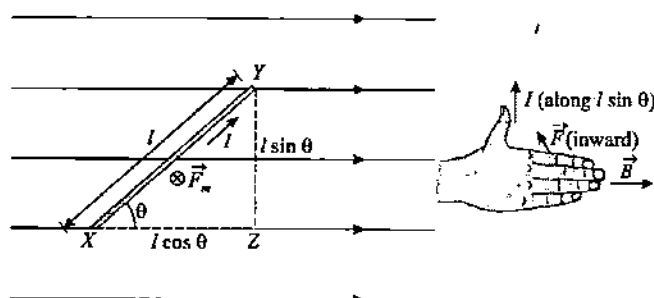


Figure 4.102

In the figure-4.102 magnitude of force on the wire can also be given by resolving its length into two components along and perpendicular to the direction of magnetic field. Here we can see that $l \cos \theta$ is parallel to magnetic field so a current or a moving charge which is parallel to magnetic field will not experience any force as discussed in article-4.5.6. Other component $l \sin \theta$ is perpendicular to magnetic field similar to

the case shown in figure-4.101 thus the force on it is given by equation-(4.111) replacing l to $l \sin \theta$ which gives equation-(4.113).

4.6.2 Direction of Magnetic Force on Currents in Magnetic Field

As discussed in previous article about direction of magnetic force that it can be given by cross product in equation-(4.112). Same can be given by right hand palm rule as already discussed for force on moving charges in magnetic field. In case of current carrying conductor we need to point right hand thumb along the direction of current which denotes the flow of positive charges. Figure-4.103 again illustrates the application of right hand palm rule for force on a current carrying conductor XY placed in a magnetic field. In this case the direction of magnetic force on conductor XY is into the plane of paper (inward).

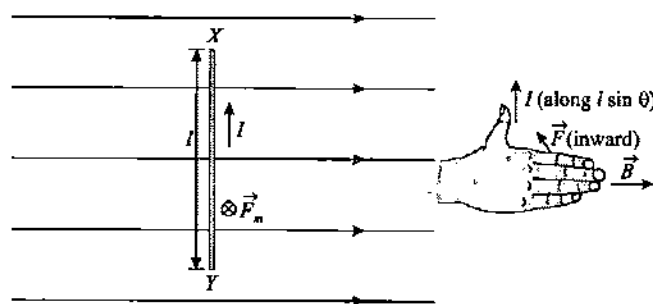


Figure 4.103

Another way of determining the direction of magnetic force on current carrying conductors in magnetic field is by 'Fleming's Left Hand Rule' stated below

"To determine the direction of magnetic force stretch your index finger, thumb and middle finger of left hand in mutually perpendicular directions as shown in figure-4.104 and orient your hand such that your index finger points in the direction of magnetic field and middle finger along the current then your left hand thumb will give you the direction of magnetic force on the conductor."

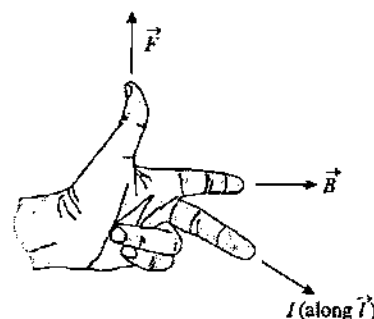


Figure 4.104

The situation shown in Figure-4.103 is redrawn again in figure-4.105 in which we can verify the direction of magnetic force on wire XY by using Fleming's left hand rule.

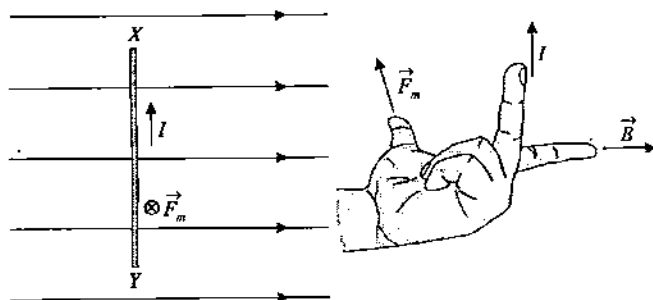


Figure 4.105

While applying Fleming's left hand rule if sometimes current is not flowing in conductor at right angle to the direction of magnetic field then resolve the components of the length of the conductor and point middle finger in the direction of the length component normal to magnetic field.

Students can also use Fleming's left hand rule to determine the direction of magnetic force on freely moving charges in magnetic field which is already discussed in previous articles.

4.6.3 Force Between Two Parallel Current Carrying Wires

Figure-4.106 shows two parallel wires at a separation r and carrying currents I_1 and I_2 in same direction. First wire is considered to be very long and second wire XY is of length l . The magnetic induction due to first wire at the location of second wire is in inward direction which is given as

$$B_1 = \frac{\mu_0 I_1}{2\pi r} \quad \dots (4.114)$$

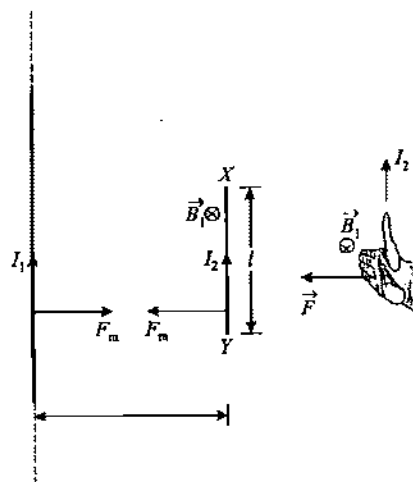


Figure 4.106

The force on second wire due to the magnetic field of first wire at its location is given by equation-(4.111) as

$$F = B_1 I_2 l$$

$$F = \left(\frac{\mu_0 I_1}{2\pi r} \right) I_2 l$$

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi r} \quad \dots (4.115)$$

Above equation-(4.115) gives the force of interaction between two parallel current carrying wires provided one wire is very long. The direction of magnetic force on wire XY can be given by right hand palm rule as shown in figure-4.106 which is toward left. According to Newton's third law the force by second wire on first will be opposite i.e. in rightward direction. Thus the two wires will attract each other when they carry currents in same direction. If the direction of currents in the two wires is opposite then they will repel each other which can be easily proven by using right hand palm rule, cross product or by using Fleming's left hand rule. Thus in general about force between two parallel current carrying wires we state that

"Two parallel wires carrying currents in same direction attract each other and those carrying currents in opposite direction repel each other because of magnetic force between them."

4.6.4 Magnetic Force on a Random Shaped Current Carrying Wire in Uniform Magnetic Field

Figure-4.107 shows a random shaped wire AB carrying a current I placed in magnetic induction B . The distance between end points of wire AB is l . To analyze the magnetic force on wire we consider coordinate axes as shown in figure with x -axis along AB and y -axis perpendicular to AB .

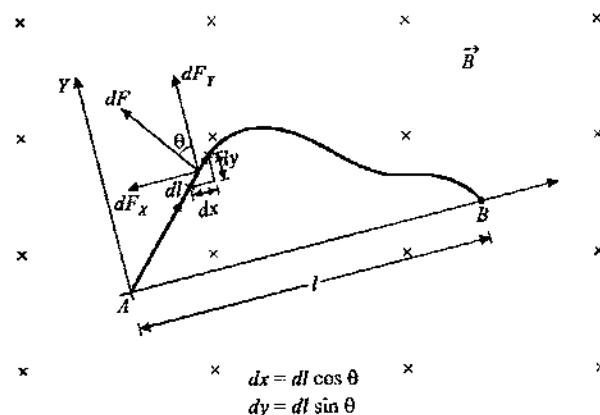


Figure 4.107

If dF is the magnetic force on an element of length dl on the

wire then it is given as $dF = BIdl$ and direction of the force on dl is shown in the figure-4.107 which is given by right hand palm rule. Now we resolve this force in two components along x and y axis as dF_x and dF_y which are given as

$$dF_x = BIdl \sin \theta = BIdy \quad \dots (4.116)$$

and
$$dF_y = BIdl \cos \theta = BIdx \quad \dots (4.117)$$

If we calculate the total force on wire AB by integrating above components, we have

$$F_x = \int dF_x = \int BIdy = BI \int_0^0 dy = 0 \quad \dots (4.118)$$

and
$$F_y = \int dF_y = \int BIdx = BI \int_0^0 dx = BI \quad \dots (4.119)$$

Thus total force on wire is along y -axis i.e. in direction perpendicular to the line joining the ends of wire AB and it is given as

$$F_{\text{wire}} = BI \quad \dots (4.120)$$

Above expression in equation-(4.120) shows that the net magnetic force on a random shaped current carrying wire which is equal to the force acting on a straight wire carrying same current and which has length equal to the line joining the end points of the random shaped wire.

If in this case we join terminals A and B of the wire and make a closed loop then $I = 0$ so $F = 0$. Thus we can also state that if there is a current carrying closed loop or coil then net magnetic force on the coil when placed in uniform magnetic field is zero. About current carrying coils or closed loops we will discuss in upcoming articles.

Illustrative Example 4.26

Two parallel horizontal conductors are suspended by light vertical threads 75cm long. Each conductor has a mass of 40g per metre, and when there is no current they are 0.5cm apart. Equal current in the two wires result in a separation of 1.5cm. Find the values and directions of currents.

Solution

The situation described in question is shown in the cross sectional view of the current carrying conductors in figure-4.108.

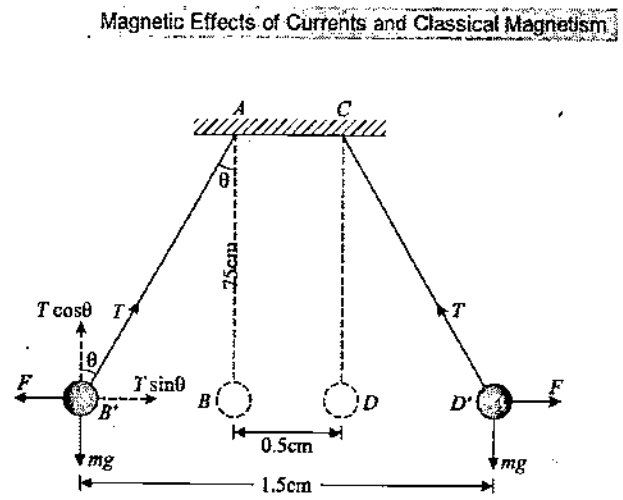


Figure 4.108

For equilibrium of conductors, we have

$$T \cos \theta = mg \quad \dots (4.121)$$

$$T \sin \theta = F = \frac{\mu_0 I^2 l}{2\pi d} \quad \dots (4.122)$$

From equations-(4.121) and (4.122) we have

$$\tan \theta = \frac{\mu_0 I^2 l}{2\pi d mg} \quad \dots (4.123)$$

When θ is small we use $\tan \theta \approx \sin \theta$ thus from figure we have

$$\sin \theta = \frac{0.5 \times 10^{-2}}{75 \times 10^{-2}} = \frac{1}{150}$$

Mass of each conductor of length l is given as

$$m = 40.0 \times 10^{-3} l \text{ kg}$$

Substituting the values in equation-(4.123), we get

$$\frac{1}{150} = \frac{2 \times 10^{-7} \times I^2}{1.5 \times 10^{-2} \times 40 \times 10^{-3} \times 10}$$

$$\Rightarrow I = 14.14 \text{ A}$$

As the two conductors are repelling each other, the currents are in opposite directions.

Illustrative Example 4.27

Two long straight parallel wires are 2m apart, perpendicular to the plane of paper as shown in figure-4.109. The wire A carries a current of 9.6A, directed into the plane of the paper. The wire B carries a current such that the magnetic induction at the point P , at the distance of 10/11m from the wire B is zero. Calculate

(a) The magnitude and direction of the current in B .

- (b) The magnitude of the magnetic induction at the point S .
 (c) The force per unit length on the wire B .

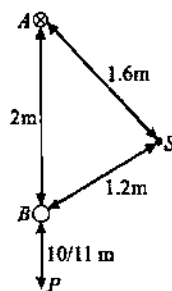


Figure 4.109

Solution

(a) As the magnetic field of induction at point P is zero, for this the magnetic induction at P produced by current carrying wires should be equal and opposite so the direction of current in B must be opposite to that of A directed upward and perpendicular to the plane of the paper.

Magnetic induction at P due to wire A is given as

$$B_1 = \frac{\mu_0 i_1}{2\pi r_1}$$

If current in wire B is taken as i_2 then magnetic induction at P due to wire B is given as

$$B_2 = \frac{\mu_0 i_2}{2\pi r_2}$$

As net magnetic induction at point P is zero, we have

$$B_1 = B_2$$

$$\Rightarrow \frac{\mu_0 i_1}{2\pi r_1} = \frac{\mu_0 i_2}{2\pi r_2}$$

$$\Rightarrow i_2 = \frac{r_2}{r_1} \times i_1$$

$$\text{We have } r_1 = AP = 2 + \frac{10}{11} = \frac{32}{11} \text{ m}$$

$$\text{and } r_2 = BP = \frac{10}{11} \text{ m}$$

$$\Rightarrow i_2 = \frac{(10/11)}{(32/11)} \times 9.6 = 3 \text{ A}$$

Thus the wire B carries a current 3A directed upward and perpendicular to the plane of paper.

- (b) The magnetic field produced by wire A at S

$$B_3 = \frac{\mu_0 i_1}{2\pi r_1'} = \frac{2 \times 10^{-7} \times 9.6}{1.6}$$

$$\Rightarrow B_3 = 12 \times 10^{-7} \text{ T}$$

The magnetic field produced by wire B at S

$$B_4 = \frac{\mu_0 i_2}{2\pi r_2'} = \frac{2 \times 10^{-7} \times 3}{1.2}$$

$$\Rightarrow B_4 = 5 \times 10^{-7} \text{ T}$$

According to figure, $\angle ASB = 90^\circ$ the distances AB , AS and BS are related as

$$(AB)^2 = (AS)^2 + (BS)^2$$

Thus net magnetic induction at point S is given as

$$\Rightarrow B_S = \sqrt{B_3^2 + B_4^2}$$

$$\Rightarrow B_S = \sqrt{[(12 \times 10^{-7})^2 + (5 \times 10^{-7})^2]}$$

$$\Rightarrow B_S = 1.3 \times 10^{-6} \text{ T}$$

- (c) Force per unit length on wire B is given as

$$F = \frac{\mu_0 i_1 i_2}{2\pi r_{AB}} = \frac{2 \times 10^{-7} \times 9.6 \times 3}{2}$$

$$\Rightarrow F = 2.88 \times 10^{-6} \text{ N/m}$$

Illustrative Example 4.28

A rod AB of mass m and length l is placed on two smooth rails P and Q in a uniform magnetic induction B as shown in figure-4.110. The rails are connected to a current source supplying a constant current I . Find the speed attained by rod AB when it leaves off the other end of rails of length L .

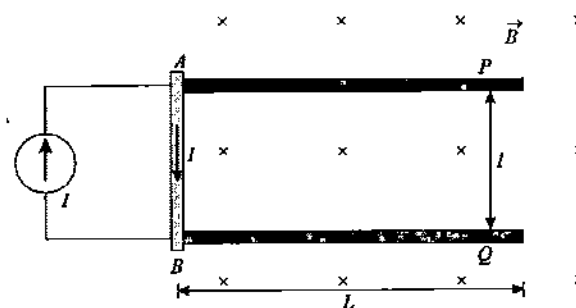


Figure 4.110

Solution

Due to magnetic force acceleration of rod AB when it slides on the rails is given as

$$a = \frac{BIl}{m}$$

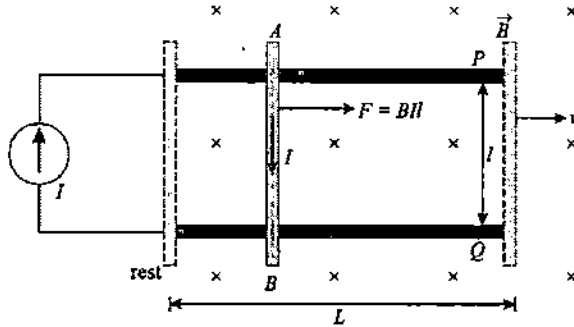


Figure 4.111

As shown in figure-4.111 after travelling a distance L on rails speed attained by rod v is given as

$$\begin{aligned} v^2 &= d + 2aL \\ \Rightarrow v &= \sqrt{2aL} \\ \Rightarrow v &= \sqrt{\frac{2BIL}{m}} \end{aligned}$$

Illustrative Example 4.29

Two long parallel wires of negligible resistance are connected at one end to a resistance R and at the other end to a constant voltage source of voltage V . The distance between the axes of the wires is η times greater than the cross-sectional radius of each wire. At what value of resistance R , does the resultant force of interaction between the wires will become zero?

Solution

Figure-4.112 shows the situation described in the question. The two wires form a wire capacitor of which capacitance is given as

$$C = \frac{\pi\epsilon_0 l}{\ln \eta}$$

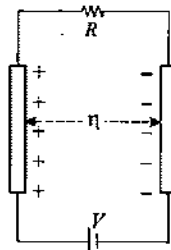


Figure 4.112

Charge on the capacitor in steady state is given as

$$q = CV = \frac{\pi\epsilon_0 l V}{\ln \eta} \quad \dots (4.124)$$

Due to voltage source a constant current flows in the wires which repel each other by a magnetic force which is nullified by the force of attraction on the two wires due to opposite charges on the two wires as shown in figure. The force of attraction is given as

$$F_1 = qE = q \left(\frac{q}{2\pi\epsilon_0 l (\eta r)} \right) \quad \dots (4.125)$$

Substituting the value of q from equation-(4.124) into (4.125),

$$F_1 = \frac{\left(\frac{\pi\epsilon_0 l V}{\ln \eta} \right)^2}{2\pi\epsilon_0 l (\eta r)} \quad \dots (4.126)$$

The current flow is in the opposite direction in the two wires so the net force of repulsion between these wires is given as

$$F_2 = \frac{\mu_0 I^2 l}{2\pi(\eta r)} = \frac{\mu_0 (V/R)^2 l}{2\pi\eta r} \quad \dots (4.127)$$

As the net force between the two wires is zero we use

$$\begin{aligned} \frac{\mu_0 (V/R)^2 l}{2\pi\eta r} &= \frac{\left(\frac{\pi\epsilon_0 l V}{\ln \eta} \right)^2}{2\pi\epsilon_0 l (\eta r)} \\ \Rightarrow R &= \sqrt{\frac{\mu_0}{\epsilon_0} \frac{\ln \eta}{\pi}} \end{aligned}$$

Illustrative Example 4.30

A rectangular loop of wire $ABCD$ is oriented with the left corner at the origin, one edge along X -axis and the other edge along Y -axis as shown in the figure-4.113. A magnetic field exist in space in direction perpendicular to the XY plane as shown and has a magnitude that is given as $B = \alpha y$ where α is a constant. Find the total magnetic force on the loop if it carries current i .

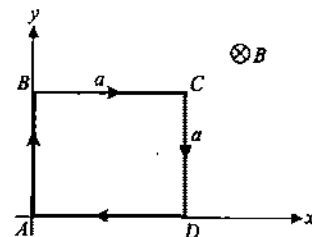


Figure 4.113

Solution

Due to the magnetic induction the four segments of the wire loop experience the forces as shown in figure-4.114. On the segment DA as $y = 0$ the magnetic induction is also zero so no force will act so we have

$$F_{AD} = 0$$

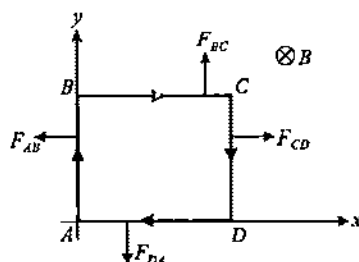


Figure 4.114

At the location of wire BC we have $y = a$ at which we have $B = \alpha a$ so the force on this segment is given as

$$F_{BC} = BIl = (\alpha a)il = \alpha ia^2$$

For the remaining two segments AB and CD the forces are opposite and equal as for every element on segment AB at a position y there is a corresponding element on CD at same y for which the force is equal and opposite so the two forces on these segments will cancel each other so total force on the wire loop is along y direction and it is given as

$$\vec{F} = \alpha ia^2 \hat{j}$$

Illustrative Example 4.31

An infinite wire, placed along z -axis, has current I_1 in positive z -direction. A conducting rod PQ placed in xy plane parallel to y -axis has current I_2 in positive y -direction. The ends of the rod subtend angles 30° and 60° at the origin with positive x -direction as shown in figure-4.115. The rod is at a distance a from the origin. Find net force on the rod.

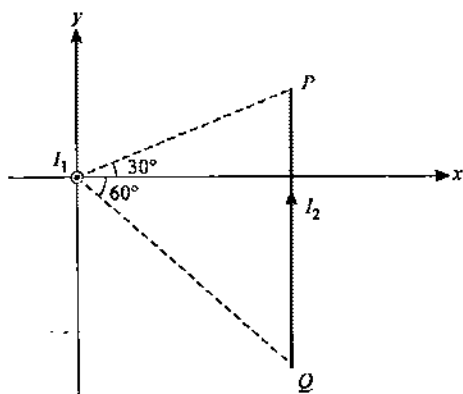


Figure 4.115

Solution

To calculate the force on rod PQ we consider an element of width dy at a distance y from the x -axis as shown in figure-4.116. The magnetic induction at the location of this element due to the wire at origin is given as

$$B = \frac{\mu_0 I_1}{2\pi\sqrt{y^2 + a^2}}$$

The force on element due to the magnetic induction of wire at origin is given as

$$dF = BI_2 dy \sin \theta$$

Net force on the rod PQ is given as

$$F = \int dF = \int BI_2 dy \sin \theta$$

$$\Rightarrow F = \int_{-a \tan 60^\circ}^{+a \tan 30^\circ} \left(\frac{\mu_0 I_1}{2\pi\sqrt{a^2 + y^2}} \right) I_2 dy \left(\frac{y}{\sqrt{a^2 + y^2}} \right)$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2}{2\pi} \int_{-\sqrt{3}a}^{+a/\sqrt{3}} \frac{y dy}{a^2 + y^2}$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2}{2\pi} \left[\frac{1}{2} \ln(a^2 + y^2) \right]_{-\sqrt{3}a}^{+a/\sqrt{3}}$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2}{4\pi} \left[\ln\left(\frac{4}{3}a^2\right) - \ln(4a^2) \right]$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2}{4\pi} \ln 3$$

In above integration we ignore a negative sign as we are calculating the magnitude of force only and the direction is given by right hand palm rule.

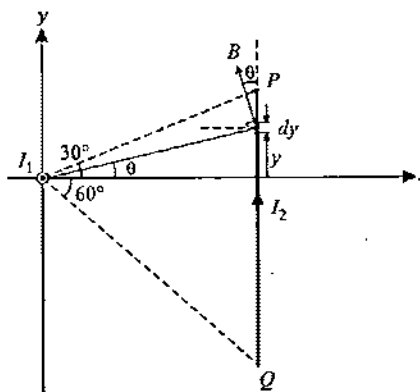


Figure 4.116

Illustrative Example 4.32

Figure-4.117 shows a long straight current carrying wire with a current I_1 and a thin strip of width b and length l placed parallel to it with a current I_2 as shown in figure. Find the magnetic force of interaction between current I_1 and I_2 .

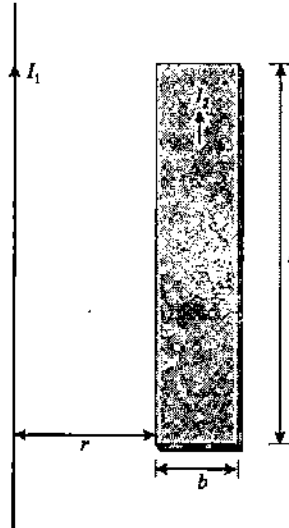


Figure 4.117

Solution

We consider an elemental strip of width dx in the strip at a distance x from the wire as shown in figure-4.118. The current in the elemental strip is given as

$$dI = \frac{I_2}{b} dx$$

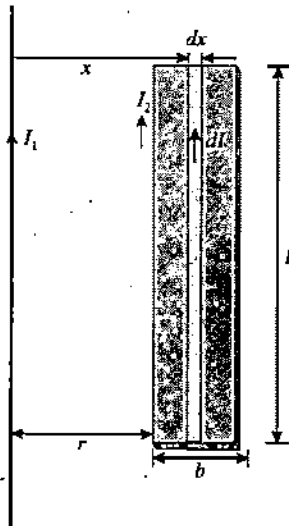


Figure 4.118

The force between I_1 and dI is given as

$$dF = \frac{\mu_0 I_1 dI l}{2\pi x}$$

Total force on the strip can be calculated by integrating the above force within limits from $x = l$ to $x = l + b$ which is given as

$$F = \int dF = \frac{\mu_0 I_1 I_2 l}{2\pi b} \int_l^{l+b} \frac{dx}{x}$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2 l}{2\pi b} [\ln x]_l^{l+b}$$

$$\Rightarrow F = \frac{\mu_0 I_1 I_2 l}{2\pi b} \ln \left(\frac{l+b}{l} \right)$$

Illustrative Example 4.33

A copper wire with density ρ with cross-sectional area S bent to make three sides of a square frame which can turn about a horizontal axis OO' as shown in figure-4.119. The wire is located in uniform vertical magnetic field. Find the magnetic induction if on passing a current I through the wire the frame deflects by an angle θ in its equilibrium position.

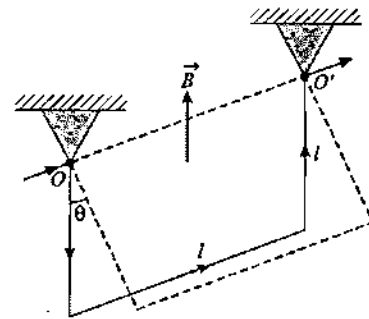


Figure 4.119

Solution

We consider the side of square wire frame is a . The horizontal wire of the frame will experience a rightward force BIl and due to this force it experiences a torque about the axis OO' and the frame will tilt. Torque on frame about OO' is given as

$$\tau = BIl \times a \cos \theta = BIl^2 \cos \theta$$

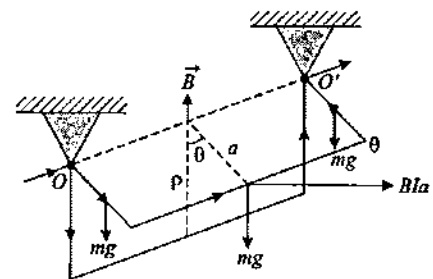


Figure 4.120

This is the deflecting torque and at equilibrium this torque is

balanced by the torque on the frame due the weight mg of the three sides of the square acting at centre of the wires as shown in figure-4.120 so at equilibrium balancing of torques gives

$$Bla^2 \cos \theta = \frac{Mg}{3} \times a \sin \theta + 2 \left(\frac{Mg}{3} \right) \times \left(\frac{a}{2} \right) \sin \theta$$

$$\Rightarrow Bla^2 \cos \theta = \frac{2}{3} Mga \sin \theta$$

$$\Rightarrow B = \frac{2Mg}{3Ia} \tan \theta$$

Mass of square wire frame can be written in terms of its density, length and cross sectional area as

$$M = 3aSp$$

$$\Rightarrow B = \frac{2(3aSp)g}{3Ia} \tan \theta$$

$$\Rightarrow B = \frac{2aSp\rho g}{Ia} \tan \theta$$

4.7 Motion of a Charged Particle in Electromagnetic Field

In previous articles we've studied the magnetic force on a moving charge and a current carrying conductor in magnetic field. There are many cases in which both electric and magnetic forces simultaneously act on a moving charge particle. We've discussed that the total electromagnetic force acting on a moving charge remain constant when observed from different frames of reference whereas electric and magnetic forces may be observed differently from different frames as these two forces are different ways to analyze a single force called electromagnetic interaction. Right now we will restrict our mathematical analysis of the two forces in ground frame only.

4.7.1 Undeflected motion of a charge particle in Electric and Magnetic Fields

When a charge particle is moving in a straight line in a gravity free region where both electric and magnetic fields are present then it is possible under two conditions. First condition is if electric and magnetic forces on the charge particle balance each other and act in opposite direction with equal magnitudes and second condition is when both electric and magnetic fields are in same direction and charge is also moving in that direction. In first condition the charge is in uniform motion whereas in second condition due to electric force charge would be accelerating.

Figure-4.121 shows such a situation in which a charge q is

moving in a region where mutually perpendicular electric and magnetic fields are present and charge is moving in direction perpendicular to both the fields.

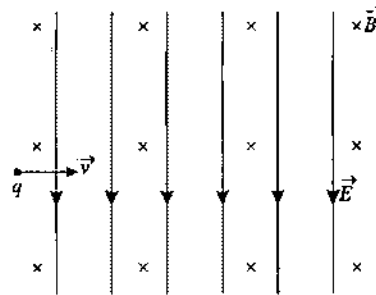


Figure 4.121

In the situation shown in figure-4.121 electric force on the charge will act in downward direction along the direction of electric field and the electric force on it is given as

$$F_e = qE$$

The magnetic force on the charge will act in upward direction which can be given by right hand palm rule. The magnitude of the magnetic force on the charge particle is given as

$$F_m = qvB$$

If charge is moving undeflected in a straight line then the two forces must balance each other and condition for undeflected motion of particle is given as

$$F_e = F_m$$

$$\Rightarrow qE = qvB$$

$$\Rightarrow v = \frac{E}{B} \quad \dots (4.128)$$

4.7.2 Cycloidal Motion of a Charge Particle in Electromagnetic Field

Figure-4.122 shows a coordinate system in which an electric field E exist along positive y -direction and a magnetic field B exist along positive z -direction. At the origin of coordinate system a point charge of mass m and charge q is placed and it is released from rest. We will analyze the motion of this charge under electromagnetic forces acting on it.

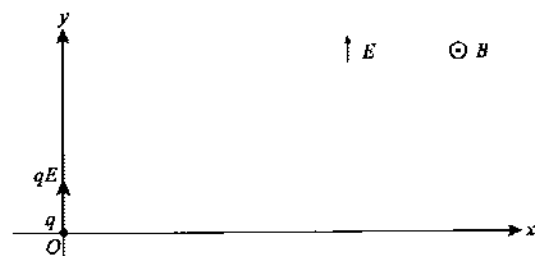


Figure 4.122

Just after release of charge particle it experiences only an electric force qE in positive y -direction. Due to this force as charge starts moving and as it gains some velocity it starts experiencing a magnetic force in positive x -direction. The direction of magnetic force can be verified using right hand palm rule. Due to the magnetic force the direction of motion of the charge particle changes and it moves along a curved path as shown in figure-4.123. If we consider the charge particle at a general point $P(x, y)$ then forces acting on it are shown in the figure.

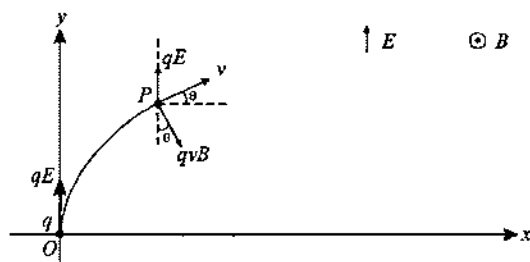


Figure 4.123

The velocity components of the particle at point P are given as

$$v_x = v \cos \theta$$

and

$$v_y = v \sin \theta$$

During motion of particle at any point the particle gains velocity only due to the work done by electric force as magnetic force does not do any work and will not cause any gain in speed of particle so at any point $P(x, y)$ the kinetic energy of particle is given as

$$qEy = \frac{1}{2}mv^2 \quad \dots (4.129)$$

If a_x and a_y are the accelerations of the particle along x and y -directions then net forces acting on the particle along x and y -directions are given as

$$F_x = ma_x = qvB \sin \theta$$

$$\text{and } F_y = ma_y = qE - qvB \cos \theta$$

$$\Rightarrow a_x = \frac{dv_x}{dt} = \frac{qvB \sin \theta}{m} = \frac{qv_y B}{m} \quad \dots (4.130)$$

$$\Rightarrow a_y = \frac{dv_y}{dt} = \frac{qE - qvB \cos \theta}{m} = \frac{qE - qv_x B}{m} \quad \dots (4.131)$$

Dividing equations-(4.130) and (4.131) gives

$$\frac{dv_x}{dv_y} = \frac{qv_y B}{qE - qv_x B}$$

$$\Rightarrow (E - v_x B) dv_x = v_y B dv_y$$

Integrating the above expression within limits from origin of coordinate system to point P gives

$$\int_0^{v_x} (E - v_x B) dv_x = B \int_0^{v_y} v_y dv_y$$

$$\Rightarrow Ev_x - \frac{1}{2}Bv_x^2 = \frac{1}{2}Bv_y^2$$

$$\Rightarrow Ev_x = \frac{1}{2}Bv^2 \quad (\text{As } v^2 = v_x^2 + v_y^2) \dots (4.132)$$

Above equation-(4.131) relates the net velocity of particle with its x -component. If we substitute the value of v_x from equation-(4.130) to equation-(4.129) we get

$$\frac{d}{dt} \left(\frac{E}{B} - \frac{m}{qB} \frac{dv_y}{dt} \right) = \frac{qv_y B}{m}$$

$$\Rightarrow \frac{d^2 v_y}{dt^2} + \left(\frac{qB}{m} \right)^2 v_y = 0 \quad \dots (4.133)$$

The expression in equation-(4.133) is the basic differential equation of SHM in v_y which indicates that the velocity component of particle in y -direction is executing SHM hence its y -coordinate will also execute SHM so the motion of particle will be oscillating in y -direction and progressive in x -direction as shown in figure-4.124. Even the x -direction motion of particle is also oscillating with moving mean position which can be verified by combining equations-(4.130) and (4.131) by eliminating v_y . This trajectory of particle in xy plane comes out to be cycloidal path which is similar to the path traced by a particle on rim of body in pure rolling on a surface. In-depth analysis of cycloidal motion is not in our scope so we will restrict our analysis upto the motion of charge particle under electromagnetic fields. Using equation-(4.132) and previous relations in the given parameters the time of periodic motion, oscillation amplitude (maximum height along y -direction), range in x -direction and other parameters can be calculated.

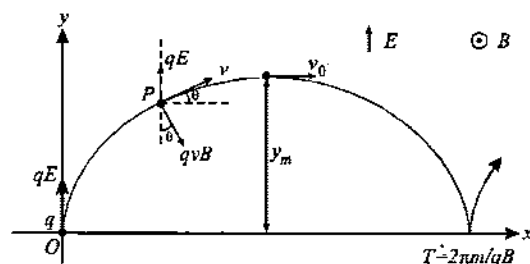


Figure 4.124

In y -direction motion of the particle, mean position is at a height where the net force on particle in y -direction becomes zero which happens at $y = y_m/2$ which would be the amplitude

of oscillations of particle in y -direction. At the topmost point of the trajectory if v_0 is the velocity of the particle then it is given by equation-(4.129) as

$$qEy_m = \frac{1}{2}mv_0^2 \quad \dots (4.134)$$

At the topmost point as $v_x = 0$, at this point net downward force on particle must be equal to the net upward force when the particle was at origin so we have

$$\begin{aligned} qE &= qv_0B - qE \\ \Rightarrow v_0 &= \frac{2E}{B} \quad \dots (4.135) \end{aligned}$$

From equation-(4.134) and (4.135) we get

$$\begin{aligned} qEy_m &= \frac{1}{2}m\left(\frac{2E}{B}\right)^2 \\ \Rightarrow y_m &= \frac{2mE}{qB} \quad \dots (4.136) \end{aligned}$$

Illustrative Example 4.34

A proton beam is fired in space where it passes without deviation through the region where there are uniform transverse mutually perpendicular electric and magnetic fields exist with electric field strength E and magnetic induction B . The beam strikes a fixed solid nonmetallic target. Find the force exerted by the beam on the target if the beam current is i . Take mass of proton m_p .

Solution

As the proton beam passes without deviation, so we use

$$\begin{aligned} evB &= eE \\ \Rightarrow v &= \frac{E}{B} \end{aligned}$$

Number of protons striking the target per second is given by using beam current as

$$n = \frac{i}{e}$$

Total mass of protons striking the target per second are

$$m = m_p \times \frac{i}{e}$$

Net force on the target is the total momentum imparted to the target per second which is given as

$$\begin{aligned} F &= mv \\ \Rightarrow F &= \frac{m_p i E}{e B} \end{aligned}$$

Illustrative Example 4.35

A particle of mass 1×10^{-26} kg and charge $+1.6 \times 10^{-19}$ C travelling with a velocity 1.28×10^6 m/s along positive x -direction in a region where uniform electric field E and a uniform magnetic field of induction B are present such that $E_x = E_y = 0$ and $E_z = 102.4$ kV/m and $B_x = B_z = 0$ and $B_y = 8 \times 10^{-2}$ T. The particle enters in this region at the origin of coordinate system at time $t = 0$. Determine the location x , y and z coordinates of the particle at $t = 5 \times 10^{-6}$ s. If only the electric field is switched off at this instant, what will be the position of the particle at $t = 7.45 \times 10^{-6}$ s?

Solution

The Electric force on the charge particle is given as

$$\begin{aligned} F_e &= qE_z = (1.6 \times 10^{-19}) (102.4 \times 10^3) \text{ N} \\ \Rightarrow F_e &= 163.84 \times 10^{-16} \text{ N} \end{aligned}$$

Above electric force acts on the particle along negative z -direction.

The Magnetic force on the charge particle is given as

$$\begin{aligned} F_m &= q v B \\ \Rightarrow F_m &= (1.6 \times 10^{-19}) (1.28 \times 10^6) (8 \times 10^{-2}) \text{ N} \\ \Rightarrow F_m &= 163.84 \times 10^{-16} \text{ N} \end{aligned}$$

Above magnetic force acts on the particle along positive z -direction. Above two forces are exactly equal and opposite so the charge particle will go undeflected in the space along x -direction.

(a) At time $t = 5 \times 10^{-6}$ second, the distance x travelled by the particle is given as

$$x = v t = (1.28 \times 10^6) (5 \times 10^{-6}) = 6.4 \text{ m}$$

Thus coordinates of the particle are given as (6.4, 0, 0).

(b) When the electric field is switched off, there will be a force 163.84×10^{-16} along $+z$ -axis acting on the particle. The particle moves in uniform magnetic field and describes a circular path in xz plane as shown in figure-4.125. The radius of the circular trajectory of particle is given as

$$\begin{aligned} r &= \frac{mv}{Bq} \\ \Rightarrow r &= \frac{(1 \times 10^{-26})(1.28 \times 10^6)}{(8 \times 10^{-2})(1.6 \times 10^{-19})} = 1 \text{ m} \end{aligned}$$

The time period t of revolution of particle due to magnetic field is given as

$$t = \frac{2\pi r}{v} = \frac{2 \times 3.14 \times 1}{1.28 \times 10^6} = 4.9 \times 10^{-6} \text{ s}$$

The time during which the magnetic field acts

$$t_1 = 7.45 \times 10^{-6} - 5 \times 10^{-6} = 2.45 \times 10^{-6} \text{ s}$$

As the time period is $4.9 \times 10^{-6} \text{ s}$ and the magnetic field acts only for a time $2.45 \times 10^{-6} \text{ s}$ which is half the time period so the particle covers a semicircle as shown in figure and displaces by 2m along z-direction. Thus final coordinates of particle are (6.4m, 0, 2m).

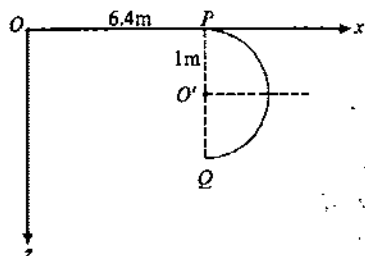


Figure 4.125

Illustrative Example 4.36

Uniform electric and magnetic fields with strength E and induction B respectively are directed along y -axis as shown in figure-126. A particle with specific charge q/m leaves the origin O in the direction of x -axis with an initial non-relativistic velocity v_0 . Find

- The coordinate y_n of the particle when it crosses the y -axis for the n^{th} time
- The angle α between the particles velocity vector and y -axis at that moment.

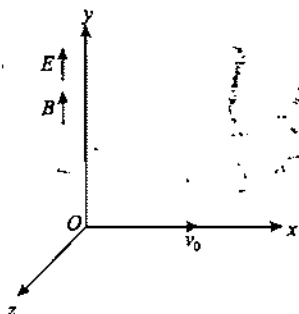


Figure 4.126

Solution

The resulting motion of the particle is helical with increasing pitch due to the electric field along y -direction. After every revolution the helical trajectory will be tangent to the y -axis when the particle crosses it.

Magnetic Effects of Currents and Classical Magnetism

- There will be an acceleration a_y on the particle due to electric force which acts along positive y -direction which is given as

$$a_y = \frac{qE}{m}$$

If the particle crosses y -axis n^{th} time after t second then we use

$$y_n = \frac{qE}{2m} t^2 \quad \dots (4.137)$$

If n be the number of revolutions, then we have

$$t = n \times T = \frac{2\pi mn}{qB} \quad \dots (4.138)$$

From equation-(4.137) and (4.138), we have

$$y_n = \frac{qE}{2m} \times \left(\frac{2\pi mn}{qB} \right)^2 = \frac{2\pi^2 mn^2 E}{qB^2}$$

- In y -direction the velocity of the particle is only due to electric field which is given as

$$v_y = a_y \times t$$

$$\Rightarrow v_y = \left(\frac{qE}{m} \right) \times \left(\frac{2\pi mn}{qB} \right) = \frac{2\pi nE}{B}$$

As along xz plane the particle velocity component will remain constant at v_0 , the angle which the velocity vector makes with y -axis is given as

$$\tan \alpha = \frac{v_0}{v_y} = \frac{v_0 B}{2\pi nE}$$

$$\Rightarrow \alpha = \tan^{-1} \left(\frac{v_0 B}{2\pi nE} \right)$$

Illustrative Example 4.37

an electron beam passes through a magnetic field of magnetic induction $2 \times 10^{-3} \text{ T}$ and an electric field of strength $3.4 \times 10^4 \text{ V/m}$ both acting simultaneously in mutually perpendicular directions. If the path of electrons remains undeviated, calculate the speed of the electrons. If the electric field is removed, what will be the radius of curvature of the trajectory of the electron path after 2s?

Solution

If motion of electron is undeviated then the magnetic and electric force must be balancing each other on electron so we have

$$qE = qvB$$

$$\Rightarrow v = \frac{E}{B} = \frac{3.4 \times 10^4}{2 \times 10^{-3}} = 1.7 \times 10^7 \text{ m/s}$$

When the electric field is removed, then only under magnetic field acting in perpendicular direction motion of electron will be along a circular path of which the radius is given as

$$r = \frac{mv}{Be}$$

$$\Rightarrow r = \frac{9 \times 10^{-31} \times (1.7 \times 10^7)}{(2 \times 10^{-3}) \times (1.6 \times 10^{-19})}$$

$$\Rightarrow r = 4.78 \times 10^{-2} \text{ m}$$

Illustrative Example 4.38

There is a constant homogeneous electric field of 100V/m within the region $x = 0$ and $x = 0.167\text{m}$ pointing along positive x -direction. There is a constant homogeneous magnetic field B within the region $x = 0.167\text{m}$ and $x = 0.334\text{m}$ along the positive z -direction. A proton at rest at the origin ($x = 0, y = 0$) is released. Find the minimum strength of the magnetic field B , so that the proton is detected again at point $x = 0, y = 0.167\text{m}$. Take mass of the proton $1.67 \times 10^{-27}\text{kg}$.

Solution

Figure-4.127 shows the resulting motion of the particle in the region where electric and magnetic fields are present as described in the question.

First the proton is accelerated in the electric field. Then it enters in magnetic field and describes a circular path of which radius is given as

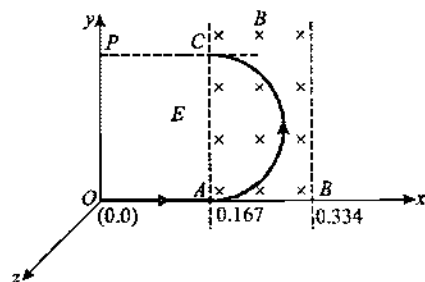


Figure 4.127

$$r = \frac{mv}{qB} \quad \dots (4.139)$$

After following a semicircular path it leaves the magnetic field in negative direction. Its motion is retarded in electric field. Finally it strikes y -axis at the same distance 0.167m.

Thus the radius of the circular path must be half of 0.167m. Using work energy theorem the velocity of the particle when it enters the magnetic field is given as

$$\frac{1}{2}mv^2 = qE(0.167) \quad \dots (4.140)$$

From equations-(4.139) and (4.140) we have

$$B = \frac{m}{e(0.167/2)} \times \sqrt{\frac{2eE}{m}}(0.167)$$

$$\Rightarrow B = 2\sqrt{\left(\frac{2m}{e} \times \frac{E}{0.167}\right)}$$

$$\Rightarrow B = 2\sqrt{\left[\frac{2(1.67 \times 10^{-27})(100)}{(1.6 \times 10^{-19})(0.167)}\right]}$$

$$\Rightarrow B = \frac{10^{-2}}{\sqrt{2}} = 7.07 \times 10^{-3} \text{ T}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Electromagnetic Force

Module Number - 14 to 23

Practice Exercise 4.4

(i) Figure-4.128 shows a wire AB of mass m placed on a rough inclined plane of inclination α and static friction coefficient μ . The wire carries a current I . Find the minimum magnitude of magnetic induction required to slide the wire up the inclined plane if direction of magnetic induction is normal to plane as shown in figure.

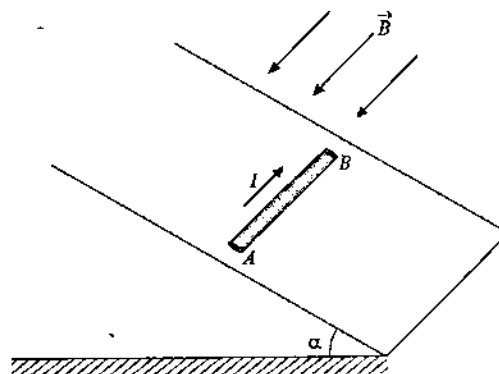


Figure 4.128

$$\left\{ \frac{mg(\sin \alpha + \mu \cos \alpha)}{I} \right\}$$

(ii) A wire of length 60cm and mass 16g is suspended by a pair of flexible supporting springs in a magnetic field of induction 0.60T. What are the magnitude and direction of the current in the wire so that the supporting springs remain unstretched?

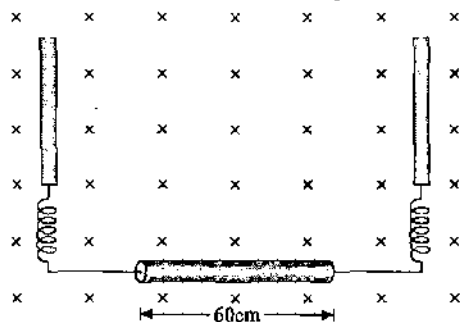


Figure 4.129

[0.41 A, left to right]

(iii) Two long parallel wires carry currents of equal magnitude but in opposite direction, these wires are suspended from ceiling by four strings of same length L as shown in figure-4.130. The mass per unit length of the wires is λ . Determine the value of the diverging angle θ assuming it to be small.

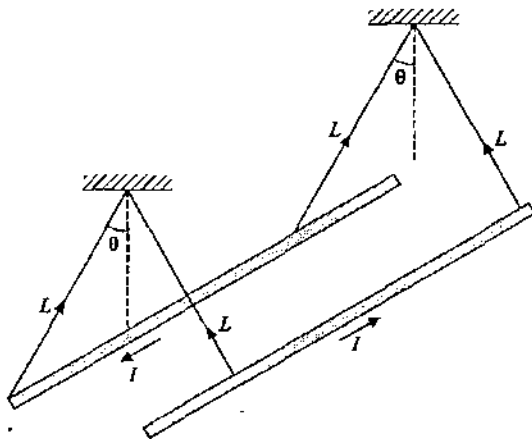


Figure 4.130

[$\sqrt{\mu_0 I^2 / 4\pi \lambda g}$]

(iv) A high speed but non-relativistic proton beam move rectilinearly in the region of space where there are uniform mutually perpendicular electric and magnetic fields with electric field strength E and magnetic induction B . The trajectory of the protons is found to be a straight line in the XY plane as shown in figure-4.131 and forms an angle ϕ with X -axis. Show that if the electric field is switched off at an instant the protons will follow helical trajectory and find the pitch of the helical trajectory.

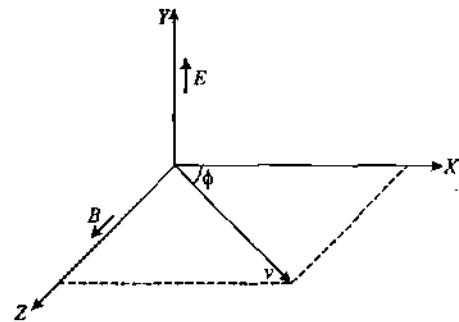


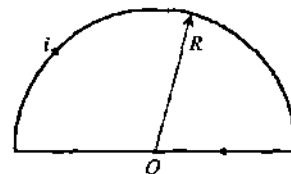
Figure 4.131

[$\frac{2\pi m E}{q B^2} \tan \phi$]

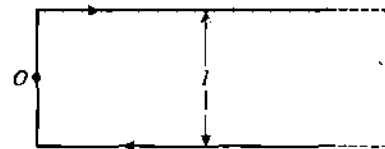
(v) A loop of flexible conducting wire of length 0.5m lies in a magnetic field of 1.0T perpendicular to the plane of the loop. Show that when a current is passed through the loop, it opens into a circle. Also calculate the tension developed in the wire if the current is of 1.57A.

[0.125 N]

(vi) Find the magnitude and direction of a force acting per unit length of a wire, carrying a current $i = 8.0\text{A}$, at a point O , if the wire is bent as shown in figures-4.132(a) and (b) as shown in figure-4.132. In figure-(a) wire is bent as semicircular loop of radius $R = 10\text{cm}$ and in figure-4.132(b) wire is bent as U-shaped with the straight segments very long with their separation equal to $l = 20\text{cm}$.



(a)



(b)

Figure 4.132

[(a) $2 \times 10^{-4} \text{ N/m}$ (b) $1280 \times 10^{-7} \text{ N/m}$]

(vii) A square cardboard of side l and mass m is suspended from a horizontal axis XY as shown in figure-4.133. A single wire is wound along the periphery of board and carrying a clockwise current I . At $t = 0$, a vertical downward magnetic

field of induction B is switched on. Find the minimum magnitude of B so that the board will be able to rotate up to horizontal level.

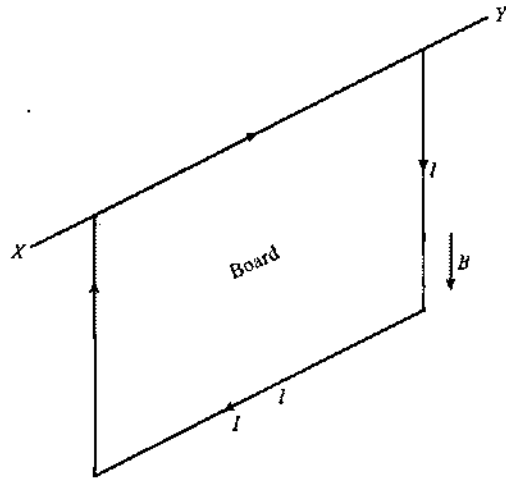


Figure 4.133

$$\left[\frac{mg}{2H} \right]$$

(viii) Figure-4.134 shows a horizontal wire MN of length l and mass m is placed in a magnetic field B . The ends of wire are bent and dipped in two bowls containing Hg which are connected to an external circuit as shown. If key is pressed for a short time δt due to which a charge q suddenly flows in the circuit. Find the maximum height above initial level the wire MN , it will jump.

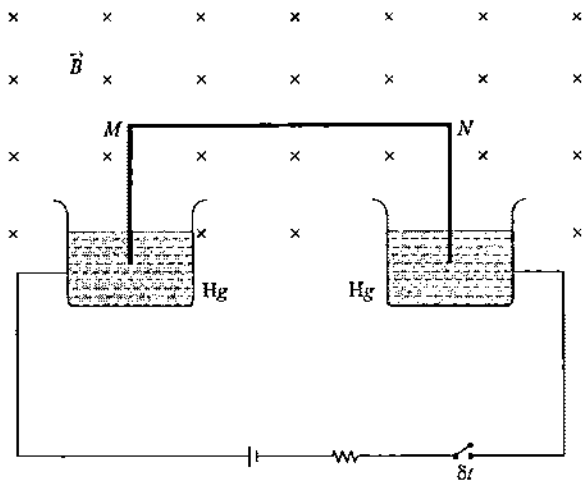


Figure 4.134

$$\left[\frac{B^2 q^2 l^2}{2m^2 g} \right]$$

(ix) A straight segment OC of length L of a circuit carrying a current I is placed along the X -axis as shown in figure-4.135. Two infinitely long straight wires A and B , each extending from $Z = -\infty$ to $+\infty$, are fixed at $y = -a$ and $y = +a$ respectively as shown in figure-4.135. If the wires A and B each carry same current I into the plane of the paper, obtain

the expression for the force acting on the segment OC . What will be the force on OC if the current in the wire B is reversed.

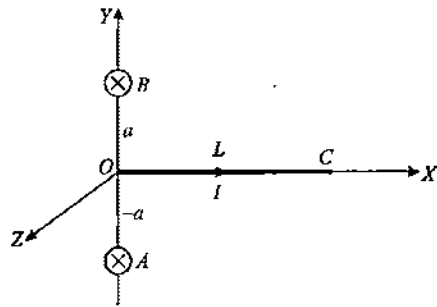


Figure 4.135

$$\left[\frac{\mu_0 I^2}{2\pi} \ln \left(\frac{L^2 + a^2}{a^2} \right), 0 \right]$$

(x) A positively charged particle having charge $q_1 = 1C$ and mass $m_1 = 40g$, is revolving along a circle of radius $R = 40cm$ with velocity $v_1 = 5m/s$ in a uniform magnetic field with centre of circle at origin O of a three dimensional system. At $t = 0$, the particle was at $(0, 0.4m, 0)$ and velocity was directed along positive X -direction. Another particle having charge $q_2 = 1C$ and mass $m_2 = 10g$ moving uniformly parallel to Z -direction with velocity $v_2 = (40/\pi)m/s$ collides with revolving particle at $t = 0$ and gets stuck to it. Neglecting gravitational force and coulomb force, calculate x , y and z coordinates of the combined particle at $t = \pi/40s$.

$$[(0.2m, 0.2m, 0.2m)]$$

(xi) From the surface of a round wire of radius a carrying a direct current i an electron is projected with a velocity v_0 perpendicular to the surface along radial direction. Find what will be the maximum distance of the electron from the axis of the wire before it turns back due to the magnetic force on the charge particle by the current flowing in the wire.

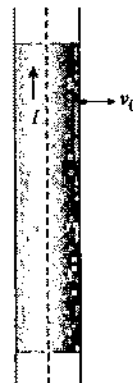


Figure 4.136

$$\left[\frac{2\pi m v_0}{ae \mu_0 i q} \right]$$

4.8 A Closed Current Carrying Coil placed in Magnetic Field

As discussed in previous article when a closed current carrying loop is placed in uniform magnetic field net magnetic force on it comes out to be zero.

Figure-4.137 shows a rectangular coil $ABCD$ carrying a current I is placed in uniform magnetic field with magnetic induction B with its plane parallel to the direction of magnetic field.

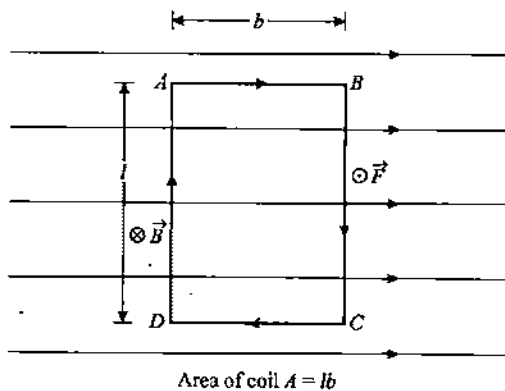


Figure 4.137

In this case if we analyze the directions of magnetic forces on different segments of the coil by using right hand palm rule then we find that

The force on AB is $F_{AB} = 0$

The force on BC is $F_{BC} = BIl$ in outward direction

The force on CD is $F_{CD} = 0$

The force on DA is $F_{DA} = BIl$ in inward direction

Thus total magnetic force on this loop is zero but the equal and opposite forces on the wire segments BC and DA on loop at different lines of actions produces a couple and coil will experience the torque which may cause it to rotate if it is free. In next article we will discuss about this torque in detail.

Consider another current carrying closed loop shown in figure-4.138(a) placed in uniform magnetic field with its plane perpendicular to the direction of magnetic field.

In this situation if we consider a small element on the loop as shown then by right hand palm rule the direction of magnetic force on this element is as shown in figure.

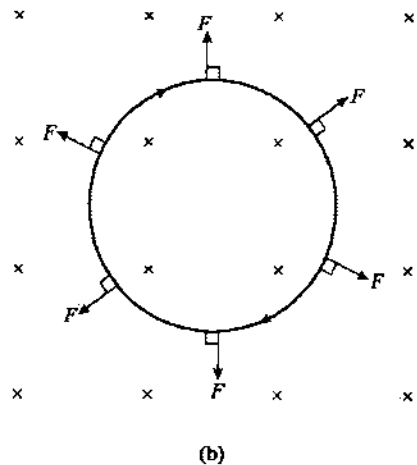
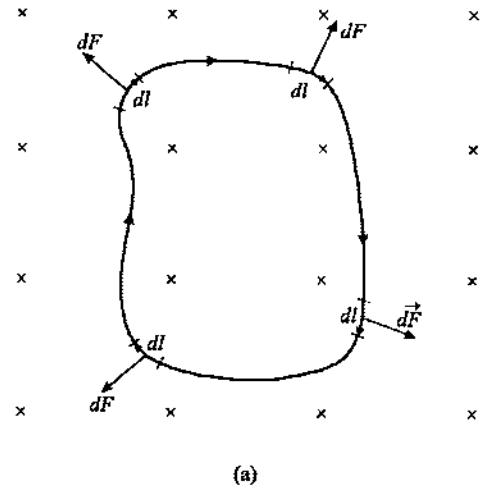


Figure 4.138

On analyzing carefully we can see that on all elements of the loop the direction of magnetic force is outward of which resultant is zero but it will have a tendency to stretch out the loop. If the wire of loop is flexible then the magnetic force will stretch it out and give a shape of a circle due to the outward force at each element as shown in figure-4.138(b).

If current direction is reversed then the force will become inward which may cause the loop to collapse if wire is flexible or build an inward stress if wire is rigid.

Above cases we analyzed for uniform magnetic field but if the closed current carrying loop is placed in a non uniform magnetic field then magnetic force on it may be non zero as at different locations of coil elements the magnetic induction magnitude can be different in non uniform magnetic field which will exert different forces on different elements of the coil which may not cancel each other.

We will see illustrations based on such situations.

4.8.1 Torque due to Magnetic Forces on a Current Carrying Loop in Magnetic Field

Consider the situation shown in figure-4.137 where we can see that the magnetic forces on wire segments AB and CD are zero as wires are parallel to magnetic induction direction and the forces on segments BC and DA are equal and in opposite direction which produce a couple. We've already studied in mechanics that torque due to couple forces is given by the product of force magnitude with the separation between lines of action of forces and this torque is independent of the location of axis of rotation. Here it is given as

$$\tau = BIl \times b = BIlb = BIA \quad \dots (4.141)$$

If number of turns in coil are N then due to N wire segments the total torque on the coil will be given as

$$\tau = BIA \times N = BINA \quad \dots (4.142)$$

Equation-(4.142) gives the net torque acting on the coil if coil is placed with its plane perpendicular to the magnetic induction direction. Consider the situation shown in figure-5.139 in which the coil is placed in such a way that its area vector is oriented at an angle θ to the direction of magnetic field or the plane of coil makes an angle $(90^\circ - \theta)$ to the magnetic field.

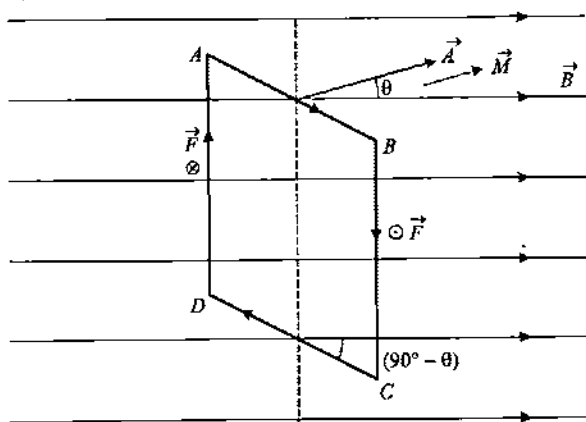


Figure 4.139

In above figure the force on wire segments AB and CD are opposite to each other along the same lines of actions so will cancel each other and the forces on wire segments BC and DA are same as before but the separation between their lines of action is now $b \sin\theta$ as shown so the torque on coil is now given as

$$\begin{aligned} \tau &= BIl \times b \sin\theta \times N \\ \Rightarrow \tau &= BINA \sin\theta \quad \dots (4.143) \end{aligned}$$

In above expression the term INA is referred as magnetic

moment of the coil which is defined as the product of current and area enclosed by all the turns of the coil. Magnetic moment of a current carrying coil is a vector quantity having direction along the area vector of the coil, given as

$$\vec{M} = NI\vec{A} \quad \dots (4.144)$$

The direction of area vector of the coil is given by right hand thumb rule for the direction of current in the coil as shown in figure-4.140

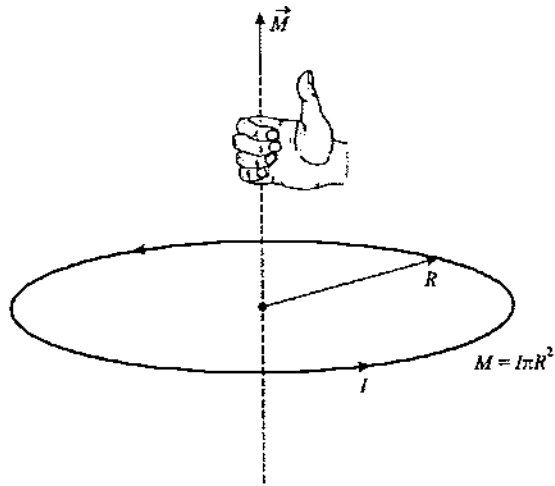


Figure 4.140

Magnetic moment of a current carrying coil is synonymous to the magnetic dipole moment of a small magnetic dipole about which we will discuss in the section of classical magnetism later in this chapter.

Using magnetic moment of the coil, equation-(4.143) can be rewritten as

$$\tau = MB \sin\theta \quad \dots (4.145)$$

Using the direction of torque, vectorially expression of torque can be given as

$$\vec{\tau} = \vec{M} \times \vec{B} \quad \dots (4.146)$$

4.8.2 Interaction Energy of a Current Carrying Loop in Magnetic Field

In classical magnetism we will discuss in detail about how a current carrying coil in magnetic field can be considered like a magnetic dipole but as of now we will use this fact for other analysis.

We will also study that a magnetic dipole placed in magnetic

field acts similar to an electric dipole placed in electric field and we've already discussed in the chapter of electrostatics that the interaction potential energy of an electric dipole with dipole moment p placed in an external electric field E is given as

$$U = -\vec{p} \cdot \vec{E} = -pE \cos \theta \quad \dots (4.147)$$

Similar to this situation for a magnetic dipole with dipole moment M placed in a magnetic field with magnetic induction B with the angle between magnetic moment and magnetic induction θ as shown in figure-5.139, its interaction potential energy is given as

$$U = -\vec{M} \cdot \vec{B} = -MB \cos \theta \quad \dots (4.148)$$

Above expression of potential energy can also be written as

$$U = -IAB \cos \theta$$

$$\Rightarrow U = -I\phi \quad \dots (4.149)$$

In above expression $\phi = BA \cos \theta$ is the magnetic flux passing through the coil.

4.8.3 Work Done in Changing Orientation of a Current Carrying Coil in Magnetic Field

When a current carrying coil is displaced in a uniform magnetic field in such a way that its orientation changes with angle between magnetic induction and magnetic moment of coil from q_1 to q_2 then the potential energy of the coil in magnetic field in initial and final state are given as

$$U_i = -MB \cos \theta_1 \quad \dots (4.150)$$

$$\text{and} \quad U_f = -MB \cos \theta_2 \quad \dots (4.151)$$

Work done in changing the orientation of the coil during its displacement is given as

$$W = U_f - U_i$$

$$\Rightarrow W = (-MB \cos \theta_2) - (-MB \cos \theta_1)$$

$$\Rightarrow W = MB(\cos \theta_1 - \cos \theta_2) \quad \dots (4.152)$$

To change the orientation of coil externally we need to apply a torque against the magnetic torque of equal magnitude thus work done in changing the orientation in above situation can also be given as

$$W = \int dW = \int \tau d\theta$$

$$\Rightarrow W = \int_{\theta_1}^{\theta_2} MB \sin \theta d\theta$$

$$\Rightarrow W = MB [-\cos \theta]_{\theta_1}^{\theta_2}$$

$$\Rightarrow W = MB (\cos \theta_1 - \cos \theta_2) \quad \dots (4.153)$$

Above equation-(4.153) is same as equation-(4.152) we obtained by using difference of interaction energies of coil in final and initial states.

Above expression of work can be rewritten as

$$W = IAB (\cos \theta_1 - \cos \theta_2)$$

$$\Rightarrow W = I(\phi_{\text{initial}} - \phi_{\text{final}})$$

$$\Rightarrow W = -I(\phi_{\text{final}} - \phi_{\text{initial}})$$

$$\Rightarrow W = -I\Delta\phi \quad \dots (4.154)$$

4.8.4 Magnetic Flux through a surface in Magnetic Field

In the region of magnetic field we can calculate the magnetic flux in the same way we did in electric field as magnetic induction is measured as magnetic flux density in a region similar to electric field.

Figure-4.141 shows a rectangular surface $ABCD$ of area S placed normal to the direction of a uniform magnetic field. The magnetic flux through this surface can be given as

$$\phi = BS \quad \dots (4.155)$$

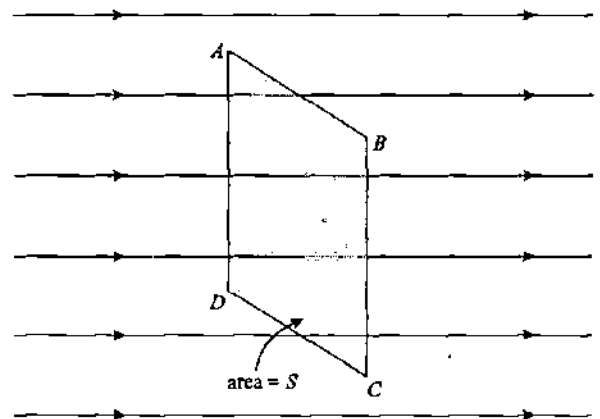


Figure 4.141

If the area is inclined to magnetic induction such that its area vector makes an angle θ with the direction of magnetic field or its plane makes an angle $(90^\circ - \theta)$ with the magnetic field as shown in figure-4.142 then magnetic flux thorough this area will be same which passes through its normal component $S \cos \theta$ as other component of area $S \sin \theta$ is parallel to the

magnetic field and no flux will pass through it as shown.

Thus magnetic flux through this surface is given as

$$\phi = BS \cos \theta \quad \dots (4.156)$$

$$\Rightarrow \phi = \vec{B} \cdot \vec{S} \quad \dots (4.157)$$

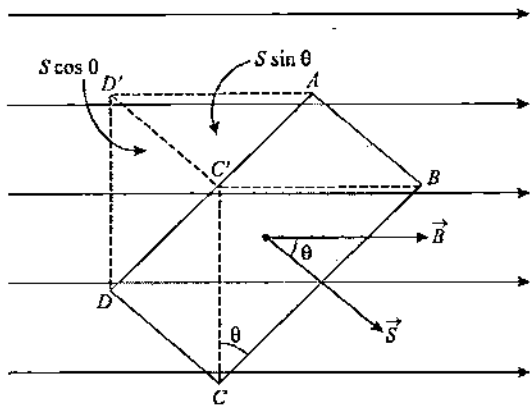


Figure 4.142

Figure-4.143 shows a region with non uniform magnetic induction in which there is a surface M placed in it.

To evaluate the magnetic flux passing through this surface, we consider an elemental area dS in this surface as shown with its area vector normal to the surface area dS at this point.

If $\vec{B}$ is the magnetic induction at the location of this area dS then magnetic flux $d\phi$ through the elemental area dS is given as

$$d\phi = B dS \cos \theta$$

$$\Rightarrow d\phi = \vec{B} \cdot \vec{dS} \quad \dots (4.158)$$

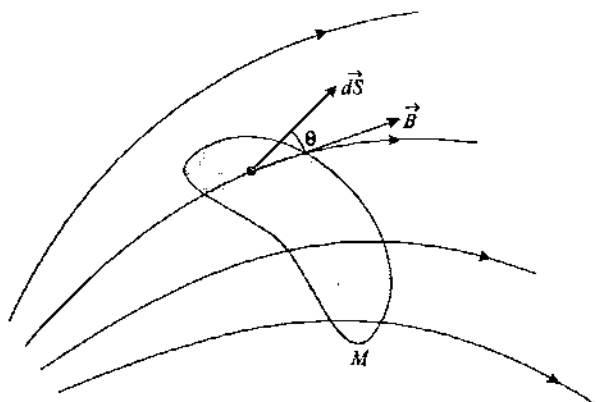


Figure 4.143

Thus total magnetic flux through the surface M is given by integrating the above elemental flux over the complete surface area given as

$$\phi = \int d\phi = \int_M \vec{B} \cdot \vec{dS} \quad \dots (4.159)$$

4.8.5 Stable and Unstable Equilibrium of a Current Carrying Loop in Magnetic Field

Figure-4.144 shows a current carrying rectangular loop placed in a plane perpendicular to the direction of magnetic field. The loop carries a current in clockwise direction due to which the direction vector of its magnetic moment is along the direction of magnetic induction due to which the angle between these two vectors is $\theta = 0$ and thus torque on loop is also $\vec{M} \times \vec{B} = 0$ and with the direction of magnetic forces on the segments of the loop we can see that all forces are acting in the plane of coil so net torque on it is zero and loop is in equilibrium.

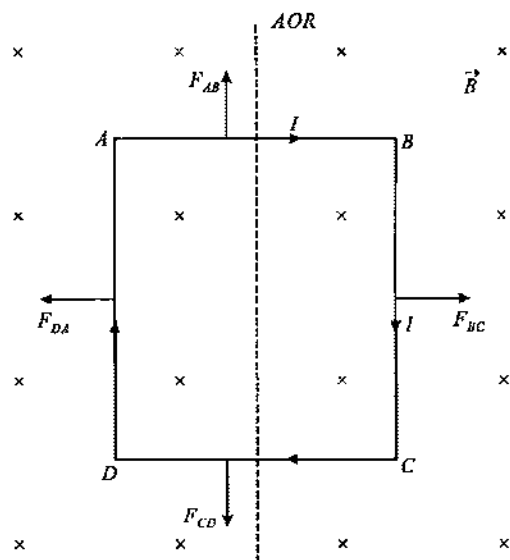


Figure 4.144

The interaction potential energy of the coil is given as

$$U = -MB \cos \theta = -MB \quad \dots (4.160)$$

If in figure-4.144 we slightly tilt the loop about the axis of rotation shown in figure then we can feel that the force acting on segments BC and DA will produce couple and it will have a tendency to bring back the coil in its initial position thus in this case these forces will develop a restoring torque hence the equilibrium of coil is stable equilibrium.

We can also see from equation-(4.160) that the potential energy of loop in this state is at its minimum value which corresponds to the case of stable equilibrium.

Figure-4.145 shows a similar situation with the current direction in the loop is in anticlockwise direction. In this case the angle between magnetic moment and magnetic induction is $\theta = 180^\circ$ and thus torque on loop is again $\vec{M} \times \vec{B} = 0$ and with the direction of magnetic forces on the segments of the loop we can see that all forces are acting in the plane of coil so net torque on it is zero and loop is in equilibrium.

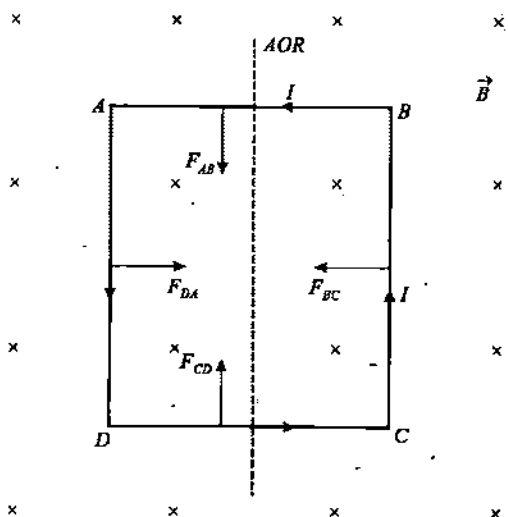


Figure 4.145

The interaction potential energy of the coil in this situation is given as

$$U = -MB\cos\theta = -MB(-1) = +MB \quad \dots (4.161)$$

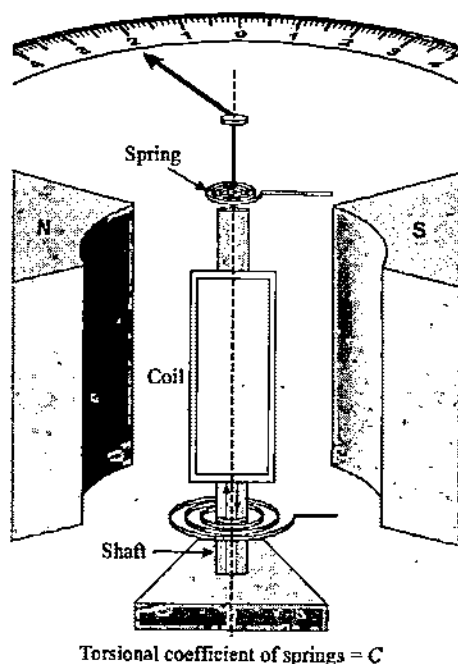
If in figure-4.145 we slightly tilt the loop about the axis of rotation shown in figure then we can feel that the force acting on segments BC and DA will produce couple and it will have a tendency to further displace the coil away from its initial position thus in this case these forces will develop a torque away from initial position hence the equilibrium of coil is unstable equilibrium.

We can also see from equation-(4.161) that the potential energy of loop in this state is at its maximum value which corresponds to the case of unstable equilibrium.

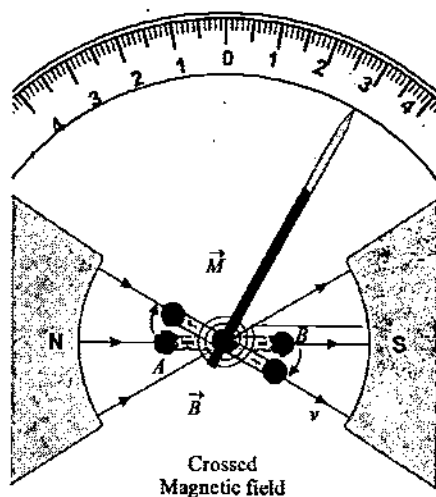
4.8.6 Moving Coil Galvanometer

We've studied in previous chapter that a galvanometer is used to measure current in electrical circuits. A moving coil galvanometer is a normal deflection type meter we discussed in different experimental setups in previous chapter in which the deflection of needle is directly proportional to the current supplied through it.

In this section we will be discussing the working mechanism of galvanometer.



(a)



(b)

Figure 4.146

Figure-4.146(a) shows two cylindrical magnetic pieces and in the region between the poles a coil is mounted on an axle which is attached with spiral springs at top and bottom mounts. The cylindrical pole pieces develop a crossed magnetic field on the coil as shown in figure-4.146(b) which is the top view of situation shown in figure-4.146(a). In this picture we can see if a current flows through the coil, the magnetic moment $\vec{M}$ of coil is perpendicular to the magnetic induction $\vec{B}$ at

initial position of coil. Due to magnetic torque in clockwise direction the coil rotates and because of crossed magnetic field. at all orientations of coil its magnetic moment is always perpendicular to the magnetic induction at that orientation of the coil so the magnetic torque on coil is independent of the orientation angle of the coil and it is given as

$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$\Rightarrow \tau = MB \quad [\text{As } \theta = 90^\circ]$$

$$\Rightarrow \tau = BINA \quad \dots (4.162)$$

Due to magnetic torque when coil rotates, spiral springs also twist which are attached at the top and bottom of the axle on which coil is mounted. These spiral springs exert a restoring torque on the shaft and have a tendency to bring the coil back to initial mean position. When a current flows through the coil, it experiences a magnetic torque given by equation-(4.162) due to which coil rotates and held at some angle where the restoring torque of spiral springs balances the magnetic torque. If overall stiffness coefficient of the spiral springs is C then at the equilibrium position if coil is deflected by an angle θ from initial position then we have

$$BINA = C\theta$$

$$\Rightarrow \theta = \left(\frac{BNA}{C} \right) I \quad \dots (4.163)$$

Here we can see that the angle of deflection of coil is directly proportional to the current supplied through the coil. In figure-4.147 we can see if a light weight needle is attached to the axle of the coil then with proper calibration the needle can be used to show the reading of current on a round scale as shown in figure. After proper calibration the whole assembly of magnets, mounts with coil is placed inside a cabinet and from outside only the scale and needle is visible. This is the system we call moving coil galvanometer which is used in our labs for current measurements.

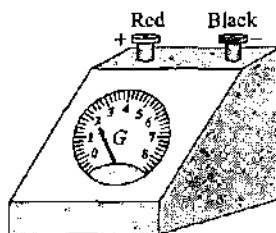


Figure 4.147

In above equation-(4.163) the proportionality constant is replaced by another constant S called 'Sensitivity of Galvanometer' given as

$$S = \frac{BNA}{C} \quad \dots (4.164)$$

$$\Rightarrow \theta = SI \quad \dots (4.165)$$

Here sensitivity of galvanometer indicates how sensitive the device is. If S is large then even for small currents flowing in the galvanometer coils deflection will be good enough to measure it and such a galvanometer is able to sense very small amounts of currents significantly and are called highly sensitive.

Illustrative Example 4.39

Figure-4.148 shows a coil of area A and N turns with a current I is placed in a uniform magnetic induction B . Find the work required to pull this coil out from magnetic field.

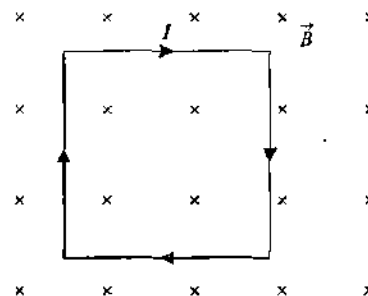


Figure 4.148

Solution

Work done in the process of pulling the coil out of field is given as

$$W = -I\Delta\phi = I(\phi_1 - \phi_2)$$

Initial flux through coil is given as

$$\phi_1 = BAN$$

Final flux through coil is given as

$$\phi_2 = 0$$

$$\text{Work done } W = I(BAN - 0) = BIAN$$

Illustrative Example 4.40

A circular coil of wire 8cm in diameter has 12 turns and carries a current of 5A. The coil is placed in a field where magnetic induction is 0.6 T.

- What is the maximum torque on the coil?
- In what position would the torque be one half as great as in (a)?

Solution

(a) The magnetic moment of the coil is given as

$$M = NiA$$

Area of coil is given as

$$A = \pi r^2 = \pi(4 \times 10^{-2})^2$$

$$\Rightarrow M = 12 \times 5 \times \pi \times (4 \times 10^{-2})^2$$

$$\Rightarrow M = 0.302 \text{ Am}^2$$

Maximum torque on coil acts when it is placed in the plane of the magnetic field or when the angle between magnetic moment and magnetic induction is 90° .

$$\tau_{\max} = MB = 0.302 \times 0.60 = 0.181 \text{ Nm}$$

(b) If θ is the angle between the area of the coil and the magnetic field direction then torque on coil is given as

$$\tau = MB \sin \theta$$

If this torque is half of the maximum torque then we have

$$MB/2 = MB \sin \theta$$

$$\Rightarrow \sin \theta = \frac{1}{2}$$

$$\Rightarrow \theta = 30^\circ$$

When the normal to the coil is at 30° to the field then the torque is half of the maximum torque on coil.

Illustrative Example 4.41

A square frame carrying a current I is located in the same plane as a long straight wire carrying a current I_0 . The frame side has a length a . The axis of the frame passing through the midpoints of the opposite sides is parallel to the wire and is separated from it by the distance which is η times greater than the side of the frame. Find

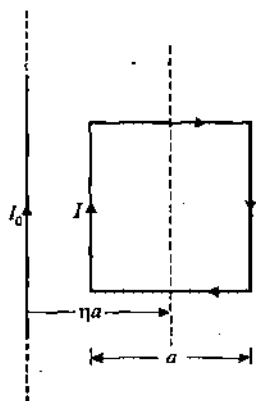


Figure 4.149

(a) Force acting on the frame

(b) The mechanical work to be performed in order to turn the frame through 180° about its axis, with the currents maintained constant.

Solution

(a) Force of attraction between parallel currents is given as

$$F_1 = \frac{\mu_0 I I_0}{2\pi \left(\eta a - \frac{a}{2} \right)} = \frac{\mu_0 I I_0}{\pi(2\eta - 1)a}$$

Similar force of repulsion between antiparallel currents is given as

$$F_2 = \frac{\mu_0 I I_0}{2\pi \left(\eta a + \frac{a}{2} \right)} = \frac{\mu_0 I I_0}{\pi(2\eta + 1)a}$$

Net force of attraction between the square frame and the long straight wire is

$$F = F_1 - F_2$$

$$\Rightarrow F = \frac{\mu_0 I I_0}{\pi(2\eta - 1)a} - \frac{\mu_0 I I_0}{\pi(2\eta + 1)a}$$

$$\Rightarrow F = \frac{2\mu_0 I I_0}{\pi(4\eta^2 - 1)}$$

(b) Work performed in turning the frame through 180° is given as

$$W = -I\Delta\phi = -I(-\phi - \phi) = 2I\phi$$

Where ϕ is the magnetic flux passing through the coil due to the current in the straight wire which is calculated by considering an elemental strip in the coil as shown in figure-4.150.

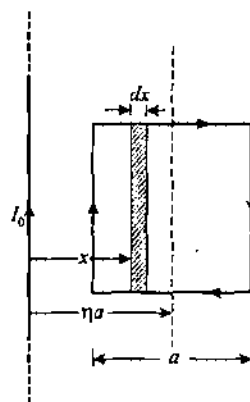


Figure 4.150

The magnetic flux through the elemental strip is given as

$$d\phi = B \cdot dS$$

$$\Rightarrow d\phi = \frac{\mu_0 I_0}{2\pi x} \cdot a dx$$

Total magnetic flux through the coil is given as

$$\phi = \int d\phi = \int_{\eta a - \frac{a}{2}}^{\eta a + \frac{a}{2}} \frac{\mu_0 I_0}{2\pi x} \cdot a dx$$

$$\Rightarrow \phi = \frac{\mu_0 I_0 a}{2\pi} \int_{\eta a - \frac{a}{2}}^{\eta a + \frac{a}{2}} \frac{dx}{x}$$

$$\Rightarrow \phi = \frac{\mu_0 I_0 a}{2\pi} \left[\ln x \right]_{\eta a - \frac{a}{2}}^{\eta a + \frac{a}{2}}$$

$$\Rightarrow \phi = \frac{\mu_0 I_0 a}{2\pi} \left[\ln \left(\eta a + \frac{a}{2} \right) - \ln \left(\eta a - \frac{a}{2} \right) \right]$$

$$\Rightarrow \phi = \frac{\mu_0 I_0 a}{2\pi} \ln \left(\frac{2\eta + 1}{2\eta - 1} \right)$$

Work done in rotation of coil is given as

$$W = 2I\phi$$

$$\Rightarrow W = \frac{\mu_0 I_0 a}{\pi} \ln \left(\frac{2\eta + 1}{2\eta - 1} \right)$$

Illustrative Example 4.42

Figure-4.151 shows a larger horizontal coil of radius R carrying a current I . Another small coil of radius r ($r \ll R$) carrying a current i & N turns is placed at the centre with its plane at an angle θ from the axis. Find the torque experienced by the smaller coil in this situation.

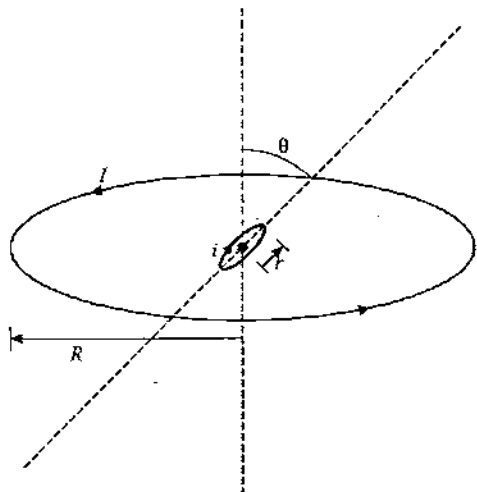


Figure 4.151

Solution

Magnetic moment of smaller coil is given as

$$M = i \times \pi r^2 N = N i \pi r^2$$

Magnetic induction due to large coil at its centre is given as

$$B = \frac{\mu_0 I}{2R}$$

Angle between magnetic induction at center of large coil and magnetic moment of smaller coil is $\frac{\pi}{2} + \theta$ as shown in figure-4.152

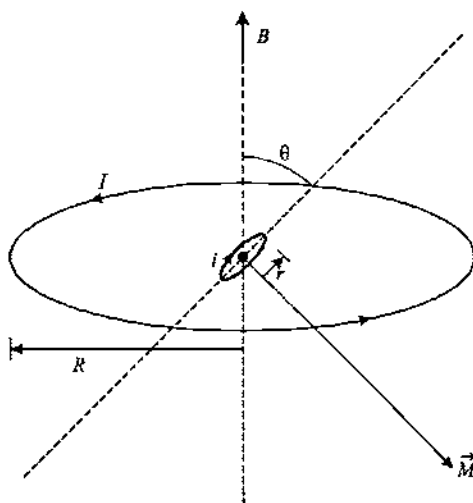


Figure 4.152

Torque on smaller coil due to the magnetic induction of larger coil at its center is given as

$$\vec{\tau} = \vec{M} \times \vec{B} = MB \sin \left(\frac{\pi}{2} + \theta \right)$$

$$\Rightarrow \tau = N i \pi r^2 \left(\frac{\mu_0 I}{2R} \right) \cos \theta$$

$$\Rightarrow \tau = \frac{\mu_0 i I N \pi r^2}{2R} \cos \theta$$

Illustrative Example 4.43

Find the magnetic moment of the current carrying loop shown in figure-4.153.

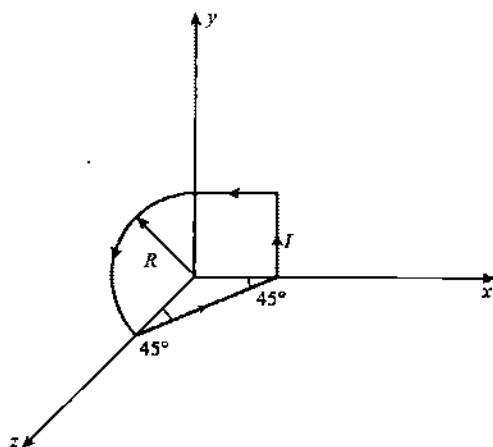


Figure 4.153

Solution

Figure-4.154 shows the splitting of the given loop in three separate loops which when overlapped at the three coordinate axes then the opposite currents in the wire segments in these three loops along the axes will cancel each other and original loop will only exist with current. The magnetic moment of the loop is given as the vector sum of the three individual loops shown in figure which is given as

$$\vec{M} = I \left(\frac{\pi R^2}{4} \right) \hat{i} + I \left(\frac{R^2}{2} \right) \hat{j} + I(R^2) \hat{k}$$

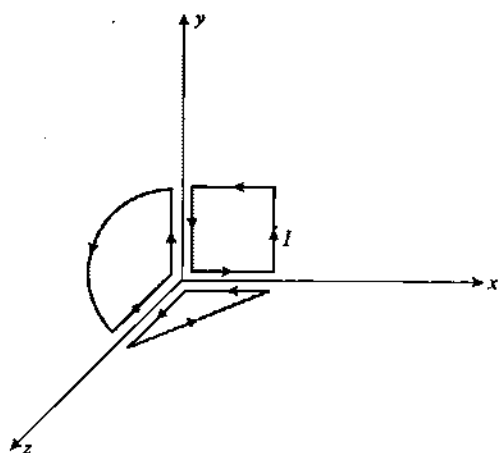


Figure 4.154

Illustrative Example 4.44

A square coil of edge l carrying a current I_2 is placed near to a long straight wire carrying current I_1 as shown in figure-4.155.

Find work required to rotate the coil $ABCD$ about the axis along edge BC by 180° to the dotted position $A'B'CD'$ as shown.

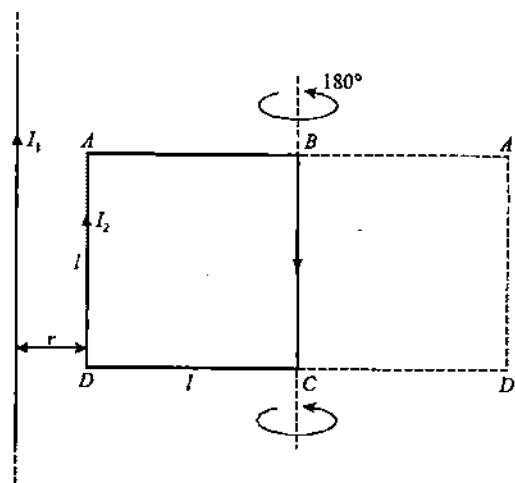


Figure 4.155

Solution

Work done in a process of changing the orientation of a coil in a magnetic field is given as

$$W = I_2(\phi_i - \phi_f)$$

We calculate the flux passing through the square coil at initial and final state by considering an elemental strip of width dx at a distance x from the long wire as shown in figure-4.156.

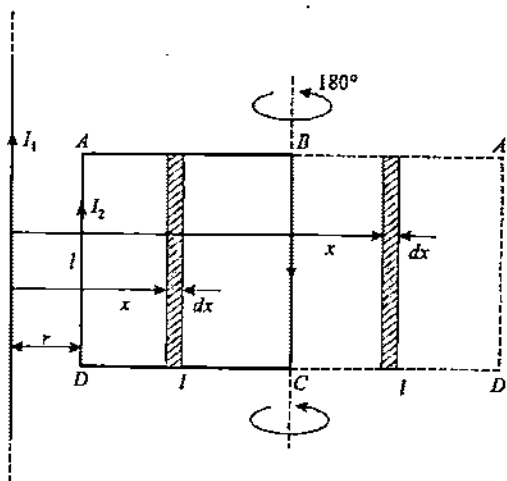


Figure 4.156

Initial flux through the coil is given as

$$\phi_i = \int \vec{B} \cdot d\vec{S} = \int_r^{r+l} \frac{\mu_0 I_1}{2\pi x} \times l dx = \frac{\mu_0 I_1 l}{2\pi} [\ln x]_r^{r+l}$$

$$\Rightarrow \phi_i = \frac{\mu_0 I_1 l}{2\pi} \left(\frac{r+l}{r} \right)$$

Final flux through the coil is given as

$$\phi_f = \int_{r+l}^{r+2l} \frac{\mu_0 I_1}{2\pi x} \cdot l dx = \frac{\mu_0 I_1}{2\pi} [\ln x]_{r+l}^{r+2l}$$

$$\Rightarrow \phi_f = \frac{\mu_0 I_1}{2\pi} \ln \left(\frac{r+2l}{r+l} \right)$$

Work done this process is given as

$$W = I_2 \left(\frac{\mu_0 I_1 l}{2\pi} \right) \left[\ln \frac{r+l}{r} - \ln \frac{r+2l}{r+l} \right]$$

$$\Rightarrow W = \frac{\mu_0 I_1 I_2 l}{2\pi} \ln \left(\frac{r+2l}{r} \right)$$

Illustrative Example 4.45

A rectangular coil of area $5.0 \times 10^{-4} \text{ m}^2$ and 60 turns is pivoted about one of its vertical sides. The coil is in a radial horizontal magnetic field of $9 \times 10^{-3} \text{ T}$. What is the torsional constant of the spring connected to the coil if a current of 0.20 mA produces an angular deflection of 18° ?

Solution

From the equation of moving coil galvanometer we have

$$i = \left(\frac{C}{NAB} \right) \theta$$

The torsional constant of the spring is given as

$$C = \frac{NABi}{\theta}$$

Substituting the values in SI units we have

$$C = \frac{(60)(5.0 \times 10^{-4})(9 \times 10^{-3})(0.2 \times 10^{-3})}{18}$$

$$\Rightarrow C = 3 \times 10^{-9} \text{ N-m/degree}$$

Illustrative Example 4.46

Figure-4.157 shows a coil bent with all edges of length 2 m and carrying a current of 2 A . There exists in space a uniform magnetic field of 2 T in positive y -direction. Find the torque on the loop.

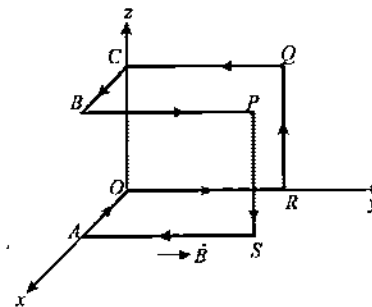


Figure 4.157

Solution

We assume equal and opposite currents in wires PQ and RS , then we split the given loop in three independent square loops and find magnetic moment vector of the loop which is only due to the square coil $PQRS$ as top and bottom square coils magnetic moments are equal and opposite so will cancel each other. Thus magnetic moment of the given loop is given as

$$\vec{M} = Ia^2 \hat{j} = 2 \times (2)^2 \hat{j} = 8 \hat{j}$$

The magnetic induction vector in space is given as

$$\vec{B} = 2 \hat{j}$$

The torque on the given loop is given as

$$\vec{\tau} = \vec{M} \times \vec{B} = 0$$

4.9 Relation in Magnetic Moment and Angular Momentum of uniformly charged and uniform dense rotating bodies

Figure-4.158 shows a thin ring of radius R with uniformly distributed charge q and mass m rotating at constant angular speed ω . The equivalent circulating current due to rotation of charge is given as

$$I = \frac{q\omega}{2\pi}$$

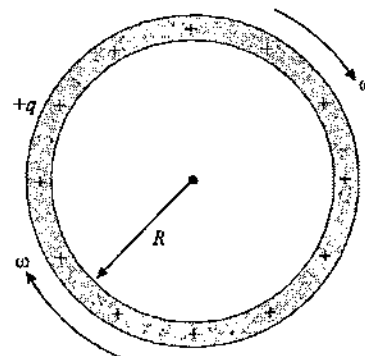


Figure 4.158

The magnetic moment of the rotating ring can be given as

$$M = I \times \pi R^2$$

$$\Rightarrow M = \frac{q\omega}{2\pi} \times \pi R^2$$

$$\Rightarrow M = \frac{1}{2} q\omega R^2 \quad \dots (4.166)$$

The angular momentum of the rotating ring can be given as

$$L = I\omega$$

$$\Rightarrow L = MR^2\omega \quad \dots (4.167)$$

Dividing equation-(4.166) and (4.167) gives

$$\frac{M}{L} = \frac{\left(\frac{1}{2} q\omega R^2\right)}{(MR^2\omega)}$$

$$\Rightarrow \frac{M}{L} = \frac{q}{2m} \quad \dots (4.168)$$

The ratio of magnetic moment to angular momentum as given in above equation-(4.117) is a standard relation which can be used for all uniformly charged and uniformly dense rotating bodies. In many cases above relation can be directly used for analysis of different situations mentioned in a question. Now we will consider one more illustration to verify the above result and its application. Figure-4.159 shows a uniform rod AB of length l , mass m , uniformly charged with a charge q is rotating at angular speed ω about one of its ends A .

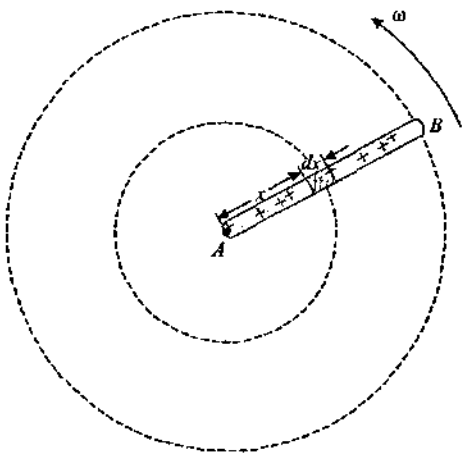


Figure 4.159

To calculate the magnetic moment of the rotating rod, we consider an element on rod of length dx at a distance x from

the fixed end as shown in figure-4.159. The charge on this element is given as

$$dq = \frac{q}{l} dx$$

During rotation of the rod, this elemental charge will revolve in circle of radius x for which the corresponding equivalent current in this circle is given as

$$dI = \frac{dq\omega}{2\pi}$$

Due to the current in element, its magnetic moment is given as

$$dM = dI \times \pi x^2$$

$$\Rightarrow dM = \left(\frac{dq\omega}{2\pi}\right) \pi x^2$$

$$\Rightarrow dM = \frac{1}{2} dq\omega x^2$$

$$\Rightarrow dM = \frac{1}{2} \left(\frac{q}{l} dx\right) \omega x^2$$

$$\Rightarrow dM = \frac{q}{2l} \omega x^2 dx$$

Total magnetic moment of the rotating rod is given by integrating above expression for the whole length of the rod within limits from 0 to l which is given as

$$M = \int dM = \int_0^l \frac{q}{2l} \omega x^2 dx$$

$$\Rightarrow M = \frac{q\omega}{2l} \int_0^l x^2 dx$$

$$\Rightarrow M = \frac{q\omega}{2l} \left[\frac{x^3}{3} \right]_0^l$$

$$\Rightarrow M = \frac{q\omega}{2l} \left(\frac{l^3}{3} \right)$$

$$\Rightarrow M = \frac{1}{6} q\omega l^2 \quad \dots (4.169)$$

The angular momentum of the rotating rod is given as

$$L = I\omega$$

$$\Rightarrow L = \left(\frac{1}{3} Ml^2 \right) \omega$$

$$\Rightarrow L = \frac{1}{3} Ml^2 \omega \quad \dots (4.170)$$

By taking the ratio of equation-(4.169) and (4.170) we can verify equation-(4.168) for this case of rotating rod as well. If we use equation-(4.168) then the magnetic moment of the rotating rod as expressed in equation-(4.169) can be obtained directly by substituting the value of angular momentum from equation-(4.170) to equation-(4.168) and this saves the time of integration process we used in obtaining equation-(4.169) but here before applying equation-(4.168) for any situation of a rotating charge students must carefully analyze the given situation that the body must be uniformly charged and uniformly dense.

4.10 Magnetic Pressure and Field Energy of Magnetic Field

Similar to the case we've studied in electrostatic that a charged metal surface always experiences an electric pressure on it in outward direction due to its surface charge density, in case of a current carrying hollow conductor also if current is flowing on its surface and parallel currents always attract each other so the surface experiences an inward magnetic pressure on its surface. To determine the magnetic pressure on the surface we consider a long straight hollow cylindrical shell carrying a current I . We know at every interior point of this hollow wire magnetic induction is zero whereas just outside the surface the magnetic induction is given as

$$B = \frac{\mu_0 I}{2\pi R} \quad \dots (4.171)$$

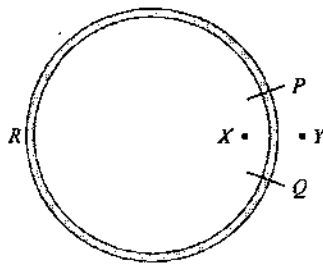


Figure 4.160

Figure-4.160 shows the cross sectional view of the hollow cylindrical wire in which we consider an elemental wire PQ of width dw and the remaining shell PRQ and consider two points X and Y just inside and outside of elemental wire PQ . At point X magnetic induction is zero and that at point Y is given by equation-(4.171). If B_1 and B_2 are the magnetic inductions at points X and Y due to the segments PQ and PRQ of the shells as shown in figure-4.160 then we have

$$B_X = B_1 - B_2 = 0 \quad \dots (4.172)$$

$$B_Y = B_1 + B_2 = B = \frac{\mu_0 I}{2\pi R} \quad \dots (4.173)$$

From equation-(4.172) and (4.173) we have

$$B_1 = B_2 = \frac{B}{2} \quad \dots (4.174)$$

Thus similar to the case of electrostatics the elemental part PQ of the shell contributes in half of the magnetic induction just outside the shell due to the whole current in it. If we consider a length l of the elemental wire PQ then force experienced by it due to the remaining section PRQ is given as

$$dF = B_2(dI)l$$

$$dF = B_2 \left(\frac{I}{2\pi R} \times dw \right) l$$

$$dF = \left(\frac{\mu_0 I}{4\pi R} \right) \left(\frac{I}{2\pi R} \right) dwl$$

The area of the elemental wire strip is $dS = dwl$ so we can calculate the inward magnetic pressure on the elemental strip which is given as

$$p_m = \frac{dF}{dS} = \frac{\mu_0 I^2}{8\pi^2 R^2}$$

$$p_m = \frac{B^2}{2\mu_0} \quad \dots (4.175)$$

Above expression given in equation-(4.175) is the expression of magnetic pressure due to a surface current. Same expression can be used for magnetic energy density in the same way it is explained in article-1.19.1 in case of electrostatic energy density.

Illustrative Example 4.47

A flat disc of radius R charged uniformly on its surface at a surface charge density σ . About its central axis of rotation it rotates at an angular speed ω . Find the magnetic moment of disc due to rotation of charges.

Solution

We consider an elemental ring of radius x and width dx in the disc as shown in figure-4.161. Equivalent current in this elemental ring is given as

$$dI = \frac{dq\omega}{2\pi}$$

Magnetic moment of the rotating elemental ring is given as

$$dM = dI \times \pi x^2 = \frac{\sigma 2\pi x dx \omega}{2\pi} \times \pi x^2$$

$$\Rightarrow dM = \sigma \omega \pi x^3 dx$$

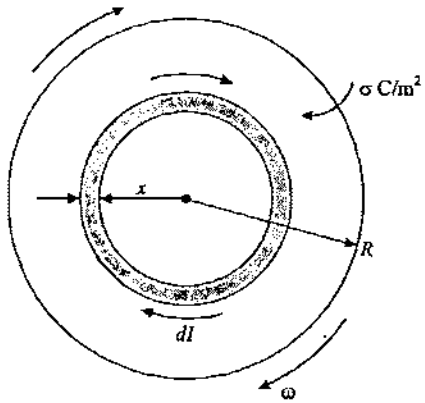


Figure 4.161

Total magnetic moment of the rotating disc is given as

$$M = \int dM = \sigma \pi \omega \int_0^R x^3 dx$$

$$\Rightarrow M = \frac{1}{4} \sigma \pi \omega R^4$$

Above result can be directly calculated by using equation-(4.168). Students are advised to verify the above result by substituting the value of angular momentum of the disc in equation-(4.178).

Illustrative Example 4.48

A flat non-conducting disc of radius R carries an excess charge on its surface. The surface charge density is σ . The disc rotates about an axis perpendicular to its plane passing through the centre with angular velocity ω . Find the torque on the disc if it is placed in a uniform magnetic field B directed perpendicular to the rotation axis.

Solution

The situation described in question is shown in figure-4.162. As explained in previous illustration the magnetic moment of the rotating disc is given as

$$M = \frac{1}{4} \sigma \pi \omega R^4$$

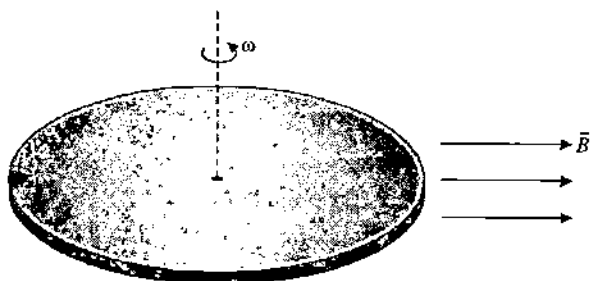


Figure 4.162

The torque on the disc due to magnetic field is given as

$$\tau = MB \sin 90^\circ$$

$$\Rightarrow \tau = \frac{\sigma \omega \pi B R^4}{4}$$

Illustrative Example 4.49

A solid sphere of radius R , uniformly charged with a charge Q is rotating about its central axis at angular speed ω . Find the magnetic moment of this rotating sphere.

Solution

The relation in magnetic moment and angular momentum of a uniformly charged and uniform body is given by equation-(4.168) as

$$\frac{M}{L} = \frac{Q}{2m}$$

The angular momentum of the rotating sphere is given as

$$L = I\omega = \frac{2}{5} m R^2 \omega$$

Thus magnetic moment of the sphere is given as

$$M = \frac{Q}{2m} \times \frac{2}{5} m R^2 \omega$$

$$\Rightarrow M = \frac{1}{5} Q R^2 \omega$$

Illustrative Example 4.50

What pressure does the lateral surface of a long straight solenoid with n turns per unit length experience when a current I flows through it.

Solution

In case of a long solenoid outside magnetic induction is zero and inside it is given as $B = \mu_0 n I$. Thus the magnetic pressure on the surface of solenoid currents is given as

$$p_m = \frac{B^2}{2\mu_0} = \frac{1}{2} \mu_0 n^2 I^2$$

Above result we've directly written by using the expression of magnetic pressure but students can note that in article-4.10 while deriving this expression we consider a surface current along the length of a hollow shell whereas in case of solenoid current flows along the curved part of its core but still the result is valid. This can be verified by following the similar derivation process as well.

Illustrative Example 4.51

A conducting current-carrying plane is placed in an external uniform magnetic field. As a result, the magnetic induction becomes equal to B_1 on one side of the plane and equal to B_2 on the other side. Find the magnetic force acting per unit area of the plane in the cases illustrated in the figure-4.163(a, b, c).

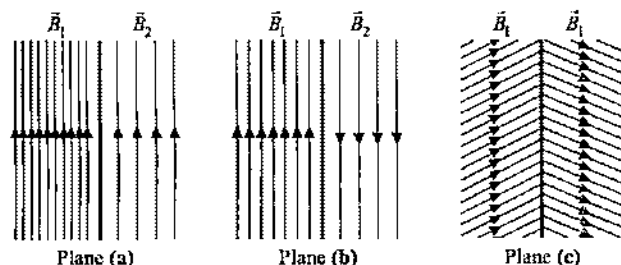


Figure 4.163

Determine the direction of the current in the plane in each case.

Solution

(a) As on the two sides of the plane fields are in same direction the external magnetic field induction B must be the average of B_1 and B_2 because on one side of the plane the external field will be added to the field of plane and on its other side it is subtracted.

So the magnetic induction due to the plane is $\frac{B_1 - B_2}{2}$ or $\frac{B_2 - B_1}{2}$ on opposite of the plane. If plane carries a linear current density I , we use

$$\frac{B_1 - B_2}{2} = \frac{1}{2} \mu_0 I$$

$$\Rightarrow I = \frac{B_1 - B_2}{\mu_0}$$

Force on the plane per unit area on the plane can be given as

$$P_m = \frac{B_1^2 - B_2^2}{2\mu_0} \quad \dots (4.176)$$

Students can derive and verify the result of magnetic pressure in equation-(4.176) in the same way it is done in article-4.10 for a long hollow current carrying cylindrical shell.

(b) As on the two sides of the plane fields are in opposite direction the external magnetic field in this case is given as

$$B_{\text{ext}} = \frac{B_1 - B_2}{2}$$

and the field due to the plane is given as

$$\frac{B_1 + B_2}{2} = \frac{1}{2} \mu_0 I$$

$$\Rightarrow I = \frac{B_1 + B_2}{2\mu_0}$$

Force on the plane per unit area on the plane can be given as

$$P_m = \frac{B_1^2 - B_2^2}{2\mu_0}$$

Students can derive and verify the result of magnetic pressure in equation-(4.176) in the same way it is done in article-4.10 for a long hollow current carrying cylindrical shell.

(c) In this case, external field is continuous across the plane and due to plane is directed parallel to the plane upward on the left and downward on the right side.

Similar to the analysis we did again the force on the plane per unit area on the plane can be given as

$$P_m = \frac{B_1^2 - B_2^2}{2\mu_0}$$

In each case, the current is in the direction directed into the plane of paper.

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Electromagnetic Force

Module Number - 24 to 38

Practice Exercise 4.5

(i) In Bohr model of hydrogen atom, the electron circulates round the nucleus in a path of radius $5.1 \times 10^{-11} \text{ m}$ at a frequency $6.8 \times 10^{15} \text{ rev/s}$.

(a) What is the magnitude of magnetic induction at the centre of the orbit in Bohr model?

(b) What is the equivalent dipole moment of the rotating electron in Bohr model?

[13.6 T, $90 \times 10^{-24} \text{ Am}^2$]

(ii) Figure-4.164 shows rectangular twenty-turn loop of wire with sides 10cm and 5cm. It carries a current of 0.10A and it is hinged at one side along y -axis as shown. Calculate the magnitude and direction of the torque acting on the loop if it is mounted with its plane at an angle of 30° to the direction of a uniform field of magnetic induction 0.50T which exist along positive x -direction.

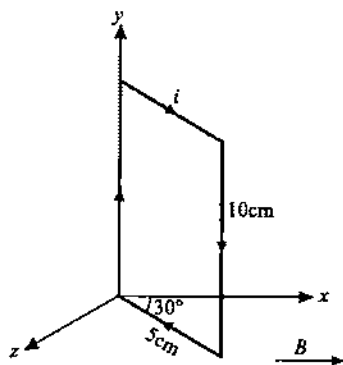


Figure 4.164

$$[4.3 \times 10^{-3} \text{ Nm}]$$

(iii) A coil in the shape of an equilateral triangle of side 0.02m is suspended from a vertex such that it is hanging in a vertical plane between the pole pieces of a permanent magnet producing a horizontal magnetic induction $5 \times 10^{-2}\text{T}$. Find the couple acting on the coil when a current of 0.1A is passed through it and the magnetic field is parallel to its plane.

$$[8.66 \times 10^{-7} \text{ Nm}]$$

(iv) A wire loop carrying a current I is placed in the x - y plane as shown in figure-4.165.

(a) If a particle with charge $+Q$ and mass m is placed at the centre P and given a velocity v along NP (see the figure), find its instantaneous acceleration of the particle at point P .

(b) If an external uniform magnetic induction field $\vec{B} = B\hat{i}$ is applied, find the force and the torque acting on the loop due to this field.

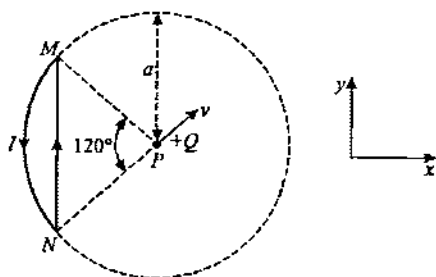


Figure 4.165

$$[(a) \frac{0.109\mu_0 IQv}{\pi a}, (b) 0.6136IBa^2\hat{j}]$$

(v) A coil of radius R carries current i_1 . Another concentric coil of radius r ($r \ll R$) carries current i_2 . Planes of two coils are mutually perpendicular and both the coils are free to rotate about common diameter. Find maximum kinetic energy of smaller coil when both the coils are released, masses of coils are M and m respectively.

$$[\frac{\mu_0 \pi r^2 i_1 i_2 MR}{2(MR^2 + mr^2)}]$$

(vi) Figure-4.166 shows a beam balance at one end of which a current carrying coil C with N turns, cross sectional area A & Current I is attached which is kept between two pole pieces as shown and on other end a pan is there in which a counter weight of mass M is kept. If in equilibrium beam remain horizontal, find the magnetic induction due to pole pieces in which coil is kept in equilibrium.

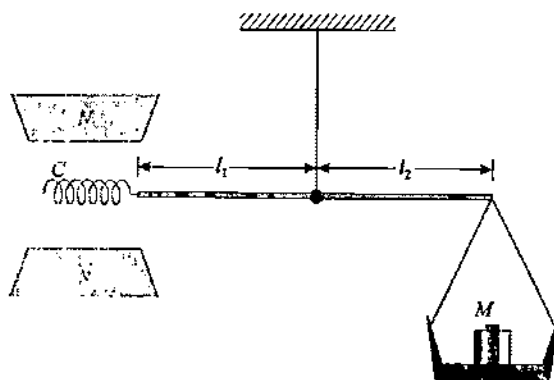


Figure 4.166

$$[\frac{Mgl_2}{INA}]$$

(vii) A uniform magnetic field of magnetic induction B is directed at an angle of 45° to the x -axis in xy plane in a coordinate system. $PQRS$ is a rigid square wire frame carrying a steady current I_0 , with its centre at the origin O . At time $t = 0$, the frame is at rest in the position shown in the figure-4.167, with its side parallel to x and y axes. Each side of the frame is of mass M and length L .

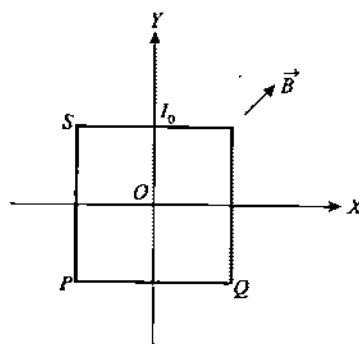


Figure 4.167

(a) Calculate the torque about O acting on the frame due to magnetic field?

(b) Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt , and the axis about which this rotation occurs. Consider that Δt is so short that any variation in the torque during this interval may be neglected.

$$[(a) \frac{I_0 L^2 B}{\sqrt{2}} (-\hat{i} + \hat{j}) ; (b) \frac{3I_0 B}{4M} (\Delta t)^2]$$

(viii) Calculate the magnetic moment of a thin wire with a current I wound tightly on a half a torus of a toroid as shown in figure-4.168. The diameter of the cross-section of the torus is d and total number of turns are N .

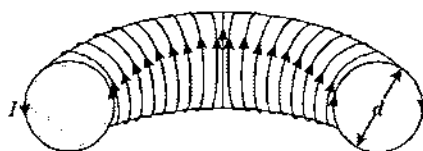


Figure 4.168

$$[\frac{1}{2} Id^2 N]$$

(ix) A flat circular coil with 10 turns of wire on it has a diameter 20mm and carries a current 0.5A. It is mounted inside a long solenoid that has 200 turns and the length of the solenoid is 0.25m. The current passing through the solenoid is 2.4A. Calculate the torque needed to hold the coil with its axis perpendicular to that of solenoid.

$$[3.78 \times 10^{-6} \text{ Nm}]$$

(x) The coil of a moving coil galvanometer twists through 90° when a current of one microampere is passed through it. If the area of the coil is 10^{-4} m^2 and it has 100 turns, calculate the magnetic field of the magnetic poles in the galvanometer. Torsional constant of the spring system used in galvanometer is given as $C = 10^{-8} \text{ N-m/degree}$.

$$[90\text{T}]$$

4.11 Classical Magnetism

About 4000 years ago in a Greek town Magnesia a shepherd named magnes was herding his sheep and suddenly he found that after he took a step some iron nails in his shoes stuck fast to a black rock on which he was standing. In state of surprise he digged the rock further and found that this rock has character to attract some of the metal objects. On the name of

the town this rock was named '*Magnetite*'. It was later analyzed and found containing a compound Fe_3O_4 because of which such properties are there in this material. Over a period of time these properties found in many other materials also are called '*Magnetic Properties*' and the force field because of which these material exert force on some specific metal objects is named '*Magnetic Field*'. Over long period of time many more discoveries and experiments evolved the knowledge of physicists in this domain and its applications helped making many industrial processes more effective and productive.

4.11.1 Pole Strength of a Magnetic Pole

Pole strength is a measurement of magnetic field produced by a magnetic pole. This is a physical quantity similar to electric charge in electrostatics like the magnitude of a charge is a measure of how strong electric field the charge can produce in its surrounding. Similarly if pole strength of a magnetic pole is high then it means this pole produces high magnetic field in its surrounding. Unit used for measurement of magnetic pole strength is '*ampere-meter*' or '*A-m*'.

Like positive and negative charges in magnetism there are two different poles exist which are called '*North and South Poles*' like positive charge from a north pole magnetic flux comes out and like a negative charge magnetic lines goes into a south pole. Similar to charges in magnetism also like poles repel each other and unlike poles attract each other.

If we consider two independent point magnetic poles, a north and a south pole as shown in figure-4.169 then magnetic field lines in their surrounding are considered radially outward and radially inward respectively as shown.

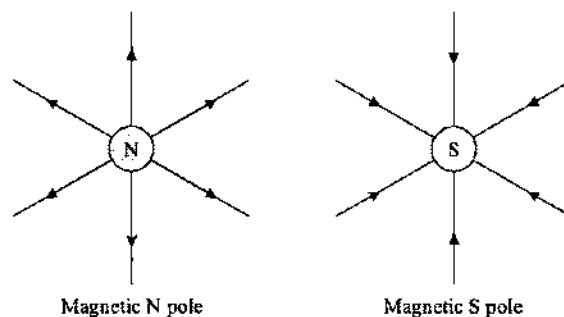


Figure 4.169

Above figure shows a theoretical configuration of magnetic field in surrounding of these point magnetic poles because practically independent poles (called '*monopoles*') never exist in nature which we will study later in upcoming articles.

4.11.2 Coulomb's Law for Magnetic Forces

This law gives the force of interaction between two magnetic poles and it is analogous to Coulomb's law for electric charges. First we will analyze this law at theoretical level. Figure-4.170 shows two magnetic monopoles with pole strengths m_1 and m_2 kept at a separation r between them. The interaction force between the two poles is

- (i) Directly proportional to the product of magnetic pole strengths of the two poles

$$F \propto m_1 m_2 \quad \dots (4.177)$$

- (ii) Inversely proportional to the square of the distance between the poles

$$F \propto \frac{1}{r^2} \quad \dots (4.178)$$

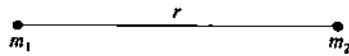


Figure 4.170

From above equations-(4.177) and (4.178) we have

$$F \propto \frac{m_1 m_2}{r^2}$$

$$\Rightarrow F = K \frac{m_1 m_2}{r^2} \quad \dots (4.179)$$

In above expression shown in equation-(4.179), K is a proportionality constant which depends upon the medium in which magnetic poles are placed. Value of K is given as

$$K = \frac{\mu}{4\pi}$$

Where μ is the magnetic permeability of the medium in which magnetic poles are kept. This is given as

$$\mu = \mu_0 \mu_r \quad \dots (4.180)$$

Where μ_0 is the permeability of free space of which numerical value is given as

$$\mu_0 = 4\pi \times 10^{-7} \text{ N-m/A}$$

and μ_r is the relative permeability of medium with respect to free space and it is given from equation-(4.180) as

$$\mu_r = \frac{\mu}{\mu_0}$$

Unlike to the value of relative permittivity of medium ϵ_r which always has numerical value greater than unity for different material medium, in magnetism depending upon the type of material medium value of μ_r can be greater or less than unity which we will discuss more in upcoming articles.

Magnetic Effects of Currents and Classical Magnetism

Thus using the value of proportionality constant expression given in equation-(4.179) can be rewritten as

$$F = \frac{\mu_0}{4\pi} \frac{m_1 m_2}{r^2} \quad \dots (4.181)$$

In above expression we considered medium to be free space or air for which we take $\mu_r = 1$.

4.11.3 Magnetic Induction in terms of Force on Poles

Every magnetic pole produces magnetic induction in its surrounding space which can exert a force on any other pole placed in surrounding. As already discussed that behaviour of magnetic poles in study of magnetism is more or less similar to that of charges in electrostatics.

Like electric field strength we can also define magnetic induction in terms of force on magnetic poles as at any point in region of magnetic field the force experienced by a unit magnetic north pole gives the magnitude of magnetic induction at that point in space. Figure-4.171 shows a region of magnetic field in which a magnetic north pole of pole strength m_0 is placed. If it experiences a force $\vec{F}$ due to magnetic induction than at the location of the pole magnetic induction is given as

$$\vec{B} = \frac{\vec{F}}{m_0} \quad \dots (4.182)$$

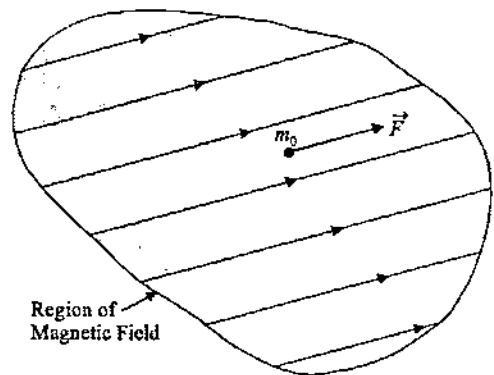


Figure 4.171

Always remember that similar to electrostatics, in magnetic field a north pole experiences force in the direction of magnetic field and a south pole experiences force opposite to the direction of magnetic field as shown in figure-4.172. The magnitude of force on a magnetic pole of strength m placed in magnetic induction B is given as

$$F = mB \quad \dots (4.183)$$

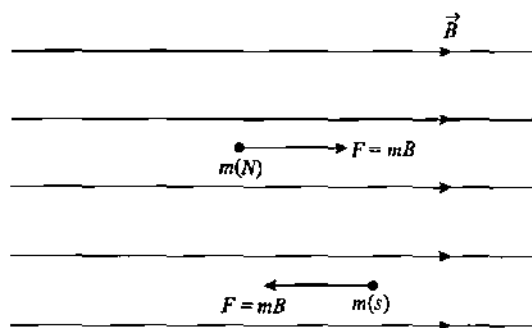


Figure 4.172

4.11.4 Magnetic Induction due to a Magnetic Pole

We've discussed in previous article that at any point magnetic induction is given as force experienced by a unit magnetic north pole placed at that point.

Thus to determine the magnetic induction in surrounding of a magnetic north pole of pole strength m , we place a test pole m_0 at a distance r from the pole at point P as shown in figure-4.173. The magnetic force on test pole placed at point P is given as

$$F = K \frac{mm_0}{r^2}$$

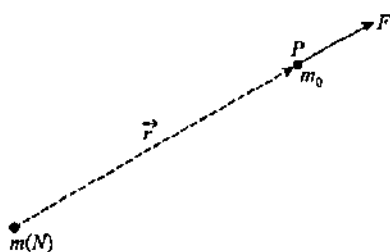


Figure 4.173

The magnetic induction at point P is given by equation-(4.182) as

$$B_P = \frac{F}{m_0}$$

$$\Rightarrow B_P = \frac{Km}{r^2} \quad \dots (4.184)$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \frac{m}{r^2} \quad \dots (4.185)$$

The expression given in equation-(4.184) is similar to the electric field produced by a point charge in its surrounding. About direction we've already discussed that due to a north pole magnetic induction is radially outward and in surrounding of a south pole magnetic induction is radially inward.

Vectorially equation-(4.185) can be written as

$$\vec{B}_P = \frac{\mu_0}{4\pi} \frac{m}{r^3} \vec{r} \quad \dots (4.186)$$

4.11.5 Bar Magnet

A bar magnet is a rod shaped metal object which produces magnetic field in its surrounding and at the two ends of it there are two magnetic poles - North Pole and South Pole. Magnetic lines originate from north pole of a magnet and terminate on south pole as shown in figure-4.174. At the magnetic poles of magnet, the strength of magnetic field is maximum.

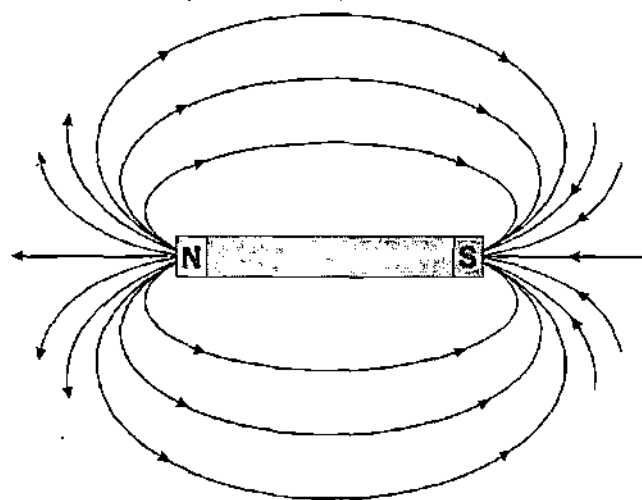


Figure 4.174

To understand how a bar magnet produces magnetic field, we first discuss about the magnetic field produced by an atom. If in an atom there are some unpaired electrons which are revolving around the nucleus then these revolving electrons can be considered as a current carrying coil which produces magnetic induction as shown in figure-4.175. Thus every atom which have unpaired electrons can be assumed like a very small coil producing its own non-zero magnetic induction.

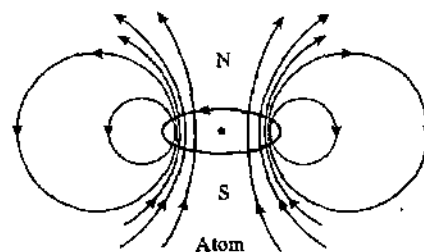


Figure 4.175

In fact every current carrying coil can be considered similar to a bar magnet. The face of coil from which the magnetic lines are emerging out can be regarded as its magnetic north pole

and the other face of coil into which the magnetic lines are getting into can be regarded as its magnetic south pole. In a bar magnet all the atoms producing magnetic induction are aligned in such a way that these all atoms produce their magnetic induction in same direction so the resulting overall magnetic field will be high in magnitude as shown in figure-4.174 above. The microscopic view of atoms inside a bar magnet is shown in figure-4.176 which explains how a bar magnet produces magnetic field.

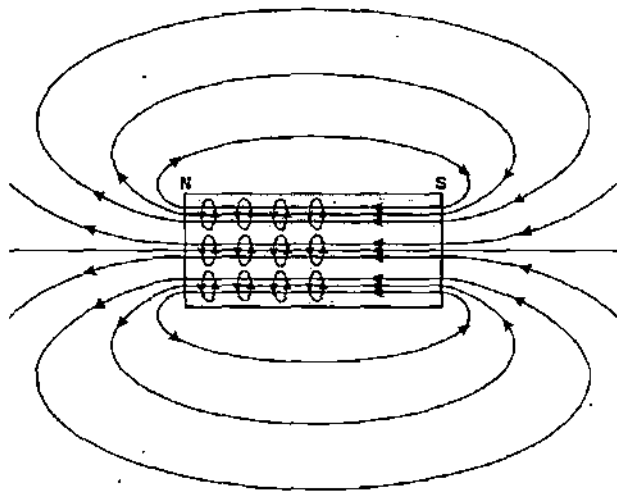


Figure 4.176

In above figure we can see that always the total magnetic flux originated from the north pole of a bar magnet is equal to the magnetic flux terminating on its south pole. Always the pole strengths of both the magnetic poles of a bar magnet are equal in magnitude.

With above explanation we can see that the magnetic lines due to a coil are closed loops thus for a bar magnet also magnetic lines are closed loops as inside the magnet the lines are in opposite direction travelling from south to north pole forming a closed loop.

Every bar magnet has a property that if it is cut or broken in parts then every part will behave as a separate bar magnet with two poles as independently all the parts of a bar magnet have internally same structure of aligned atoms producing magnetic field as shown in figure-4.177.

With the above discussion we can say that at microscopic level magnetic field is produced by atoms of substances which behave like current carrying coils. Thus in case of a current carrying coil always both the magnetic poles north and south exist together as magnetic lines are closed loop lines so we can also state that in nature magnetic unipole can never exist as the two poles can never be isolated in any situation of a device producing magnetic field.

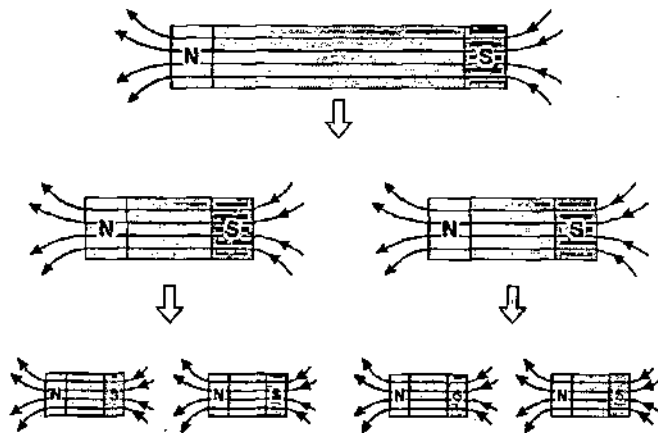


Figure 4.177

The strength of a bar magnet is characterized by its magnetic dipole moment. If the pole strength of either pole of a bar magnet is m and the separation between its poles is d then the magnetic dipole moment of a bar magnet is given as

$$\mu = md \quad \dots (4.187)$$

The symbol used for magnetic dipole moment is either μ or M . Like electric dipole moment, magnetic dipole moment is also a vector quantity with direction taken from south pole to north pole. The unit used for measurement of magnetic dipole moment is 'ampere - meter square' or ' $A \cdot m^2$ '. Magnetic dipole moment of a bar magnet in general also called as '*Magnetic Moment*' of the magnet.

4.11.6 Magnetic Induction due to a Bar Magnet on its Axis

Figure-4.178 shows a bar magnet with pole strength m and length $2d$. P is a point located at a distance r from the center and on the axis of magnet. The two magnetic poles produce magnetic induction at point P in opposite directions as shown in figure.

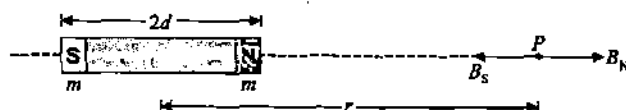


Figure 4.178

The net magnetic induction at point P is given as

$$\begin{aligned} B_P &= B_N - B_S \\ \Rightarrow B_P &= \frac{\mu_0}{4\pi} \frac{m}{(r-d)^2} - \frac{\mu_0}{4\pi} \frac{m}{(r+d)^2} \\ \Rightarrow B_P &= \frac{\mu_0 m}{4\pi} \left[\frac{1}{(r-d)^2} - \frac{1}{(r+d)^2} \right] \end{aligned}$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \frac{4rd}{(r^2 - d^2)^2}$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \frac{2Mr}{(r^2 - d^2)^2} \quad \dots (4.188)$$

The expression in equation-(4.188) gives the magnetic induction due to a bar magnet on its axial point in which we used magnetic moment of the magnet as $M = 2md$. If the bar magnet is replaced with a magnetic dipole in which $d \ll r$ as shown in figure-4.179 then the above result will be modified as given below

$$B_P = \frac{\mu_0}{4\pi} \frac{2Mr}{(r^2 - d^2)^2} \approx \frac{\mu_0}{4\pi} \frac{2M}{r^3}$$

$$\Rightarrow B_P = \frac{2KM}{r^3} \quad \dots (4.189)$$

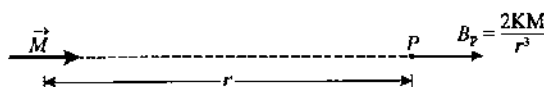


Figure 4.179

The result given in Equation-(4.189) is similar to the electric field due to an electric dipole at a point on its axis so this is easy to remember also.

4.11.7 Magnetic Induction due to a Bar Magnet on its Equatorial Line

Figure-4.180 shows a bar magnet with pole strength m and length $2d$. P is a point located at a distance r from the center and on the perpendicular bisector of magnet called its equatorial line. The two magnetic poles produce magnetic induction of equal magnitude at point P due to symmetry along the directions as shown in figure.

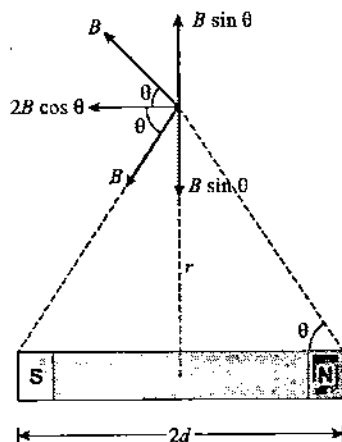


Figure 4.180

Thus the net magnetic induction at point P is given as

$$B_P = 2B \cos \theta$$

$$\Rightarrow B_P = 2 \left(\frac{\mu_0}{4\pi} \frac{m}{(r^2 + d^2)^{3/2}} \right) \cdot \frac{d}{\sqrt{r^2 + d^2}}$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \frac{M}{(r^2 + d^2)^{3/2}} \quad \dots (4.190)$$

Similar to previous article if we calculate the magnetic induction at equatorial line of a magnetic dipole as shown in figure-4.181 then it is given by modifying expression in equation-(4.190) using $d \ll r$ as

$$B_P = \frac{\mu_0}{4\pi} \frac{M}{(r^2 + d^2)^{3/2}} \approx \frac{\mu_0}{4\pi} \frac{M}{r^3}$$

$$\Rightarrow B_P = \frac{KM}{r^3} \quad \dots (4.191)$$

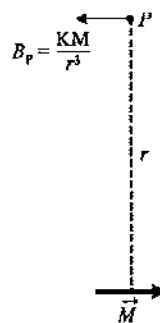


Figure 4.181

The result in equation-(4.191) is also similar to the electric field due to an electric dipole at a point on its equator.

4.11.8 Analogy between Electric and Magnetic Dipoles

In previous articles we've discussed that the expressions of magnetic induction due to a magnetic dipole at its axial and equatorial line are similar to those we've already studied for electric field produced by an electric dipole and with the analogy of these results we can analyze the magnetic induction due to a magnetic dipole in its surrounding along radial and transverse directions. Figure-4.182 shows a point P at a distance r from a magnetic dipole of dipole moment M . The line joining point P to the center of dipole is making an angle θ with the axis of dipole. With the analogy from electric dipole the radial and transverse components of magnetic induction at point P due to this magnetic dipole are given below.

Radial component of magnetic induction is given as

$$B_r = \frac{2KM \cos \theta}{r^3} \quad \dots (4.192)$$

Transverse component of magnetic induction is given as

$$B_\theta = \frac{KM \sin \theta}{r^3} \quad \dots (4.193)$$

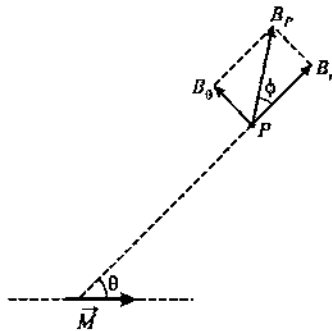


Figure 4.182

Using equation-(4.192) and (4.193), the resultant magnetic induction at point P due to magnetic dipole is given as

$$B_P = \sqrt{B_r^2 + B_\theta^2}$$

$$\Rightarrow B_P = \frac{KM}{r^3} \sqrt{1 + 3 \cos^2 \theta} \quad \dots (4.194)$$

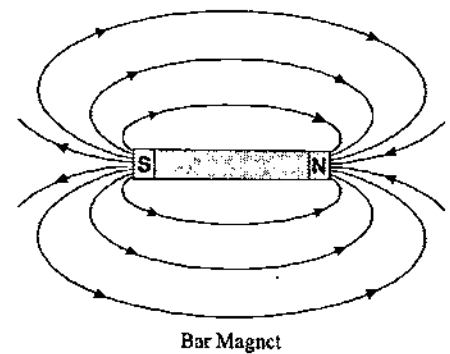
The resulting magnetic induction at point P is aligned at an angle ϕ from the radial direction as shown in figure-4.182 which is given as

$$\phi = \tan^{-1} \left(\frac{B_\theta}{B_r} \right)$$

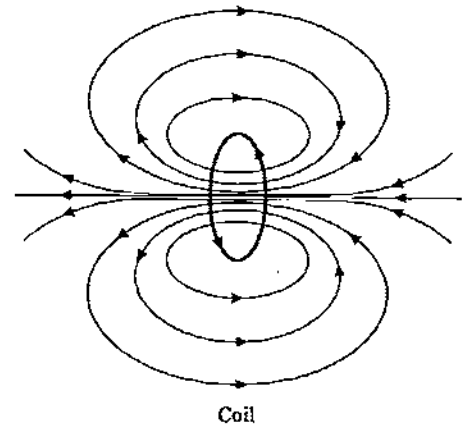
$$\Rightarrow \phi = \tan^{-1} \left(\frac{1}{2} \tan \theta \right) \quad \dots (4.195)$$

4.11.9 A Small Current Carrying Coil as a Magnetic Dipole

Figure-4.183 shows the magnetic field produced by a bar magnet and that of a current carrying coil which looks similar except the length of bar magnet but if we look at the figure-4.184 which shows the magnetic field in surrounding of a small magnetic dipole and that of a small current carrying coil. Both the configurations of magnetic field lines look almost identical thus a very small current carrying coil can be regarded as a magnetic dipole and all the results we've obtained for a magnetic dipole can be used for such small current carrying loops with magnetic dipole moment of dipole replaced with the magnetic moment of the coil.

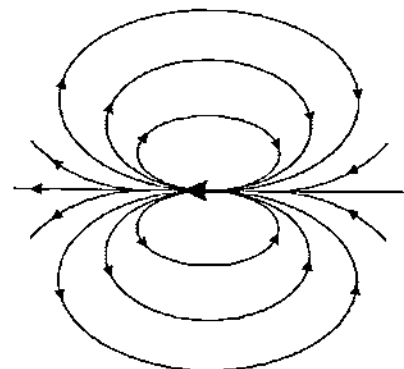


Bar Magnet

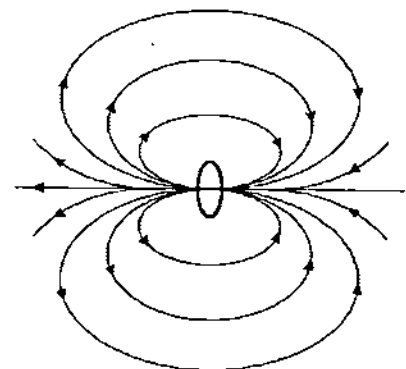


Coil

Figure 4.183



Magnetic Dipole



Small Current Coil

Figure 4.184

Figure-4.185 shows a tightly wound long solenoid and the magnetic induction produced by it by the magnetic field lines configuration. This magnetic field lines configuration is almost same as that produced by a bar magnet thus a current carrying solenoid can be considered to be behaving like a bar magnet with magnetic moment given as

$$M = NI A \quad \dots (4.196)$$

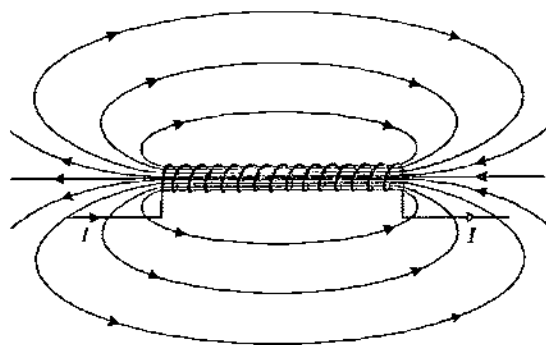
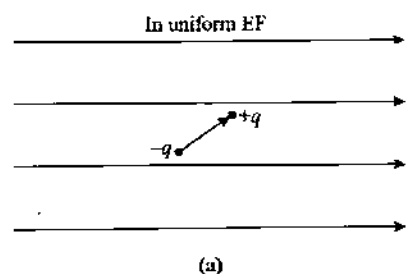


Figure 4.185

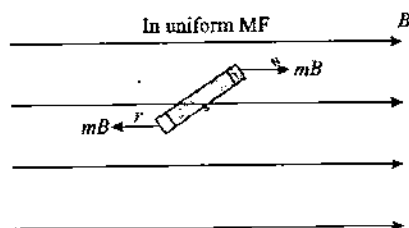
4.11.10 Force on a Magnetic Dipole in Magnetic Field

Figure-4.186 shows a magnetic dipole of dipole moment M placed in a uniform magnetic induction B . As both of its poles will experience equal and opposite forces as shown in figure, the net magnetic force on the dipole will be zero similar to an electric dipole placed in uniform electric field. Due to equal and opposite forces acting at different lines of action a couple will be produced and the torque on magnetic dipole due to these couple forces can be given by its analogy with the case of electric dipole already studied in the chapter of electrostatics as

$$\vec{\tau} = \vec{M} \times \vec{B} \quad \dots (4.197)$$



(a)



(b)

Figure 4.186

If the dipole is placed in a non uniform magnetic field then the force on it will also be non-zero which can also be given by the analogy with the case of electric dipole, given as

$$\vec{F} = \vec{M} \cdot \frac{d\vec{B}}{dx} \hat{n} \quad \dots (4.198)$$

In above case $\hat{n}$ is the unit vector along the direction of magnetic induction. Similar to the case of an electric dipole placed in non uniform electric field above equation-(4.198) gives the force on dipole along the direction of magnetic induction.

Above results obtained for a magnetic dipole can be directly applied to a small current carrying coil in different situations.

Illustrative Example 4.52

A bar magnet is 0.1m long and its pole strength is 12Am. Find the magnetic induction at a point on its axis at a distance of 0.2m from its centre.

Solution

Net magnetic induction due to bar magnet at point P is given by the vector sum of the magnetic inductions at P due to the two poles given as

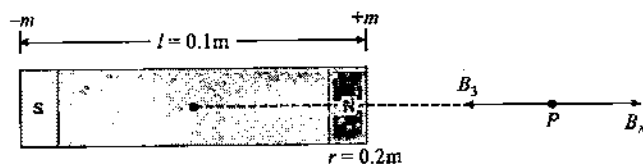


Figure 4.187

$$B_P = B_N - B_S$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \cdot \frac{m}{(r-l/2)^2} - \frac{\mu_0}{4\pi} \cdot \frac{m}{(r+l/2)^2}$$

$$\Rightarrow B_P = \frac{\mu_0}{4\pi} \left[\frac{2rl}{(r^2 - l^2/4)^2} \right] = \frac{32\mu_0 m r l}{4\pi(4r^2 - l^2)^2}$$

$$\Rightarrow B_P = \frac{32 \times 10^{-7} \times 12 \times 0.2}{(4(0.2)^2 - (0.1)^2)^2}$$

$$\Rightarrow B_P = 3.4 \times 10^{-4} \text{ T}$$

Illustrative Example 4.53

A magnetic dipole of magnetic moment M is suspended by a string in a uniform horizontal magnetic field as shown in figure-4.188. If in horizontal plane this dipole is slightly tilted

and released, show that it will execute simple harmonic motion and find its oscillation period. Consider the dipole as a uniform rod of mass m and length l .

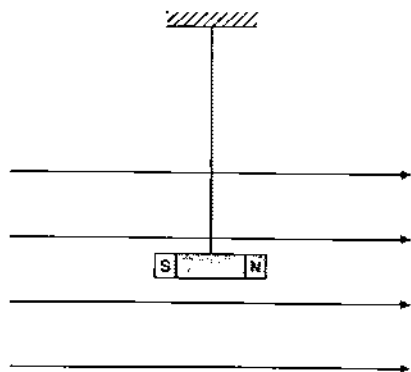


Figure 4.188

Solution

If dipole is tilted from its equilibrium position by a small angle θ , in horizontal plane, restoring torque on magnetic dipoles is given as

$$\tau_R = MB \sin \theta \approx MB\theta$$

The angular acceleration of dipole is given as

$$\alpha = -\frac{\tau_R}{I}$$

$$\Rightarrow \alpha = -\frac{MB}{(ml^2/12)}\theta$$

$$\Rightarrow \alpha = -\frac{12MB}{ml^2}\theta \quad \dots (4.199)$$

Equation-(4.199) shows that restoring angular acceleration of the dipole is directly proportional to the angular displacement hence it will execute SHM. Comparing this with angular acceleration of standard angular SHM equation given as $\alpha = -\omega^2\theta$ we get

$$\omega = \sqrt{\frac{12MB}{ml^2}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{ml^2}{12MB}}$$

Illustrative Example 4.54

A small magnetic dipole of magnetic moment $\pi \times 10^{-3} \text{ A}\cdot\text{m}^2$ is placed on the Y -axis at a distance of 0.1m from the origin with its axis parallel to X -axis. A coil having 169 turns and radius 0.05m is placed on the X -axis at a distance of 0.12m

from the origin with the axis of the coil coinciding with X -axis. Find the magnitude and direction of the current in the coil for a compass needle placed at the origin, to point in the north-south direction.

Solution

The situation described in question is shown in figure-4.189.

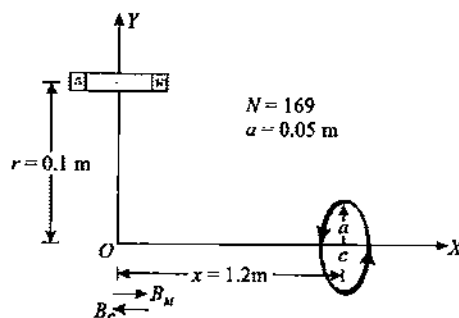


Figure 4.189

The magnetic induction at O due to magnetic dipole moment M is given as

$$B_M = \frac{\mu_0}{4\pi} \frac{M}{r^3} \text{ N/A}\cdot\text{m} \quad \dots (4.200)$$

The direction of above magnetic induction is along positive X -direction. The magnetic induction at O due to the current carrying coil of radius a and having N turns is given as

$$B_C = \frac{\mu_0 N i a^2}{2(a^2 + x^2)^{3/2}} \quad \dots (4.201)$$

The direction of above magnetic induction is along negative or positive X -direction depending upon the direction of current in the coil. As the deflection of the compass needle at O has to remain north-south that means resultant field at O other than Earth's magnetic field should be zero or in this case B_M should be equal and opposite to B_C so as to nullify it thus we have

$$\frac{\mu_0}{4\pi} \frac{M}{r^3} = \frac{\mu_0 N i a^2}{2(a^2 + x^2)^{3/2}}$$

$$\Rightarrow i = \frac{2M(a^2 + x^2)^{3/2}}{4\pi r^3 \times N a^2} \quad \dots (4.202)$$

$$\Rightarrow i = \frac{2 \times (\pi \times 10^{-3}) [(0.05)^2 + (0.12)^2]^{3/2}}{4\pi \times (0.1)^3 \times 169 \times (0.05)^2}$$

$$\Rightarrow i = 2.6 \text{ mA}$$

For the field B_C to be along negative X -direction, the current in the coil should be in anticlockwise direction.

Illustrative Example 4.55

Centres of two similar coils P and Q having same number of turns are located at the coordinates $(0.4, 0)$ and $(0, 0.3)$ such that the plane of coils are perpendicular to X and Y axes respectively. The areas of cross sections of coils P and Q are in the ratio $4 : 3$. Coil P has 16A current in clockwise direction and coil Q has $9\sqrt{3}\text{A}$ current in anti-clockwise direction as seen from the origin. A small compass needle is placed at the origin. Find the deflection of the needle, assuming the Earth's magnetic field negligible and the radii of the coils very small compared to their distance from the origin.

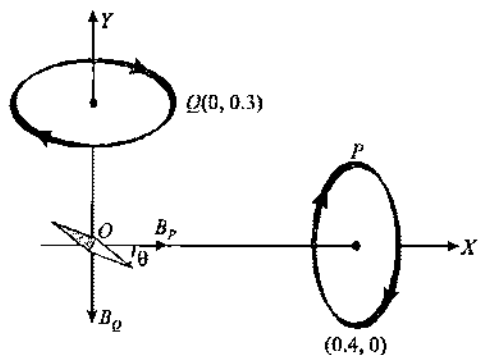


Figure 4.190

Solution

As the radius of coils is considered to be very small we can consider these coils as magnetic dipoles so the magnetic induction due to coil at a distance x along its axis is given as

$$B = \frac{2KM}{x^3} = \frac{\mu_0 NiA}{2\pi x^3}$$

Magnetic induction at O due to coil P is given as

$$B_P = \frac{\mu_0 Ni_P A_P}{2\pi x^3} \quad (\text{along } OX)$$

Similarly the magnetic induction due to coil Q at O is given as

$$B_Q = \frac{\mu_0 Ni_Q A_Q}{2\pi y^3} \quad (\text{along } YO)$$

If the compass needle makes an angle θ with X -axis, then it is along the direction of net magnetic induction at point O which is given as

$$\tan \theta = \frac{B_Q}{B_P} = \left(\frac{i_Q}{i_P} \right) \left(\frac{A_Q}{A_P} \right) \left(\frac{x^3}{y^3} \right)$$

$$\Rightarrow \tan \theta = \left(\frac{9\sqrt{3}}{16} \right) \left(\frac{3}{4} \right) \left(\frac{0.4}{0.3} \right)^3 = \sqrt{3}$$

$$\Rightarrow \theta = \tan^{-1} \sqrt{3} = 60^\circ$$

Illustrative Example 4.56

A magnetic dipole with a dipole moment of magnitude 0.020Am^2 is released from rest in a uniform magnetic field of magnetic induction 52mT . When the dipole rotates due to magnetic couple on it through the orientations where its dipole moment is aligned with the magnetic field, its kinetic energy is 0.80 mJ .

(a) What is the initial angle between the dipole moment and the magnetic field?

(b) What is the angle when the dipole be at rest again next time?

Solution

(a) By energy conservation, we have

$$U_\theta + K_\theta = U_{0^\circ} + K_{0^\circ}$$

$$\Rightarrow (-MB \cos \theta) + 0 = (-MB \cos 0^\circ) + K_{0^\circ}$$

$$\Rightarrow MB(1 - \cos \theta) = K_{0^\circ}$$

$$\Rightarrow \cos \theta = 1 - \frac{K_{0^\circ}}{MB}$$

$$\Rightarrow \cos \theta = 1 - \frac{0.8 \times 10^{-3}}{0.02 \times 52 \times 10^{-3}} = 0.23$$

$$\Rightarrow \theta = \cos^{-1}(0.23) = 76.67^\circ$$

(b) As already discussed that dipole will execute oscillatory motion so on the other side of equilibrium position it will come to rest at the same angle 76.67° .

Illustrative Example 4.57

Figure-4.191 shows two small bar magnets having dipole moments M_1 and M_2 placed at separation r . Find the magnetic interaction energy of this system of dipoles for $d \ll r$.

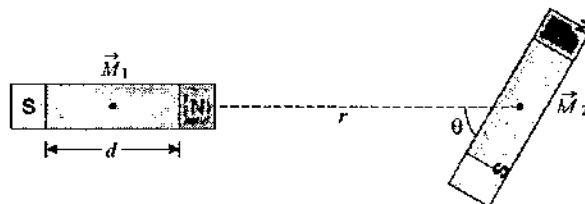


Figure 4.191

Solution

Magnetic induction at the location of M_2 due to M_1 is given as

$$B_1 = \frac{2KM_1}{r^3} = \frac{\mu_0 M_1}{2\pi r^3}$$

Interaction energy of M_2 in the field of M_1 is given as

$$U = -\vec{M}_2 \cdot \vec{B}_1$$

$$\Rightarrow U = -M_2 B_1 \cos \theta = -\frac{\mu_0 M_1 M_2}{2\pi r^3} \cos \theta$$

4.12 Terrestrial Magnetism

It has been known from centuries that on suspending a magnet freely it aligns with its axis along north-south direction. This leads physicists to study more about the magnetic field produced by Earth and magnetic properties associated with it. The study of Earth's magnetism and its magnetic properties is called '*Terrestrial Magnetism*'. Earth's magnetic field is also called '*Geomagnetic Field*'. Cause of Earth's magnetic field is the rotation of ionized particles in molten core of earth which constitutes the convection currents at the Earth's core and behave like a coil which produces magnetic field in surrounding and it extends from inner core to outer space. The configuration of Earth's magnetic field is shown in figure-4.192.

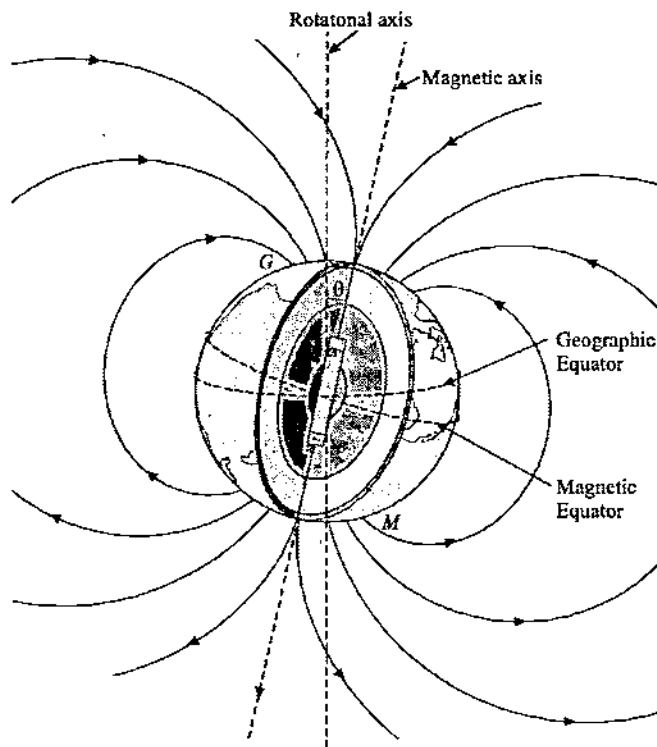


Figure 4.192

As shown in above figure, we can see that the axis of Earth's magnetic field is slightly tilted from its axis of rotation (Geographic Axis) at about 11° . This angle changes with time

very slowly due to some geophysical factors not in scope of this book. This axis along which Earth's magnetic coil is considered is called '*Magnetic Axis*' of Earth.

The points on Earth surface where this magnetic axis meet are called magnetic poles. Due to the direction of convection current inside the core of Earth its magnetic induction comes out from the geographic south pole and it gets into the Earth at geographic north pole. Thus the magnetic poles of Earth are located near to the opposite geographic poles as shown in figure-4.192.

4.12.1 Direction of Earth's Magnetic Field on Surface

As discussed in previous article the magnetic poles of earth are located near to the opposite geographic poles shown in figure-4.192. With the configuration of magnetic lines of forces in surrounding we can see that in northern hemisphere of Earth magnetic field direction is inward into the Earth surface and in southern hemisphere of Earth direction of magnetic field is outward from Earth surface.

Figure-4.193(a) shows the Earth's magnetic induction B_E at surface of Earth in northern hemisphere at point G in figure-4.192. The angle θ which the magnetic induction makes with the horizontal at this point is called '*Dip Angle*'. Similarly in southern hemisphere of Earth at point H shown in figure-4.193(a) the magnetic induction is shown in figure-4.193(a). The horizontal and vertical components of Earth's magnetic induction at point G are given as

$$B_H = B_E \cos \theta \quad \dots (4.203)$$

and
$$B_V = B_E \sin \theta \quad \dots (4.204)$$

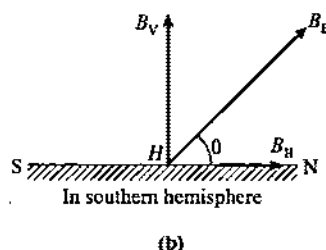
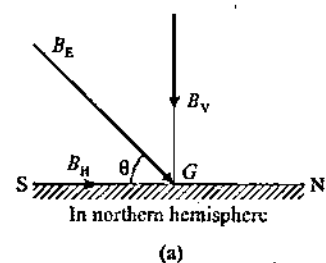


Figure 4.193

Dividing equations-(4.203) and (4.204), we have

$$\tan \theta = \left(\frac{B_V}{B_H} \right)$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{B_V}{B_H} \right) \quad \dots (4.205)$$

At a point in southern or northern hemisphere we can see that always the horizontal component of Earth's magnetic induction points from approximately south to north at that point that's why when we suspend a magnet freely at any point due to the magnetic torque on it due to horizontal component of Earth's magnetic induction the magnet aligns along north-south direction.

A magnetic compass also works because of the torque on the magnetic needle due to horizontal component of Earth's magnetic induction. As we've discussed that magnetic poles of Earth do not exactly coincide with the geographic poles but still the direction of B_H approximately points from south to north direction at that point which fairly works well for the purpose of navigation as shown in figure-4.194.

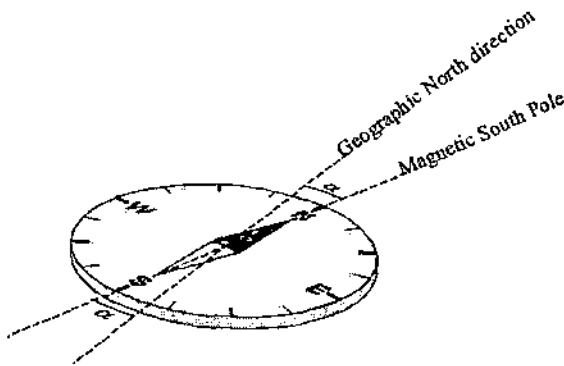


Figure 4.194

4.12.2 Some Definitions and Understanding Earth's Magnetic Field

While analyzing Earth's magnetism there are some terms we need to understand clearly which helps in various applications related to Earth's magnetism. We will discuss and understand all these terms one by one.

Geographic Meridian : At any point on Earth surface this is a vertical plane containing north-south direction at that point. Figure-4.195 shows the geographic meridian at a point A on Earth Surface. This can also be defined as a vertical plane at any point on earth surface such that every horizontal line along east-west direction passing through this plane will cross the plane normally.

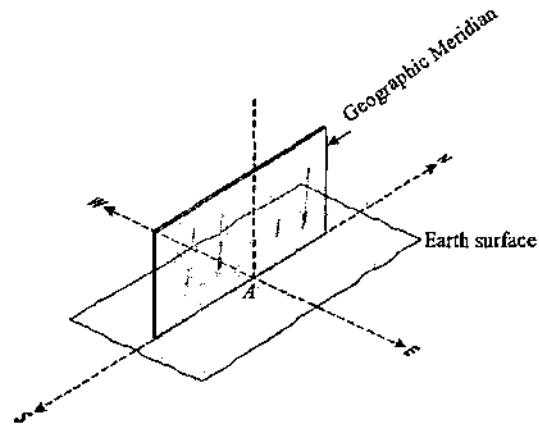


Figure 4.195

Magnetic Meridian : At any point on Earth surface this is a vertical plane containing the direction of net magnetic induction of Earth's field. If a surface is placed in this vertical plane then the magnetic flux of Earth's magnetic field through this surface is always zero. Figure-4.196 shows both Geographic and Magnetic Meridian at a point A on Earth surface.

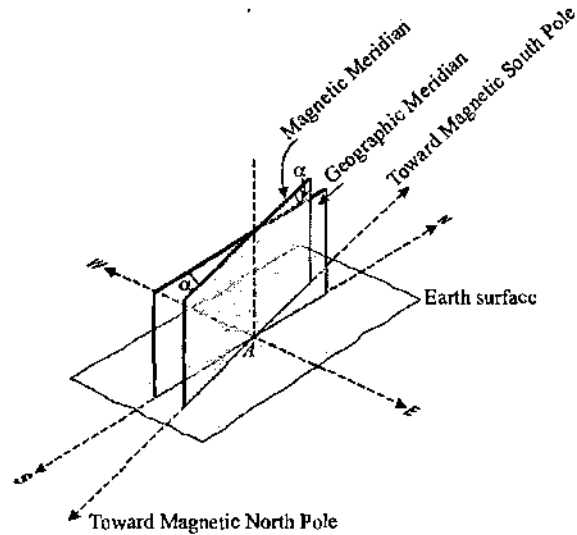


Figure 4.196

Geographic and Magnetic Equator : Geographic equator is the circular line on earth which divides the Earth in two equal hemispheres at a plane normal to the axis of rotation of Earth whereas magnetic equator is also a circular line on Earth surface at which Earth's magnetic field is horizontal or dip angle is 0° . Both geographic and magnetic equators are shown in figure-4.192.

Magnetic Declination : At any point on Earth surface it is the angle between the magnetic north-south direction and true north-south direction at that point. This is also defined as the angle between geographic and magnetic meridian at a point on Earth surface.

Agonic Line : It is the circular line on Earth surface which passes through the geographic as well as magnetic poles of Earth. Figure-4.197 shows the agonic line.

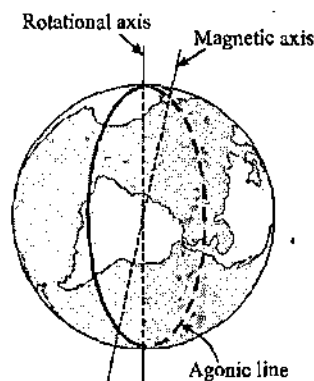


Figure 4.197

Isogonic Lines : These are lines on Earth surface joining the points on Earth surface where magnetic declination is same. Further details about these lines are not in our scope of discussion.

Aclinic Line : It is the circular line on Earth surface at every point of which dip angle is 0° . This line is same as magnetic equator already discussed.

Isoclinic Lines : These are lines on Earth surface joining the points on Earth surface where dip angle has same values.

Dip Needle : There is a device similar to compass needle which is used to measure the dip angle at a point on earth surface.

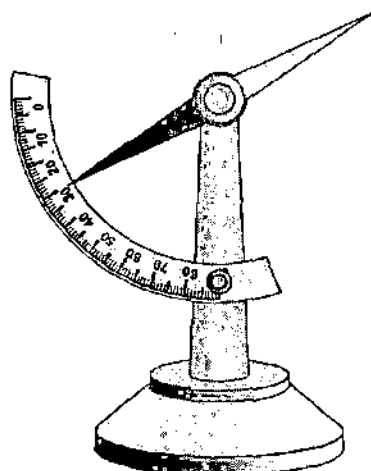


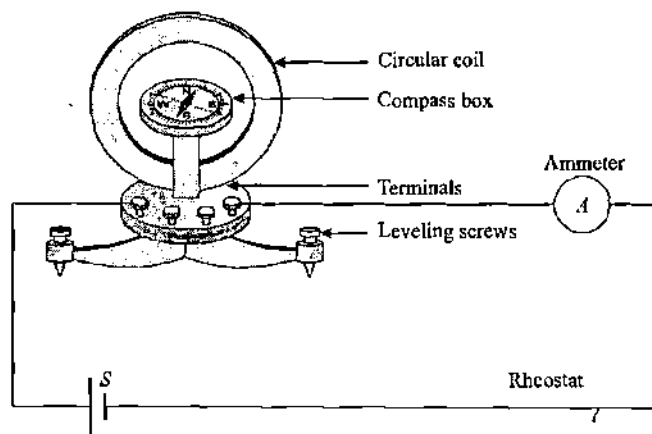
Figure 4.198

This is called '*Dip Needle*' shown in figure-4.198. This needle is mounted to rotate freely in a vertical plane. First the needle plane is placed in magnetic meridian so that the needle gets aligned along the direction of net magnetic induction of Earth

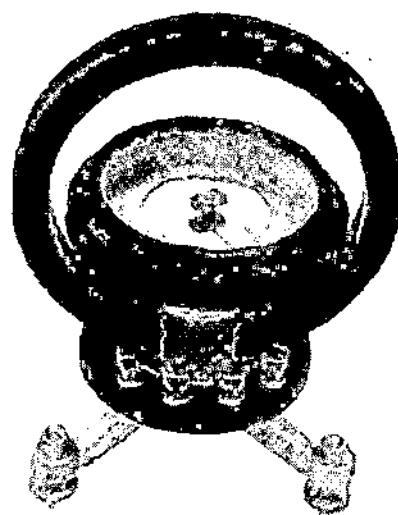
as shown in figure then on the pre-calibrated scale dip angle can be measured.

4.12.3 Tangent Galvanometer

Tangent galvanometer is a device used to calculate horizontal component of Earth's magnetic induction. Figure-4.199(a) shows a tangent galvanometer in which there is a circular coil C mounted on a stand in vertical plane which can be rotated on the mount. A horizontal compass needle is fixed at the center of coil as shown. On the mount of the device the ends of the coil wire are connected to terminals at which external lead wires can be joined and current is passed in the coil C . Figure-4.199(b) shows the picture of an actual tangent galvanometer used in labs.



(a)



(b)

Figure 4.199

To setup the tangent galvanometer for experiment first the coil C is rotated and aligned along the magnetic meridian. For this coil C is to be rotated till the compass needle comes in the vertical plane of coil as the vertical plane in which compass needle naturally floats is the magnetic meridian at a point on earth surface.

Now a current is passed through the coil C and it is gradually increased by sliding the rheostat. Due to current in coil it produces a magnetic induction at its center due to which the magnetic needle deflects. If B_C is the magnetic induction at center of coil due to coil current and B_H is the horizontal component of earth's magnetic induction at this point then deflection of compass needle from its initial direction is given as

$$\theta = \tan^{-1} \left(\frac{B_C}{B_H} \right) \quad \dots (4.206)$$

Figure-4.200 shows the deflection of compass needle at the center of coil C due to magnetic induction of the coil current.

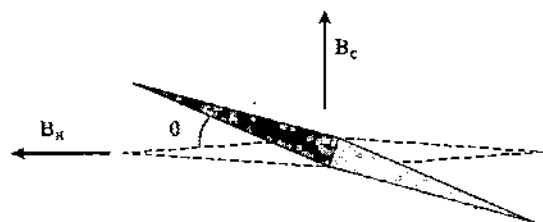


Figure 4.200

While executing the experiment we change the current in coil till the deflection angle of needle becomes 45° and at this state we measure the reading of ammeter, say this is I . Thus at $\theta = 45^\circ$ we use

$$\begin{aligned} B_H &= B_C \\ \Rightarrow B_H &= \frac{\mu_0 IN}{2R} \quad \dots (4.207) \end{aligned}$$

By substituting the values on RHS of equation-(4.207) we can calculate the value of B_H . The experiment can be repeated for different values of N by selecting different terminals on the mount of tangent galvanometer and calculate several values of B_H of which average can be obtained along with error analysis and standard deviation of the experimental value of B_H .

4.12.4 Deflection Magnetometer

Deflection magnetometer is an experimental setup used to measure the magnetic moment and pole strength of a bar magnet. It is also used to compare the pole strengths of two bar magnets. A deflection magnetometer consists of a small magnetic compass needle pivoted at the center of a scale as

shown in figure-4.201. The compass is fixed at the scale in such a way that its circular scale can be rotated which is marked with $0^\circ-90^\circ-0^\circ-90^\circ$ in perpendicular direction as shown which are kept exactly perpendicular to the scale as shown in figure-4.201.

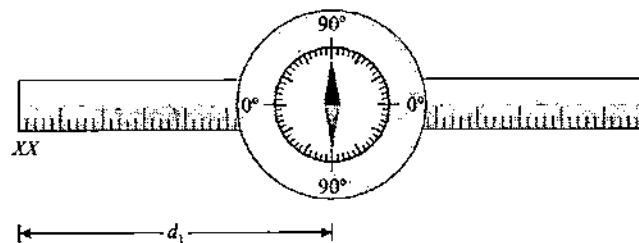


Figure 4.201

There are two positions of the magnetometer for measuring the magnetic moment of a bar magnet or a magnetic dipole. These are called 'tanA' and 'tanB' positions. In tanA position of magnetometer, the scale of magnetometer is placed at a point along east-west line as shown in figure-4.202 and it is adjusted so that the compass needle points exactly along N-S direction of compass which is perpendicular to the scale. The magnetic dipole of which magnetic moment is to be measured is also placed along the scale line as shown at some distance d_1 from the compass needle. The magnetic induction due to the magnetic dipole at the location of compass is given as

$$B_r = \frac{2KM}{d_1^3}$$

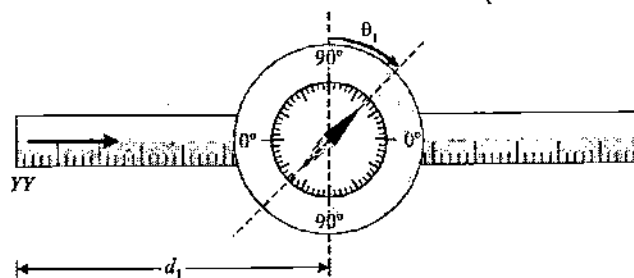


Figure 4.202

If horizontal component of earth's magnetic field is B_H then in tanA position deflection of compass needle is given as

$$\tan \theta_1 = \frac{B_r}{B_H} \quad \dots (4.208)$$

$$\Rightarrow B_r = B_H \tan \theta_1$$

$$\Rightarrow \frac{2KM}{d_1^3} = B_H \tan \theta_1$$

$$\Rightarrow M = \frac{2\pi B_H d_1^3 \tan \theta_1}{\mu_0} \quad \dots (4.209)$$

Now the magnetometer scale is rotated by 90° and it is placed along north-south direction and its circular scale is also rotated with 90° - 90° line along N-S direction and it is adjusted so that magnetic needle exactly points parallel to the scale. This is called $\tan B$ position of the magnetometer which is shown in figure-4.203. At the same distance from the compass the magnetic dipole is now placed along east-west line perpendicular to the scale as shown in figure. At the location of compass needle the magnetic induction due to the magnetic dipole is given as

$$B_\theta = \frac{KM}{d_1^3}$$

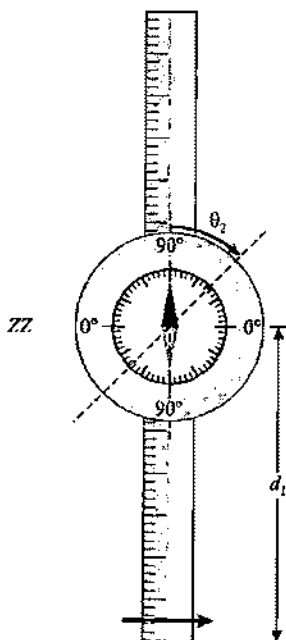


Figure 4.203

Now again the compass needle deflects under the influence of two mutually perpendicular magnetic fields so its deflection is given as

$$\tan \theta_2 = \frac{B_\theta}{B_H} \quad \dots (4.210)$$

$$\Rightarrow B_r = B_H \tan \theta_2$$

$$\Rightarrow \frac{KM}{d_1^3} = B_H \tan \theta_2$$

$$\Rightarrow M = \frac{4\pi B_H d_1^3 \tan \theta_2}{\mu_0} \quad \dots (4.211)$$

Using equation-(4.209) and (4.211) magnetic moment of magnetic dipole can be calculated. If we use a bar magnet then instead of results of magnetic dipole we need to use the

result of magnetic induction due to a bar magnet along its axial point in $\tan A$ position and result of magnetic induction due to a bar magnet at its equatorial point in $\tan B$ position.

4.12.5 Earth's Magnetic Field in Other Units

Earth's magnetic field is produced by the convection currents at the core of earth and the field which is observed outside the earth's surface magnetizes different materials found on earth surface and inside as well. Over a long period of time such materials transform into magnets as these are magnetised by earth's magnetic field. So it was considered that earth's magnetic field is independent of the type of material which is being magnetized and it was used to be referred as '*Earth's Magnetizing Field*' and it was denoted by the symbol $\bar{H}$ which is measured in units of '*Ampere per meter*' or '*A/m*'. After the detailed study of magnetic induction over a period of time it was understood that magnetic induction changes when the field enters inside a material medium due to polarization of medium similar to the case of electric field when it enters in a dielectric medium. The only difference in magnetic induction and magnetizing field is that of medium dependency and magnetic induction is measured in units of '*Tesla*' or '*T*'. Still in many question sometimes earth's magnetic field is given in units of '*A/m*' which can be converted into '*T*' by using the relation in $\bar{B}$ and $\bar{H}$ as

$$\bar{B} = \mu \bar{H}$$

$$\Rightarrow \bar{B} = \mu_0 \mu_r \bar{H} \quad \dots (4.212)$$

About the above relation given in equation-(4.212) and magnetizing field we will study in detail under the topic of magnetic properties of material in next chapter.

4.12.6 Apparent angle of dip at a point on Earth's Surface

When at a point on earth's surface dip needle is placed in magnetic meridian then it aligns along the direction of earth's magnetic field and on the vertical circular scale we can measure the dip angle at that location.

When the dip needle is placed in a plane which is at an angle θ to the magnetic meridian then in this plane the horizontal component of earth's magnetic field is given as

$$B'_H = B_H \cos \theta$$

If θ'_D is the dip angle measured by the dip needle which is called apparent dip angle at this location then it is given as

$$\tan \theta'_D = \frac{B_V}{B_H \cos \theta} \quad \dots (4.213)$$

True dip angle θ_D at this location is given as

$$\tan \theta_D = \frac{B_V}{B_H} \quad \dots (4.214)$$

From above equations-(4.213) and (4.214) we have

$$\tan \theta_D = \tan \theta'_D \cos \theta \quad \dots (4.215)$$

Above equation-(4.215) relates the true dip angle, apparent dip angle and the angle between the vertical plane and the magnetic meridian in which the dip angle is apparently measured.

Illustrative Example 4.58

At a point on earth surface horizontal component of earth's magnetic field is $40 \mu\text{T}$ and dip angle is 30° . Find the total magnetic field of earth at this point.

Solution

In this case we use

$$B_H = B_E \cos \theta$$

$$\Rightarrow B_E = \frac{B_H}{\cos \theta} = \frac{40 \times 10^{-6}}{\sqrt{3}/2}$$

$$\Rightarrow B_E = 46 \mu\text{T}$$

Illustrative Example 4.59

The radius of tangent galvanometer coil is 16cm. Find the number of turn in its coil if a current of 40mA is required to produce a deflection of 30° in it from magnetic meridian. Horizontal component of earth's magnetic field is $36 \times 10^{-6} \text{ T}$.

Solution

In a tangent galvanometer if needle deflection is θ from magnetic meridian as shown in figure-4.204 we use

$$B_C = B_H \tan \theta$$

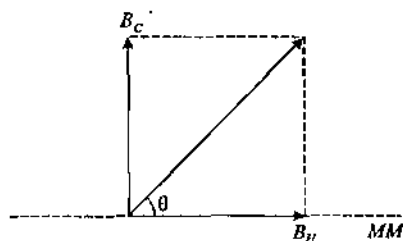


Figure 4.204

$$\Rightarrow \frac{\mu_0 NI}{2R} = B_H \tan \theta$$

$$\Rightarrow N = \frac{2RB_H \tan \theta}{\mu_0 I}$$

$$\Rightarrow N = \frac{2 \times 0.16 \times 36 \times 10^{-6} \times 1/\sqrt{3}}{4\pi \times 10^{-7} \times 40 \times 10^{-3}}$$

$$\Rightarrow N = 0.0132 \times 10^4 = 132 \text{ turns}$$

Illustrative Example 4.60

In the magnetic meridian of a certain place at the center of the coil of a tangent galvanometer, the horizontal component of earth's magnetic field is 0.26 G and the dip angle is 60° . Find :

- Vertical component of earth's magnetic field
- The net magnetic field at this place.
- If a current is passed in tangent galvanometer, its coil produces a magnetic induction $3.47 \times 10^{-3} \text{ T}$. Calculate the deflection in compass needle of tangent galvanometer.

Solution

- Vertical component of earth's field is given as

$$B_V = B_H \tan \theta$$

$$\Rightarrow B_V = (0.26) \tan 60^\circ = 0.45 \text{ G}$$

- Horizontal component and total field of earth are related as

$$B_H = B_E \cos \theta$$

$$\Rightarrow B_E = \frac{B_H}{\cos \theta} = \frac{0.26}{\cos 60^\circ} = 0.52 \text{ G}$$

- At the center of coil the horizontal component of earth's magnetic induction in tesla is given as

$$B_H = 0.26 \times 10^{-4} \text{ T}$$

The deflection in compass needle of a tangent galvanometer is given as

$$\theta = \tan^{-1} \left(\frac{B_C}{B_H} \right)$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{3.47 \times 10^{-3}}{2.6 \times 10^{-3}} \right)$$

$$\Rightarrow \theta = \tan^{-1}(1.33) = 63^\circ$$

Electromagnetic Induction and Alternating Current

FEW WORDS FOR STUDENTS

After the discovery of Faraday's law of Electromagnetic Induction only generation of electricity became easier and cheaper. In magnetism we studied that variation of electric field induces magnetic field and its converse is the phenomenon of electromagnetic induction. In this chapter we will study that according to the law of conservation of energy the work which is done in changing magnetic field associated with a conductor transforms into generation of electricity by induction and whenever work is done always there is some opposition due to which negative work is done and energy is extracted from some body. This negative work is also accounted in determining the direction of induced current and emf which is studied under Lenz's law. Overall this chapter is going to cover the most useful applications of electromagnetism in real life.

CHAPTER CONTENTS

5.1	<i>Faraday's Law of Electromagnetic Induction</i>	5.7	<i>Mutual Induction</i>
5.2	<i>Time Varying Magnetic Fields (TVMF)</i>	5.8	<i>LC Oscillations</i>
5.3	<i>Self Induction</i>	5.9	<i>Magnetic Properties of Matter</i>
5.4	<i>Growth of Current in an Inductor</i>	5.10	<i>Alternating Current</i>
5.5	<i>Energy Stored in an Inductor</i>	5.11	<i>AC Circuit Components</i>
5.6	<i>Decay of Current in RL Circuit</i>	5.12	<i>Phasor Analysis</i>
		5.13	<i>Power in AC Circuits</i>
		5.14	<i>Transformer</i>

COVER APPLICATION

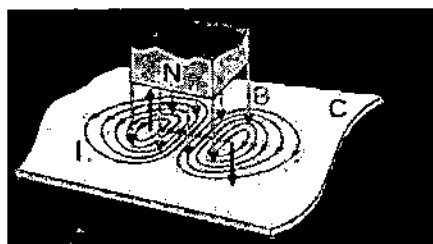


Figure-(a)

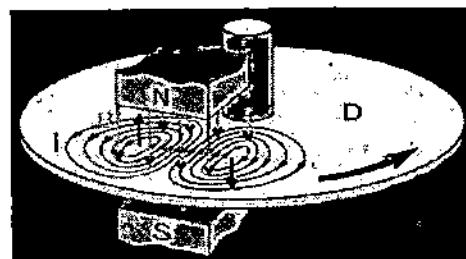


Figure-(b)

Figure-(a) shows a moving metal strip in the magnetic field of a magnetic pole and due to motion of free electrons in the strip these electrons experience a magnetic force and start flowing in closed loops within strip as shown. Such currents are called eddy currents and the magnetic field exerts an opposing magnetic force on these currents and the motion of metal body is opposed. Figure-(b) shows a rotating metal disc and a magnetic field due to magnetic poles passes through the disc as shown and it develops eddy currents and an opposing force on these currents which causes an opposite torque on the rotating disc. This phenomenon is used in design eddy current braking system for automobiles.

We've studied in previous chapter that magnetic field never does any work on charges as magnetic force acts in direction normal to their motion. We can modify this statement as net work done by magnetic force on moving charges is always zero but it is possible that there are work done involved by magnetic forces in a system but equal positive and negative work causes no net work done. In the phenomenon of electromagnetic induction magnetic force does negative work on mechanical agents who are displacing a body which contains charges and extract energy from mechanical work and does positive work on charges in inducing EMF in the conducting bodies which may cause induced current to flow if circuit is closed. Always total work by magnetic forces is zero in system but it acts as a mediator in transformation of mechanical energy to electrical energy.

5.1 Faraday's Law of Electromagnetic Induction

Based on several experiments Michael Faraday observed that due to relative motion between a magnetic field and a conductor if magnetic lines are cut by the conductor or total magnetic flux through a coil changes then an EMF is induced in the conductor or coil. This phenomenon is called Electromagnetic Induction.

It is analyzed that the magnitude of induced EMF is equal to the rate at which the magnetic flux associated with the conductor or coil changes or being cut. If in time dt the magnetic flux cut or change by $d\phi$ then the magnitude of induced EMF in the conductor or coil is given as

$$e = \left| \frac{d\phi}{dt} \right| \quad \dots (5.1)$$

As already discussed above that the EMF is induced with the energy extracted from mechanical energy causing the relative motion between magnetic field and conductor so the induced EMF always opposes the mechanical motion causing the change of magnetic flux thus equation-(5.1) can be rewritten without modulus as

$$e = - \frac{d\phi}{dt} \quad \dots (5.2)$$

Above equation-(5.2) is commonly known as 'Faraday's Law of Electromagnetic Induction'. There are different situations in which electromagnetic induction can take place which are shown in figure-5.1. Figure-5.1(a) shows a fixed circular coil connected with a galvanometer and a bar magnet is coming close to the coil along its axis as shown. Due to the motion of magnet the magnetic flux through the coil changes and it induces an EMF in the coil. As wire of coil is forming a closed loop, an induced current also flows and galvanometer shows some deflection. Figure-5.1(b) shows a similar situation but in this case magnet is fixed and coil setup is moving. Again due to continuous change in magnetic flux through the coil an EMF is induced in the coil causing an induced current to flow through

galvanometer. In both of these cases if the speed of motion between coil and magnet increases the rate of flux change increases causing higher EMF to be induced more induced current will flow.

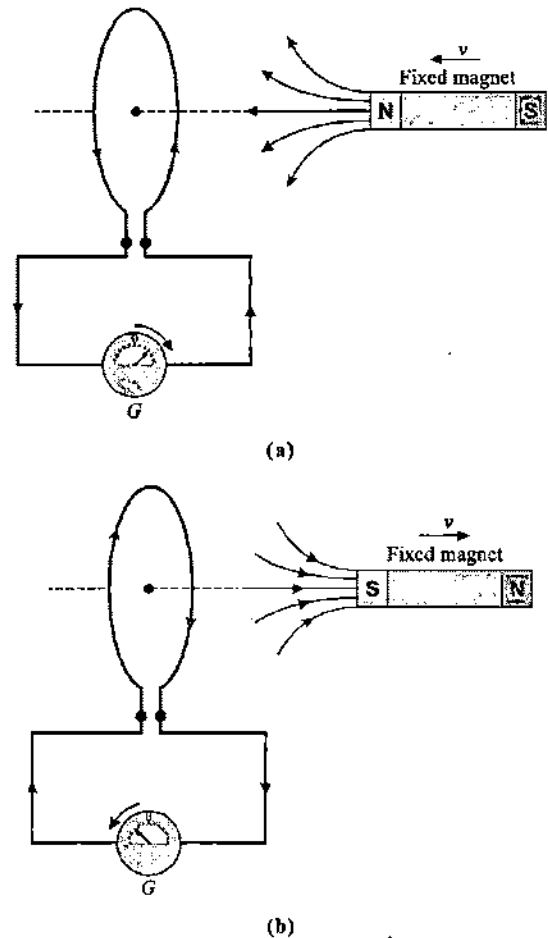


Figure 5.1

Figure-5.2 shows a system of two parallel rails connected to a galvanometer and this setup is placed in a region having a normal magnetic field. A sliding wire AB is placed on the rails which moves at some speed maintaining contacts with the rails. In this case the sliding wire is cutting the magnetic lines or the magnetic flux on the left part of wire which is passing through the closed loop is changing due to which an EMF is induced in the sliding wire causing a current flow through the circuit.

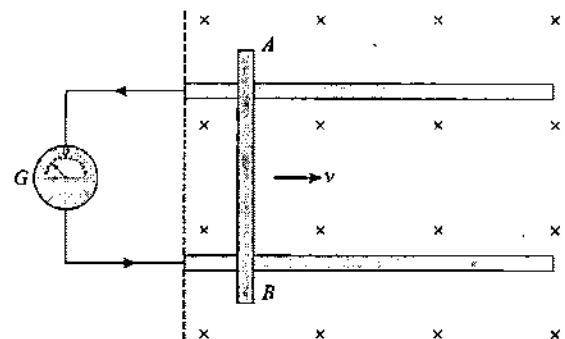


Figure 5.2

There can be many such practical situations in which the magnetic flux associated with a closed loop or a coil changes or flux is cut by a conductor then an EMF is induced.

5.1.1 Lenz's Law

We've already discussed in previous article that the direction of induced EMF in electromagnetic induction is such that it opposes the causes of change of magnetic flux so that negative work done in processes causing change of flux will be used in generation of electrical EMF that's why in equation-(5.2) there is a negative sign which denotes that sign of induced EMF is opposite to the flux change.

Lenz's Law is a way to understand the application of Newton's third law and conservation of energy in circuits when electromagnetic induction takes place. According to Newton's third law equal and opposite forces cause both positive and negative work to be done simultaneously and by conservation of energy the amount of energy extracted from the processes causing flux variation is equal to the electrical energy generated in form of EMF. In upcoming articles we will mathematically verify this fact also.

In different cases of electromagnetic induction Lenz's law can be directly used to determine the direction of induced EMF or induced current in conductors or coils through which magnetic flux is changing. The Lenz's law can be stated as

"The direction of induced EMF or induced current in a conductor or coil due to electromagnetic induction is such that the effects produced by the induced EMF or current opposes the cause of change in magnetic flux or the cause due to which electromagnetic induction is taking place."

Figure-5.3 shows a circular coil placed in plane perpendicular to a magnetic field. If magnetic field starts increasing with time then magnetic flux through the coil area starts increasing. This induces an EMF in the coil and an induced current also. The direction of induced current is such that it opposes the cause of induction.

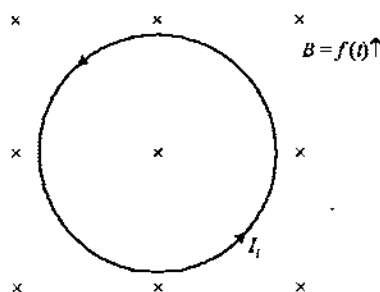
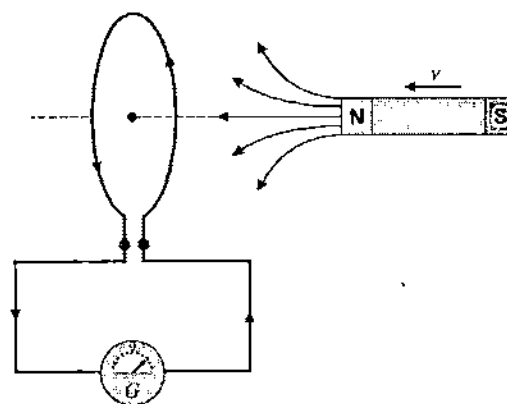


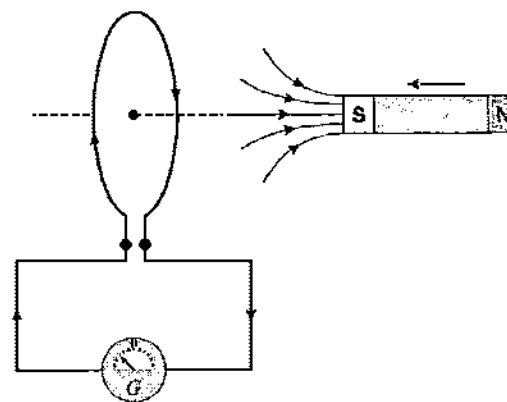
Figure 5.3

In this case the cause of induction is increase in magnetic flux so direction of induced current is such that it produces its own magnetic field in outward direction to oppose the increase in external magnetic field thus the direction of induced current must be anticlockwise in this case as shown.

Figure-5.4(a) and (b) shows a fixed coil toward which a bar magnet is moving along its axis with its north and south pole facing the coil. In these two cases we can see that current induced in coil by electromagnetic induction is such that it produces its own magnetic field in opposition to the fields of the bar magnet which was increasing due to motion of magnet toward the coil.



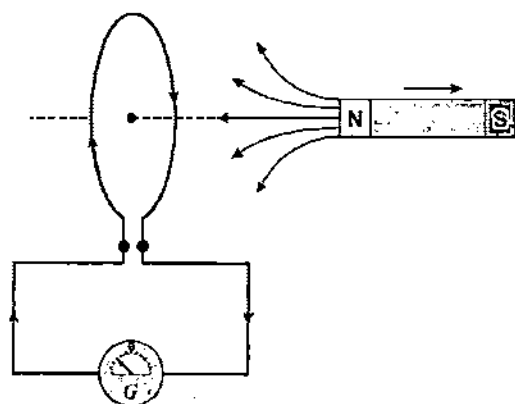
(a)



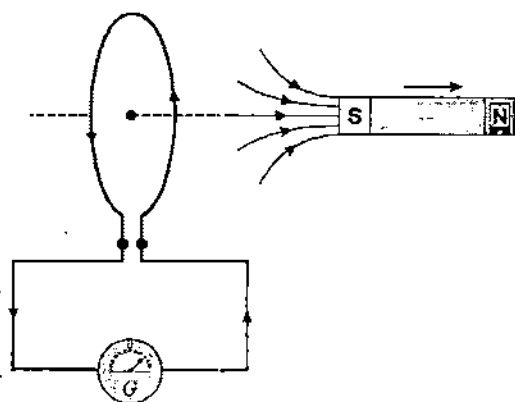
(b)

Figure 5.4

Figure-5.5(a) and (b) shows a situation similar to that shown in figure-5.4 but in this case bar magnet is moving away from the coil due to which the magnetic flux through the coil is decreasing. The current induced in coil in this case should be such that the magnetic induction due to induced current in coil is in the same direction as that of magnet because with the motion of magnet away from the coil magnetic flux through the coil decreases and the induced current should oppose the decrease of magnetic flux.



(a)



(b)

Figure 5.5

Another direct method to find the induced current using Lenz's law in above cases of a magnet moving in front of a coil is to consider coil as a magnet with opposite pole inducing on coil if magnet is moving away from it to attract it and same pole is induced on coil if magnet is moving toward it to repel it. Figure-5.6 shows the coil faces with currents in it behaving like a south or a north pole. If we look at coil and a clockwise current is flowing in it then this face is behaving like a south pole and if anticlockwise current is flowing in it then this face is behaving like a north pole. Using this logic also the direction of current in coils shown in figure-5.4 and 5.5 can be determined quickly.

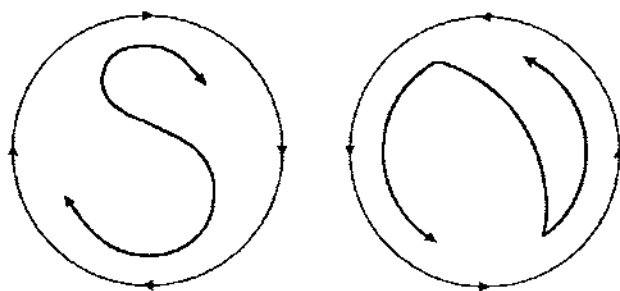
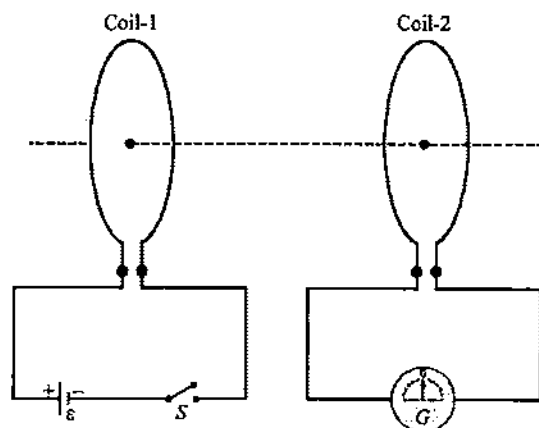
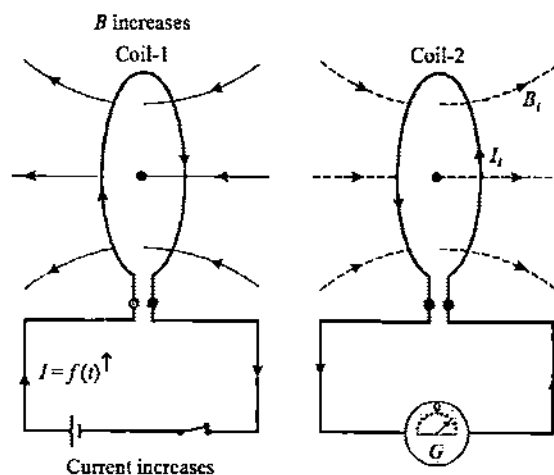


Figure 5.6

Figure-5.7(a) shows a setup of two coaxial coils. One connected with a battery and a switch and other with a galvanometer. When the switch is closed as shown in figure-5.7(b) just after closing the switch a current flows in coil 1 due to which it generates a magnetic induction and a part of its magnetic flux passes through coil 2 due to which a current is induced in coil 2 which opposes the flux of coil 1 because initially no flux was there in coil 2 and it grows because of closing the switch.



(a)



(b)

Figure 5.7

When current in coil 1 becomes steady then flux through coil 2 also becomes constant and its induced current drops to zero as shown in figure-5.8(a) because induction takes place when flux changes which happens again when we open the switch as shown in figure-5.8(b). The direction of induced current in coil 2 as shown in figure-5.8(b) is self explanatory as it should oppose the change in flux in coil 2 which decreases on opening the switch.

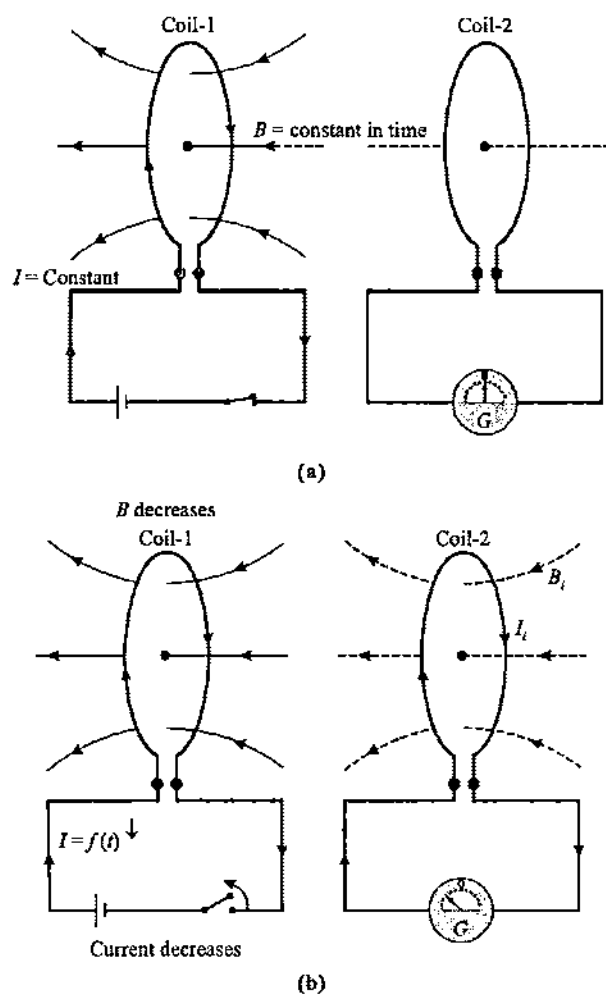


Figure 5.8

We will consider one more illustration to understand application of Lenz's law. Figure-5.9 shows a circular coil placed near to a long wire connected with a battery and a rheostat. Due to the current in long wire magnetic field is passing through the circular coil. If current in wire is steady, flux through the coil remain constant and thus no current is induced in coil.

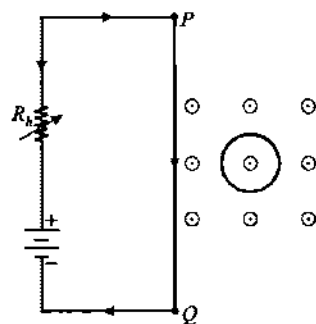


Figure 5.9

If in above case resistance of rheostat is gradually decreased then current in wire increases which also increases the magnetic flux through the coil and due to this a clockwise current is induced in coil as shown in figure-5.10(a) to oppose the outward increasing magnetic flux. If resistance of rheostat is increased

gradually then current in wire decreases which also decreases the magnetic flux through coil and due to this an anticlockwise current is induced in coil as shown in figure-5.10(b) to oppose the outward decreasing magnetic flux.

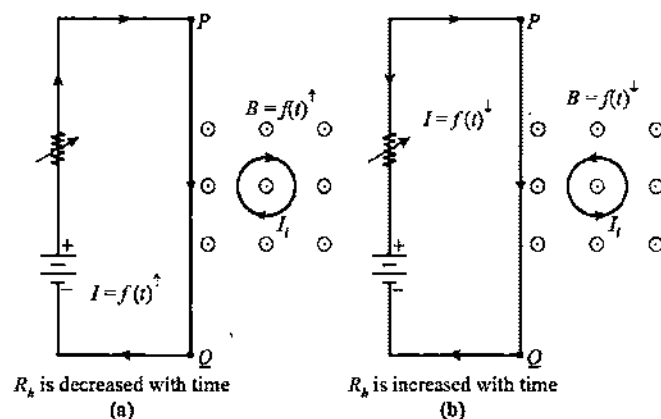


Figure 5.10

5.1.2 Motion EMF in a Straight Conductor Moving in Uniform Magnetic Field

Figure-5.11 shows a conductor PQ of length l moving in a uniform magnetic induction B at constant speed v . Along with conductor its free electrons inside are also moving so these free electrons experience a magnetic force $F = evB$ of which direction is given by right hand palm rule which comes in downward direction as shown in figure.

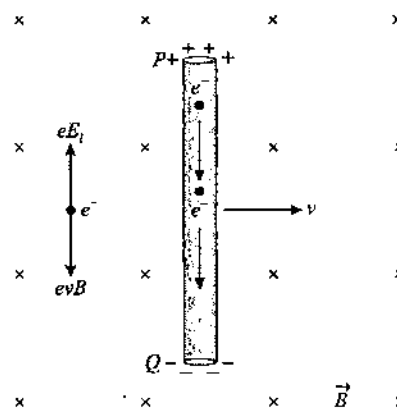


Figure 5.11

Due to this magnetic force electrons will drift to the end Q of the conductor and other end P will become slightly positive due to deficiency of electrons. This establishes an induced electric field from P to Q inside the conductor due to this separation of charges along the length of conductor because of magnetic force. Due to this induced electric field free electrons will experience an upward force eE_i as shown. These electrons will drift until the upward electric force balances the magnetic force on these electrons and then across the length of conductor electric field becomes steady and balances magnetic forces on free electrons. This electric field which is induced due to motion

of a conductor in magnetic field is given as

$$eE_i = evB$$

$$\Rightarrow E_i = vB \quad \dots (5.3)$$

Across the length of conduction during motion the potential difference due to the induced electric field is given as

$$V_P - V_Q = E_i l$$

$$\Rightarrow V_P - V_Q = Bvl \quad \dots (5.4)$$

As the above potential difference is developed due to magnetic forces which is non-electrostatic so this potential difference can be called as EMF and as it is developed due to motion of a conductor in magnetic field it is also termed as 'Motional EMF'.

5.1.3 Motional EMF by Faraday's Law

In previous article we discussed about EMF induced in a conductor moving in uniform magnetic field given by equation-(5.4). As due to motion of conductor magnetic lines were being cut according to Faraday's law the EMF induced in the conductor can be given as

$$e = \left| \frac{d\phi}{dt} \right| \quad \dots (5.5)$$

In figure-5.12 when the conductor moves by a distance vdt in time dt , it sweeps an area $lvdt$ which is shown by shaded part, the magnetic flux $d\phi$ in this shaded region is cut by the conductor in time dt is given as

$$d\phi = BdS$$

$$\Rightarrow d\phi = B(lvdt)$$

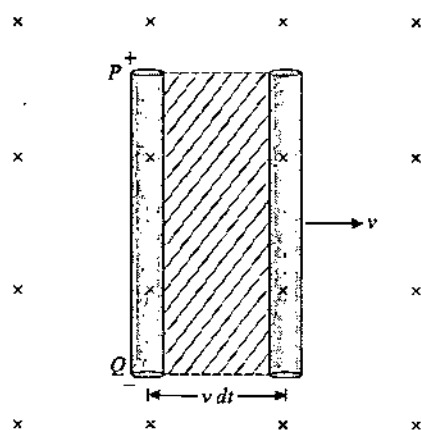


Figure 5.12

Thus from equation-(5.5) the induced EMF in the conductor is given as

$$e = \left| \frac{d\phi}{dt} \right| = Bvl \quad \dots (5.6)$$

The expression of induced EMF in conductor in equation-(5.6) is same which we obtained by induced electric field in previous article as given by equation-(5.4).

In previous article the direction of induced EMF we determined by finding the force on free electrons using right hand palm rule by which we determined that end P of conductor is positive (at high potential) and Q is negative (at low potential). The direction of induced EMF can also be directly determined by Fleming's Right Hand Rule as given below.

"According to Fleming's Right Hand Rule if we stretch the index finger, middle finger and thumb of right hand in mutually perpendicular directions as shown in figure-5.13 with index finger pointing toward the direction of magnetic induction and thumb along the velocity of conductor then middle finger will point toward the high potential end of the conductor."

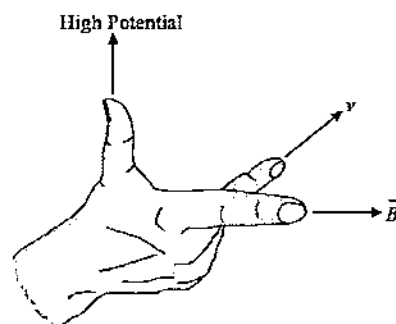


Figure 5.13

Above direction can also be given by Right Hand Palm Rule as shown in figure-5.14.

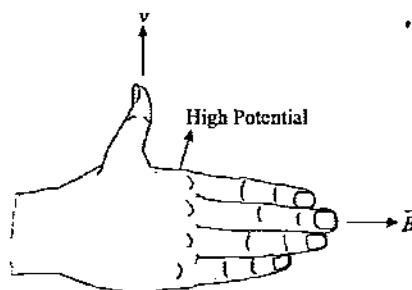


Figure 5.14

As till now we were using RHPR for analysis of direction of magnetic field, now onward we prefer to use the same rule so that all cases we'll be able to analyze by a single right hand only. However students can use either rule for any case.

5.1.4 Motional EMF as an Equivalent Battery

In different situations when a conductor moves in magnetic field, it can be considered like an equivalent battery or a source of potential difference with internal resistance equal to the resistance of the conductor which can supply current as shown

in figure-5.15 in which a conductor of length l , resistance r moving at a velocity v in a uniform magnetic field B with l , v and B are perpendicular to each other can be replaced by an equivalent battery of EMF Bvl and internal resistance r as shown in figure.

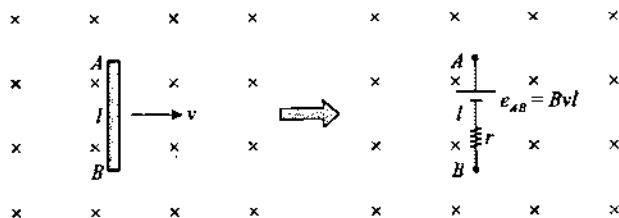
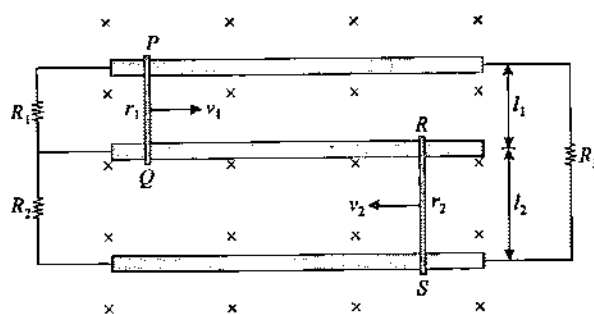
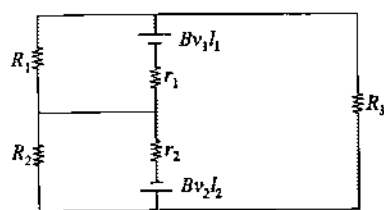


Figure 5.15

To understand the above situation better we consider an illustration shown in figure-5.16(a) in which two sliding wires PQ and RS are sliding on three conducting rails which are connected to some resistances at their ends as shown. In this case we can determine the currents through resistances and different sections of the setup by making an equivalent circuit of this setup as shown in figure-5.16(b).



(a)



(b)

Figure 5.16

5.1.5 EMF Induced in a Rotating Conductor in Uniform Magnetic Field

Figure-5.17 shows a uniform conducting rod PQ of length l which is rotating in a plane perpendicular to the direction of a uniform magnetic field as shown about an axis of rotation passing through point P at an angular speed ω . As the conductor's motion is cutting the magnetic lines an EMF is induced in it and by right hand palm rule we can see that point P is at higher potential.

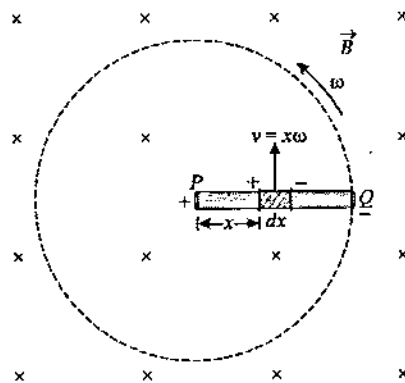


Figure 5.17

To calculate the induced EMF in the conductor we consider an elemental segment of width dx at a distance x from the end P of the rod as shown in figure-5.17. The speed of motion of this element is $v = x\omega$. The EMF induced in this element can be calculated by the expression of motional EMF given in equation-(5.4) as

$$de = Bvdx$$

$$\Rightarrow$$

$$de = B(x\omega)dx$$

$$\Rightarrow$$

Thus total EMF induced across the length of the conductor PQ is given by integrating above expression for the total length of the conductor from 0 to l as

$$e_{PQ} = \int de = \int_0^l B\omega x dx$$

$$\Rightarrow$$

$$e_{PQ} = B\omega \left[\frac{x^2}{2} \right]_0^l$$

$$\Rightarrow$$

$$e_{PQ} = \frac{1}{2} B\omega l^2 \quad \dots (5.7)$$

Alternative Analysis by Faraday's law :

Figure-5.18 shows the area swapped by the rotating conductor in time dt . The angle by which the rod rotates in this time is given as

$$d\theta = \omega dt$$

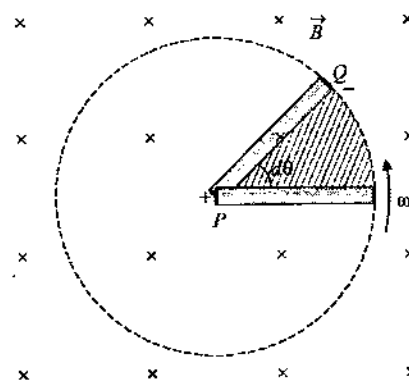


Figure 5.18

Area of the sector of which the flux is cut by the conductor in time dt is given as

$$dS = \frac{1}{2} r^2 d\theta = \frac{1}{2} r^2 (\omega dt)$$

Amount of flux cut by the conductor in time dt is given as

$$d\phi = B dS$$

$$\Rightarrow d\phi = B \left(\frac{1}{2} r^2 \omega dt \right)$$

Thus EMF induced in the conductor is given as

$$e = \left| \frac{d\phi}{dt} \right| = \frac{1}{2} B \omega l^2 \quad \dots (5.8)$$

Equation-(5.8) is same as that of equation-(5.7) but that was obtained by integrating motional EMF in small elements in the conductor whereas equation-(5.8) is obtained directly by Faraday's law.

5.1.6 EMF induced in a Conductor in Magnetic Field which is Moving in Different Directions

In previous articles we've discussed when a conductor moves in a magnetic field such that it cuts magnetic lines then an EMF is induced in it which we call motional EMF. When the conductor's length is oriented normal to the magnetic field direction and its velocity is also normal to the direction of its length as well as magnetic field as explained in figure-5.11 and 5.12 the motional EMF induced in the conductor is given by equation-(5.4) or (5.6) but if the conductor is moving in such a way that its length and/or velocity are not normal to the direction of magnetic field then we need to take the components of length as well as velocity normal to the magnetic field.

Figure-5.19(a) shows the situation, we've already discussed in which the conductor is moving normal to the field with its length also perpendicular to the field thus EMF induced in this case is given as

$$e_{PQ} = Bvl$$

Figure-5.19(b) shows a situation in which conductor is moving with a velocity along its length but length is perpendicular to the direction of magnetic field. In this case due to motion along the length conductor does not cut any flux as the area swapped out by its length perpendicular to magnetic field is zero thus no EMF is induced in it so we have

$$e_{PQ} = 0$$

In this case if conductor is thick then an EMF is induced across its width. Figure-5.19(c) shows a situation in which the velocity of conductor and its length both are perpendicular to magnetic field but velocity is not perpendicular to its length so the area

swapped out by the conductor in magnetic field will be less and to calculate the induced EMF in conductor we consider the component of the length normal to its velocity which is $l \sin \theta$ here. Thus the motional EMF induced in the conductor in this case is given as

$$e_{PQ} = Bv(l \sin \theta) \quad \dots (5.9)$$

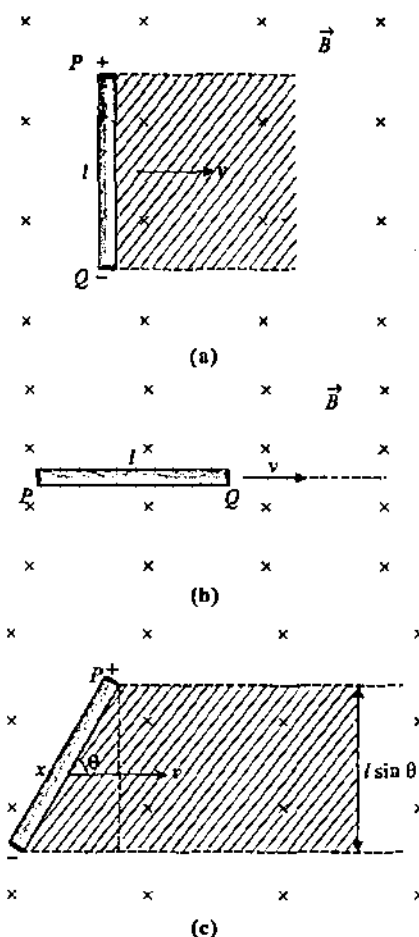


Figure 5.19

In general students should keep in mind that in case when length of rod is not perpendicular to magnetic field then consider the component of length perpendicular to magnetic field as well as velocity of rod in expression of motional EMF in equation-(5.4) and if velocity of rod is not perpendicular to magnetic field consider the velocity component which is perpendicular to magnetic field in the same expression.

5.1.7 Motional EMF in a Random Shaped Wire Moving in Magnetic Field

Figure-5.20 shows a random shaped wire AB in which the length between its ends A and B is l is moving with a velocity v perpendicular to line AB in a uniform magnetic field as shown. In this case we can consider a small element dl in the wire as shown in figure which will have its two components, one along the line AB and other perpendicular to the line AB . Due to motion

of wire EMF will only be induced in the component of dl along line AB due to the reason discussed in previous article thus total motional EMF between ends AB of the wire can be directly given as

$$e_{AB} = Bvl$$

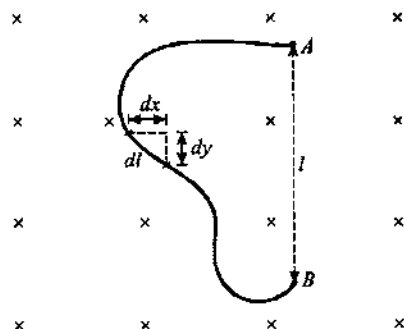
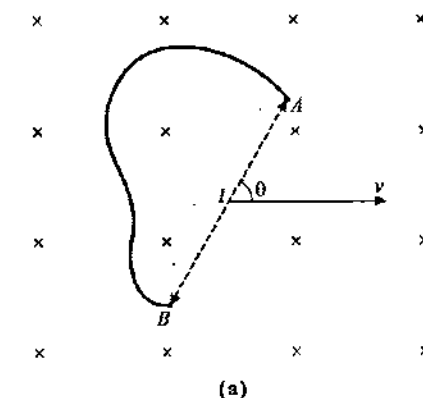
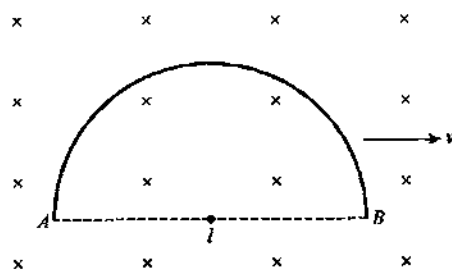


Figure 5.20

Similar to the previous article figures-5.21(a) and (b) shows the situations in which the random shaped wires are moving in different directions for which the EMF induced can be calculated by using the component of velocity perpendicular to the line AB which are given as $Bvl\sin\theta$ for figure-5.21(a) and zero for figure-5.21(b).



(a)



(b)

Figure 5.21

5.1.8 EMF induced in a Rotating Coil in Uniform Magnetic Field

Figure-5.22 shows a rectangular coil of enclosed area A and N turns mounted on an axis OO' which is initially in a plane perpendicular to the direction of magnetic induction B . It starts rotating at an angular velocity ω as shown. The initial magnetic

flux linked with the coil at $t = 0$ is given as

$$\phi = BAN$$

After time $t = t$ the coil will rotate by an angle $\theta = \omega t$ so flux linked with all the turns of the coil is given as

$$\phi = (\vec{B} \cdot \vec{A})N$$

$$\phi = BAN \cos\theta$$

$$\phi = BAN \cos\omega t \quad \dots (5.10)$$

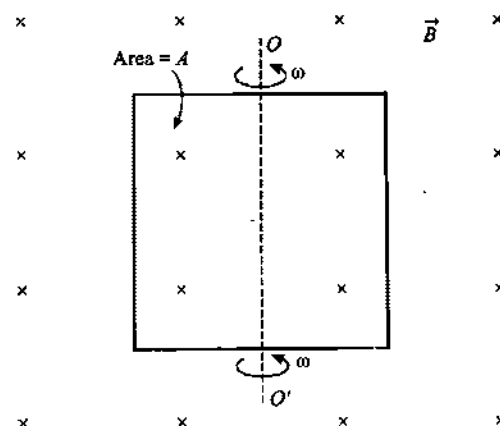


Figure 5.22

As the flux linked with the coil is changing with time, EMF induced in coil at time $t = t$ is given as

$$e = \left| \frac{d\phi}{dt} \right|$$

$$e = BAN \sin\omega t \cdot \omega$$

$$e = BAN\omega \sin\omega t \quad \dots (5.11)$$

$$e = e_0 \sin\omega t \quad \dots (5.12)$$

Where $e_0 = BAN\omega$ which is considered as the amplitude of the sinusoidal induced EMF given by equation-(5.11). Such a sinusoidal function of EMF is also called 'Alternating EMF' which is a source of alternating current in electrical circuits and this EMF which is produced by rotating a coil in magnetic field is the basis of working for an AC (Alternating Current) Generator.

The time function of alternating EMF induced in coil is shown in figure-5.23.

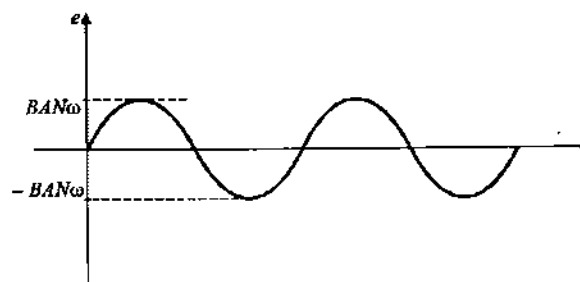


Figure 5.23

5.1.9 Motional EMF under External Force and Power Transfer

Figure-5.24 shows a setup with two long conducting rails with separation l in a plane perpendicular to a uniform magnetic field of magnetic induction B . A sliding wire PQ of mass m is placed on rails as shown. The rails are connected with a resistance R and the sliding wire is pulled with a constant external force F . If initially the sliding wire is at rest at $t = 0$ then at a general instant of time t it is considered to be moving at instantaneous velocity v then at this instant motional EMF induced in this wire is given as

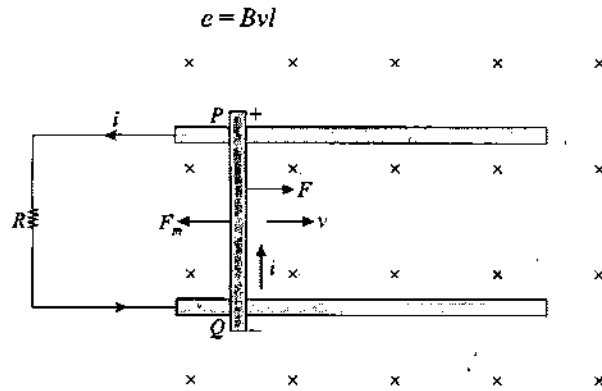


Figure 5.24

The current through the resistance due to induced EMF is given as

$$i = \frac{e}{R} = \frac{Bvl}{R} \quad \dots (5.13)$$

Due to the direction of induced EMF current flows from Q to P in the sliding wire because of which magnetic field exerts a magnetic force on the sliding wire which is given as

$$F_m = Bil = \frac{B^2 l^2 v}{R} \quad \dots (5.14)$$

By right hand palm rule we can analyze that the direction of this magnetic force on sliding wire PQ is toward left which is in opposition to the external force. This is also validating Lenz's law that the effects produced by induction oppose the causes of induction.

If at a general instant of time t acceleration of wire PQ is a then we have

$$F - F_m = ma \quad \dots (5.15)$$

$$\Rightarrow a = \frac{F - F_m}{m}$$

$$\Rightarrow a = \frac{F - \frac{B^2 l^2 v}{R}}{m} \quad \dots (5.16)$$

$$\Rightarrow \frac{dv}{dt} = a = \frac{FR - B^2 l^2 v}{mR}$$

$$\Rightarrow \frac{dv}{FR - B^2 l^2 v} = \frac{dt}{mR}$$

To find the velocity of wire PQ as a function of time we integrate the above expression from initial instant $t = 0$ to a general time instant $t = t$ given as

$$\begin{aligned} \int_0^v \frac{dv}{FR - B^2 l^2 v} &= \int_0^t \frac{dt}{mR} \\ \Rightarrow -\frac{1}{B^2 l^2} [\ln (FR - B^2 l^2 v)]_0^v &= \frac{1}{mR} [t]_0^t \\ \Rightarrow -\frac{1}{B^2 l^2} [\ln (FR - B^2 l^2 v) - \ln (FR)] &= \frac{1}{mR} [t - 0] \\ \Rightarrow \ln \left(\frac{FR - B^2 l^2 v}{FR} \right) &= -\frac{B^2 l^2 t}{mR} \\ \Rightarrow \frac{FR - B^2 l^2 v}{FR} &= e^{-\frac{B^2 l^2 t}{mR}} \\ \Rightarrow FR - B^2 l^2 v &= FR e^{-\frac{B^2 l^2 t}{mR}} \\ \Rightarrow B^2 l^2 v &= FR \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right) \\ \Rightarrow v &= \frac{FR}{B^2 l^2} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right) \quad \dots (5.17) \end{aligned}$$

With the above expression given in equation-(5.17) it can be seen that with time velocity of sliding wire approaches to a steady value and at $t \rightarrow \infty$ velocity approaches to the steady velocity given by equation-(5.17) as

$$v = \frac{FR}{B^2 l^2} \quad \dots (5.18)$$

As external force is acting on the wire PQ , from equation-(5.15) we can state that due to acceleration, velocity of wire PQ increases and with increase in velocity from equation-(5.14) we can see that the opposing magnetic force on wire PQ increases due to which acceleration decreases and when the opposing magnetic force on it becomes equal to external force then acceleration of wire becomes zero and velocity becomes steady which is given by equation-(5.18).

When after start velocity of wire PQ increases the instantaneous power supplied by the external force also increases which is given by equation-(5.17) as

$$\begin{aligned} P &= Fv \\ \Rightarrow P &= F \left(\frac{FR}{B^2 l^2} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right) \right) \\ \Rightarrow P &= \frac{F^2 R}{B^2 l^2} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right) \quad \dots (5.19) \end{aligned}$$

From equation-(5.19) we can analyze that the power supplied by external force increases with time and after a long time this power becomes steady which is given as

$$P_s = \frac{F^2 R}{B^2 l^2} \quad \dots (5.20)$$

Initially when wire is accelerating, the supplied power by the external force is being used in two parts. One part of supplied power is increasing the kinetic energy of the wire and other part is being dissipated as heat in the resistance R which is given as

$$H = i^2 R$$

In above expression, we substitute the value of current from equation-(5.13) as

$$H = \left(\frac{Blv}{R} \right)^2 R = \frac{B^2 l^2 v^2}{R}$$

Substituting the value of velocity from equation-(5.17) in above expression gives

$$H = \frac{B^2 l^2}{R} \left(\frac{FR}{B^2 l^2} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right) \right)^2$$

$$\Rightarrow H = \frac{F^2 R}{B^2 l^2} \left(1 - e^{-\frac{B^2 l^2 t}{mR}} \right)^2 \quad \dots (5.21)$$

After a long time above expression approaches to a steady value given as

$$H_s = \frac{F^2 R}{B^2 l^2} \quad \dots (5.22)$$

Above expression in equation-(5.22) is same as equation-(5.20) so we can state that after wire PQ attains steady velocity then power supplied by the external agent is fully dissipated as heat in the resistor connected across the rails upto the time wire PQ attains steady velocity the a part of power supplied increases the kinetic energy of wire and remaining is dissipated as heat.

In different cases of electromagnetic induction in mechanical setups when external forces are present, due to induction some opposition is developed which causes the effect of external force to reduce and at some point of time the system attains a steady state when external force is balanced by the opposing effects. Under this state the power supplied by external force to the system is transformed into some other form. This happens due to magnetic field as magnetic forces does equal and opposite work to transform energy from one form to another. This phenomenon can be used in design of many electromechanical system for efficient transformation of energy from one form to another.

Illustrative Example 5.1

A copper rod of length L is moving at a uniform speed v parallel to a long straight wire carrying a current of I as shown in figure-5.25. The rod is perpendicular to the wire with its ends at distance a and b from it. Calculate the motional EMF induced in the rod.

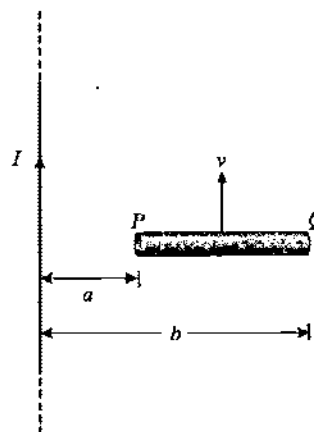


Figure 5.25

Solution

The magnetic field at a distance x from straight wire is given as

$$B = \frac{\mu_0 I}{2\pi x}$$

We consider an element of length dx in rod at a distance x from the straight wire as shown in figure-5.26. Due to the motion of the element dx in the rod in magnetic field B of straight wire, the motional induced EMF in the element is given as

$$de = Bvdx$$

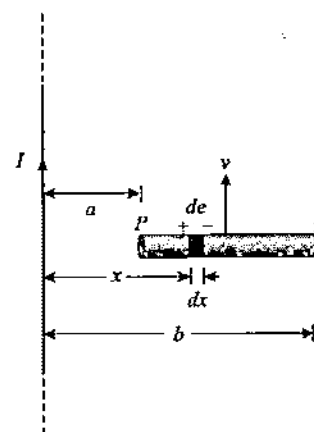


Figure 5.26

The direction of de is shown in figure by right hand palm rule. As all such de in all the elements in the rod are in series so

induced EMF in whole rod PQ is given by integrating above expression within limits from a to b given as

$$\begin{aligned}
 e &= \int de = \int_a^b Bv dx \\
 \Rightarrow e &= \int_a^b \frac{\mu_0 I v}{2\pi x} \\
 \Rightarrow e &= \frac{\mu_0 I v}{2\pi} [\ln x]_a^b \\
 \Rightarrow e &= \frac{\mu_0 I v}{2\pi} [\ln b - \ln a] \\
 \Rightarrow e &= \frac{\mu_0 I v}{2\pi} \ln\left(\frac{b}{a}\right)
 \end{aligned}$$

Illustrative Example 5.2

A circular copper disc of radius 10cm rotates at 1800rpm about an axis through its centre and at perpendicular to the disc. A uniform field of magnetic induction 1T exist in region perpendicular to the disc. What potential difference develops between the axis of the disc and its rim?

Solution

The disc can be considered to be made of several elemental sectors as shown in figure-5.27. Each such thin sectors can be considered like rotating rods across which the EMF induced is given as

$$e = \frac{1}{2} B \omega R^2$$

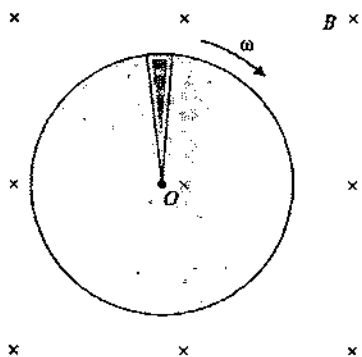


Figure 5.27

All such sectors can be considered as EMFs connected in parallel across the center of the disc and its rim and we know when identical EMFs are connected in parallel the equivalent EMF remains the same so the net induced EMF across the center of the disc and its rim is given as

$$\begin{aligned}
 e &= \frac{1}{2} \times 1 \times 2 \times 3.14 \times \frac{1800}{60} \times (0.1)^2 \\
 e &= 0.942 \text{ V}
 \end{aligned}$$

Illustrative Example 5.3

A copper rod PQ of mass m slides down two smooth copper bars which are set at an angle α to the horizontal as shown in figure-5.28. At the top of the bars these are interconnected through a resistance R . The separation between the bars is equal to l . The system is located in a uniform magnetic field of induction B in vertically upward direction as shown in figure. The resistances of the bars, the rod and the sliding contacts are considered to be negligible. If the rod is released from rest, find the velocity of rod as a function of time and its steady velocity attained.

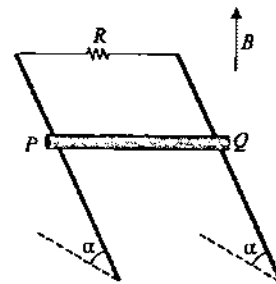


Figure 5.28

Solution

When the rod is released from rest and as it starts moving, it cuts the magnetic flux and an EMF is induced in it with polarity given by right hand palm rule as shown in figure-5.29. If at any instant conductor velocity is v and as it is moving perpendicular to the magnetic field component $B \cos \alpha$, the induced EMF in the rod is given as

$$e = (B \cos \alpha) v l = B v l \cos \alpha$$

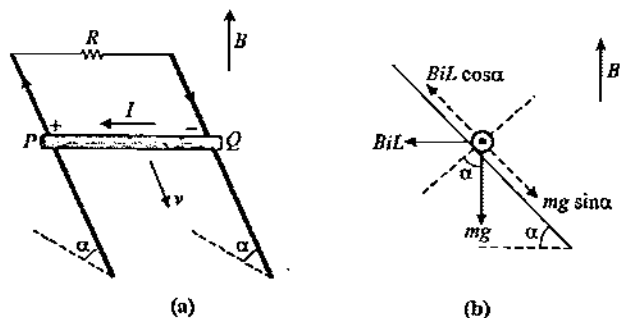


Figure 5.29

Thus current induced due to induced EMF in the loop containing resistance as shown is given as

$$i = \frac{B v l \cos \alpha}{R}$$

Figure-5.29(b) shows the side view of the sliding rod on which

by right hand palm rule, we can see that magnetic force will act along leftward direction and if rod slides with an acceleration a then equation of motion of the rod is given as

$$mg \sin \alpha - BiL \cos \alpha = ma$$

$$\Rightarrow m \left(\frac{dv}{dt} \right) = mg \sin \alpha - \frac{B^2 L^2 v \cos^2 \alpha}{R}$$

$$\Rightarrow dv = \left(g \sin \alpha - \frac{B^2 L^2 v \cos^2 \alpha}{mR} \right) dt$$

$$\Rightarrow \int_0^v \frac{dv}{mgR \sin \alpha - B^2 L^2 v \cos^2 \alpha} = \int_0^t \frac{dt}{mR}$$

$$\Rightarrow -\frac{1}{B^2 L^2 \cos^2 \alpha} \left[\ln (mgR \sin \alpha - B^2 L^2 v \cos^2 \alpha) \right]_0^v = \frac{t}{mR}$$

$$\Rightarrow \ln \left(\frac{mgR \sin \alpha - B^2 L^2 v \cos^2 \alpha}{mgR \sin \alpha} \right) = -\frac{B^2 L^2 t \cos^2 \alpha}{mR}$$

$$\Rightarrow v = \frac{mgR \sin \alpha}{B^2 L^2 \cos^2 \alpha} \left(1 - e^{-\frac{B^2 L^2 t \cos^2 \alpha}{mR}} \right)$$

Steady state is theoretically attained after a long time when velocity becomes

$$v_s = \frac{mgR \sin \alpha}{B^2 L^2 \cos^2 \alpha}$$

In steady state the component of gravitational force is balanced by magnetic force on rod which gives

$$mg \sin \alpha = BiL \cos \alpha$$

$$\Rightarrow mg \sin \alpha = \frac{B^2 L^2 v \cos^2 \alpha}{R}$$

$$\Rightarrow v_s = \frac{mgR \sin \alpha}{B^2 L^2 \cos^2 \alpha}$$

Illustrative Example 5.4

A straight horizontal conductor PQ of length l , and mass m slides down on two smooth conducting fixed parallel rails, set inclined at an angle θ to the horizontal as shown in figure-5.30.

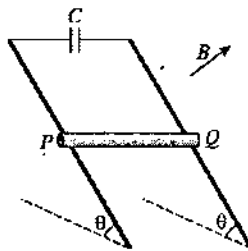


Figure 5.30

The top end of the bar are connected with a capacitor of capacitance C . The system is placed in a uniform magnetic field, in the direction perpendicular to the inclined plane formed

by the rails as shown in figure. If the resistance of the bars and the sliding conductor are negligible calculate the acceleration of sliding conductor as a function of time if it is released from rest at $t = 0$.

Solution

When the conductor is released from rest and if at time $t = t$ its speed is v then induced EMF in the conductor is given as

$$e_{PQ} = Blv$$

This potential difference is across the capacitor so instantaneous charge on capacitor is given as

$$q = CV = CBlv$$

Current through the capacitor is given as

$$i = \frac{dq}{dt} = CBl \frac{dv}{dt} = CBla$$

In magnetic field the current carrying conductor experiences a magnetic force in upward direction by right hand palm rule as shown in figure-5.31(b) which shows the side view of sliding conductor and its free body diagram. This gives

$$F_{mag} = Bil = B^2 l^2 Ca$$

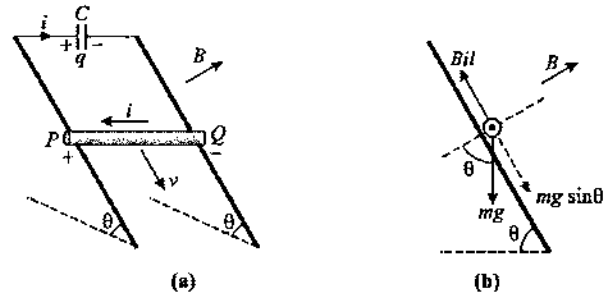


Figure 5.31

If a is the acceleration of conductor then its equation of motion is given as

$$ma = mg \sin \theta - B^2 l^2 Ca$$

$$\Rightarrow a[m + B^2 l^2 C] = mg \sin \theta$$

$$\Rightarrow a = \frac{mg \sin \theta}{m + B^2 l^2 C}$$

Here we can see that a is constant in time.

Illustrative Example 5.5

In a region of space a horizontal magnetic induction exist along $+Z$ -direction which is into the plane of paper and magnitude of magnetic induction varies along Y -direction (vertically downward), is given as

$$\vec{B} = \frac{B_0 y}{a} \hat{k}$$

Where B_0 and a are positive constants. A square loop $EFGH$ of side a , mass m and resistance R is placed in X - Y plane and start falling under gravity from rest as shown in figure-5.32. Find

- The induced current in the loop as a function of instantaneous speed of the loop and indicate its direction
- The total Lorentz force acting on the loop as a function of instantaneous speed of the loop and indicate its direction
- An expression for the speed of the loop, $v(t)$ and its terminal velocity.

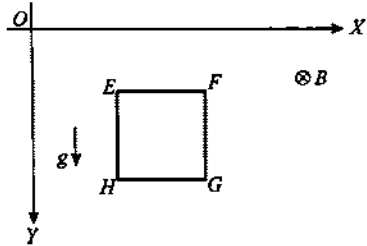


Figure 5.32

Solution

- (a) If v be the instantaneous velocity of the loop at any time t . Induced EMF in segment HG of the loop is given as

$$e_{HG} = \frac{B_0(y+a)}{a} va = B_0(y+a)v \quad \dots (5.23)$$

The point G will be at higher potential in above EMF. Similarly the EMF induced in segment EF is given as

$$e_{EF} = \frac{B_0 y}{a} va = B_0 yv \quad \dots (5.24)$$

Here F is at higher potential and the EMF induced in remaining two segments are zero as these are not cutting any magnetic flux.

$$e_{EH} \text{ and } e_{FG} = 0$$

So, the current in the loop is given as

$$i = \frac{B_0(y+a)v - B_0 yv}{R} = \frac{B_0 av}{R}$$

The current will be in anticlockwise direction due to higher EMF in segment HG .

- (b) The magnetic forces on EF and HG are given as

$$F_{EF} = \frac{B_0 y}{a} \times i \times a$$

$$F_{HG} = \frac{B_0(y+a)}{a} \times i \times a$$

Different elements along the vertical segments will experience

different forces but in opposite direction as shown in figure-5.33 so will get cancelled out.

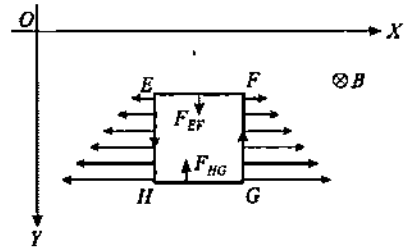


Figure 5.33

Net Lorentz force on loop is given as

$$F = B_0(y+a)i - B_0 yi = B_0 ai$$

$$\Rightarrow \vec{F} = \frac{B_0^2 a^2 v}{R} (-\hat{j})$$

- (c) Net force acting on loop in downward direction is

$$F = mg - \frac{B_0^2 a^2 v}{R}$$

$$\Rightarrow m \frac{dv}{dt} = mg - \frac{B_0^2 a^2 v}{R}$$

$$\Rightarrow \int_0^v \frac{dv}{mg - \frac{B_0^2 a^2 v}{R}} = \int_0^t \frac{dt}{m}$$

$$\Rightarrow -\frac{R}{B_0^2 a^2} \left[\ln \left(mg - \frac{B_0^2 a^2 v}{R} \right) \right]_0^v = \frac{1}{m} [t]_0^t$$

$$\Rightarrow \left[\ln \left(mg - \frac{B_0^2 a^2 v}{R} \right) - \ln(mg) \right] = -\frac{B_0^2 a^2 t}{mR}$$

$$\Rightarrow \frac{mgR - B_0^2 a^2 v}{mgR} = e^{-\frac{B_0^2 a^2 t}{mR}}$$

$$\Rightarrow v = \frac{mgR}{B_0^2 a^2} \left(1 - e^{-\frac{B_0^2 a^2 t}{mR}} \right)$$

After a long time terminal velocity of conductor is given as

$$v_{\text{terminal}} = \frac{mgR}{B_0^2 a^2}$$

Illustrative Example 5.6

Two parallel vertical metallic rails AB and CD are separated by 1 m . They are connected at the two ends by resistances R_1 and R_2 as shown in figure-5.34. A horizontal metallic bar PQ of mass

0.2kg slides without friction with maintaining contacts with the rails, vertically downward under the action of gravity. There is a uniform horizontal magnetic field of 0.6T perpendicular to the plane of the rails in the region. It is observed when the terminal velocity is attained, the powers dissipated in R_1 and R_2 are 0.75W and 1.2W respectively. Find the terminal velocity of the bar and the values of R_1 and R_2 .

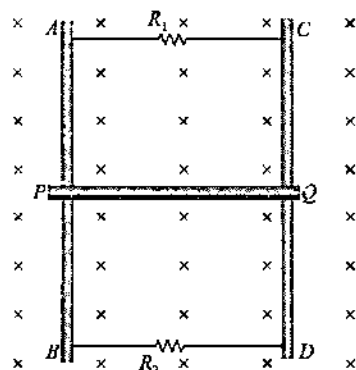


Figure 5.34

Solution

Due to sliding bar an EMF is induced in it which causes currents to flow in the resistances as shown in figure-5.35 the forces acting on the bar are the weight of rod in vertically downward direction and the magnetic force Bil in vertically upwards direction by right hand palm rule as shown.

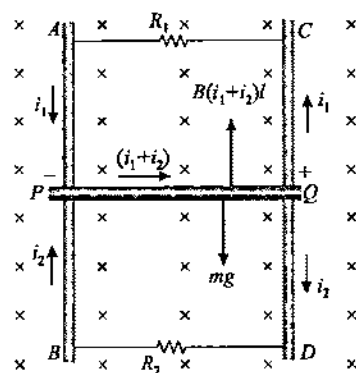


Figure 5.35

When the two forces are equal then the bar acquires terminal velocity v_T which happens when

$$mg = Bil$$

$$\Rightarrow i = \frac{mg}{Bl}$$

$$\Rightarrow i = \frac{0.2 \times 9.8}{0.6 \times 1} = \frac{49}{15} \text{ A}$$

If i_1 and i_2 be the currents in R_1 and R_2 respectively. Then above current is distributed in inverse ratio of resistances so we use

$$i = i_1 + i_2$$

$$\Rightarrow \frac{49}{15} = i_1 + i_2 \quad \dots (5.25)$$

$$\text{and } e = i_1 R_1 = i_2 R_2 \quad \dots (5.26)$$

We have

$$P_1 = ei_1$$

$$\Rightarrow 0.76 = ei_1 \quad \dots (5.27)$$

$$\text{and } P_2 = ei_2$$

$$\Rightarrow 1.2 = ei_2 \quad \dots (5.28)$$

From equations-(5.27) and (5.28) we have

$$\frac{i_1}{i_2} = \frac{0.76}{1.2}$$

$$\Rightarrow i_1 = \frac{19}{30} i_2 \quad \dots (5.29)$$

Solving equations-(5.25) and (5.29), we get

$$i_2 = 2\text{A}$$

$$\text{and } i_1 = \frac{19}{15} \text{ A}$$

From equation-(5.28) we have

$$1.2 = e \times 2$$

$$\Rightarrow e = 0.6\text{V} \quad \dots (5.30)$$

At terminal speed EMF is given as

$$e = Bv_T l$$

$$\Rightarrow v_T = (e/Bl)$$

$$\Rightarrow v_T = \frac{0.6}{0.6 \times 1} = 1\text{m/s}$$

Resistances are given as

$$R_1 = \frac{e}{i_1} = \frac{0.6 \times 15}{19} = 0.47\Omega$$

$$\text{and } R_2 = \frac{e}{i_2} = \frac{0.6}{2} = 0.3\Omega$$

Illustrative Example 5.7

Figure-5.36 shows a conductor OA of length l placed along y -axis with one end at origin. In this region a non-uniform magnetic field exist along $+Z$ -direction of which magnitude depends only on its Y coordinate which is given as

$$B = B_0 \left(1 + \frac{y^2}{l^2} \right) T$$

If the conductor OA starts translating with velocity $\vec{v} = v_0 \hat{i}$, find the EMF induced in conductor.

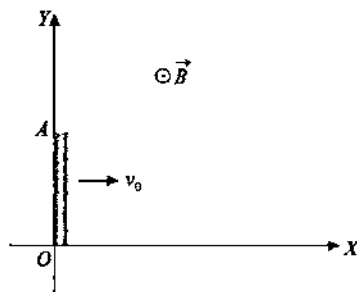


Figure 5.36

Solution

We consider an element of width dy at a distance y from origin in the conductor as shown in figure-5.37. Motional EMF induced in element dy is given as

$$de = Bv_0 dy = B_0 v_0 \left(1 + \frac{y^2}{l^2} \right) dy$$

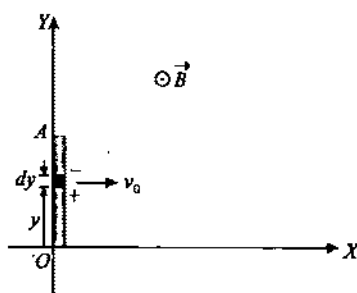


Figure 5.37

Total EMF induced in OA is given by integrating the above induced EMF in the element which is given as

$$e_{OA} = \int de = \int_0^l B_0 v_0 \left(1 + \frac{y^2}{l^2} \right) dy$$

$$\Rightarrow e_{OA} = B_0 v_0 \left[y + \frac{y^3}{3l^2} \right]_0^l$$

$$\Rightarrow e_{OA} = \frac{4}{3} B_0 v_0 l$$

Illustrative Example 5.8

A coil A of radius R and number of turns n carries a current i and it is placed in a horizontal plane. A small conducting ring P of radius r ($r \ll R$) is placed at a height y_0 above the centre of the coil A as shown in figure-5.38. Calculate the induced EMF in the ring when the ring is allowed to fall freely. Express the induced EMF as a function of instantaneous speed of the

falling ring and its height above the center of the coil A .

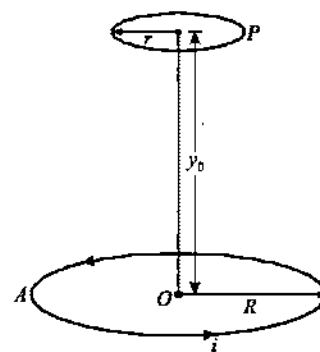


Figure 5.38

Solution

The magnetic induction at a point on the axis of a current carrying coil at a distance y from its centre is given by

$$B = \frac{\mu_0 N i R^2}{2(R^2 + y^2)^{3/2}} \quad \dots (5.31)$$

Due to the coil A the magnetic flux linked with ring P is given as

$$\phi = BA$$

$$\Rightarrow \phi = \frac{\mu_0 N i R^2}{2(R^2 + y^2)^{3/2}} \times \pi r^2$$

$$\Rightarrow \phi = \frac{\mu_0 N i \pi R^2 r^2}{2(R^2 + y^2)^{3/2}} \quad \dots (5.32)$$

If the ring P falls with instantaneous velocity v at any instant of time. The induced EMF induced in the ring P is given as

$$e = \frac{d\phi}{dt} = \frac{\mu_0 N i \pi R^2 r^2}{2} \times \frac{d}{dt} [(R^2 + y^2)^{-3/2}]$$

$$\Rightarrow e = -\frac{3}{4} \mu_0 N i \pi R^2 r^2 (R^2 + y^2)^{-5/2} \times 2y \frac{dy}{dt}$$

$$\Rightarrow e = -\frac{3}{2} \frac{\mu_0 \pi n i R^2 r^2}{(R^2 + y^2)^{5/2}} y(-v)$$

$$\Rightarrow e = \frac{3}{2} \frac{\mu_0 \pi n i R^2 r^2 y v}{(R^2 + y^2)^{5/2}}$$

Illustrative Example 5.9

Figure-5.39 shows two vertical smooth rails AB and CD separated by a distance l . Ends A and C are connected with capacitor of capacitance C . A rod PQ of mass m is horizontally kept in touch with both rails as shown. If it is released at $t=0$ and it remains in contact with rails during its fall, find the charge on capacitor

as a function of time. Neglect resistance of connecting wires and rails.

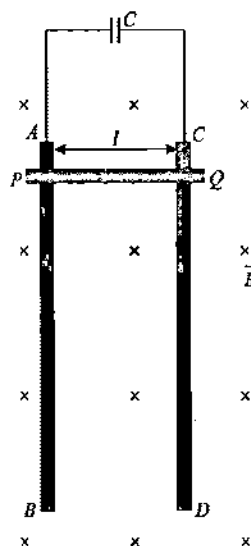


Figure 5.39

Solution

When the rod falls then at any instant when its speed is v , the motional emf across PQ is given as

$$e = Blv$$

As we can neglect the resistance of all connecting wires and rails the charge on capacitor is given as

$$q = Ce = CBlv \quad \dots (5.33)$$

Current through the capacitor is given as

$$i = \frac{dq}{dt} = CBla$$

The rod experiences an upward magnetic force and downward its weight as shown in figure-5.40.

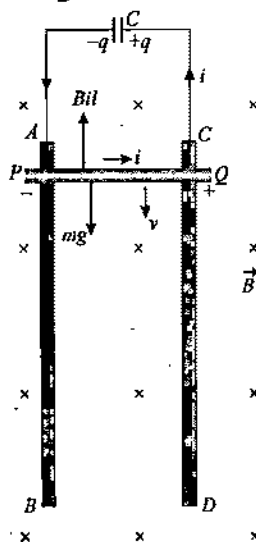


Figure 5.40

If rod has an acceleration a , it can be given as

$$a = \frac{mg - Bil}{m} = \frac{mg - B^2 l^2 Ca}{m}$$

$$\Rightarrow a = \frac{mg}{m + B^2 l^2 C}$$

As acceleration is constant, after time t its velocity is given as

$$v = at = \frac{mgt}{m + B^2 l^2 C}$$

Thus charge on capacitor after time t is given by equation-(5.33) as

$$q = CBl \left(\frac{mgt}{m + B^2 l^2 C} \right) = \frac{CBlmgt}{m + B^2 l^2 C}$$

Illustrative Example 5.10

Figure-5.41 shows a rectangular wire loop $ABCD$ with length l and breadth b . The wire is having a resistance λl per unit length. If the loop is pulled out from the magnetic field at a uniform speed v as shown in figure-5.00, find the potential difference across points of the loop $V_B - V_C$ and $V_A - V_D$.

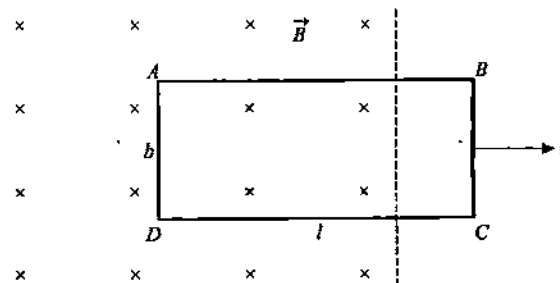


Figure 5.41

Solution

Due to motion of loop motional EMF is only induced across segment AD which is given as

$$e_{AD} = Bvb$$

The equivalent circuit of the loop with motional EMF is shown in figure-5.42

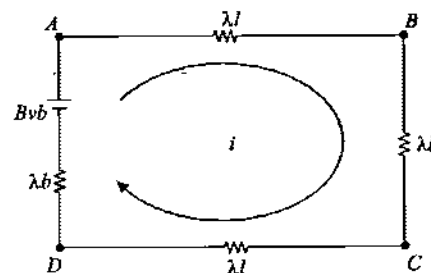


Figure 5.42

Induced current in above loop is given as

$$i = \frac{Bvb}{2\lambda(b+l)}$$

Now we can calculate the required potential differences as

$$V_B - V_C = i \times \lambda b = \frac{Bv \times b^2}{2 \times (b+l)} = \frac{Bvb^2}{2(b+l)}$$

$$V_A - V_D = i \times \lambda(b+2l) = \frac{Bvb(b+2l)}{2(b+l)}$$

Illustrative Example 5.11

A metal rod of mass m can rotate about a horizontal axis O , sliding along a circular wire ring mounted in a vertical plane of radius a as shown in figure-5.43. The arrangement is located in a uniform horizontal magnetic field of induction B directed perpendicular to the plane of ring. The axis and the ring are connected to an EMF to form a circuit of resistance R . Neglecting the friction and ring resistance, find the time function according to which the source EMF must vary to make the rod rotate at a constant angular velocity ω . Given that at $t=0$ rod was vertical.

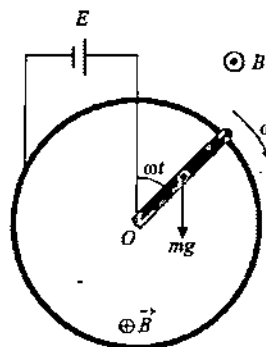


Figure 5.43

Solution

If rod is rotating at constant angular speed then motional EMF induced in it is given as

$$e = \frac{1}{2} B\omega a^2$$

If a current I flows in the circuit then force on the rod at the instant shown in figure-5.44 is given as

$$F = BIl$$

For a constant angular velocity, torque about O must be zero which happens when torque due to magnetic force balances the torque on rod due to its weight so we have

$$mg(a/2)\sin \omega t = BIl(a/2)$$

$\Rightarrow$

$$I = \frac{mg \sin \omega t}{Ba}$$

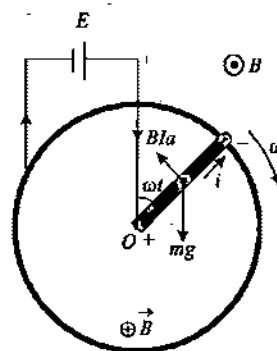


Figure 5.44

This must be equal to the current due to total EMF in the circuit which is given as

$$I = \frac{E - \frac{1}{2} B\omega a^2}{R} = \frac{mg \sin \omega t}{aB}$$

$\Rightarrow$

$$E = \frac{1}{2Ba} (2mgR \sin \omega t + B^2 \omega a^3)$$

Illustrative Example 5.12

Figure-5.45 shows a semicircular wire loop of resistance R and radius a hinged at point O and rotating at an angular speed ω . Point O is located on the boundary of a uniform magnetic induction B . Plot the current as a function of time in the loop taking clockwise direction as positive. Take $\theta = 0$ at $t = 0$.

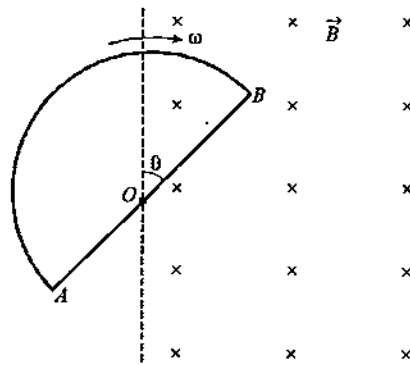


Figure 5.45

Solution

In the rotating loop shown in figure EMF will only be induced in the segment of OB and OA of the loop when these are inside the magnetic field region and cutting the magnetic lines. When segment OB is in the magnetic field then by right hand palm rule point B will be at higher potential and EMF induced across OB is given as

$$e = \frac{1}{2} B \omega a^2$$

This causes an anticlockwise current in the loop which is given as

$$i = \frac{e}{R} = \frac{B \omega a^2}{2R} \quad \dots (5.34)$$

After half revolution when segment OB comes out of the region of magnetic field, segment OA enters into the region and same EMF is induced in it and a clockwise current of same magnitude flows in the loop. Thus the plot of time function of current is shown in figure-5.46.

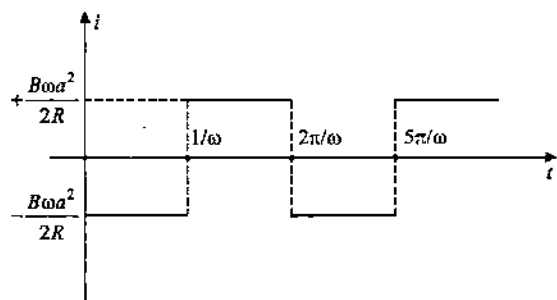


Figure 5.46

Illustrative Example 5.13

P and Q are two vertical infinite conducting plates kept parallel to each other and separated by a distance $2r$. A conducting ring of radius r falls vertically between the planes such that planes are always tangential to the ring. Both the planes are connected by a resistance R . There exists a uniform horizontal magnetic field of strength B perpendicular to the plane of ring. The arrangement is shown in figure-5.47. Plane Q is smooth and friction between the plane P and the ring is enough to prevent slipping between ring and plane P . At $t = 0$, the ring was at rest and neglect the resistance of the planes and the ring. Find

- The current through resistance R as a function of time
- Terminal velocity of the ring
- State the difference in analysis of situation if ring is also made up of a wire of resistance R .

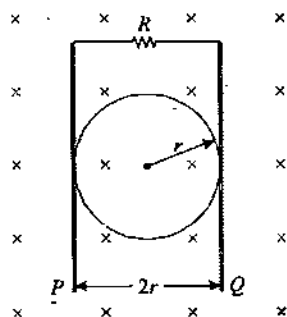


Figure 5.47

Solution

If v be the velocity of center of mass of the ring at any time t , the EMF across the diameter of ring is given as

$$e = 2Bvr$$

Current through resistance is given as

$$i = \frac{2Bvr}{R} \quad \dots (5.35)$$

The forces acting on the ring are shown in figure-5.48. For rolling motion of ring its equation of motion for rotational motion is

$$fr = Ia$$

$$f = \frac{I\alpha}{r} = \frac{Ia}{r^2} \quad \dots (5.36)$$

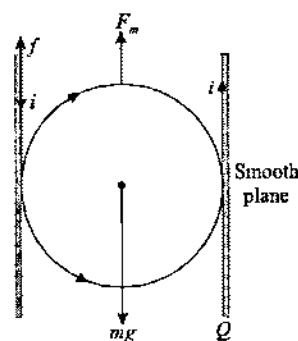


Figure 5.48

The upward magnetic force on ring is given as

$$F_m = Bi(2r)$$

$$\Rightarrow F_m = \frac{4B^2 r^2 v}{R} \quad \dots (5.37)$$

Equation of motion for translational motion of the ring is

$$mg - f - F_m = ma \quad \dots (5.38)$$

$$\Rightarrow mg - \frac{Ia}{r^2} - \frac{4B^2 r^2}{R} v = ma$$

$$\Rightarrow mg - \frac{4B^2 r^2}{R} v = ma + \frac{Ia}{r^2}$$

$$\Rightarrow mg - \frac{4B^2 r^2}{R} v = a \left[m + \frac{I}{r^2} \right]$$

$$\Rightarrow \left(mg - \frac{4B^2 r^2}{R} v \right) = \frac{dv}{dt} \left[m + \frac{I}{r^2} \right]$$

$$\Rightarrow \left(\frac{dv}{mg - \frac{4B^2 r^2}{R} v} \right) \times \left(m + \frac{I}{r^2} \right) = dt$$

For a ring we use its moment of inertia is given as $I = mr^2$ which gives

$$\Rightarrow \frac{2mdv}{\left(mg - \frac{4B^2r^2}{R}v\right)} = dt \quad \dots (5.39)$$

Integrating equation-(5.39) within proper limits, we get

$$\begin{aligned} -\frac{mR}{2B^2r^2} \left[\ln \left(mg - \frac{4B^2r^2}{R}v \right) \right]_0^v &= t \\ \Rightarrow -\frac{mR}{2B^2r^2} \left[\ln \left(mg - \frac{4B^2r^2}{R}v \right) - \ln(mg) \right] &= t \\ \Rightarrow -\frac{mR}{2B^2r^2} \ln \left(\frac{mgR - 4B^2r^2v}{mgR} \right) &= t \\ \Rightarrow v = \frac{mgR}{4B^2r^2} \left(1 - e^{-\frac{2B^2r^2t}{mR}} \right) &\quad \dots (5.40) \end{aligned}$$

From equations-(5.35) and (5.40), we get

$$i = \frac{2Bvr}{R} = \frac{mg}{2Br} \left(1 - e^{-\frac{2B^2r^2t}{mR}} \right)$$

(b) Terminal velocity of the ring can be calculated by equation-(5.40) after a long time which is given as

$$v = \frac{mgR}{4B^2r^2}$$

(c) If ring is made up of a wire of resistance R then EMF induced in the two half rings will remain same but as both the half rings will have resistance $R/2$ and the two EMFs in these half rings are considered in parallel so the total resistance of the circuit will become $R + R/4 = 5R/4$. Thus the analysis done in part (a) and (b) remain same with the resistance value replaced with $5R/4$ instead of R .

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11-12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Electromagnetic Induction

Module Number - 1 to 19

Practice Exercise 5.1

(i) A coil of mean area 500cm^2 and having 1000 turns is held perpendicular to a uniform field of 0.4G . The coil is turned through 180° in $1/10\text{s}$. Calculate the average induced EMF.

[0.04 V]

(ii) Two long parallel conducting horizontal rails are connected by a conducting wire at one end. A uniform magnetic field B directed vertically downwards exists in the region of space. A light uniform ring of diameter d which is practically equal to separation between the rails, is placed over the rails as shown in figure-5.49.

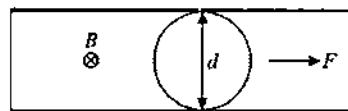


Figure 5.49

If resistance of ring be λ per unit length, calculate force required to pull the ring with uniform velocity.

$$\left[\frac{4B^2vd}{\pi\lambda} \right]$$

(iii) A very small circular loop of area $5 \times 10^{-4}\text{m}^2$, resistance 2Ω and negligible inductance is initially coplanar and concentric with a much larger fixed circular loop of radius 0.1m . A constant current of 1A is passed in the bigger loop and the smaller loop is rotated with angular velocity ω rad/s about a diameter. Calculate

- Maximum flux linked with the smaller loop
- Average induced EMF in one rotation of the loop
- Induced current in the smaller loop, as a function of time.

[(a) $10\pi \times 10^{-10}\text{Wb}$; (b) $10^{-9}\omega$; (c) $2.5 \times 10^{-10}\omega \sin\omega t$]

(iv) A plane spiral with a great number N of turns wound tightly to one another is located in a uniform magnetic field perpendicular to the spiral's plane. The outside radius of the spiral's turns is equal to a . The magnetic induction varies with time as $B = B_0 \sin \omega t$, where B_0 and ω are constants. Find the amplitude of induced EMF in the spiral.

$$\left[\frac{1}{3} \pi N a^2 B_0 \omega \right]$$

(v) Two long parallel horizontal rails, a distance d apart and each having a resistance λ per unit length, are joined at one end by a resistance R . A perfectly conducting rod MN of mass m is free to slide along the rails without friction as shown in figure-5.50. There is a uniform magnetic field of induction B normal to the plane of the paper and directed into the paper. A variable force F is applied to the rod MN such that, as the rod moves, a constant current i flows through R .

- Find the velocity of the rod and the applied force F as function of the distance x of the rod from R .
- What fraction of the work done per second by F is converted into heat?

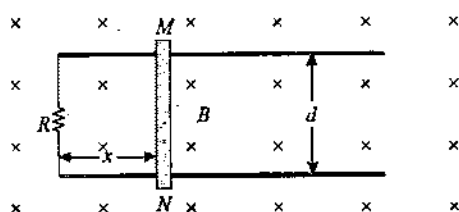


Figure 5.50

$$[(a) \frac{i(r+2\lambda x)}{Bd}, m \left(\frac{2i^2\lambda(R+2\lambda x)}{B^2d^2} \right) (b) \frac{B^3d^3}{2m\lambda(R+2\lambda x)}]$$

(vi) A wire frame of area $3.92 \times 10^{-4} \text{ m}^2$ and resistance 20Ω is suspended freely from a 0.392 m long thread. There is a uniform magnetic field of 0.784 T and the plane of wire-frame is perpendicular to the magnetic field. The frame is made to oscillate under gravity by displacing it through $2 \times 10^{-2} \text{ m}$ from its initial position along the direction of magnetic field. The plane of the frame is always along the direction of thread and does not rotate about it. What is the induced EMF in wire-frame as a function of time? Also find the maximum current in the frame.

$$[2 \times 10^{-6} \text{ V}, 10^{-7} \text{ A}]$$

(vii) A square frame with side a and a long straight wire carrying a current i are located in the same plane as shown in figure-5.51. The frame translates to the right with a constant velocity v . Find the EMF induced in the frame as a function of distance x .

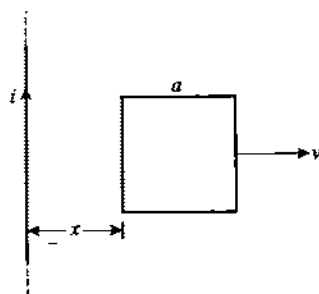


Figure 5.51

$$[\frac{\mu_0 i a^2 x v}{2\pi x(x+a)}]$$

(viii) A rod of length $2a$ is free to rotate in a vertical plane, about a horizontal axis O passing through its mid-point. A long straight, horizontal wire is in the same plane and is carrying a constant current i as shown in figure-5.52.

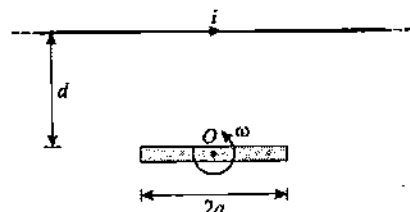


Figure 5.52

At initial moment of time. The rod is horizontal and starts to rotate with constant angular velocity ω , calculate EMF induced in rod as a function of time.

$$[\frac{\mu_0 i \omega}{2\pi \sin^2 \omega t} \left[d \ln \left(\frac{d - a \sin \omega t}{d + a \sin \omega t} \right) - 2a \sin \omega t \right]]$$

(ix) Two long wires are placed on a pair of parallel rails perpendicular to the wires. The spacing between the rails d is large compared with x , the distance between the wires. Both wires and rails are made of a material of resistivity ρ per unit length. A magnetic flux density B is applied perpendicular to the rectangle by the wires and rails. One wire is moved along the rails with a uniform speed v while the other is held stationary. Determine how the force on the stationary wire varies with x and show that it vanishes for a value of x approximately equal to $(\mu_0 v/4\pi\rho)$.

(x) A uniform rod AB of mass m and length l is placed over two smooth conducting rails P and Q . If the switch shown as closed at $t = 0$, find the velocity of rod AB as a function of time.

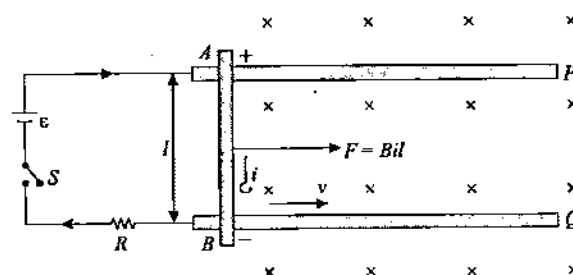


Figure 5.53

$$[\frac{e}{Bl} (1 - e^{-B^2 l^2 t / mR})]$$

(xi) In figure-5.54 a wire ring of radius R is in pure rolling on a surface. Find the EMF induced across the top and bottom points of the ring at any instant.

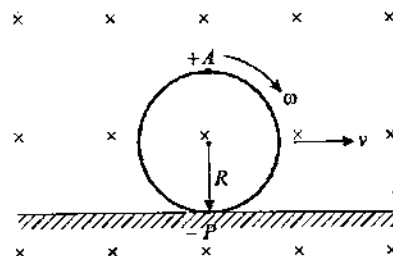


Figure 5.54

$$[2B\omega R^2]$$

(xii) Figure-5.55 shows a small circular coil of area A suspended from a point O by a string of length l in a uniform magnetic induction B in horizontal direction. If the coil is set into oscillations like a simple pendulum

by displacing it a small angle θ_0 as shown, find EMF induced in coil as a function of time. Assume the plane of coil is always in plane of string.

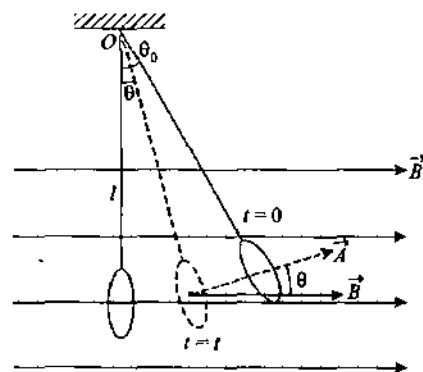


Figure 5.55

$$\left[\frac{1}{2} B A \theta_0^2 \omega \sin \left(2 \sqrt{\frac{g}{l}} t \right) \right]$$

5.2 Time Varying Magnetic Fields (TVMF)

Whenever a coil is placed in a magnetic field which varies with time the magnetic flux linked with the coil changes with time. As already studied in previous chapter that variation in magnetic field causes an electric field to be induced which can cause charge inside coil to flow and can cause induced current. There are several electrical parameters which can be calculated in any electrical circuit when magnetic flux associated with the circuit changes with time. In this section we'll discuss some of specific cases related to time varying magnetic fields.

5.2.1 Charge Flown through a Coil due to Change in Magnetic Flux

If magnetic flux associated with a coil changes with time then according to Faraday's law the EMF induced in the coil is given as

$$e = \left| \frac{d\phi}{dt} \right|$$

If coil resistance is R then the induced current in coil is given as

$$i = \frac{e}{R} = \frac{1}{R} \left| \frac{d\phi}{dt} \right|$$

If in time dt a charge dq flows through the coil when flux change is $d\phi$ in this time then we have

$$\frac{dq}{dt} = \frac{1}{R} \left| \frac{d\phi}{dt} \right|$$

$$\Rightarrow dq = \frac{|d\phi|}{R}$$

Integrating above expression we get

$$\Delta q = \frac{|\Delta\phi|}{R} \quad \dots (5.41)$$

Using above equation-(5.41) we can calculate the total charge flown through a coil if the flux linked with the coil changes by $\Delta\phi$. In above expression modulus for change in magnetic flux is used to avoid the sign for the charge flown as direction in which charge flows can be calculated by Lenz's law which we've already discussed.

5.2.2 Induced EMF in a Coil in Time Varying Magnetic Field

When a coil is placed in time varying magnetic field, as discussed in previous article the changing magnetic flux causes an EMF to be induced in the coil. If the coil area is A through which the magnetic flux is passing as shown in figure-5.56 then at any instant magnetic flux passing through the coil is given as

$$\phi = BA$$

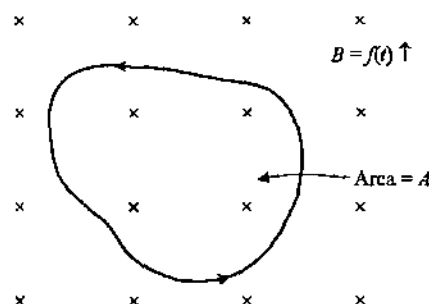


Figure 5.56

If there are N turns in the coil then all these turns are considered in series one after another at the same place so in such condition the magnetic flux 'linked' with the coil is given as

$$\phi = BA \times N \quad \dots (5.42)$$

The EMF induced in coil (in all turns) is given as

$$e = \left| \frac{d\phi}{dt} \right| = AN \frac{dB}{dt} \quad \dots (5.43)$$

Above EMF is called 'Loop EMF' induced in the coil which is distributed in every element of the coil.

Figure-5.57(a) shows a single turn circular coil placed in time varying magnetic field in inward direction of which magnitude is increasing with the function $B = f(t)$. According to Lenz's law due to increase in magnetic flux through the coil an anticlockwise current is induced in it to oppose the increasing flux. Actually this induced current is caused by the EMF induced in every element of the coil as shown in figure-5.57(b). Every small element of the coil behaves like an elemental EMF de and all such

elemental EMFs are in series. Sum of all these EMFs in the coil is called loop EMF given by equation-(5.43).

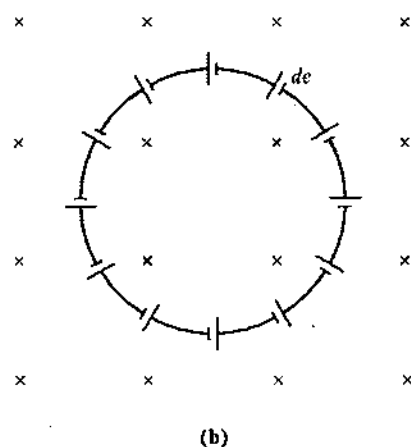
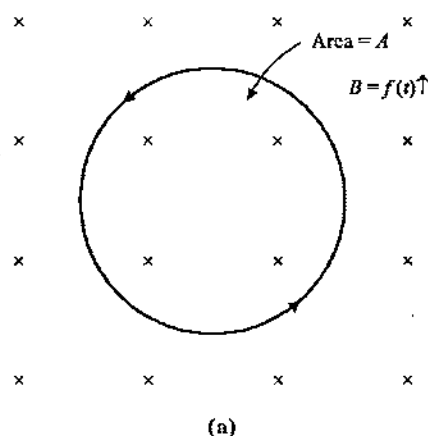


Figure 5.57

5.2.3 Induced Electric Field in Cylindrical Region Time Varying Magnetic Field

Figure-5.58 shows magnetic field along the axis of a cylindrical region of radius R . A coil of radius r is placed coaxially in the region as shown. If magnetic induction starts increasing with time an EMF is induced in the coil wire which causes an electric field to be induced in the wire due to which induced current flows.

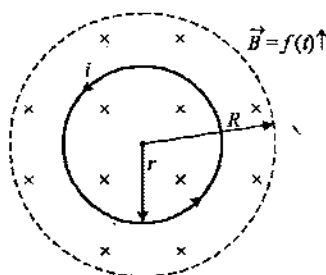


Figure 5.58

By symmetry of the region in which magnetic field exist, the direction of induced electric field in the coil wire must be same

as that of the induced current so at every point of the coil we can consider electric field to be along the tangent as shown in figure-5.59(a). Even if coil is not placed here, induced electric field lines will be closed circles as shown in figure-5.59(b).

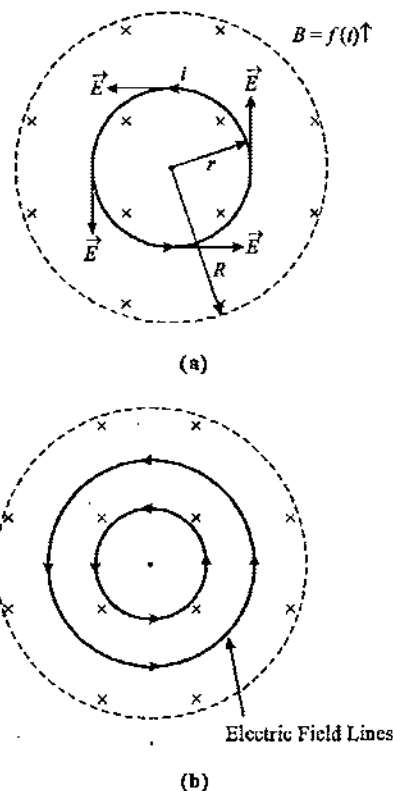


Figure 5.59

The induced EMF in the coil shown in figure-5.58 can be given as

$$e_{\text{loop}} = \pi R^2 \frac{dB}{dt} \quad \dots (5.44)$$

If we consider an electric field line of radius x as shown in figure-5.59(a) then along this circular line total loop EMF induced is given as

$$e_{\text{loop}} = \pi x^2 \frac{dB}{dt}$$

If $\vec{E}$ be the strength of induced electric field along the above line of force in consideration then we can write

$$e_{\text{loop}} = \int_0^{2\pi} \vec{E} \cdot d\vec{l} = \pi x^2 \frac{dB}{dt}$$

By symmetry we can state that magnitude of electric field along the coaxial line of force should remain same and it is tangential to the line of force so we can write

$$E(2\pi x) = \pi x^2 \frac{dB}{dt}$$

$$E = \frac{1}{2} x \frac{dB}{dt} \quad \dots (5.45)$$

Equation-(5.45) shows that electric field increases with distance from axis of region thus the density of electric lines of forces increases as we move away from the central axis. The configuration of electric lines is shown in figure-5.60.

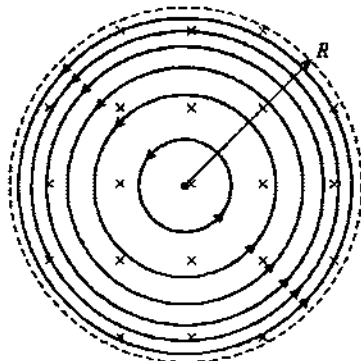


Figure 5.60

5.2.4 Induced Electric Field outside a Cylindrical Region Time Varying Magnetic Field

Figure-5.61 shows a time varying magnetic field confined in a cylindrical region of radius R and we consider a closed circular path of radius x concentric with the region of magnetic field as shown by dotted path. Within this path the magnetic flux passing is given as

$$\phi = \pi R^2 B$$

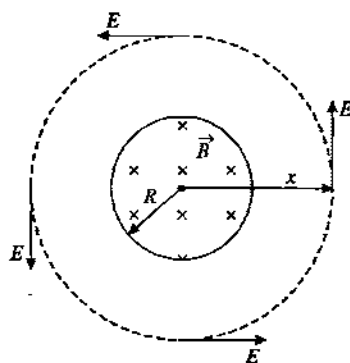


Figure 5.61

If we place a coil along this path then EMF induced in the coil is given as

$$e_{\text{loop}} = \pi R^2 \frac{dB}{dt}$$

By symmetry the direction of induced electric field will be tangential to the path as discussed in previous article and using induced electric field loop EMF can be given as

$$e_{\text{loop}} = \int_0^{2\pi} \vec{E} \cdot d\vec{l} = \pi R^2 \frac{dB}{dt}$$

$$\Rightarrow E(2\pi x) = \pi R^2 \frac{dB}{dt}$$

$$\Rightarrow E = \frac{1}{2} \left(\frac{R^2}{x} \right) \frac{dB}{dt} \quad \dots (5.46)$$

From equation-(5.46) we can see with distance from the axis of the cylindrical region magnitude of electric field decreases. The configuration of electric lines of forces inside and outside the region of time varying magnetic field is shown in figure-5.62.

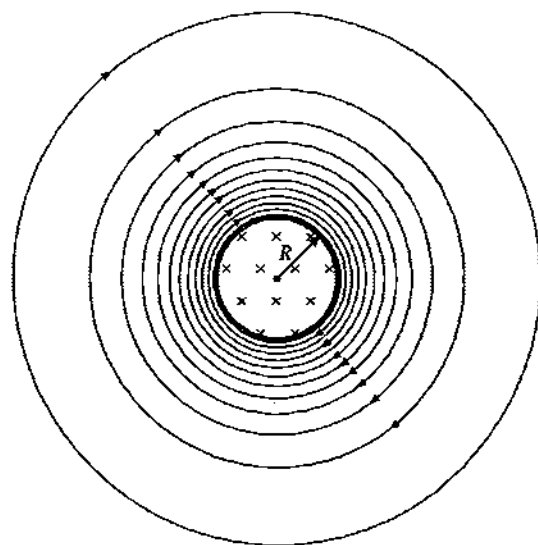


Figure 5.62

5.2.5 Electric Potential in Region of Time Varying Magnetic Field

We've discussed in previous article that induced electric field in time varying magnetic field have closed loop electric lines of forces as there are no static charges or source of electric field is present in this situation for which the field is non conservative in nature as for a charge going round the loop electric force acts in same direction and hence work done is non zero. Thus for the induced electric field in time varying magnetic field we have for a general closed path

$$\oint \vec{E} \cdot d\vec{l} \neq 0 \quad \dots (5.47)$$

Thus in regions of time varying magnetic fields at any point in space we cannot define potential as electric field in this region is non conservative in nature.

When a metal body is placed in time varying magnetic field then due to induced electric field free electrons of the metal are displaced which causes an opposing static electric field to be developed inside the metal due to static charges and at every point inside the metal its magnitude is equal to that of non conservative induced electric field of the region. To understand

this we will discuss on an illustration shown in figure-5.63.

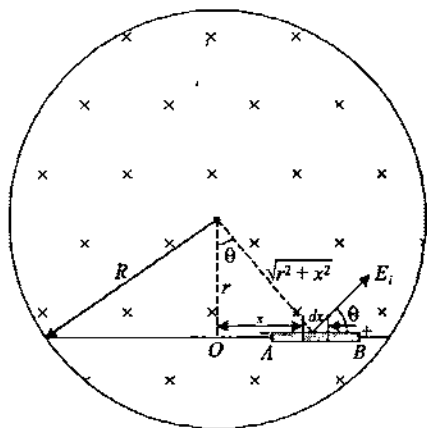


Figure 5.63

Inside a time varying magnetic field, a rod AB of length L is placed such that the axis of rod is at a distance r from the rod. We consider an element of length dx on rod at a distance x from point O as shown in figure above. The induced electric field at the location of element dx can be obtained by equation-(5.45), given as

$$E_i = \frac{1}{2} \sqrt{r^2 + x^2} \frac{dB}{dt} \quad \dots (5.48)$$

Due to the component of this electric field along the length of the rod its free electrons will drift toward left and left end A of rod becomes negative and due to deficiency of electrons end B becomes positive. These charges establish another electric field inside the rod from B to A . In steady condition this electric field balances the component of E_i along the rod so that no further drift of free electrons take place. Thus the electric field inside the rod due to static charges which is conservative in nature is given as

$$E = E_i \cos \theta \quad \dots (5.49)$$

$$\Rightarrow E = \frac{1}{2} \sqrt{r^2 + x^2} \frac{dB}{dt} \times \frac{r}{\sqrt{r^2 + x^2}}$$

$$\Rightarrow E = \frac{1}{2} r \frac{dB}{dt} \quad \dots (5.50)$$

The expression in above equation-(5.50) shows that the static electric field inside the rod is uniform and depends only upon the distance of axis of rod from the center of the cylindrical region in which magnetic field is confined.

If this electric field is uniform, the potential difference across the rod is given as

$$V_B - V_A = EL$$

$$\Rightarrow V_B - V_A = \left(\frac{1}{2} r \frac{dB}{dt} \right) L$$

$$\Rightarrow V_B - V_A = \frac{1}{2} r L \frac{dB}{dt} \quad \dots (5.51)$$

5.2.6 Eddy Currents

When a conductor moves in a magnetic field or there exist some relative motion between a conductor and magnetic field then localized currents are induced within the body of conductor in plane perpendicular to magnetic field if magnetic flux passing through the conductor changes. These currents are called 'Eddy Currents' or 'Foucaults Currents'.

Figure-5.64 shows a closed loop which is being pulled out from a magnetic field due to which the magnetic flux through the loop is decreasing and according to Faraday's law a motional EMF Bvl is induced in the segment DA of the conductor and a clockwise induced current flows in it. Due to this current segment DA experiences an opposing force Bil which opposes the loop to be pulled out from the magnetic field which is in accordance of Lenz's law.

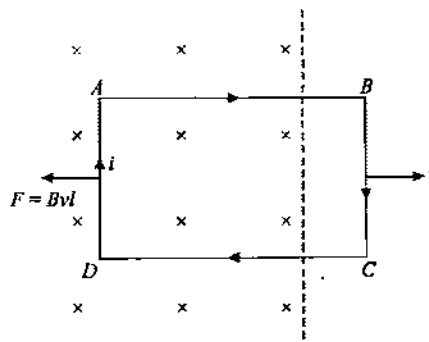


Figure 5.64

See figure-5.65 in which a sheet of metal is being pulled through a magnetic field. In this case there is no wire loop like figure-5.64 but the dotted segments of the sheets as shown in below figure will develop an induced EMF and tend to supply clockwise induced currents as shown in whichever path is available in the body of conductor. Due to these induced current these all segments within the magnetic fields experience a backward force by magnetic field and opposes the motion of sheet or develop a braking action for the moving conductor. These induced currents are 'Eddy Currents'.

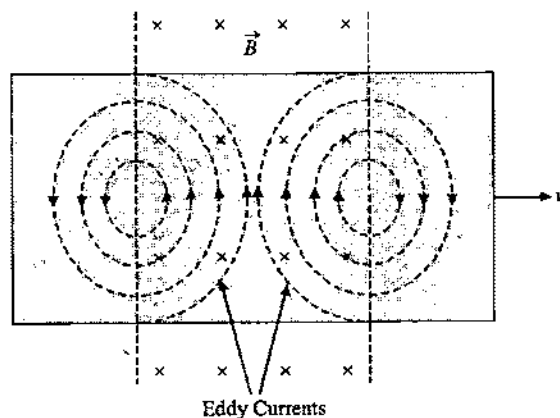


Figure 5.65

Figure-5.66 shows another illustration of braking of a rotating conducting disc by eddy currents. On the surface of a rotating metal disc we apply magnetic field on a part of its surface area as shown. Those segments of the metal disc which are moving through this region of magnetic field develop an induced EMF and eddy currents flow as shown in figure-5.66(b) which is the top view of the rotating disc. These currents are concentrated mainly in the part of disc where magnetic field is applied and outside these are scattered to form close loops. In this part of disc these currents will experience an opposing magnetic force to rotation of disc and hence it provides a braking action on rotating disc by the torque due to this magnetic force.

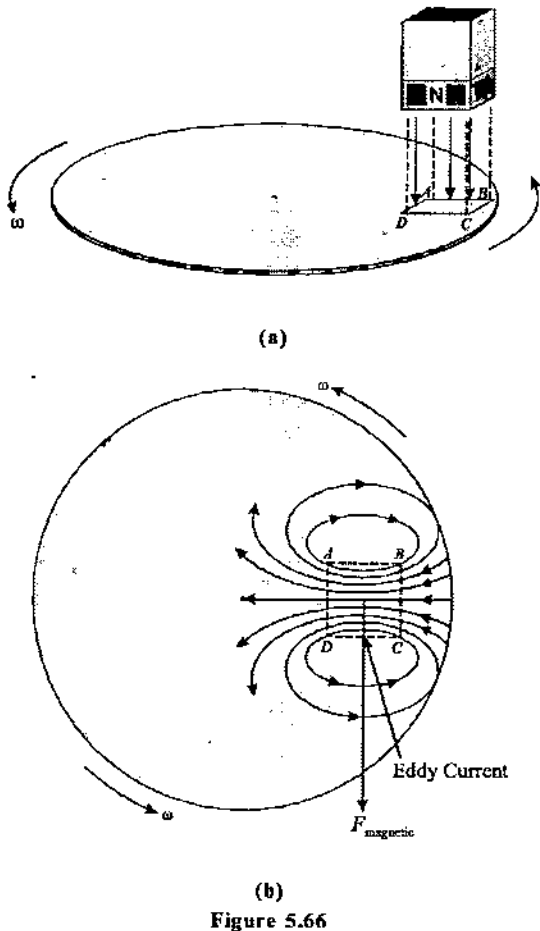


Figure 5.66

When a metal body is placed in a time varying magnetic field then localized currents are induced in metal body due to swirling of free electrons of metal in small closed paths. These are also eddy currents which produce their localized magnetic field in opposition to the change in external magnetic field. Figure-5.67 shows a metal sheet placed in a time varying magnetic field into the plane of sheet which increases with time. This causes several miniature anticlockwise currents induced in the body of sheet due to swirling of free electrons in small closed path at their local positions. Such currents cause continuous dissipation of heat in the metal body.

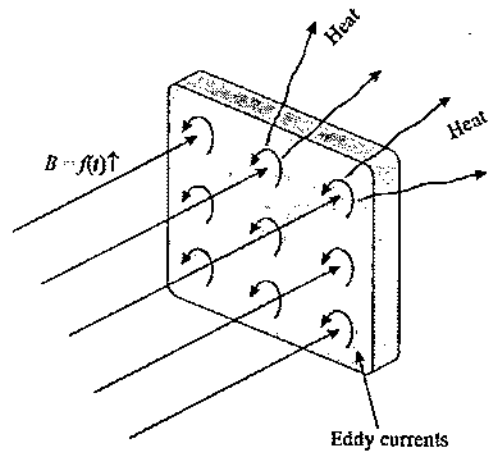


Figure 5.67

As discussed above whenever eddy currents are produced in a metal body these have two main effects on the specific situation under which these eddy currents are produced. One is the braking action on the relative motion of conductor and magnetic field and other is the heat dissipation in the body of conductor. Due to these actions there are many useful applications of eddy currents are developed at industrial and laboratory level.

Illustrative Example 5.14

Two concentric coplanar circular loops made of wire with resistance per unit length $10^{-4}\Omega/\text{m}$, have diameters 0.2m and 2m. A time varying potential difference $(4 + 2.5t)$ volt is applied to the larger loop. Calculate the current in the smaller loop.

Solution

The situation described in question is shown in figure-5.68

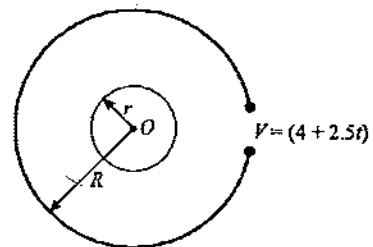


Figure 5.68

The magnetic induction at O , due to the current in large loop is given as

$$B_0 = \frac{\mu_0 i}{2R}$$

where

$$i = \frac{V}{R} = \frac{(4 + 2.5t)}{(2\pi R)\rho}$$

$\Rightarrow$

$$B_0 = \frac{\mu_0}{2R} \left[\frac{(4 + 2.5t)}{(2\pi R)\rho} \right]$$

Here we consider that the central loop is very small i.e., $r \ll R$, so the field may be considered as constant within the loop. Thus within the loop the field is B_0 so the flux through smaller loop is given as

$$\phi = B_0 \times (\pi r^2) = \frac{\mu_0}{2R} \left[\frac{(4+2.5t)}{(2\pi R)\rho} \right] \cdot \pi r^2$$

Induced EMF in the smaller loop is given as

$$e = \left| \frac{d\phi}{dt} \right| = \frac{\mu_0 r^2}{4R^2 \rho} \times (2.5)$$

Thus current i in the smaller loop is given as

$$i = \frac{e}{R} = \frac{\mu_0 r^2}{4R^2 \rho} \times 2.5 \times \left(\frac{1}{2\pi r \rho} \right)$$

$$\Rightarrow i = \frac{2.5 \mu_0 r}{8\pi R^2 \rho^2}$$

$$\Rightarrow i = \frac{2.5 \times (4\pi \times 10^{-7}) (0.1)}{8\pi \times 1 \times 10^{-8}} = 1.25 \text{ A}$$

Illustrative Example 5.15

A current $i = 3.36(1 + 2t) \times 10^{-2} \text{ A}$ increases at a steady rate in a long straight wire. A small circular loop of radius 10^{-3} m has its plane parallel to the wire and is placed at a distance of 1 m from the wire. The resistance of the loop is $8.4 \times 10^{-4} \Omega$. Find the magnitude and direction of induced current in the loop.

Solution

Situation described in question is shown in figure-5.69

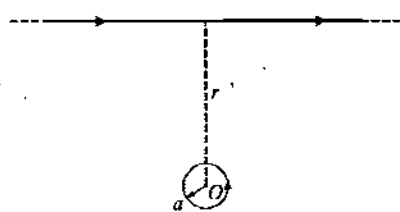


Figure 5.69

The magnetic flux passing through the loop is given as

$$\phi = B \cdot A = \left(\frac{\mu_0 i}{2\pi r} \right) \pi a^2 = \frac{\mu_0 a^2 i}{2r}$$

$$\Rightarrow \phi = \frac{\mu_0 a^2}{2 \times 1} [3.36(1 + 2t) \times 10^{-2}]$$

The induced e.m.f. is given by

$$e = \frac{d\phi}{dt} = \frac{\mu_0 a^2}{2} \times [3.36 \times 2 \times 10^{-2}]$$

$$\Rightarrow e = 3.36 \times 10^{-2} \times \mu_0 a^2$$

So, the induced current i is given by

$$i = \frac{e}{R} = \frac{1}{R} \times 3.36 \times 10^{-2} \times \mu_0 a^2$$

$$\Rightarrow i = \frac{(3.36 \times 10^{-2}) \times (4\pi \times 10^{-7}) (10^{-3})^2}{(8.4 \times 10^{-4})}$$

$$\Rightarrow i = \frac{4.22 \times 10^{-14}}{8.4 \times 10^{-4}} = 5 \times 10^{-11} \text{ A}$$

Illustrative Example 5.16

A long straight solenoid of cross-sectional diameter d and with n turns per unit of its length has a round turn of copper wire of cross-sectional area A and density ρ is tightly put on its winding. Find the current flowing in the turn if the current in the solenoid winding is increased with a constant rate I ampere per second.

Solution

The magnetic field inside the solenoid is given as

$$B = \mu_0 n i$$

The flux through its cross section of copper winding is given as

$$\phi = BA = (\mu_0 n i) \frac{\pi d^2}{4}$$

EMF induced in the copper wire is given as

$$e = \left| \frac{d\phi}{dt} \right| = \mu_0 n \pi \frac{d^2}{4} \frac{di}{dt}$$

Current in the copper wire is given as

$$I_C = \frac{e}{R} = \frac{\mu_0 n \pi d^2 I}{4 \cdot \rho \left(\frac{\pi d}{A} \right)} = \frac{\mu_0 n d A I}{4 \rho}$$

Illustrative Example 5.17

For the situation described in figure-5.70 the magnetic field changes with time according to,

$$B = (2.00t^3 - 4.00t^2 + 0.8) \text{ T and } r_2 = 2R = 5.0 \text{ cm}$$

(a) Calculate the force on an electron located at point P_2 at time instant $t = 2 \text{ s}$

(b) What are the magnitude and direction of the electric field at P_1 when $t = 3 \text{ s}$ and $r_1 = 0.02 \text{ m}$.

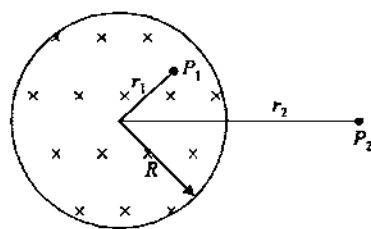


Figure 5.70

Solution

(a) At point P_2 the induced electric field strength is given as

$$E = \frac{R^2}{2r_2} \left(\frac{dB}{dt} \right)$$

$$F = qE = \frac{eR^2}{2r_2} (6t^2 - 8t)$$

Substituting the values we get

$$F = \frac{(1.6 \times 10^{-19})(2.5 \times 10^{-2})}{2 \times 5 \times 10^{-2}} [6(2)^2 - 8(2)]$$

$$\Rightarrow F = 8.0 \times 10^{-21} \text{ N}$$

(b) At P_1 being a point inside of the magnetic field, the induced electric field at this point is given as

$$E = \frac{r_1}{2} \left(\frac{dB}{dt} \right)$$

$$\Rightarrow E = \frac{r_1}{2} [6t^2 - 8t]$$

$$\Rightarrow E = \frac{0.02}{2} [6(3)^2 - 8(3)]$$

$$\Rightarrow E = 0.3 \text{ V/m}$$

As discussed in the above part, direction of electric field is in the direction of induced current (anticlockwise) in an imaginary circular conducting loop passing through P_1 .

Illustrative Example 5.18

A plane loop shown in figure-5.71 is shaped in the form of two squares with sides a and b and it is introduced into a uniform magnetic field at right angles, to the loop's plane. The magnetic induction varies with time as $B = B_0 \sin \omega t$. Find the amplitude of the current induced in the loop if its resistance per unit length is equal to r . The induction of the loop is negligible.

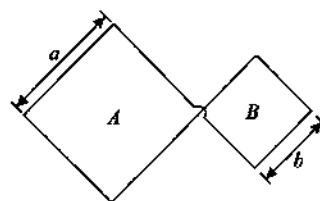


Figure 5.71

Solution

The loops are connected in such a way that EMFs in loops will oppose each other. The EMF induced in first loop is given as

$$e_A = \frac{d}{dt} (a^2 B) = a^2 \frac{d}{dt} (B_0 \sin \omega t)$$

$$\Rightarrow e_A = a^2 B_0 \omega \cos \omega t$$

EMF induced in the second loop is given as

$$e_B = -b^2 B_0 \omega \cos \omega t$$

Resistance of the circuit is given as

$$R = 4(a+b)r$$

Amplitude of current is given as

$$i_0 = \frac{(a^2 - b^2) B_0 \omega}{4(a+b)r}$$

$$\Rightarrow i_0 = \frac{1}{4} \frac{(a-b) B_0 \omega}{r}$$

Illustrative Example 5.19

Figure-5.72 shows a conducting loop of which semi-circular part lies in the magnetic induction B which varies with time, given as

$$B = at^3 + ct^2 + ft$$

The wire is having a resistance $R\Omega/\text{m}$. Find current in loop at time $t = 2\text{ s}$.

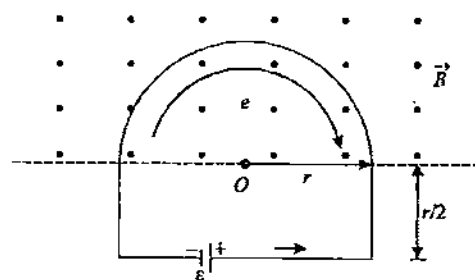


Figure 5.72

Solution

Induced emf in the loop is given as

$$e = A \frac{dB}{dt} = \frac{\pi r^2}{2} (3at^2 + 2ct)$$

Total resistance of the loop is given as

$$R_T = R \left[\frac{r}{2} + \frac{r}{2} + 2\pi + \pi r \right] = rR(3 + \pi)$$

Current in loop is given as .

$$i = \frac{\varepsilon - e}{R_T}$$

$$\Rightarrow i = \frac{\varepsilon - \frac{\pi r^2}{2} (3at^2 + 2ct)}{rR(3 + \pi)}$$

Illustrative Example 5.20

A thin non-conducting ring of mass m carrying a charge q can freely rotate about its axis. At the initial moment, the ring was at rest and no magnetic field was present. Then a practically uniform magnetic field was switched on, which was perpendicular to the plane of the ring and increased with time according to a certain law $B(t)$. Find the angular velocity attained by the ring as a function of the magnetic induction $B(t)$.

Solution

A changing magnetic field sets up an induced electric field due to which the torque on ring causes it to accelerate, we use

$$\tau = mR^2 \frac{d\omega}{dt} \quad \dots (5.52)$$

$$\text{But } \tau = q_0 ER = q \cdot \frac{1}{2} R^2 \frac{dB}{dt} \quad \dots (5.53)$$

From equation-(5.52) and (5.53) we have

$$\frac{1}{2} R^2 q \frac{dB}{dt} = mR^2 \frac{d\omega}{dt}$$

$$\Rightarrow \frac{1}{2} q dB = m d\omega$$

Integrating above expression upto a time t we get

$$\frac{1}{2} q B(t) = m\omega$$

$$\Rightarrow \omega = \frac{q}{2m} B(t)$$

The direction of ω will be opposite to the direction of magnetic induction.

Illustrative Example 5.21

An infinitesimally small bar magnet of dipole moment M is pointing and moving with the speed v is in the X -direction. A small closed circular conducting loop of radius a is placed in the Y - Z plane with its centre at $x = 0$, and its axis coinciding with the X -axis. Find the force opposing the motion of the magnet, if the resistance of the loop is R . Assume that the distance x of the magnet from the centre of the loop is much greater than a .

Solution

The varying magnetic field due to magnetic dipole will produce induced current in the coil. Due to induced current in circular conducting loop, it will also behave like a small bar magnet. This bar magnet will produce a force opposing the motion of the magnet. The situation is shown in figure-5.73.

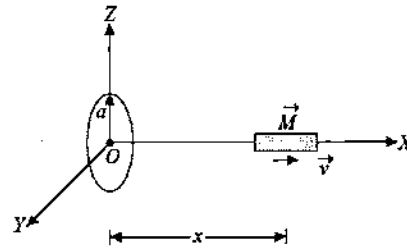


Figure 5.73

The magnetic field at the centre O of the coil due to bar magnet is given as

$$B = \frac{\mu_0}{4\pi} \times \frac{2M}{x^3}$$

The magnetic flux through the loop due to bar magnet is given as

$$\phi = \frac{\mu_0}{4\pi} \times \frac{2M}{x^3} \times (\pi a^2)$$

Induced EMF induced in circular loop is given as

$$e = -\frac{d\phi}{dt} = -\frac{d}{dt} \left[\frac{\mu_0}{4\pi} \times \frac{2M}{x^3} \times (\pi a^2) \right]$$

$$\Rightarrow e = -\frac{\mu_0 M a^2}{2} \left(\frac{-3}{x^4} \times \frac{dx}{dt} \right)$$

$$\Rightarrow e = \frac{3\mu_0 M a^2 v}{2x^4} \text{ where } v = \frac{dx}{dt}$$

Induced current in the circular loop is given as

$$i = \frac{e}{R} = \frac{3\mu_0 M a^2 v}{2x^4 R}$$

If M_{coil} be the magnetic moment of the coil. Then we have

$$M_{\text{coil}} = i \times A = \frac{3\mu_0 Ma^2 v}{2x^4 R} \times \pi a^2$$

$$\Rightarrow M_{\text{coil}} = \frac{3\pi\mu_0 Ma^4 v}{2x^4 R}$$

Now, opposing force on bar magnet is given as

$$F = M \frac{dB}{dx} = \frac{\mu_0}{4\pi} \times \frac{6MM_{\text{coil}}}{x^4}$$

$$\Rightarrow F = \frac{\mu_0}{4\pi} \frac{6M}{x^4} \left[\frac{3\pi\mu_0 Ma^4 v}{2x^4 R} \right]$$

$$\Rightarrow F = \frac{9\mu_0^2 M^2 a^4 v}{4Rx^8}$$

Illustrative Example 5.22

A closed coil having 50 turns, area 300cm^2 , is rotated from a position where its plane makes an angle of 45° with a magnetic field of flux density 2.0T to a position perpendicular to the field in a time of 0.1s . What is the average EMF induced in the coil?

Solution

Initially the flux linked with each turn of the coil is given as

$$\phi = \vec{B} \cdot \vec{A} = BA \cos \theta = BA \cos 45^\circ$$

Substituting the values, we get

$$\phi = 2.0 \times 300 \times 10^{-4} \times 0.7071$$

$$\Rightarrow \phi = 4.24 \times 10^{-2} \text{ Wb}$$

Finally flux linked with each turn of the coil is given as

$$\phi' = BA \cos 0^\circ = BA$$

$$\Rightarrow \phi' = 2.0 \times 300 \times 10^{-4} = 6.0 \times 10^{-2} \text{ Wb}$$

Change in flux in this process is given as

$$\Delta\phi = \phi' - \phi$$

$$\Rightarrow \Delta\phi = (6.0 \times 10^{-2}) - 4.24 \times 10^{-2}$$

$$\Rightarrow \Delta\phi = 1.76 \times 10^{-2} \text{ Wb}$$

This change is carried out in 0.1s . The magnitude of the average EMF induced in the coil is given as

$$e = N \frac{d(\phi' - \phi)}{dt} = 50 \times \frac{1.76 \times 10^{-2}}{0.1}$$

$$\Rightarrow e = 8.8\text{V}$$

Illustrative Example 5.23

A long solenoid has n turns per unit length, radius R and carries a current I is kept in gravity free region. From its axis at a distance twice that of its radius a charge $+q$ and mass m is placed. If the current in solenoid is suddenly switched off, find the velocity attained by the charge.

Solution

Magnetic induction inside the solenoid is given as

$$B_i = \mu_0 n I$$

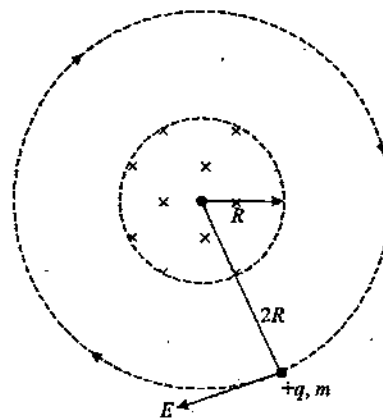


Figure 5.74

If the duration in which field drops to zero after switching off then electric field induced at a distance x outside in this duration δt is given as

$$E = \frac{1}{2} \frac{R^2}{x} \cdot \frac{dB}{dt} = \frac{R}{4} \cdot \frac{(\mu_0 n I - 0)}{\delta t}$$

Due to the induced electric field, impulse on the charge $+q$ in this duration is given as

$$J = qE\delta t = mv$$

$$\Rightarrow v = \frac{qE\delta t}{m}$$

$$\Rightarrow v = \frac{q}{m} \left(\frac{\mu_0 n I R}{4} \right) = \frac{\mu_0 n I R q}{4m}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Electromagnetic Induction

Module Number - 20 to 34

Practice Exercise 5.2

(i) A magnetic flux through a stationary loop with a resistance R varies during the time interval τ as $\phi = a t(\tau - t)$. Find the amount of heat generated in the loop during that time. The inductance of the loop is to be neglected.

$$\left[\frac{a^2 \tau^3}{3R} \right]$$

(ii) A long solenoid having 1000 turns per cm carries an alternating current of peak value 1A. A search coil having a cross-sectional area of $1 \times 10^{-4} \text{ m}^2$ and 20 turns is kept in the solenoid so that its plane is perpendicular to the axis of the solenoid. The search coil registers a peak voltage of $2.5 \times 10^{-2} \text{ V}$. Find the frequency of the current in the solenoid.

$$[15.8 \text{ s}^{-1}]$$

(iii) Two infinite long straight parallel wires A and B are separated by 0.1 m distance and carry equal currents in opposite directions. A square loop of wire C of side 0.1 m lies in the plane of A and B . The loop of wire C is kept parallel to both A and B at a distance of 0.1 m from the nearest wire. Calculate the EMF induced in the loop C while the currents in A and B are increasing at the rate of 10^3 A/s . Also indicate the direction of current in the loop C .

$$[2 \times 10^{-5} \ln\left(\frac{4}{3}\right) \text{ volt, clockwise}]$$

(iv) A flat circular coil of 200 turns of diameter 25 cm is laid on a horizontal table and connected to a ballistic galvanometer. The complete circuit is having resistance of 800Ω . When the coil is quickly turned over, the spot of light swings to a maximum reading of 30 divisions. When a $0.1 \mu\text{F}$ capacitor charged to 6 V is discharged through the same ballistic galvanometer, a maximum reading of 20 divisions is obtained. Calculate the vertical component of the earth's magnetic induction.

$$[0.37 \times 10^{-4} \text{ T}]$$

(v) A rectangular frame $ABCD$ made of a uniform metal wire has a straight connection between E and F made of the same wire as shown in figure-5.75. $AEFD$ is a square of side 1 m and $EB = FC = 0.5 \text{ m}$. The entire circuit is placed in a steadily increasing uniform magnetic field directed into the plane of the paper and normal to it. The rate of change of magnetic field is 1 T/s . The resistance per unit length of the wire is $1 \Omega/\text{m}$. Find the magnitudes and directions of the currents in the segment AE , BE and EF .

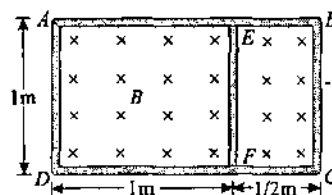


Figure 5.75

$$[\frac{3}{11} \text{ A}, \frac{3}{11} \text{ A}, \frac{1}{22} \text{ A}]$$

(vi) In a long straight solenoid with cross-sectional radius a and number of turns per unit length n a current varies with a constant velocity I ampere per second. Find the magnitude of the induced electric field strength as a function of the distance r from the solenoid axis. Draw the approximate plot of this function.

$$[\text{For } r < a, \frac{1}{2} r \mu_0 n I \text{ and for } r > a, \frac{1}{2} \frac{a^2 \mu_0 n I}{r}]$$

(vii) A thin non-conducting ring of mass m radius a carrying a charge q rotates freely about its own axis which is vertical. At the initial moment, the ring was at rest and no magnetic field was present. At instant $t = 0$, a uniform magnetic field is switched on which is vertically downwards and increases with time according to the law $B = B_0 t$. Neglecting magnetism induced due to rotational motion of the ring, calculate

(a) The angular acceleration of the ring and its direction of rotation as seen from above.

(b) Power developed by the forces acting on the ring as function of time.

$$[(a) \frac{B_0 q}{2m} \quad (b) \frac{q^2 B_0^2 a^2 t}{4m}]$$

(viii) In the middle of a long solenoid, there is a coaxial ring of square cross-section, made of conducting material with resistivity ρ . The thickness of the ring is equal to h , its inside and outside radii are equal to a and b respectively. Find the current induced in the ring if the magnetic induction produced by the solenoid varies with time as $B = \beta t$, where β is a constant.

$$[\frac{h\beta}{4\rho} (b^2 - a^2)]$$

(ix) A long solenoid of diameter 0.1 m has 2×10^4 turns per meter. At the centre of the solenoid a 100 turn coil of radius 0.01 m is placed with its axis coinciding with the solenoid axis. The current in the solenoid is decreased at a constant rate from $+2 \text{ A}$ to -2 A in 0.05 s. Find the EMF induced in the coil. Also find the total charge flowing through the coil during this time if the resistance of the coil is $10\pi^2 \Omega$.

$$[6.4\pi^2 \times 10^{-3} \text{ V}, 32\mu\text{C}]$$

(x) A long solenoid of cross-sectional radius a has a thin insulated wire ring tightly put on its winding; one-half of the ring has the resistance η times that of the other half. The magnetic induction produced by the solenoid varies with time as $B = bt$, where b is a constant. Find the magnitude of the electric field strength in the ring.

$$\left[\frac{ab}{2} \left(\frac{\eta-1}{\eta+1} \right) \right]$$

(xi) A magnetic field induction is changing in magnitude at a constant rate dB/dt . A given mass m of copper is drawn into a wire of radius a and formed into a loop of radius r is placed perpendicular to the field. Show that the induced current in the loop is given by

$$i = \frac{m}{4\pi\rho\delta} \frac{dB}{dt}$$

where ρ is the specific resistance and δ , the density of copper.

(xii) A long solenoid of cross-sectional area 5.0cm^2 is wound with 25 turns of wire per centimeter. It is placed in the middle of a closely wrapped coil of 10 turns and radius 25cm as shown in figure-5.76.

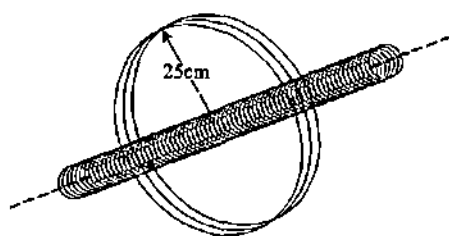


Figure 5.76

(a) What is the EMF induced in the coil when the current through the solenoid is decreasing at a rate -0.20A/s ?

(b) What is the electric field induced in the coil?

[(a) $3.14 \times 10^{-6}\text{V}$; (b) $2.0 \times 10^{-6}\text{V/m}$]

(xiii) In a coil of resistance R , magnetic flux due to an external magnetic field varies with time as $\phi = k(C - t^2)$. Where k and C are positive constants. Find the total heat produced in coil in time $t = 0$ to $t = C$.

$$\left[\frac{4k^2 C^2}{3R} \right]$$

(xiv) Figure-5.77 shows a fixed coil of N turns and radius a carrying a current I . At a distance x from its centre another small coaxial coil of radius b ($b \ll a$) and resistance R is moving toward the first coil at a uniform speed v . Find the induced current in smaller coil.

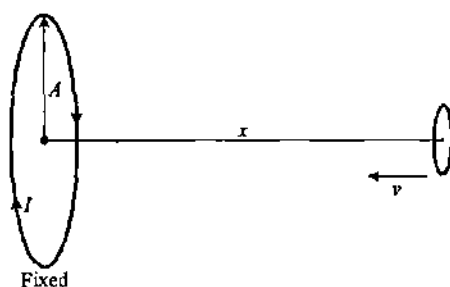


Figure 5.77

$$\left[\frac{3\mu_0\pi N I a^2 b^2 v x}{2R(x^2 + a^2)^{3/2}} \right]$$

5.3 Self Induction

We've studied that current flows in a closed circuit only and a circuit may consist of one or more loops. Whenever current flowing through a conductor (part of a closed circuit) or a coil changes then magnetic field produced by that current also changes and it changes the magnetic flux through the loop or loops in that circuit of which the conductor or coil is a part.

Due to change in magnetic flux linked with the circuit of conductor or coil an EMF is induced in the coil which opposes the causes of induction i.e. the effects of induced EMF opposes the change in current. Thus every conductor or a coil when current flowing in it is changed due to the flux linked with the circuit of its own current, electromagnetic induction takes place which opposes the change in current in itself. This property of a conductor or coil by which it opposes change in current in itself is called 'Self Induction'.

If in a coil or a conductor a current I is flowing then magnetic flux ϕ which is linked with the same coil or conductor is directly proportional to the current in it so we have

$$\phi \propto I$$

$$\Rightarrow \phi = LI \quad \dots (5.54)$$

In above equation-(5.54), L is a proportionality constant which is called 'Coefficient of Self Induction' for the coil or conductor and its associated circuit in which current I is flowing.

If current changes at a rate of dI/dt then EMF induced in the circuit of coil or conductor is given as

$$e = \left| \frac{d\phi}{dt} \right| = L \left| \frac{dI}{dt} \right| \quad \dots (5.55)$$

According to Lenz's law the direction of induced EMF is such that it opposes the causes of induction which is change in

current thus signs of e and dl/dt are opposite in above equation which can be written without modulus as

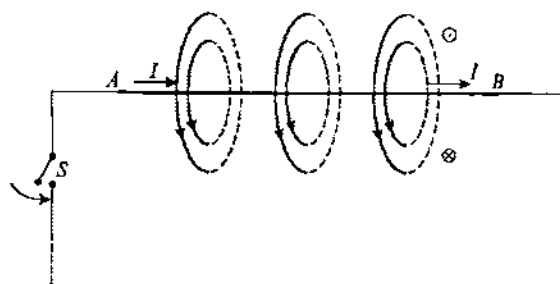
$$e = -L \frac{dl}{dt} \quad \dots (5.56)$$

Unit used for measurement of coefficient of Self Induction is 'Henery' or 'H' which is given as

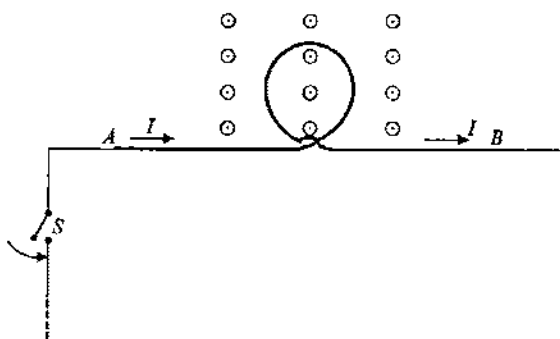
$$1H = 1 \text{ V-s/A}$$

5.3.1 Understanding Self Induction

Consider two situations of a wire segment AB in a circuits as shown in figure-5.78. In first case shown in figure-5.78(a) its just a straight wire AB in which a current I flows when switch is closed. Here we consider that remaining part of circuit is far away from the wire and not affected by the magnetic induction produced by this wire AB . In second case shown in figure-5.78(b) in the middle of wire AB a loop is made by twisting the wire through which magnetic flux passes when current flows through wire AB .



(a)



(b)

Figure 5.78

When the switch is closed in first case the current in wire quickly grows to a value I but in second case when switch is closed current grows but with this magnetic flux through the loop also increases which causes an EMF to be induced in the loop of which the direction is such that it opposes the change in flux. We've discussed that in time varying magnetic fields every element of the conductor or coil acts as an EMF thus the direction

of such elemental EMFs in the loop are as shown in figure-5.79. Due to the induced EMF in the loop the growth of current in wire AB after closing the switch is opposed and it grows slowly as compared to the first case.

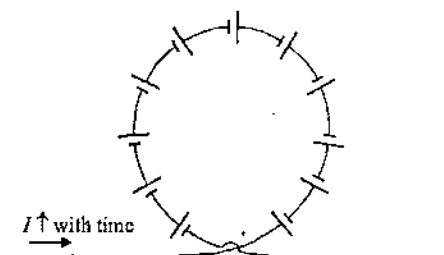
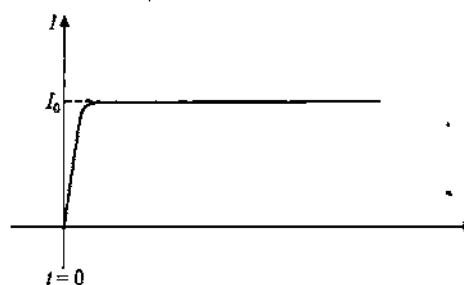
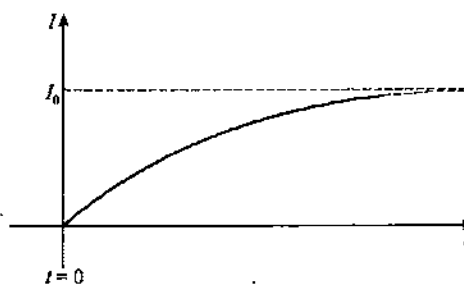


Figure 5.79

Figure-5.80(a) and (b) shows the time variation of current in wire AB in the above two cases. In second case the growth of current is slow due to the induced EMF in the loop while current increases.



(a)



(b)

Figure 5.80

We can imagine a situation if instead of one we make three loops in wire AB as shown in figure-5.81. In this case EMF will be induced in all three loops which opposes the growth of current in wire AB hence opposition is more and growth of current will be even slower compared to the case with one loop in wire AB . Here we can say that with three loops self induction of wire AB is higher compared to the situation with one loop and that is more than the situation without a loop. With the above illustrations we've taken it is evident that self induction of a conductor or a coil depends upon the shape and size of the conductor.

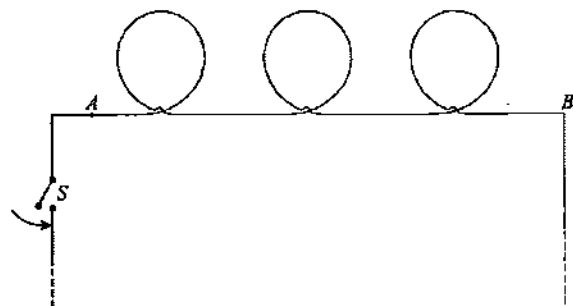


Figure 5.81

5.3.2 Self Inductance of a Solenoid

A solenoid is a coil with multiple turns wound on a cylindrical core. When a steady current I is flowing through a long solenoid of length l , total turns N with cross sectional area S , the magnetic induction inside is given as

$$B = \mu_0 n I = \mu_0 \left(\frac{N}{l} \right) I$$

The magnetic flux through any cross section of the solenoid is given as

$$\phi_s = BS = \left(\frac{\mu_0 N I}{l} \right) S \quad \dots (5.57)$$

Magnetic flux linked with all the turns of the solenoid is given as

$$\begin{aligned} \phi &= \phi_s \times N \\ \Rightarrow \phi &= \frac{\mu_0 N^2 S I}{l} \\ \Rightarrow \phi &= \left(\frac{\mu_0 N^2 S}{l} \right) I \quad \dots (5.58) \end{aligned}$$

Comparing above equation-(5.58) with equation-(5.54) we get the coefficient of self induction for a solenoid given as

$$L = \frac{\mu_0 N^2 S}{l} \quad \dots (5.59)$$

Due to multiple turns connected in series in a solenoid when current flowing through it changes the EMF induced in it due to self induction is significant high compared to normal conductors and this property makes a solenoid very useful in different types of electrical circuits. When a solenoid is used in circuits it is called an 'Inductor'.

Whenever current flowing through an inductor changes with time then the inductor behaves like a battery of EMF given as

$$e = L \frac{di}{dt} \quad \alpha$$

In next article we'll discuss about the behaviour of an inductor when used in electrical circuits.

5.3.3 Behaviour of an Inductor in Electrical Circuits

For use of a solenoid in electrical circuits it is made in miniature form with close packed turns and cross sectional radius is kept very small compared to its length. Figure-5.82(a) shows a common commercial inductor which is used in electrical circuits. To save the space instead of keeping it as a straight solenoid the wire is wound in form of a toroid as shown. Figure-5.82(b) and (c) shows different types of inductors used in a circuit on a printed circuit board (PCB).

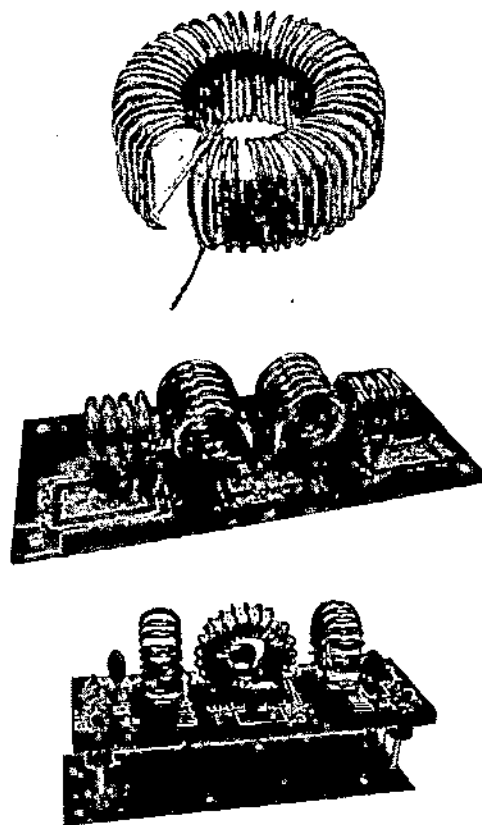


Figure 5.82

When a steady current flows through an inductor the magnetic flux due to the current remains constant and no EMF is induced due to self induction. Always remember that self induction plays its role of opposition only when current through a coil changes. If current is steady then the inductor behaves like a straight wire as shown in figure-5.83(b)

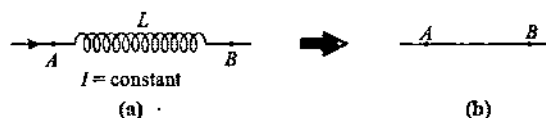


Figure 5.83

Consider the case shown in figure-5.84 when current flowing through the inductor increases with time. In this case due to self induction an EMF is induced in the inductor so as to oppose the change in current. In this case as current is increasing and

flowing to the right then left end of inductor will have high potential and right end of inductor will be at low potential so that the EMF induced in the inductor tend to supply its current in direction opposite to the flowing current so that increase in current is opposed. In this situation the inductor behaves like an equivalent EMF across its terminals A and B as shown in figure-5.84(b).

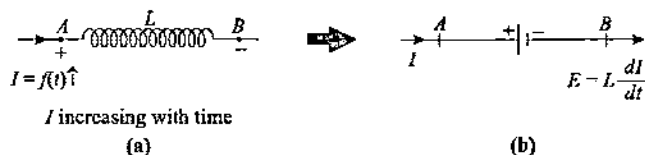


Figure 5.84

Figure-5.85 shows a case when current flowing through an inductor decreases with time. In this case due to self induction an EMF is induced so as to oppose the change in current. As current is decreasing, right end of inductor will be at higher potential and left end at lower potential so that the self induced EMF will tend to supply its current in the same direction as that of the existing current to oppose the decrement in current. In this situation the inductor behaves like an equivalent EMF across its terminals A and B as shown in figure-5.85(a) and (b).

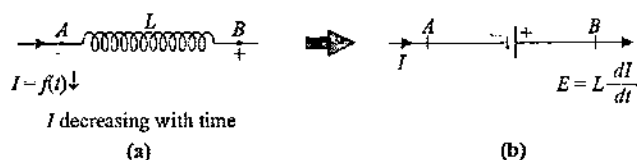


Figure 5.85

Illustrative Example 5.24

How many meters of a thin wire are required to manufacture a solenoid of length $l_0 = 100$ cm and inductance $L = 1.0$ mH, if the solenoid's cross-sectional diameter is considerably less than its length?

Solution

Self induction coefficient of a solenoid is given as

$$L = \frac{\mu_0 N^2 \pi r^2}{l_0}$$

Total length of wire is given as

$$l = N \times 2\pi r$$

$$\Rightarrow L = \frac{\mu_0 l^2}{4\pi l_0}$$

$$\Rightarrow l = \sqrt{\frac{4\pi l_0 L}{\mu_0}} = 0.1 \text{ km}$$

Illustrative Example 5.25

The inductor shown in figure-5.86 has inductance 0.54 H and carries a current in the direction shown that is decreasing at a uniform rate, $di/dt = -0.030$ A/s.

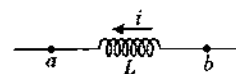


Figure 5.86

- Find the self-induced emf.
- Which end of the inductor, a or b , is at a higher potential?

Solution

- Self induced EMF is given as

$$e = L \left| \frac{di}{dt} \right| = 0.54 \times 0.03 \text{ V}$$

$$\Rightarrow e = 1.62 \times 10^{-2} \text{ V}$$

- Current in the shown direction is decreasing. Hence induced EMF should produce current in the same direction. Hence induced battery polarity will be as shown in figure-5.87 below.

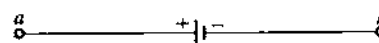


Figure 5.87

So we have

$$V_a > V_b$$

Illustrative Example 5.26

Calculate the inductance of a unit length of a double tape line if the tapes are separated by a distance h which is considerably less than their width b . Current flows in the two sheets of the tape line in opposite direction as shown in figure-5.88. Such a tapeline is used as connecting wires to industrial appliances where additional inductance is required for supply of current.

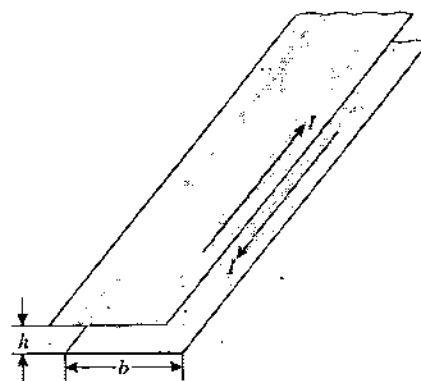


Figure 5.88

Solution

The current per unit length in the width of the sheets of tape line is I/b due to which each sheet of the line produces a magnetic induction between the sheets which is given as

$$B = \frac{1}{2} \mu_0 \left(\frac{I}{b} \right)$$

In between the sheets of line the total magnetic induction due to both the sheets is given as

$$B = \frac{\mu_0 I}{b}$$

The magnetic flux passing through the region between the two sheets of tape line of length l is given as

$$\phi = \frac{\mu_0 I}{b} lh$$

$$\Rightarrow \frac{\phi}{I} = \frac{\mu_0}{b} lh$$

If self inductance of the tape line is L then we have

$$\Rightarrow L = \frac{\phi}{I} = \frac{\mu_0}{b} lh$$

Self inductance per unit length of the line is given as

$$\frac{L}{l} = \frac{\mu_0 h}{b}$$

Illustrative Example 5.27

Calculate the inductance of a toroid whose inside radius is equal to b and cross-section has the form of a small square of side a . The solenoid winding consists of N turns. The space inside the solenoid is filled with uniform paramagnetic material having relative permeability μ_r .

Solution

If we consider an elemental ring of radius r and width dr as shown in the figure-5.89. The number of turns per unit length in this ring shaped element is given as

$$n = \frac{N}{2\pi r}$$

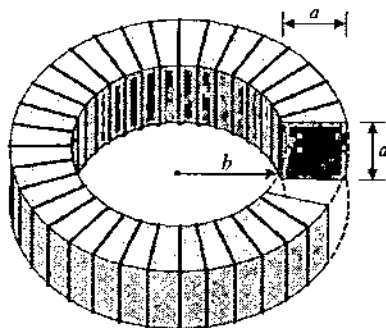


Figure 5.89

The magnetic induction inside the elemental toroid is given as

$$B = \mu_r \mu_0 n I = \frac{\mu_r \mu_0 N I}{2\pi r}$$

The magnetic flux through a cross section of the elemental ring shaped toroid is given as

$$d\phi = \frac{\mu_r \mu_0 N I}{2\pi r} \times a dr$$

Magnetic flux linked with all N turns of the toroid in the elemental ring shaped element is given as

$$Nd\phi = \frac{\mu_r \mu_0}{2\pi} N^2 I a \frac{dr}{r}$$

Total magnetic flux linked with the given toroid is given as

$$\phi = \int d\phi = \frac{\mu_r \mu_0}{2\pi} N^2 I a \int_b^{b+a} \frac{dr}{r}$$

$$\Rightarrow \phi = \frac{\mu_r \mu_0}{2\pi} N^2 I a \ln \left(\frac{b+a}{b} \right)$$

Self inductance of the toroid is given as

$$L = \frac{\phi}{I}$$

$$\Rightarrow L = \frac{\mu_r \mu_0}{2\pi} N^2 a \ln \left(\frac{b+a}{b} \right)$$

Illustrative Example 5.28

An iron core is inserted into a solenoid 0.5m long with 400 turns per unit length. The area of cross-section of the solenoid is 0.001 m^2 . (a) Find the relative permeability of the core when a current of 5A flows through the solenoid winding. Under these conditions, the magnetic flux through the cross-section of the solenoid is $1.6 \times 10^{-3} \text{ Wb}$. (b) Find the inductance of the solenoid under these conditions.

Solution

We know that the magnetic induction on the axis of a solenoid is given by

$$B = \mu \left(\frac{\mu_0}{4\pi} \right) 4\pi n i \quad \dots (5.60)$$

where μ is the permeability of the medium.

(a) As magnetic flux $\phi = BA$ we use $B = \phi/A$ which gives

$$B = \frac{1.6 \times 10^{-3}}{10^{-3}} = 1.6 \text{ T}$$

Substituting the values in equation-(5.60), we get

$$1.6 = \mu(10^{-7}) \times 4\pi \times 400 \times 5$$

$$\Rightarrow \mu \approx 637.$$

(b) Total number of turns in the solenoid are given as

$$N = n \times l = 400 \times 0.5 = 200$$

Total magnetic flux linked to the solenoid is given as

$$\phi_p = N \times \phi = 200 \times 1.6 \times 10^{-3} = 0.32 \text{ Wb}$$

$$\Rightarrow L = \frac{N\phi}{i} = \frac{0.32}{5} = 0.064 \text{ H} = 64 \text{ mH.}$$

5.4 Growth of Current in an Inductor

In previous articles we've discussed that an inductor behaves like an equivalent battery when current through it changes. Due to the self induction of an inductor it opposes the change in current in itself and depending upon the value of coefficient of self induction of the inductor the opposition to change in external current is different. In next article we will discuss and analyze the growth of current in an inductor when it is connected across a battery.

5.4.1 Growth of Current in an Inductor Connected Across a Battery

Figure-5.90(a) shows an inductor of self induction coefficient L is connected across an ideal battery of emf E . When the switch is closed a current starts growing in circuit which is opposed by the induced EMF in inductor as shown in figure-5.90(b).

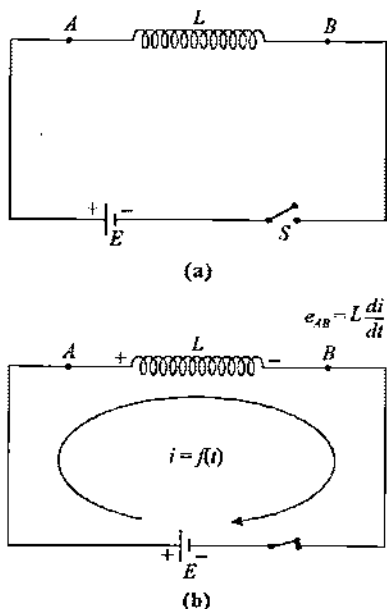


Figure 5.90

In figure-5.90(b) we can see that the inductor is connected across the battery so potential difference of battery is always equal to the induced EMF across the inductor. If at time $t = 0$ switch is closed and current in circuit at time $t = t$ is i then we have

$$E = L \frac{di}{dt}$$

$$\Rightarrow L di = E dt$$

$$\Rightarrow \int_0^i L di = \int_0^t E dt$$

$$\Rightarrow L [i]_0^i = E [t]_0^t$$

$$\Rightarrow Li = Et$$

$$\Rightarrow i = \frac{Et}{L} \quad \dots (5.61)$$

Above expression in equation-(5.61) shows that current in circuit linearly grows with time and variation curve is shown in figure-5.91.

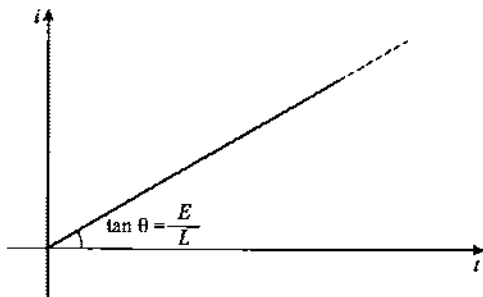


Figure 5.91

5.4.2 Growth of Current in RL Circuit across a battery

Figure-5.92 shows a circuit with an inductor of inductance L connected in series with a resistor of resistance R across a battery of EMF E . In this circuit when the switch is closed at $t = 0$, a current starts flowing and if at $t = t$ current flowing in circuit is considered to be i .

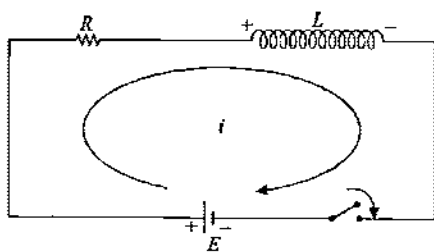


Figure 5.92

We write KVL equation for the above loop as

$$+E - iR - L \frac{di}{dt} = 0$$

$$\Rightarrow \frac{di}{dt} = \frac{E - iR}{L}$$

$$\Rightarrow \int_0^i \frac{di}{E - iR} = \int_0^t \frac{dt}{L}$$

$$\Rightarrow -\frac{1}{R} [\ln(E - iR)]_0^i = \frac{1}{L} [t]_0^t$$

$$\Rightarrow [\ln(E - iR) - \ln(E)] = -\frac{R}{L}[t - 0]$$

$$\Rightarrow \ln\left(\frac{E - iR}{E}\right) = -\frac{Rt}{L}$$

$$\Rightarrow \frac{E - iR}{E} = e^{-\frac{Rt}{L}}$$

$$\Rightarrow i = \frac{E}{R} \left(1 - e^{-\frac{Rt}{L}}\right) \quad \dots (5.62)$$

$$\Rightarrow i = i_s \left(1 - e^{-\frac{Rt}{L}}\right) \quad \dots (5.63)$$

Here $i_s = E/R$ is the steady state current of RL circuit.

Expression in equation-(5.62) shows that the current in circuit grows exponentially and the variation curve for the growth of current in RL circuit is shown in figure-5.93.

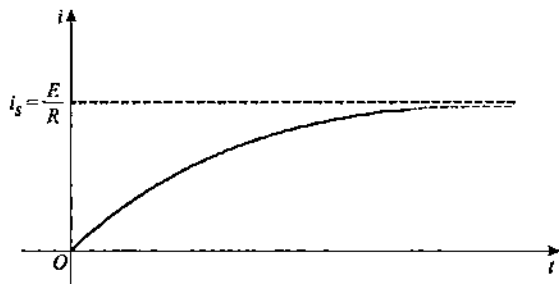


Figure 5.93

5.4.3 Time Constant of RL Circuit

From equation-(5.62) we can analyze that current in RL circuit approaches to steady value $i_s = E/R$ when $t \rightarrow \infty$ and theoretically it takes very long time. In above equation-(5.62) we analyze the circuit current at time $t = \tau = L/R$ which is called 'Time Constant' of RL circuit. Using this time $t = L/R$ in equation-(5.62) we get

$$i = \frac{E}{R} (1 - e^{-1})$$

$$\Rightarrow i = \frac{E}{R} \left(1 - \frac{1}{e}\right)$$

$$\Rightarrow i = \frac{E}{R} \left(1 - \frac{1}{2.718}\right)$$

$$\Rightarrow i = \frac{E}{R} (1 - 0.37)$$

$$\Rightarrow i = 0.63 \frac{E}{R} \quad \dots (5.64)$$

From equation-(5.64) we can state that time constant of a RL circuit is that time duration in which current in RL circuit grows to 63% of the steady state value. If in equation-(5.62) we analyze the current after five times the time constant then we have

$$i = \frac{E}{R} (1 - e^{-5})$$

$$\Rightarrow i = 0.993 \frac{E}{R} \quad \dots (5.65)$$

From equation-(5.65) it can be seen that in duration equal to five times of time constant current attains 99.3% of steady state value and remaining 0.7% will be achieved theoretically in infinite time which can be ignored practically. Thus for analysis of RL circuits we consider that in approximately five times the time constant duration steady state arrives practically so transient period of growth of current in a RL circuit is approximately equal to $5L/R$. The growth of current is shown in the plot of current vs time in figure-5.94. In the below graph transient period and steady state are also mentioned.

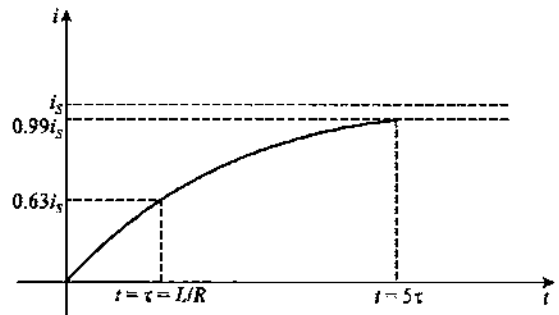


Figure 5.94

5.4.4 Instantaneous Behaviour of an Inductor

We've already discussed that at an instant inductor behaves like a battery of EMF $L(di/dt)$ when current through it changes at a rate di/dt . From equation-(5.64) we can analysis the EMF across the inductor at an instant of time t which is given as

$$e = L \frac{di}{dt} = L \times \frac{E}{R} \times \left(-\frac{R}{L}\right) e^{-\frac{Rt}{L}}$$

$$\Rightarrow e = -E e^{-\frac{Rt}{L}} \quad \dots (5.66)$$

From equation-(5.64) we can see that at $t = 0$ current $i = 0$ and from equation-(5.62) at $t = 0$ the EMF induced in inductor is $e = -E$ that means the EMF in inductor is in opposition to battery EMF and it is equal so current in circuit is zero at $t = 0$.

In other words we can state that just after closing the switch at $t = 0$ inductor in circuit behaves like open circuit through which no current can pass as shown in figure-5.95(a). After a long time when steady state is arrived we know that current becomes E/R and becomes steady due to which EMF induced in the inductor

becomes zero so we can state that in steady state an inductor behaves like a straight wire or short circuit as shown in figure-5.95(b).

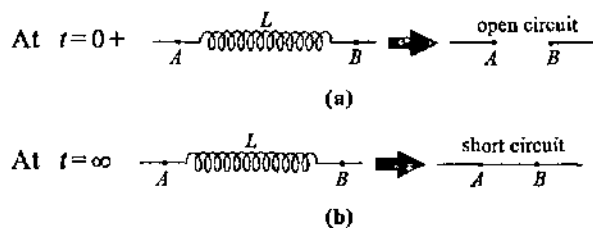


Figure 5.95

5.4.5 Transient Analysis of Advance RL Circuits

We've already discussed in the topic of capacitance that in case of a circuit if multiple resistors and batteries are present in a circuit then across the capacitor we can replace the remaining circuit by Thevenin's equivalent circuit and then we can use the results derived for simple RC circuit. In the same way for RL circuits also if multiple resistances and batteries are present then we can replace the circuit across inductor by a single equivalent battery and its internal resistance and use the results of simple RL circuit. Will analyze such cases in upcoming illustrations.

Illustrative Example 5.29

A solenoid has an inductance of 10H and a resistance of 2Ω . It is connected to a 10V battery. How long will it take for the magnetic energy to reach one fourth of its maximum value?

Solution

(a) The L - R circuit is shown in figure-5.96 Let at a time instant t , after shorting of switch, current through the circuit is i .

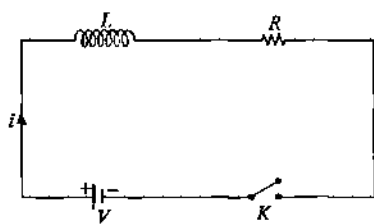


Figure 5.96

For the given loop of circuit current as a function of time is given as

$$i = \frac{V}{R} (1 - e^{-Rt/L}) = i_0 (1 - e^{-Rt/L}) \quad \dots (5.67)$$

Where $i_0 = V/R$ is the maximum current in circuit. The magnetic energy stored in inductor is given as

$$U = \frac{1}{2} Li^2$$

Given that at an instant the magnetic energy attains one fourth of the maximum gives

$$\frac{1}{2} Li^2 = \frac{1}{4} \left(\frac{1}{2} Li_0^2 \right)$$

$$\Rightarrow i = \frac{i_0}{2} = \frac{V}{2R}$$

Substituting the values in equation-(5.67), we get

$$\frac{V}{2R} = \frac{V}{R} (1 - e^{-2t/10}) \text{ or } 0.5 = 1 - e^{-0.2t}$$

or $e^{-0.2t} = 0.5$ which gives $t = 3.478$ s

Illustrative Example 5.30

Calculate the time constant τ of a straight solenoid of length l having a single layer winding of copper wire whose total mass is equal to m . The cross-sectional diameter of the solenoid is assumed to be considerably less than its length. Take resistivity of copper to be ρ and density is equal to δ .

Solution

Time constant of LR circuit is given as

$$\tau = \frac{L}{R}$$

Inductance of a solenoid is given as

$$L = \frac{\mu_0 N^2 A}{l} = \frac{\mu_0 N^2}{l} \times \pi r^2$$

If l_0 is the total length of wire making the solenoid then we can rewrite above expression as

$$L = \frac{\mu_0 (2\pi N r)^2}{4\pi l} = \frac{\mu_0 l_0^2}{4\pi l}$$

The resistance of the wire is given as

$$R = \frac{\rho l_0}{A} = \frac{\rho l_0^2}{A l_0} = \frac{\rho l_0^2}{V} = \frac{\rho l_0^2}{(m/\delta)}$$

$$\Rightarrow R = \frac{\delta \rho l_0^2}{m}$$

$$\Rightarrow l_0^2 = \frac{mR}{\sigma \rho}$$

$$\Rightarrow L = \frac{\mu_0 mR}{4\pi \sigma \rho l}$$

$$\Rightarrow \tau = \frac{L}{R} = \frac{\mu_0 m}{4\pi \delta \rho l}$$

Illustrative Example 5.31

A battery of EMF E and of negligible internal resistance is connected in an LR circuit as shown in figure-97. The inductor has a piece of soft iron inside it. When steady state is reached the piece of soft iron is abruptly pulled out suddenly so that the inductance of the inductor decreases to nL with $n < 1$ with battery remaining connected. Find :

- Current as a function of time if at $t = 0$ is the instant when the soft iron piece is pulled out.
- The work done in pulling out the piece.
- Thermal power generated in the circuit as a function of time.
- Power supplied by the battery as a function of time.

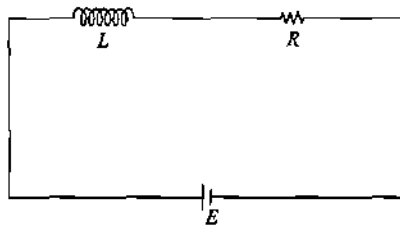


Figure 5.97

Solution

(a) At $t = 0$, steady state current in the circuit is $i_0 = E/R$. Suddenly L reduces to nL ($n < 1$), so to keep the flux constant, the current in the circuit at time $t = 0$ will increase to $\frac{i_0}{n} = \frac{E}{nR}$. If i be the current at time t in circuit as shown in figure-5.98 then by using Kirchhoff's law in the loop we have,

$$E - nL \left(\frac{di}{dt} \right) - iR = 0$$

$$\Rightarrow \frac{di}{E - iR} = \frac{1}{nL} dt$$

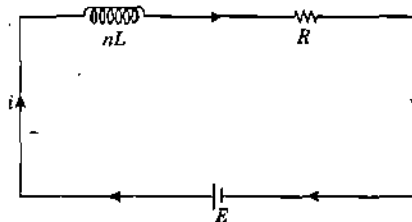


Figure 5.98

Integrating the above expression, gives

$$\int_{i_0/n}^i \frac{di}{E - iR} = \frac{1}{nL} \int_0^t dt$$

$$i = \frac{E}{R} - \frac{E}{R} \left(1 - \frac{1}{n} \right) e^{-Rt/nL}$$

- (b) Work done in pulling out the piece is given as

$$W = U_f - U_i = \frac{1}{2} L_f i_f^2 - \frac{1}{2} L_i i_i^2$$

$$\Rightarrow W = \frac{1}{2} (nL) \left(\frac{E}{nR} \right)^2 - \frac{1}{2} (L) \left(\frac{E}{R} \right)^2$$

$$\Rightarrow W = \frac{1}{2} L \left(\frac{E}{R} \right)^2 \left(\frac{1}{n} - 1 \right)$$

$$\Rightarrow W = \frac{1}{2} L \left(\frac{E}{R} \right)^2 \left(\frac{1-n}{n} \right)$$

- (c) Thermal power generated in the circuit as a function of time is given as

$$P_1 = i^2 R$$

Here, i the time function of current obtained in part (a).

- (d) Power supplied by the battery as a function of time is,

$$P_2 = Ei$$

Here, i the time function of current obtained in part (a).

Illustrative Example 5.32

A thin wire ring of radius a and resistance r is located inside a long solenoid so that their axes coincide. The length of the solenoid is equal to l its cross-section radius to b . At a certain moment, the solenoid was connected to a source of constant voltage V . The total resistance of the circuit is equal to R . Assuming the inductance of the ring to be negligible, find the maximum value of the radial force acting per unit length of the ring.

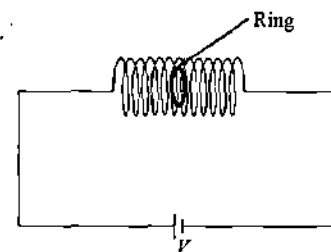


Figure 5.99

Solution

The inductance L of the solenoid is given as

$$L = \frac{\mu_0 N^2 A}{l} = \frac{\mu_0 (nl)^2 (\pi b^2)}{l} = \mu_0 n^2 \pi b^2 l \quad \dots (5.68)$$

The current through the solenoid varies with time as

$$i = \frac{V}{R}(1 - e^{-Rt/L})$$

Magnetic induction inside the solenoid varies with time as

$$B = \mu_0 ni$$

Magnetic flux passing through the ring is given as

$$\Rightarrow \phi = B\pi a^2$$

$$\Rightarrow \phi = \mu_0 ni\pi a^2$$

$$\Rightarrow \phi = \frac{\mu_0 n V \pi a^2}{R} (1 - e^{-Rt/L})$$

EMF induced in the ring is given as

$$e = \frac{d\phi}{dt}$$

$$\Rightarrow e = \frac{\mu_0 n V \pi a^2}{L} e^{-Rt/L}$$

Current induced in the ring is given as

$$i_R = \frac{e}{r} = \frac{\mu_0 n V \pi a^2}{rL} e^{-Rt/L}$$

Radial force per unit length on the ring circumference is given as

$$F_r = \frac{dF}{dl} = Bi_R$$

$$\Rightarrow F_r = \mu_0 n I \left(\frac{\mu_0 n V \pi a^2}{rL} e^{-Rt/L} \right)$$

$$\Rightarrow F_r = \left(\frac{\mu_0^2 n^2 \pi a^2 V}{rL} e^{-Rt/L} \right) \left(\frac{V}{R} (1 - e^{-Rt/L}) \right)$$

$$\Rightarrow F_r = \frac{\mu_0^2 n^2 \pi a^2 V^2}{rRL} e^{-Rt/L} (1 - e^{-Rt/L}) \quad \dots (5.69)$$

Substituting the value of L from equation-(5.68) to equation-(5.69), we get

$$F_r = \frac{\mu_0 a^2 V^2}{rRb^2 l} e^{-Rt/L} (1 - e^{-Rt/L}) \quad \dots (5.70)$$

In above equation we can see that at $t = 0$, $F_r = 0$ and also $F_r = 0$ at $t = \infty$. Thus at some intermediate time value of radial force per unit length is maximum for this we differentiate F_r with respect to time t and equate it to zero which gives the time at which this occurs and it is obtained as

$$e^{-Rt/L} = 1/2$$

Thus maximum value of F_r is given as

$$F_{\max} = \frac{\mu_0 a^2 V^2}{4rRl b^2}$$

Illustrative Example 5.33

A section of an electrical circuit XY is shown in figure-5.100 which carries a current of 5A which is decreasing at the rate of 10 A/s. Find the potential difference $V_y - V_x$.

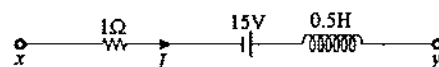


Figure 5.100

$$e = L \frac{di}{dt}$$

Solution

In the above circuit section as current is decreasing with time, the polarity of self induced EMF is shown in figure-5.101. In this branch of circuit we write equation of potential drop from point x to y which is given as

$$V_x - iR + E + L \frac{di}{dt} = V_y$$

$$\Rightarrow V_x - 5 \times 1 + 15 + 0.5 \times 10 = V_y$$

$$\Rightarrow V_y - V_x = 15V$$

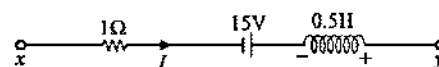


Figure 5.101

Illustrative Example 5.34

Figure-5.102 shows a RL circuit. If at $t = 0$ switch is closed, find the current in the inductor as a function of time.

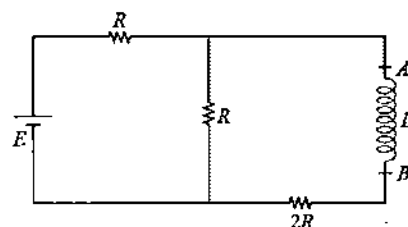


Figure 5.102

Solution

In the above circuit, we can remove inductor and calculate the Thevenin's equivalent battery across the terminals A and B as shown in figure-5.103. The internal resistance across terminals A and B can be calculated by shorting the battery which gives

$$r = 2R + \frac{R}{2} = \frac{5R}{2}$$

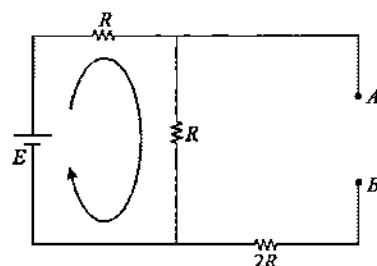


Figure 5.103

Due to removal of inductor right loop of circuit is now open and a current flows in the left loop which is given as

$$i = \frac{E}{2R}$$

The open circuit potential difference across terminals A and B is given as

$$V_{AB} = iR = \frac{E}{2}$$

Thus the reduced circuit for the given circuit is shown in figure-5.104. This circuit is now reduced to simple RL circuit and in this case the current as a function of time can be given as

$$i = \frac{(E/2)}{(5R/2)} \left(1 - e^{-\frac{5Rt}{2L}} \right)$$

$$i = \frac{E}{5R} \left(1 - e^{-\frac{5Rt}{2L}} \right)$$

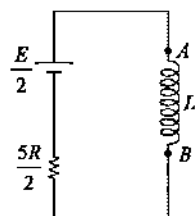


Figure 5.104

5.5 Energy Stored in an Inductor

When current grows in an inductor then the EMF induced across its terminals opposes the current flowing into it and the high potential of induced EMF is at that end of inductor from which the current is entering into it as shown in figure-5.105 and the inductor behaves like a battery as shown.

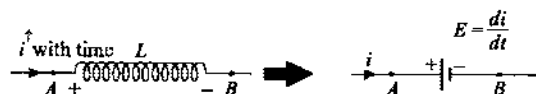


Figure 5.105

In the chapter of current electricity we've studied if current enters into a battery from its high potential end then power is

absorbed by the battery and chemical energy of battery increases. Similarly in above case the EMF absorbs the power from the current flowing into it. As with current magnetic field increases in the inductor, the energy of magnetic field also increases. At any instant of time t the power absorbed by the inductor is given as

$$P = ei = \left(L \frac{di}{dt} \right) i \quad \dots (5.71)$$

In time dt , energy absorbed by the inductor is

$$dU = Pdt$$

$$\Rightarrow dU = Lidi$$

When current in inductor increases from 0 to I , total energy absorbed upto this instant by the inductor is given as

$$U = \int dU = \int_0^I Lidi$$

$$\Rightarrow U = L \left[\frac{i^2}{2} \right]_0^I$$

$$\Rightarrow U = \frac{1}{2} LI^2 \quad \dots (5.72)$$

Above expression of energy stored in an inductor carrying a current I as given in equation-(5.72) is a standard result which can be used in different cases. Whenever in a given physical situation an inductor carrying a current I is given that means when this current was grown to I in the inductor that time this energy was stored by the inductor and it is maintained along with the current in it. This energy is actually stored in inductor in form of field energy of magnetic field within the volume of inductor which is an energy similar to the field energy of electric field we studied in the chapter of electrostatics. In next section same will be discussed in detail.

5.5.1 Magnetic Field Energy Density

In the chapter of electrostatics we discussed that in whole region wherever electric field is present there exist electric field energy of which the volume density is given as

$$u_e = \frac{1}{2} \epsilon_0 E^2 \text{ J/m}^3$$

Similarly in region of magnetic field also there exist magnetic field energy which we've already discussed in article-4.10. The volume density of magnetic field energy in space is given as

$$u_m = \frac{B^2}{2\mu_0} \text{ J/m}^3 \quad \dots (5.73)$$

5.5.2 Field Energy Stored in a Current Carrying Inductor

In an inductor of length l , cross sectional area S with N turns carrying a current I , the magnetic induction inside is given as

$$B = \mu_0 \left(\frac{N}{l} \right) I$$

Total magnetic energy stored in the volume of inductor is given as

$$U = u_m \times Sl$$

$$\Rightarrow U = \frac{B^2}{2\mu_0} \times Sl$$

$$\Rightarrow U = \frac{Sl}{2\mu_0} \left(\frac{\mu_0 NI}{l} \right)^2$$

$$\Rightarrow U = \frac{1}{2} \left(\frac{\mu_0 N^2 S}{l} \right) I^2$$

$$\Rightarrow U = \frac{1}{2} LI^2 \quad \dots (5.74)$$

Above equation-(5.74) is same as equation-(5.72) which verifies that the energy absorbed by the inductor during the time its current grows is stored in form of magnetic field energy in the volume of inductor.

When an inductor in which a current I_0 is flowing and it is suddenly cut from the circuit then the energy stored in it does not dissipate immediately as under ideal conditions it is considered to remain intact by magnetic polarization of molecules in the volume of inductor. This phenomenon of retention of magnetic field is called hysteresis about which we will discuss in magnetic properties of material later in this chapter. Such an isolated inductor which contains magnetic energy is called 'Active Inductor' which is indicated by the direction of last current flowing in it as shown in figure-5.106. In this figure 'dot' at one end of the inductor indicates the end which behaves like its north pole. This dot convention is used because on paper in two dimensional picture of inductor it is difficult to show the actual direction of current by which the magnetic field direction can be determined. Like in the below figure magnetic field inside solenoid exist from left to right.

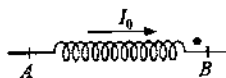


Figure 5.106

Illustrative Example 5.35

A solenoid of resistance 50Ω and inductance $80H$ are connected to a $200V$ battery. How long will it take for the current to reach

50% of its final equilibrium value? Calculate the maximum energy stored in inductor.

Solution

The current i in RL circuit is given as

$$i = \frac{E}{R} (1 - e^{-Rt/L})$$

When $t \rightarrow \infty$ the final steady state or equilibrium value of current is given as

$$i_0 = \frac{E}{R}$$

If the current has half of this value at time t_0 then we have

$$\frac{1}{2} \frac{E}{R} = \frac{E}{R} (1 - e^{-Rt_0/L})$$

$$\Rightarrow \frac{1}{2} = (1 - e^{-Rt_0/L})$$

$$\Rightarrow e^{-Rt_0/L} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow e^{Rt_0/L} = 2$$

$$\Rightarrow t_0 = \frac{L}{R} \ln(2)$$

$$\Rightarrow t_0 = \frac{80}{50} \times 0.693 = 1.104s$$

The maximum energy stored in inductor is given as

$$E_{\max} = \frac{1}{2} Li_0^2 = \frac{1}{2} \times 80 \times \left(\frac{200}{50} \right)^2 = 640J$$

Illustrative Example 5.36

The current in a coil of self inductance $2.0H$ is increasing according to $I = 2\sin(t^2)$ ampere. Find the amount of energy spend during the period when the current changes from zero to $2A$.

Solution

The energy stored in a coil of self inductance L when the current changes by $2A$ is given as

$$E = \frac{1}{2} LI^2 = \frac{1}{2} \times (2.0) \times (2)^2 = 4J$$

This is equal to the amount of energy spent by the source which is supplying the current.

Illustrative Example 5.37

Find the magnetic field energy in a cubical region of edge 'a' above a large current carrying sheet carrying a uniform linear current density $I \text{ A/m}$.

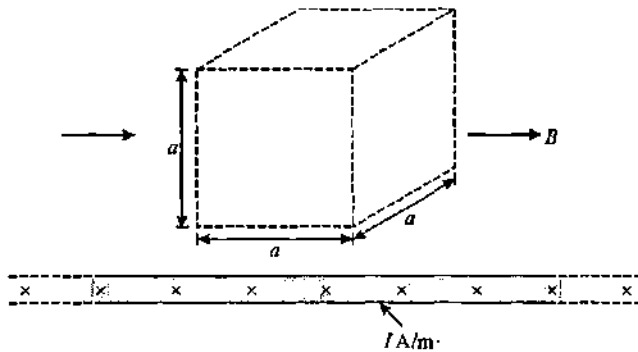


Figure 5.107

Solution

Magnetic induction due to a large current sheet in its surrounding is given as

$$B = \frac{1}{2} \mu_0 I$$

Magnetic field energy density in space is given as

$$u = \frac{B^2}{2\mu_0} = \frac{\left(\frac{1}{2} \mu_0 I\right)^2}{2\mu_0} = \frac{1}{8} \mu_0 I^2 \text{ J/m}^3$$

Field energy in cubical region is given as

$$U = 4 \times a^3 = \frac{1}{8} \mu_0 a^3 I^2$$

Illustrative Example 5.38

What inductance would be needed to store 1.0 kWh of energy in a coil carrying a 200A current.

Solution

Magnetic energy stored in an inductor of inductance L and carrying a current i is given as

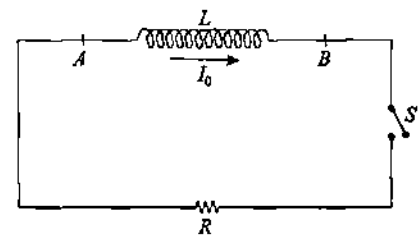
$$U = \frac{1}{2} L i^2$$

$$\Rightarrow L = \frac{2U}{i^2}$$

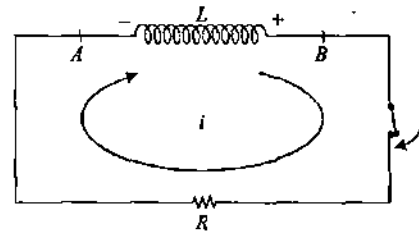
$$\Rightarrow L = \frac{2(3.6 \times 10^6)}{(200)^2} = 180 \text{ H}$$

5.6 Decay of Current in RL Circuit

Figure-5.108 shows an active inductor carrying a current I_0 is connected to a resistor of resistance R via a switch S . As already discussed in previous article that I_0 was the last current flowing through the inductor when it was disconnected from a circuit. The magnetic flux through the inductor that time was $\phi = L I_0$. Because of hysteresis (also called magnetic inertia which we will study in article-5.9.17), the magnetic flux through a coil or inductor cannot change suddenly so if this active inductor is again connected in a circuit, the total flux linked in the circuit remain same. In figure below when switch is closed then immediately after closing the switch current through this inductor will be I_0 to keep the magnetic flux same.



(a)



(b)

Figure 5.108

Due to current heat started dissipating in the resistor and the energy for this comes from the magnetic energy in inductor so current through the inductor gradually decreases and an EMF is induced across inductor with high and low potential terminals as shown in above figure. As current is decreasing for the inductor and resistor in parallel we can write

$$-L \frac{di}{dt} = iR$$

$$\Rightarrow \frac{di}{i} = -\frac{R}{L} dt$$

In this situation when the switch was closed at $t = 0$ the current in inductor was I_0 which decreases with time. If after some time at $t = t$ current in circuit is i then integrating above expression from $t = 0$ to t , we have

$$\int_{I_0}^i \frac{di}{i} = -\frac{R}{L} \int_0^t dt$$

$$\Rightarrow [\ln(i)]_{I_0}^t = -\frac{R}{L} [t]_0^t$$

$$\Rightarrow [\ln(i) - \ln(I_0)] = -\frac{R}{L} [t - 0]$$

$$\Rightarrow \ln\left(\frac{i}{I_0}\right) = -\frac{Rt}{L}$$

$$\Rightarrow i = I_0 e^{-\frac{Rt}{L}} \quad \dots (5.75)$$

Expression in equation-(5.75) gives the current in RL circuit during its decay. Figure-5.109 shows the variation of decay current in inductor with time. We can also see that in one time constant current drops to 37% of the initial value. Substituting $t = L/R$ in above equation-(5.75), we have

$$i = I_0 e^{-1}$$

$$\Rightarrow i = \frac{I_0}{e}$$

$$\Rightarrow i = \frac{I_0}{2.718} = 0.37 I_0 \quad \dots (5.76)$$

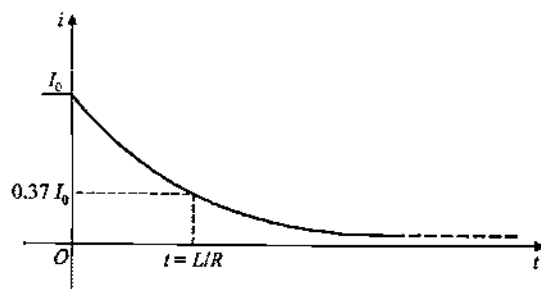


Figure 5.109

As current decays exponentially, after a long time in steady state current becomes zero and in this period whole of the magnetic energy stored in inductor is dissipated as heat which can also be verified by calculating the heat dissipation in resistor. Using expression of current in equation-(5.75), we can write the thermal power produced in resistor R is given as

$$P_{th} = i^2 R$$

$$\Rightarrow P_{th} = I_0^2 R e^{-\frac{2Rt}{L}} \quad \dots (5.77)$$

Heat dissipated in resistor during an elemental time dt is given as

$$dH = P_{th} dt$$

$$\Rightarrow dH = I_0^2 R e^{-\frac{2Rt}{L}} dt$$

To calculate the total heat dissipated in resistance we integrate the above expression from $t = 0$ to $t \rightarrow \infty$ as theoretically it takes infinite time after which current will drop to zero.

$$\Rightarrow H = \int dH = \int_0^{\infty} I_0^2 R e^{-\frac{2Rt}{L}} dt$$

$$\Rightarrow H = I_0^2 R \left[-\frac{L}{2R} \right] \left[e^{-\frac{2Rt}{L}} \right]_0^{\infty}$$

$$\Rightarrow H = -\frac{1}{2} L I_0^2 [0 - 1]$$

$$\Rightarrow H = \frac{1}{2} L I_0^2 \quad \dots (5.78)$$

Equation-(5.78) is the total magnetic energy initially stored in the inductor before closing the switch. This verifies that all the magnetic energy of inductor is dissipated as heat in resistor during the time current decays.

Illustrative Example 5.39

A thin charged ring of radius $a = 10\text{cm}$ rotates about its axis with an angular velocity $\omega = 100\text{ rad/s}$. Find the ratio of volume energy densities of magnet and electric field on the axis of the ring at a point removed from its centre by a distance $l = a$.

Solution

We know the magnetic field at P a distance l from the centre O of the coil on the axis

$$B = \frac{\mu_0 I a^2}{2\pi(l^2 + a^2)^{3/2}}$$

Current due to revolving charge is given as

$$I = \frac{q\omega}{2\pi}$$

$$\Rightarrow B = \frac{\mu_0}{4\pi} \frac{q\omega a^2}{(l^2 + a^2)^{3/2}}$$

Energy density of magnetic field is given as

$$u_M = \frac{B^2}{2\mu_0}$$

The electric field strength at an axial point is given as

$$E = \frac{1}{4\pi\epsilon_0} \frac{ql}{(l^2 + a^2)^{3/2}}$$

The energy density of the electric field is given as

$$u_E = \frac{1}{2} \epsilon E^2$$

Ratio of energy densities of magnetic and electric field are given as

$$\frac{u_M}{u_E} = \frac{(B/E)^2}{\mu_0 \epsilon_0} = \left(\frac{\omega a^2}{l} \right)^2 \mu_0 \epsilon_0$$

$$\frac{u_M}{u_E} = \frac{\left(\frac{\omega a^2}{l} \right)^2}{c^2} = 1.1 \times 10^{-15}$$

Illustrative Example 5.40

A 10H inductor carries a current of 20A. How much ice at 0°C could be melted by the energy stored in the magnetic field of the inductor? Latent heat of ice is $22.6 \times 10^3 \text{ J/kg}$.

Solution

Energy stored in inductor is given as

$$U = \frac{1}{2} Li^2$$

This energy is completely used in melting the ice. If L_f is the latent heat of fusion, we use

$$\frac{1}{2} Li^2 = mL_f$$

$$\Rightarrow m = \frac{Li^2}{2L_f}$$

Substituting the values we get

$$m = \frac{(10)(20)^2}{2(2.26 \times 10^3)} = 0.88 \text{ kg}$$

Illustrative Example 5.41

A coil of inductance $L = 2.0 \mu\text{H}$ and resistance $R = 1.0\Omega$ is connected to a source of constant e.m.f. $E = 3.0\text{V}$. A resistance $R_0 = 2.0\Omega$ is connected in parallel with the coil. Find the amount of heat generated in the coil after the switch Sw is disconnected. The internal resistance of the source is negligible.

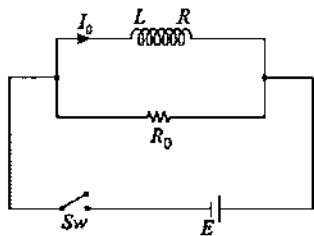


Figure 5.110

Solution

When the switch is disconnected the decay of current I_0 starts where I_0 is the steady state current in the inductor. KVL equation for the loop in which current decays at an instant when a current I exist in loop is given as

$$IR_0 + IR = -L \frac{dI}{dt}$$

$$\Rightarrow \frac{dI}{I} = -\left(\frac{R_0 + R}{L}\right) dt$$

$$\Rightarrow \int_{I_0}^I \frac{dI}{I} = \int_0^t \frac{R_0 + R}{L} dt$$

Solving the above integration gives

$$I = I_0 e^{-(R+R_0)t/L}$$

Heat generated in time dt is given as

$$dQ = I^2 R dt$$

$$\Rightarrow Q = \int dQ = \int_0^\infty I^2 R dt$$

$$\Rightarrow Q = I_0^2 R \frac{L}{2(R_0 + R)} = \frac{LE^2}{2R(R + R_0)} \quad [\text{As } I_0 R = E]$$

Substituting the values we get

$$Q = 3 \times 10^{-6} \text{ J} = 3 \mu\text{J}$$

Alternative way : Above result can be directly obtained as on disconnecting the switch current decays in resistance R and R_0 in series combination so the total heat produced is divided in same ratio of the two resistances which is equal to the energy stored in the inductor in steady state. Thus the heat generated in the coil can be given as

$$H = \frac{1}{2} LI_0^2 \times \frac{R}{R + R_0}$$

Illustrative Example 5.42

In circuit shown in figure-5.111, find the current through battery just after closing the switch and after a long time in steady state.

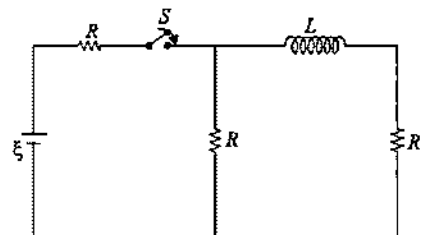


Figure 5.111

Solution

Just after closing the switch inductor behaves like open circuit so the circuit just after closing the switch is shown in figure-5.112 below.

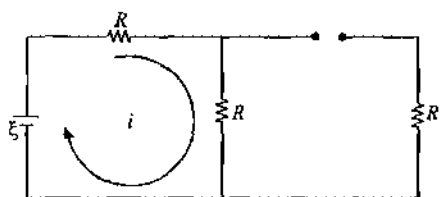


Figure 5.112

The current only flows in the left loop as shown which is given as

$$i_B = \frac{\xi}{2R}$$

In steady state after a long time the inductor behaves like a straight wire thus equivalent circuit in steady state is shown in figure-5.113 below. The two resistances on the right side can be considered in parallel and thus across the battery equivalent resistance will be $R + R/2 = 3R/2$.

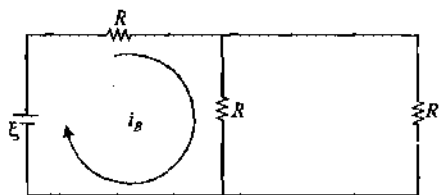


Figure 5.113

The current flowing through battery in steady state is given as $\Rightarrow$

$$i_B = \frac{\xi}{3R/2} = \frac{2\xi}{3R}$$

Illustrative Example 5.43

For the circuit shown in figure-5.114. $E = 50\text{V}$, $R_1 = 10\Omega$, $R_2 = 20\Omega$, $R_3 = 30\Omega$ and $L = 2.0\text{mH}$. Find the current through R_1 and R_2 .

- Immediately after switch S is closed
- A long time after S is closed
- Immediately after S is reopened
- A long time after S is reopened

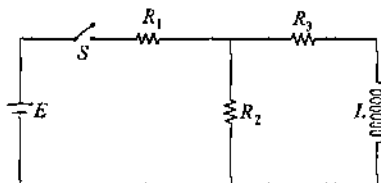


Figure 5.114

Solution

- Resistance offered by inductor immediately after switch is closed will be infinite as it behaves like open circuit. Therefore,

current through R_3 will be zero and current through R_1 is given as

$$I_1 = \frac{E}{R_1 + R_2}$$

$$\Rightarrow I_1 = \frac{50}{10+20} = \frac{5}{3} \text{ A}$$

- After a long time of closing the switch, resistance offered by inductor will be zero as in steady state it behaves like short circuit. In that case R_2 and R_3 are in parallel, and the resultant of these two is then in series with R_1 thus equivalent resistance across the battery in steady state is given as

$$R_{\text{net}} = R_1 + \frac{R_2 R_3}{R_2 + R_3} = 10 + \frac{(20)(30)}{20+30} = 22\Omega$$

Current through the battery is given as

$$I_2 = \frac{E}{R_{\text{net}}} = \frac{50}{22} \text{ A}$$

This current will distribute in R_2 and R_3 in inverse ratio of resistance so current through R_2 is given as

$$R_2 = \left(\frac{50}{22}\right) \left(\frac{R_3}{R_2 + R_3}\right)$$

$$R_2 = \left(\frac{50}{22}\right) \left(\frac{30}{30+20}\right) = \frac{15}{11} \text{ A}$$

- Immediately after switch is reopened, the current through R_1 will become zero.

But current through R_2 will be equal to the steady state current through R_3 , which is given as

$$I_3 = \left(\frac{50}{22} - \frac{15}{11}\right) = 0.91 \text{ A}$$

- A long after S is reopened, current through all resistors will become zero.

Illustrative Example 5.44

In the circuit shown in figure-115, find the current in inductor as a function of time if switch is closed at $t = 0$.

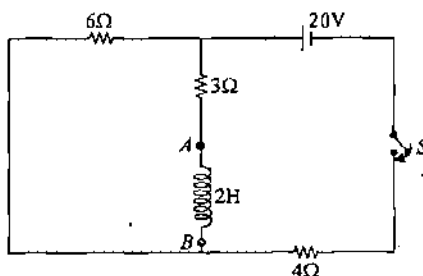


Figure 5.115

Solution

In above given circuit we remove the inductor and across terminals A and B we reduce the circuit by its Thevenin's equivalent battery as explained in figures-5.116 below.

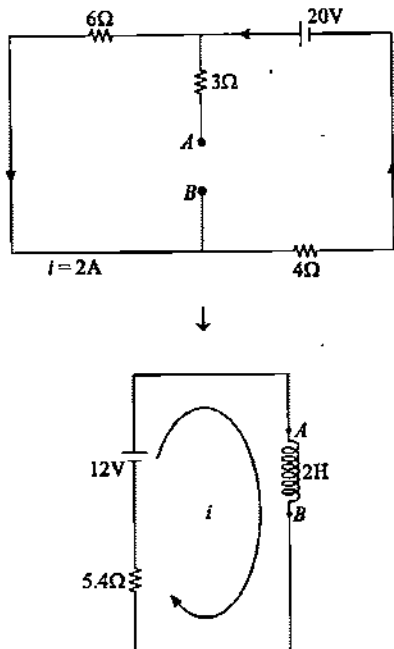


Figure 5.116

Internal resistance of circuit across A & B is given as

$$r = 3 + \frac{6 \times 4}{6 + 4} = 5.4\Omega$$

Open circuit potential difference across A & B is given as

$$V_A - V_B = 12V$$

Growth of current in LR circuit can be directly given by the relation of simple LR circuit as

$$i = \frac{\mathcal{E}}{R} (1 - e^{-Rt/L})$$

$$\Rightarrow i = \frac{12}{5.4} \left(1 - e^{-\frac{5.4t}{2}} \right)$$

$$\Rightarrow i = 2.22(1 - e^{-2.7t}) \text{ A}$$

Illustrative Example 5.45

In a process of current decay in an inductor through a resistance, if in time t , current falls to η times ($\eta < 1$) the initial value, find the time constant of circuit.

Solution

For decay of current in an inductor in LR circuit current is given as

$$i = I_0 e^{-Rt/L}$$

In above situation time constant of circuit is given as

$$\tau = L/R$$

$$\Rightarrow i = I_0 e^{-t/\tau}$$

Given that at $t = t$, $i = \eta I_0$ which gives

$$\eta I_0 = I_0 e^{-t/\tau}$$

$$\Rightarrow \ln\left(\frac{1}{\eta}\right) = \frac{t}{\tau}$$

$$\Rightarrow \tau = \frac{t}{\ln(1/\eta)}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Self & Mutual Inductance

Module Number - 1 to 21

Practice Exercise 5.3

(i) Find the self inductance of a unit length of a cable consisting of two thin walled coaxial metallic cylinders if the radius of the outside cylinder is η ($\eta > 1$) times that of the inside one. The permeability of the medium between the cylinders is assumed to be equal to unity.

$$\left[\frac{\mu_0}{2\pi} \ln(\eta) \right]$$

(ii) Find the inductance of a unit length of a double line if the radius of each wire is η times less than the distance between the axes of the wires. The field inside the wires is to be neglected, the permeability is assumed to be equal to unity throughout, and $\eta \gg 1$.

$$\left[\frac{\mu_0}{\pi} \ln(\eta) \right]$$

(iii) In the circuit shown in figure-5.117, find potential of point C at the instant shown. The rate of increase of current in resistance is 2.5 A/s .

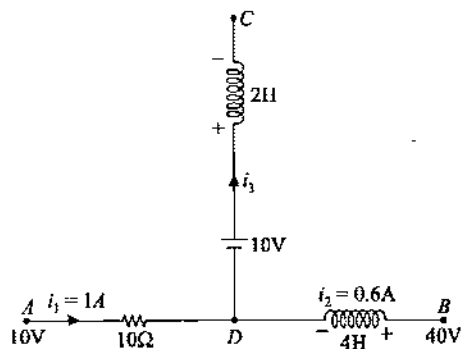


Figure 5.117

[- 15V]

(iv) A long cylinder of radius a carrying a uniform surface charge rotates about its axis with an angular velocity ω . Find the magnetic field energy per unit length of the cylinder inside of it if the linear charge density equals λ and consider permeability of the medium is equal to unity.

$$\left[\frac{\mu_0 \lambda^2 \omega^2 a^2}{8\pi} \right]$$

(v) A long coaxial cable consists of two concentric cylinders of radii a and b . The central conductor of the cable carries a steady current i and the outer conductor provides the return path of the current. Calculate the energy stored in the magnetic field of length l of such a cable

$$\left[\frac{\mu_0 i^2 l}{4\pi} \ln\left(\frac{b}{a}\right) \right]$$

(vi) In a branch of an electrical circuit shown in figure-5.118, $R = 10\Omega$, $L = 5H$, $E = 20V$, $i = 2A$. This current is decreasing at a rate of $1.0A/s$. Find V_{ab} at this instant.

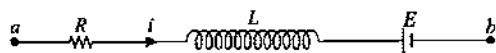


Figure 5.118

[35V]

(vii) The potential difference across a 150mH inductor as a function of time is shown in figure-5.119. Assume that the initial value of the current in the inductor is zero. What is the current when (a) $t = 2.0ms$ (b) $t = 4.0ms$?

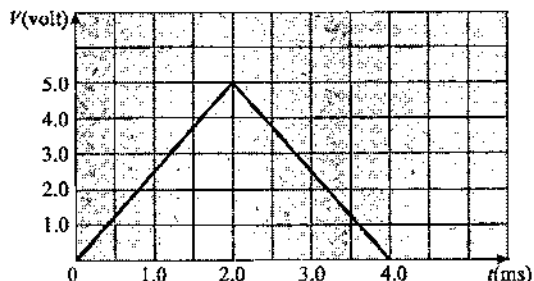


Figure 5.119

[(a) $3.33 \times 10^{-2} A$ (b) $6.67 \times 10^{-2} A$]

(viii) A metal rod OA of mass m and length l is kept rotating with a constant angular speed ω in a vertical plane about a horizontal axis at the end O as shown in figure-5.120. The free end A is arranged to slide without friction along a fixed conducting circular ring in the same plane as that of rotation. A uniform and constant magnetic induction B is applied perpendicular and into the plane of rotation as shown in figure. An inductor and an external resistance R are connected through a switch S between the point O and a point C on the ring to form an electrical circuit. Neglect the resistance of the ring and the rod. Initially, the switch is open.

(a) What is the induced EMF across the terminals of the switch?

(b) The switch S is closed at time $t = 0$. Obtain an expression for the current as a function of time. In the steady state, obtain the time dependence of the torque required to maintain the constant angular speed, given that the rod OA was along the positive X -axis at $t = 0$.

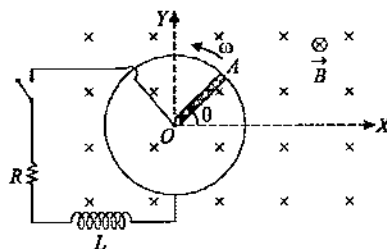


Figure 5.120

$$[(a) \frac{1}{2} B r^2 \omega \quad (b) \frac{B r^2 \omega}{2R} (1 - e^{-Rt/L}); \frac{B^2 r^2 \omega}{4R} (1 - e^{-Rt/L})]$$

(ix) A solenoid of inductance L with resistance r is connected in parallel to a resistance R . A battery of EMF E and of negligible internal resistance is connected across the parallel combination as shown in the figure-5.121.

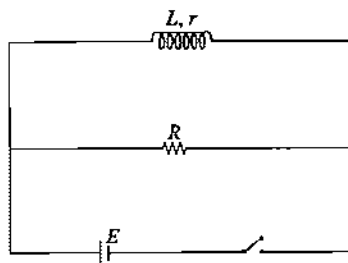


Figure 5.121

(a) Current through the solenoid after the switch is opened at $t = 0$.

(b) Amount of heat generated in the solenoid

$$[(a) \frac{E}{r} e^{-\left(\frac{R+r}{L}\right)t}; (b) \frac{E^2 L}{2r(R+r)}]$$

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(x) In the circuit diagram shown, initially there is no energy in the inductor and the capacitor. The switch is closed at $t = 0$. Find the current I as a function of time if $R = \sqrt{L/C}$

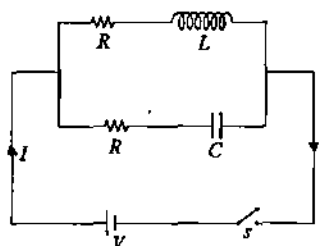


Figure 5.122

$$\left[\frac{V}{R} \right]$$

(xi) A closed circuit consists of a source of constant EMF ξ and a choke coil of inductance L connected in series. The active resistance of the whole circuit is equal to R . At the moment $t = 0$, the choke coil inductance was decreased η ($\eta < 1$) times suddenly. Find the current in the circuit as a function of t .

$$\left[\frac{\xi}{R} \left[1 - (\eta - 1)e^{-\frac{\eta R t}{L}} \right] \right]$$

(xii) In the circuit shown in figure-5.123, switch S is closed at time $t = 0$. Find the current through the inductor as a function of time t .

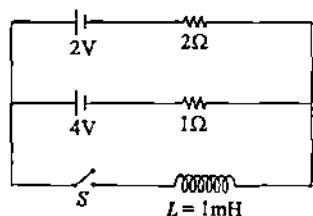


Figure 5.123

$$[i = 5(1 - e^{-2000t/3})\text{A}]$$

(xiii) Find the total field energy of magnetic field stored per unit length inside a long cylindrical wire of radius R and carrying a current I .

$$\left[\frac{\mu_0 I^2}{16\pi} \right]$$

(xiv) Figure-5.124 shows an active inductor with current I_0 connected in series with a resistance and a battery. If at $t = 0$ the switch is closed, find the current in inductor as a function of time.

Electromagnetic Induction and Alternating Current

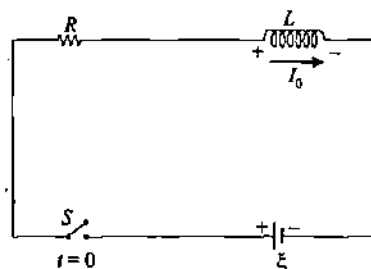


Figure 5.124

$$\left[\frac{1}{R} [\xi - (\xi_0 - I_0 R)e^{-Rt/L}] \right]$$

5.7 Mutual Induction

It is a property of a pair of coils due to which change in current in one coil is opposed by the EMF induced in other coil because of magnetic flux linkage in the two coils. Figure-5.125 shows two coils kept close to each other along same axis. When a current i_1 flows in coil-1 it develops a flux ϕ_1 of which some fraction passes through coil-2 as shown which is linked with it. When current in coil-1 is changed, the flux due to its current which is linked with coil-2 changes and induces an EMF in coil-2. This is called 'Mutually Induced EMF' as its due to flux change in coil-1.

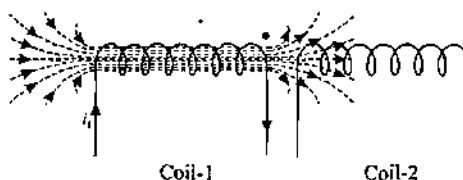


Figure 5.125

If ϕ_2 is the magnetic flux linked with coil-2 due to the current i_1 in coil-1 then we have

$$\phi_2 \propto i_1$$

$$\Rightarrow$$

$$\phi_2 = M i_1 \quad \dots (5.79)$$

Where M is the proportionality constant called 'Coefficient of Mutual Induction' which depends upon the dimensions of the coil and separation between the two coils. Remember that M is bidirectional coefficient which remains same for a given pair of coils and can be used in calculation of flux linked with either coil due to the current in other coil. If ϕ_{12} is the magnetic flux linked with coil-1 due to a current i_2 in coil-2 then we can also write

$$\phi_{12} = M i_2 \quad \dots (5.80)$$

Thus mutually induced EMF in coils will depend upon the rate at which the current changes in other coils which are given as

$$e_2 = M \frac{di_1}{dt}$$

and

$$e_1 = M \frac{di_2}{dt}$$

The direction of mutually induced EMF in either coil will be such that it opposes the change in current in the other coil.

5.7.1 Total Flux linked to a Coil with Mutual Induction

When there is a pair of coils kept fixed at some separation having their self induction coefficients L_1 and L_2 and mutual induction coefficient for the pair M then total magnetic flux linked with either coil will be the sum of magnetic flux linked with the coil due to both self induction and mutual induction if both the coils are connected to closed circuits and carry currents.

Figure-5.126 shows two coils which carry currents i_1 and i_2 and the direction of their magnetic flux is denoted by the dot at one end of the coil according to the dot convention we already discussed in previous article-5.5.2.

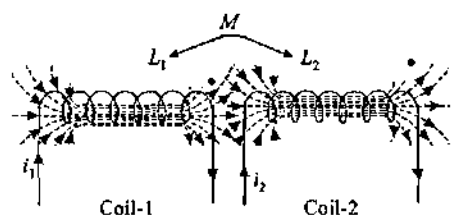


Figure 5.126

In figure-5.126 we can see that magnetic flux due to both coils are from left to right so the part of flux due to one coil which is linked with other coil is in same direction as that of their own flux so net flux linked with coil-1 and coil-2 are given as

$$\phi_1 = L_1 i_1 + M i_2 \quad \dots (5.81)$$

$$\phi_2 = L_2 i_2 + M i_1 \quad \dots (5.82)$$

If current in above coils starts changing then EMF induced in the two coils are given as

$$e_1 = \frac{d\phi_1}{dt} = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

and

$$e_2 = \frac{d\phi_2}{dt} = L_2 \frac{di_2}{dt} + M \frac{di_1}{dt}$$

Now consider a pair of coils shown in figure-5.127 in which the direction of magnetic flux due to the two coils are in opposite direction so the net magnetic flux linked with the two coils are given as

$$\phi_1 = L_1 i_1 - M i_2 \quad \dots (5.83)$$

$$\phi_2 = L_2 i_2 - M i_1 \quad \dots (5.84)$$

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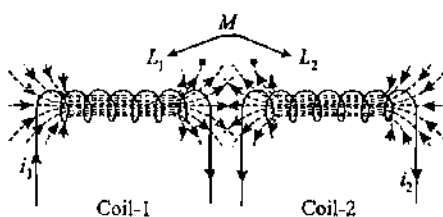


Figure 5.127

If current in above coils starts changing then EMF induced in the two coils are given as

$$e_1 = \frac{d\phi_1}{dt} = L_1 \frac{di_1}{dt} - M \frac{di_2}{dt}$$

and

$$e_2 = \frac{d\phi_2}{dt} = L_2 \frac{di_2}{dt} - M \frac{di_1}{dt}$$

For a given pair of coils the coefficient of mutual induction is used for calculation of magnetic flux through one coil due to the current in either coil. This property of mutual induction for a pair of coils is called '*Reciprocity Property*' of mutual induction.

5.7.2 Relation between Coefficients of Self and Mutual Induction for a given Pair of Coils

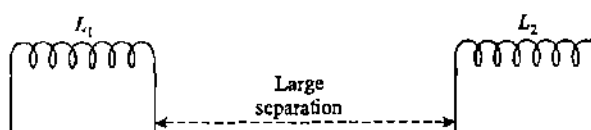
Figure-5.128 shows a pair of coils with their coefficient of self induction L_1 and L_2 and coefficient of mutual induction M . The relationship between these coefficients is given as

$$M = k \sqrt{L_1 L_2} \quad \dots (5.85)$$



Figure 5.128

Where k is called '*Coupling factor*' which depends upon the orientation of coils and the separation between the two coils. If the fraction of magnetic flux produced by one coil which is linked with other is high then coupling factor between the two coils will be higher. In general $0 \leq k \leq 1$ for any set of coils. If there is no flux linkage between the two coils then $k = 0$ and for full flux linkage between the coils $k = 1$. No flux linkage means no part of flux of one coil passes through the other coil. This happens when the coils are separated by a very large distance as shown in figure-5.129(a) or when the coils are mounted at right angle to each other as shown in figure-5.129(b) in which flux of one coil does not link with turns of other coil.



(a)

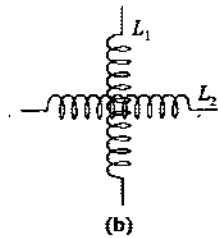


Figure 5.129

When coils are wound over each other as shown in figure-5.130 then complete flux of one coil is linked with the other coil so in this case we can consider the value of k to be equal to unity.

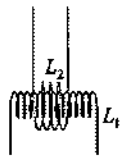


Figure 5.130

For partial linkage of flux between the two coils the coefficient of mutual induction is given by equation-(5.85) with value of k between 0 and 1. Students need not think about derivation of equation-(5.85) which is considered as a standard empirical relation here.

5.7.3 Inductors in Series Combination

There are two ways in which inductors can be connected in series combination with their flux in same direction or in opposite direction. We will discuss the two ways as different cases one by one.

Case I : When flux of the two inductors are in same direction

Figure-5.131 shows two inductor coils in series with their flux in same directions as shown by the dots on the right end of the two coils. If a current I flows through the coils the magnetic flux linked with each coil are given as

$$\phi_1 = L_1 I + MI$$

and

$$\phi_2 = L_2 I + MI$$

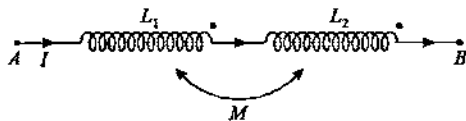


Figure 5.131

When the flowing current starts increasing at a rate dI/dt then the induced emf in the two coils are given as

$$e_1 = \frac{d\phi_1}{dt} = L_1 \frac{dI}{dt} + M \frac{dI}{dt} \quad \dots (5.86)$$

and

$$e_2 = \frac{d\phi_2}{dt} = L_2 \frac{dI}{dt} + M \frac{dI}{dt} \quad \dots (5.87)$$

As current is increasing with time the EMF induced in the two coils will oppose the increment in current and have their high

and low potential ends across the two coils as shown in figure-5.131. Writing KVL equation from A to B in figure-5.131 we have

$$\begin{aligned} V_A - \left(L_1 \frac{dI}{dt} + M \frac{dI}{dt} \right) - \left(L_2 \frac{dI}{dt} + M \frac{dI}{dt} \right) &= V_B \\ \Rightarrow V_A - V_B &= (L_1 + L_2) \frac{dI}{dt} + 2M \frac{dI}{dt} \\ \Rightarrow V_A - V_B &= (L_1 + L_2 + 2M) \frac{dI}{dt} \\ \Rightarrow V_A - V_B &= L_{eq} \frac{dI}{dt} \quad \dots (5.88) \end{aligned}$$

In above equation-(5.88) the equivalent self inductance of the two inductors in series is given as

$$L_{eq} = L_1 + L_2 + 2M \quad \dots (5.89)$$

Case II : When the flux of the two inductors are in opposite direction

Figure-5.132 shows two inductor coils in series with their flux in opposite directions as shown by the dots on the different ends of the two coils. If a current I flows through the coils the magnetic flux linked with each coil are given as

$$\phi_1 = L_1 I - MI$$

and

$$\phi_2 = L_2 I - MI$$

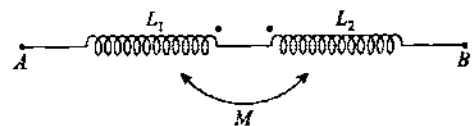


Figure 5.132

When the flowing current starts increasing at a rate dI/dt then the induced emf in the two coils are given as

$$e_1 = \frac{d\phi_1}{dt} = L_1 \frac{dI}{dt} - M \frac{dI}{dt} \quad \dots (5.90)$$

and

$$e_2 = \frac{d\phi_2}{dt} = L_2 \frac{dI}{dt} - M \frac{dI}{dt} \quad \dots (5.91)$$

As current is increasing with time the EMF induced in the two coils will oppose the increment in current and have their high and low potential ends across the two coils as shown in figure-5.131. Writing KVL equation from A to B in figure-5.131 we have

$$\begin{aligned} V_A - \left(L_1 \frac{dI}{dt} - M \frac{dI}{dt} \right) - \left(L_2 \frac{dI}{dt} - M \frac{dI}{dt} \right) &= V_B \\ \Rightarrow V_A - V_B &= (L_1 + L_2) \frac{dI}{dt} - 2M \frac{dI}{dt} \\ \Rightarrow V_A - V_B &= (L_1 + L_2 - 2M) \frac{dI}{dt} \\ \Rightarrow V_A - V_B &= L_{eq} \frac{dI}{dt} \quad \dots (5.92) \end{aligned}$$

In above equation-(5.88) the equivalent self inductance of the two inductors in series is given as

$$L_{eq} = L_1 + L_2 - 2M \quad \dots (5.93)$$

5.7.4 Inductors in Parallel Combination

There are several ways in which inductors can be connected in parallel combination depending upon the orientation of inductors but here we will discuss simple combination without considering mutual induction between them.

Figure-5.133 shows the two inductors having self inductances L_1 and L_2 connected in parallel combination. If a current i flows through the combination and it divides as i_1 and i_2 in the two inductors which are changing with time then for parallel combination of inductors we use

$$L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$$

$$\Rightarrow L_1 di_1 = L_2 di_2$$

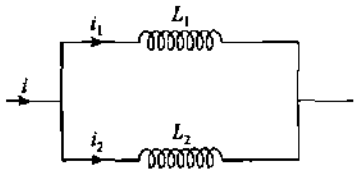


Figure 5.133

Integrating the above expression for any time limits when currents are i_1 and i_2 in the two inductors, we have

$$L_1 i_1 = L_2 i_2 \quad \dots (5.94)$$

$$\text{and } i_1 + i_2 = i \quad \dots (5.95)$$

From equation-(5.94) and (5.95) we can calculate the current distribution in two parallel inductors at any time which gives

$$i_1 = \frac{L_2 i}{L_1 + L_2} \quad \dots (5.96)$$

$$\text{and } i_2 = \frac{L_1 i}{L_1 + L_2} \quad \dots (5.97)$$

From equation-(5.95) we have

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} \quad \dots (5.98)$$

If L_{eq} is the equivalent self induction for the combination then for the terminals A and B across the inductors we have

$$V_{AB} = L_{eq} \frac{di}{dt} = L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt} \quad \dots (5.99)$$

From equation-(5.98) and (5.99) we have

$$\begin{aligned} \frac{V_{AB}}{L_{eq}} &= \frac{V_{AB}}{L_1} + \frac{V_{AB}}{L_2} \\ \Rightarrow \frac{1}{L_{eq}} &= \frac{1}{L_1} + \frac{1}{L_2} \quad \dots (5.100) \end{aligned}$$

Equation-(5.100) gives the formula for calculation of equivalent self inductance for the inductors connected in parallel which can be generalized to multiple inductors if mutual induction is neglected between these inductors.

Illustrative Example 5.46

A coil of 100 turns and 1cm radius is kept coaxially within a long solenoid of 8 turns per cm and 5cm radius. Find the mutual inductance.

Solution

The magnetic field B inside the solenoid is given as

$$B = \mu_0 n_s i_s$$

The magnetic flux linked with the coaxial coil placed inside the solenoid is given as

$$\phi_c = N_c B A_c = N_c (\mu_0 n_s i_s) A_c$$

Where A_c is the cross sectional area of coil.

The mutual inductance between solenoid and coil can be given as

$$M = \frac{\phi_c}{i_s} = \mu_0 n_s N_c A_c$$

Substituting the given values, we get

$$M = 4\pi \times 10^{-7} \times 800 \times 100 \times \pi \times 10^{-4} \text{ H}$$

$$\Rightarrow M = 3.15 \times 10^{-5} \text{ H}$$

Illustrative Example 5.47

Calculate the mutual inductance of a long straight wire and a rectangular frame with side a and b as shown in figure-5.134. The wire and the frame lie in the same plane, with side b being closest to the wire, separated by a distance l from it and oriented parallel to it.

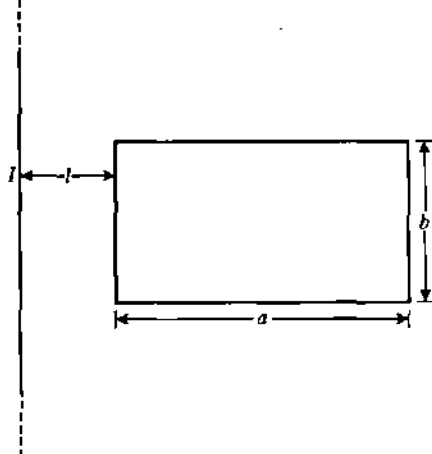


Figure 5.134

Solution

To calculate the mutual inductance between the wire and the frame we supply a current I through the wire and calculate the magnetic flux through the frame. For this we consider an elemental strip in the frame of width dr at a distance r from wire as shown in figure-5.135. The magnetic flux through the elemental strip is given as

$$d\phi = \left(\frac{\mu_0 I}{2\pi r} \right) b dr$$

$$\Rightarrow \phi = \int_l^{a+l} \left(\frac{\mu_0 I}{2\pi r} \right) b dr$$

$$\Rightarrow \phi = \frac{\mu_0 I b}{2\pi} \int_l^{a+l} \frac{dr}{r}$$

$$\Rightarrow \phi = \frac{\mu_0 I b}{2\pi} [\ln r]_l^{a+l}$$

$$\Rightarrow \phi = \frac{\mu_0 I b}{2\pi} \ln \left(1 + \frac{a}{l} \right)$$

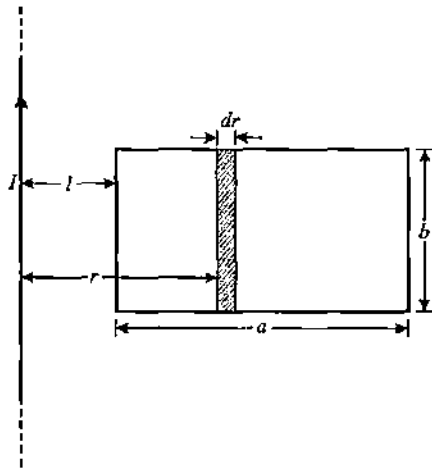


Figure 5.135

Thus mutual inductance between the two is given as

$$M = \frac{\phi}{I} = \frac{\mu_0 b}{2\pi} \ln \left(1 + \frac{a}{l} \right)$$

Illustrative Example 5.48

An inductor of inductance L is cut in three equal parts and two of these parts are interconnected (a) in series, (b) in parallel. Assuming the mutual inductance between the parts to be negligible, calculate the inductance of the combination in both the cases.

Solution

(a) When the given inductor is cut in identical parts then each will have an inductance $L/3$. When two such coils are connected in series then equivalent self inductance of the combination is given as

$$L_s = L_1 + L_2$$

$$\Rightarrow L_s = \frac{L}{3} + \frac{L}{3} = \frac{2L}{3}$$

(b) In parallel combination we use

$$\frac{1}{L_p} = \frac{1}{L_1} + \frac{1}{L_2}$$

$$\Rightarrow \frac{1}{L_p} = \frac{1}{L/3} + \frac{1}{L/3}$$

$$\Rightarrow L_p = \frac{L}{6}$$

Illustrative Example 5.49

Figure-5.136 shows two coaxial coils of radii r and R ($R \gg r$) kept at large separation x ($x \gg R$). Calculate the magnetic flux passing through coil A due to a current I in coil B .

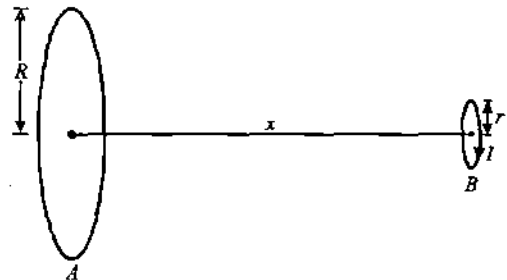


Figure 5.136

Solution

If M is the coefficient of mutual induction between the two coils then magnetic flux through coil A due to current in coil B is given as

$$\phi_A = M I_B \quad \dots (5.101)$$

First we will calculate the mutual induction coefficient between the two coils and for this we find the flux through coil B if a current I flows in coil A as shown in figure-5.136.

The magnetic induction at the location of coil B due to current in coil A is given as

$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

The magnetic flux through coil B due to above magnetic induction is given as

$$\phi_B = B \cdot \pi r^2$$

$$\Rightarrow \phi_B = \frac{\mu_0 \pi r^2 R^2}{2(x^2 + R^2)^{3/2}} \cdot I$$

As coefficient of mutual inductance is a bidirectional coefficient, we can calculate it as

$$M = \frac{\phi_B}{I} = \frac{\mu_0 \pi r^2 R^2}{2(x^2 + R^2)^{3/2}}$$

Above is the mutual inductance coefficient for the pair of these coil which can be used in calculation of flux from either coil due to current in other coil so from reciprocity property, we have

$$\phi_A = MI_B = \frac{\mu_0 \pi r^2 R^2 I}{2(x^2 + R^2)^{3/2}}$$

Illustrative Example 5.50

In the electric circuit shown in figure-5.137 an EMF E , a resistance R and coils of inductances L_1 and L_2 are used. The internal resistance of the source and the coil resistances are negligible. Find the steady state currents in the coils long time after the switch Sw was shorted.

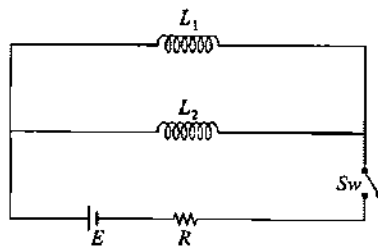


Figure 5.137

Solution

In steady state inductors behave like short circuits so the current through battery is given as

$$I = I_1 + I_2 = \frac{E}{R}$$

As already discussed in article-5.7.4 that current in inductors connected in parallel combination is divided in inverse ratio of their inductances so we have currents through inductors given as

$$I_1 = \frac{EL_2}{R(L_1 + L_2)}$$

and

$$I_2 = \frac{EL_1}{R(L_1 + L_2)}$$

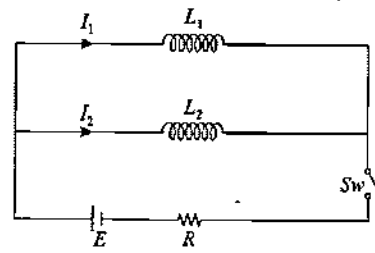


Figure 5.138

Illustrative Example 5.51

Two coaxial circular loops of radii 0.5m and 0.05m are separated by a distance 0.5m and carry currents 2A and 1A respectively. Calculate the mutual inductance for this pair of coils and the force between the two coils.

Solution

Magnetic field at the location of smaller loop due to larger loop is given as

$$B = \frac{\mu_0 i_1 r_1^2}{2(x^2 + r_1^2)^{3/2}}$$

Flux through the smaller loop is given as

$$\phi_2 = B \times (\pi r_2^2)$$

$$\Rightarrow \phi_2 = \frac{\mu_0 i_1 r_1^2}{2(x^2 + r_1^2)^{3/2}} \times \pi r_2^2$$

If M is the coefficient of mutual induction between the pair of coils, we use

$$\phi_2 = Mi_1$$

$$\Rightarrow M = \frac{\phi_2}{i_1} = \frac{\mu_0 \pi r_1^2 r_2^2}{2(x^2 + r_1^2)^{3/2}}$$

$$\Rightarrow M = \frac{(4\pi \times 10^{-7}) \pi \times 0.25 \times (25 \times 10^{-4})}{2(0.25 + 0.25)^{3/2}}$$

$$\Rightarrow M = 3.5 \times 10^{-9} \text{ H}$$

If m is the magnetic moment of the coil then force on this coil is given as

$$F = m \left(\frac{dB}{dx} \right)$$

$$\Rightarrow F = (\pi r^2 i_2) \frac{dB}{dx}$$

$$\text{As } B = \frac{\mu_0 i_1 r_1^2}{2(x^2 + r_1^2)^{3/2}}$$

$$\Rightarrow \frac{dB}{dx} = -\frac{\mu_0 i_1^2}{2} \times \frac{3}{2} (x^2 + r_1^2)^{-5/2} \cdot 2x$$

$$\Rightarrow \frac{dB}{dx} = -\frac{3}{2} \frac{\mu_0 i_1^2 x}{(x^2 + r_1^2)^{5/2}}$$

$$\Rightarrow F = -\frac{3 \pi \mu_0 i_1 i_2 r_1^2 r_2^2 x}{2 (x^2 + r_1^2)^{5/2}}$$

$$\Rightarrow F = \frac{-3 i_1 i_2 x}{(x^2 + r_1^2)} \times M_{12}$$

$$\Rightarrow F = \frac{-3 \times 2 \times 1 \times 0.5}{(0.25 + 0.25)} \times 3.5 \times 10^{-9}$$

$$\Rightarrow F = 2.1 \times 10^{-8} \text{ N}$$

Illustrative Example 5.52

There are two stationary loops with mutual inductance M . The current in one of the loops starts varying as $I_1 = \alpha t$, where α is a constant and t is any time instant starting from $t = 0$. Find the time dependence $I_2(t)$ of the current in the other loop which has self inductance L and resistance R .

Solution

The induced EMF in the second loop when the current changes in the first loop is given as

$$e_2 = M \frac{dI_1}{dt}$$

$$\Rightarrow e_2 = M\alpha$$

For the second loop we use KVL equation which gives

$$I_2 R + L \frac{dI_2}{dt} = M\alpha$$

$$\Rightarrow \int_0^{I_2} \frac{dI_2}{\left(I_2 - \frac{M\alpha}{R}\right)} = -\frac{R}{L} \int_0^t dt$$

$$\Rightarrow I_2 = \frac{\alpha M}{R} (1 - e^{-Rt/L})$$

5.8 LC Oscillations

In article-5.6 we studied that on connecting an active inductor across a resistor current in inductor decays and its magnetic energy decays in resistor as heat. Figure-5.139 shows an active inductor connected across a capacitor via a switch. In this situation when switch is closed (Stage-I) then current flows in the loop with initial current I_0 and the current charges the capacitor due to which electrical energy of capacitor increases

which comes from the magnetic energy of inductor thus current in loop decreases with time as capacitor energy increases and inductor energy decreases as shown in stage-II.

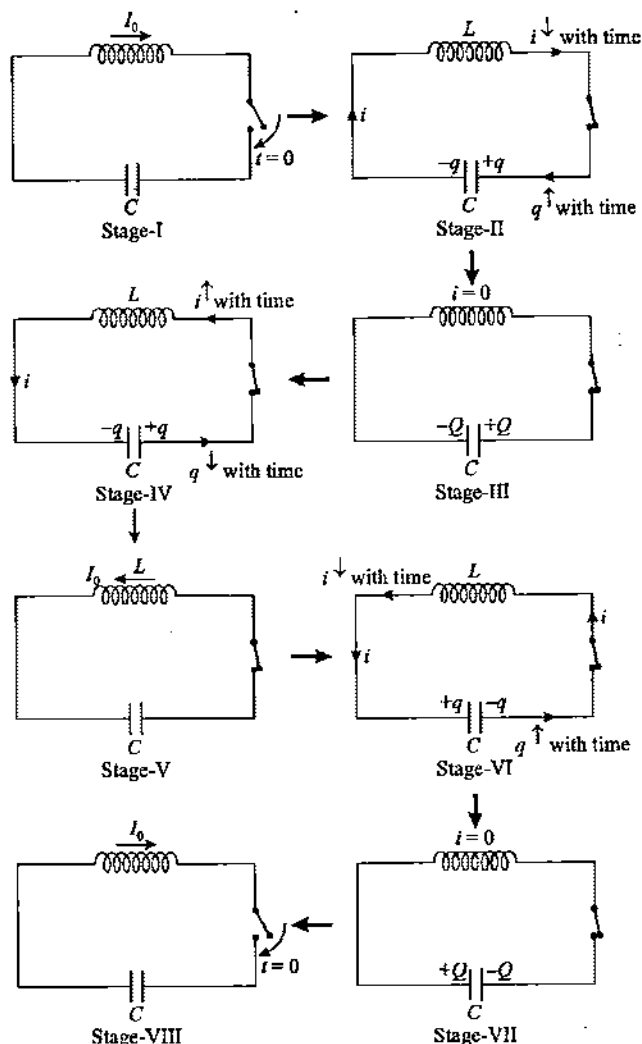


Figure 5.139

After sometime capacitor charge increases to maximum when current in loop becomes zero and at this instant magnetic energy in inductor becomes zero as shown in stage-III. We can state that whole initial magnetic energy of inductor is transformed into electrical energy of capacitor that's why its charge is maximum. After this instant capacitor starts discharging through the inductor for which a current flows in loop in anticlockwise direction and current increases and charge on capacitor starts decreasing and again electrical energy of capacitor starts transforming into magnetic energy of inductor as shown in stage-IV. After sometime stage-V comes when whole energy is transformed as magnetic and current in inductor becomes I_0 again and charge on capacitor becomes zero. After this further stages-VI, VII and VIII are self explanatory and finally in stage-VIII loop attains the initial state as shown in stage-I. With this cycle of oscillation of charge in the loop in above

explained stages is called '*LC Oscillations*'. Under ideal conditions resistance of the circuit is considered zero with inductor and capacitor are considered ideal, no energy dissipation is considered so charge in an inductor and capacitor connected in parallel oscillates indefinitely.

5.8.1 Oscillation Period of LC Oscillations

As discussed in previous article when an inductor and a capacitor with some energy in either of these connected in parallel, charge and energy starts oscillating in the loop. At any instant if q is the charge on capacitor and i is the current in circuit then total energy of circuit can be given as

$$E = \frac{q^2}{2C} + \frac{1}{2}Li^2 \quad \dots (5.102)$$

During oscillations energy periodically transforms from electrical to magnetic and then back to electric and vice versa and under ideal conditions of zero resistance no energy loss take place so the above energy of circuit E will remain constant in time. Differentiating the above equation-(5.102) gives

$$\frac{dE}{dt} = \frac{1}{2C} (2q) \frac{dq}{dt} + \frac{1}{2}L(2i) \frac{di}{dt} = 0$$

$$\Rightarrow \frac{q}{C} \frac{dq}{dt} + Li \frac{di}{dt} = 0$$

$$\Rightarrow \frac{d^2q}{dt^2} + \left(\frac{1}{LC}\right)q = 0 \quad \dots (5.103)$$

Above equation-(5.103) is similar to the basic differential equation of SHM which is for displacement and it is written as

$$\frac{d^2x}{dt^2} + \omega^2x = 0 \quad \dots (5.104)$$

This shows that in *LC* oscillations charge executes SHM of which angular frequency can be given by comparing equations-(5.103) and (5.104) which gives

$$\omega = \frac{1}{\sqrt{LC}} \quad \dots (5.105)$$

Thus time period of oscillating charge in *LC* circuit is given as

$$T = \frac{2\pi}{\omega}$$

$$\Rightarrow T = 2\pi\sqrt{LC} \quad \dots (5.106)$$

From equation-(5.103) we've seen that charge on capacitor is executing SHM so by the general solution of SHM from equation-(5.103) the charge on capacitor plates as a function of time can be given as

$$q = q_0 \sin(\omega t + \alpha) \quad \dots (5.107)$$

In above equation q_0 is the charge amplitude which is the maximum charge on capacitor plates during oscillations, ω is

the angular frequency of oscillations which is given by equation-(5.105) for the case when one capacitor and one inductor are connected in parallel. In other cases it can be obtained by differentiating energy equation for the given circuit and α is the initial phase of oscillating charge which depends upon the initial condition of the circuit when switch is closed. Initial phase can also be calculated by the uniform circular motion analysis like the way we did it in different illustrations while analyzing SHM. This will be further explained with the help of illustrations on *LC* oscillations.

During oscillations the current in circuit can be given as

$$i = \frac{dq}{dt} = q_0 \omega \cos(\omega t + \alpha) \quad \dots (5.108)$$

Above equation is similar to the velocity of a particle executing SHM thus current in circuit in terms of instantaneous charge on capacitor can be given as

$$i = \omega \sqrt{q_0^2 - q^2} \quad \dots (5.109)$$

As already discussed that circuit current becomes maximum when capacitor charge is zero thus maximum current in circuit is related to the maximum charge on capacitor as

$$i_{\max} = \omega q_0 \quad \dots (5.110)$$

$$\Rightarrow i_{\max} = \frac{q_0}{\sqrt{LC}} \quad \dots (5.111)$$

Above expression of maximum current can also be obtained by conservation of energy. If maximum charge on capacitor is known then at the instant charge on capacitor is maximum current in circuit is zero as whole energy of circuit is electrical energy in capacitor and at the instant of maximum current whole energy is transformed into magnetic energy thus by energy conservation we have

$$\frac{q_0^2}{2C} = \frac{1}{2}Li_{\max}^2$$

$$\Rightarrow i_{\max} = \frac{q_0}{\sqrt{LC}}$$

Illustrative Example 5.53

A capacitor of capacitance $25\mu\text{F}$ is charged to 300V . It is then connected across a 10mH inductor. The resistance in the circuit is negligible.

(a) Find the frequency of oscillation of the circuit.

(b) Find the potential difference across capacitor and magnitude of circuit current 1.2ms after the inductor and capacitor are connected

- (c) Find the magnetic energy and electric energy at $t = 0$ and $t = 1.2 \text{ ms}$.

Solution

- (a) The frequency of oscillation of the LC circuit is given as

$$f = \frac{1}{2\pi\sqrt{LC}}$$

Substituting the given values we have

$$f = \frac{1}{2\pi\sqrt{(10 \times 10^{-3})(25 \times 10^{-6})}}$$

$$\Rightarrow f = 318.3 \text{ Hz}$$

- (b) Charge across the capacitor at time t will be given as

$$q = q_0 \cos \omega t$$

$$\text{and } i = -q_0 \omega \sin \omega t$$

Given that initial maximum charge on capacitor is

$$q_0 = CV_0 = (25 \times 10^{-6})(300)$$

$$\Rightarrow q_0 = 7.5 \times 10^{-3} \text{ C}$$

The charge in the capacitor after time $t = 1.2 \times 10^{-3} \text{ s}$ is given as

$$q = (7.5 \times 10^{-3}) \cos(2\pi \times 318.3)(1.2 \times 10^{-3}) \text{ C}$$

$$\Rightarrow q = -5.53 \times 10^{-3} \text{ C}$$

Potential difference across capacitor is given as

$$V = \frac{|q|}{C} = \frac{5.53 \times 10^{-3}}{25 \times 10^{-6}} = 221.2 \text{ V}$$

The magnitude of current in the circuit at $t = 1.2 \times 10^{-3} \text{ s}$ is given as

$$|i| = q_0 \omega \sin \omega t$$

$$\Rightarrow |i| = (7.5 \times 10^{-3})(2\pi)(318.3) \sin[(2\pi)(318.3)(1.2 \times 10^{-3})] \text{ A}$$

$$\Rightarrow |i| = 10.13 \text{ A}$$

- (c) At $t = 0$ current in the circuit is zero so magnetic energy in inductor is also zero and charge in the capacitor is maximum so total energy is magnetic energy which is given as

$$U_C = \frac{1}{2} \frac{q_0^2}{C}$$

$$\Rightarrow U_C = \frac{1}{2} \times \frac{(7.5 \times 10^{-3})^2}{(25 \times 10^{-6})} = 1.125 \text{ J}$$

Total energy in circuit at any instant which is stored in inductor and capacitor is also given as

$$E = U_L + U_C = 1.125 \text{ J}$$

At $t = 1.2 \text{ ms}$, we use

$$U_L = \frac{1}{2} Li^2 = \frac{1}{2} (10 \times 10^{-3})(10.13)^2$$

$$\Rightarrow U_L = 0.513 \text{ J}$$

$$\text{and } U_C = E - U_L = 1.125 - 0.513$$

$$\Rightarrow U_C = 0.612 \text{ J}$$

Otherwise U_C can be calculated as

$$U_C = \frac{1}{2} \frac{q^2}{C} = \frac{1}{2} \times \frac{(5.53 \times 10^{-3})^2}{(25 \times 10^{-6})}$$

$$\Rightarrow U_C = 0.612 \text{ J}$$

Illustrative Example 5.54

In circuit shown in figure-5.140 charge on capacitor is $Q = 100 \mu\text{C}$. If switch is closed at $t = 0$ find current in circuit when charge on capacitor reduces to $50 \mu\text{C}$. Also find the maximum current in circuit.

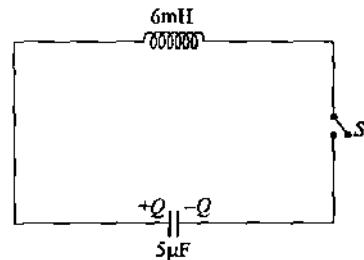


Figure 5.140

Solution

At $t = 0$ when switch is closed capacitor starts discharging through the inductor and current in circuit increases. When charge on capacitor is q and current in circuit is i by conservation of energy we have

$$\frac{q^2}{2C} + \frac{1}{2} Li^2 = \frac{Q^2}{2C}$$

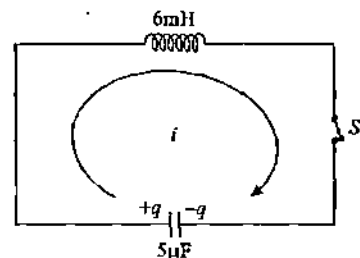


Figure 5.141

Current in circuit in terms of charge on capacitor is given as

$$i = \frac{1}{\sqrt{LC}} \sqrt{Q^2 - q^2} = \frac{\sqrt{(100\mu)^2 - (50\mu)^2}}{\sqrt{3 \times 10^{-8}}}$$

$$\Rightarrow i = \frac{10^{-6}}{\sqrt{3 \times 10^{-8}}} \sqrt{7500}$$

$$\Rightarrow i = 0.5 \text{ A}$$

Maximum energy stored in inductor is given as

$$\frac{1}{2} L I_{\max}^2 = \frac{Q^2}{2L}$$

$$\Rightarrow I_{\max} = \frac{Q}{\sqrt{LC}} = \frac{100 \times 10^{-6}}{\sqrt{3 \times 10^{-8}}} = \frac{1}{\sqrt{3}} \text{ A}$$

Illustrative Example 5.55

In an L - C circuit $L = 3.3 \text{ H}$ and $C = 840 \text{ pF}$. At $t = 0$ charge on the capacitor is $105 \mu\text{C}$ and maximum. Compute the following quantities at $t = 2.0 \text{ ms}$

- The energy stored in the capacitor
- The total energy in the circuit
- The energy stored in the inductor

Solution

The angular frequency of L - C oscillations is,

$$\omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{3.3 \times 840 \times 10^{-12}}}$$

$$\Rightarrow \omega = 1.9 \times 10^4 \text{ rad/s}$$

Charge stored in the capacitor at time t , is given as

$$q = q_0 \cos \omega t$$

$$(a) \text{ At } t = 2 \times 10^{-3} \text{ s}$$

$$q = (105 \times 10^{-6}) \cos [1.9 \times 10^4] [2 \times 10^{-3}]$$

$$\Rightarrow q = 100.3 \times 10^{-6} \text{ C}$$

Energy stored in the capacitor at this instant is given as

$$U_C = \frac{1}{2} \frac{q^2}{C}$$

$$\Rightarrow U_C = \frac{(100.3 \times 10^{-6})^2}{2 \times 840 \times 10^{-12}}$$

$$\Rightarrow U_C = 6.0 \text{ J}$$

- (b) Total energy in the circuit is the energy in capacitor at maximum charge which is given as

$$U = \frac{1}{2} \frac{q_0^2}{C}$$

$$\Rightarrow U = \frac{(105 \times 10^{-6})^2}{2 \times 840 \times 10^{-12}}$$

$$\Rightarrow U = 6.56 \text{ J}$$

- (c) Energy stored in inductor in the given time is given by conservation of energy as

$$U_L = (6.56 - 6.0) \text{ J}$$

$$\Rightarrow U_L = 0.56 \text{ J}$$

Illustrative Example 5.56

In an oscillating LC circuit in which $C = 4.00 \mu\text{F}$, the maximum potential difference across the capacitor during the oscillations is 1.50 V and the maximum current through the inductor is 50.0 mA .

- What is the inductance L ?
- What is the frequency of the oscillations?
- How much time does the charge on the capacitor take to rise from zero to its maximum value?

Solution

- (a) By conservation of energy we have

$$\frac{1}{2} L i_0^2 = \frac{1}{2} C V_0^2$$

$$\Rightarrow L = \frac{C V_0^2}{i_0^2}$$

$$\Rightarrow L = \frac{(4 \times 10^{-6})(1.5)^2}{(50 \times 10^{-3})^2}$$

$$\Rightarrow L = 3.6 \times 10^{-3} \text{ H}$$

- (b) Oscillation frequency of the LC circuit is given as

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$\Rightarrow f = \frac{1}{2\pi\sqrt{(3.6 \times 10^{-3})(4 \times 10^{-6})}}$$

$$\Rightarrow f = 0.133 \times 10^4 \text{ Hz}$$

$$\Rightarrow f = 1.33 \text{ kHz}$$

(c) During oscillations the time taken for charge to rise from zero to maximum is one fourth of the oscillation period which is given as

$$t = \frac{T}{4} = \frac{1}{4f}$$

$$\Rightarrow t = \frac{1}{4 \times 1.33 \times 10^3} \text{ s}$$

$$\Rightarrow t = 0.188 \times 10^{-3} \text{ s}$$

$$\Rightarrow t = 0.188 \text{ ms}$$

Illustrative Example 5.57

In the circuit shown in the figure-5.142, $E = 50.0 \text{ V}$, $R = 250 \Omega$ and $C = 0.500 \mu\text{F}$. The switch S is closed for a long time, and no voltage is measured across the capacitor. After the switch is opened, the voltage across the capacitor reaches a maximum value of 150 V . What is the inductance L ?

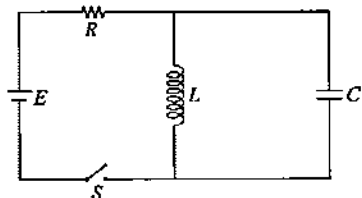


Figure 5.142

Solution

In steady state when switch was closed,

$$i_0 = \frac{E}{R} = 0.2 \text{ A}$$

After switch is opened, it becomes LC circuit in which peak current is 0.2 A . By energy conservation we have

$$\frac{1}{2} L i_0^2 = \frac{1}{2} C V_0^2$$

$$\Rightarrow L = \frac{V_0^2}{i_0^2} \cdot C$$

$$\Rightarrow L = \frac{(150)^2}{(0.2)^2} \times 0.5 \times 10^{-6}$$

$$\Rightarrow L = 0.28 \text{ H}$$

Illustrative Example 5.58

An inductor of inductance 2.0 mH is connected across a charged capacitor of capacitance $5.0 \mu\text{F}$, and resulting $L-C$ circuit is set oscillating at its natural frequency. Let Q denote the

instantaneous charge on the capacitor, and I the current in the circuit. It is found that the maximum value of Q is $200 \mu\text{C}$.

(a) When $Q = 100 \mu\text{C}$, what is the value of $\left| \frac{dI}{dt} \right|$?

(b) When $Q = 200 \mu\text{C}$, what is the value of I ?

(c) Find the maximum value of I .

(d) When I is equal to one half its maximum value, what is the value of $|Q|$?

Solution

Charge stored in the capacitor oscillates simple harmonically and charge as a function of time is given as

$$q = q_0 \sin(\omega t \pm \phi)$$

Here we use $q_0 = 200 \mu\text{C} = 2 \times 10^{-4} \text{ C}$

$$\text{and } \omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{(2 \times 10^{-3})(5.0 \times 10^{-6})}} = 10^4 \text{ s}^{-1}$$

If at $t = 0$, $q = q_0$

$$q = q_0 \cos \omega t \quad \dots (5.112)$$

$$\Rightarrow I = \frac{dq}{dt} = -q_0 \omega \sin \omega t \quad \dots (5.113)$$

$$\text{and } \frac{dI}{dt} = -q_0 \omega^2 \cos \omega t \quad \dots (5.114)$$

(a) At $q = 100 \mu\text{C}$ we have

$$\cos \omega t = \frac{1}{2}$$

$$\Rightarrow \omega t = \frac{\pi}{3}$$

At $\cos(\omega t) = \frac{1}{2}$ and from equation-(5.114) we have

$$\left| \frac{dI}{dt} \right| = (2.0 \times 10^{-4} \text{ C}) (10^4 \text{ s}^{-1})^2 \left(\frac{1}{2} \right)$$

$$\Rightarrow \left| \frac{dI}{dt} \right| = 10^4 \text{ A/s}$$

(b) At $q = 200 \mu\text{C}$ we have

$$\cos \omega t = 1$$

$$\Rightarrow \omega t = 0, 2\pi, \dots$$

At this time instant we have

$$I = -q_0 \omega \sin \omega t$$

$$\Rightarrow I = -0$$

(c) Current is given by equation-(5.113) as

$$I = -Q_0 \omega \sin \omega t$$

Maximum value of current is $Q_0 \omega$

$$\Rightarrow I_{\max} = Q_0 \omega$$

$$\Rightarrow I_{\max} = (2.0 \times 10^{-4})(10^4)$$

$$\Rightarrow I_{\max} = 2.0 \text{ A}$$

(d) From energy conservation we have

$$\frac{1}{2} L I_{\max}^2 = \frac{1}{2} L I^2 + \frac{1}{2} \frac{Q^2}{C}$$

$$\Rightarrow Q = \sqrt{LC(I_{\max}^2 - I^2)}$$

$$\text{Using } I = \frac{I_{\max}}{2} = 1.0 \text{ A}$$

$$\Rightarrow Q = \sqrt{(2.0 \times 10^{-3})(5.0 \times 10^{-6})(2^2 - 1^2)}$$

$$\Rightarrow Q = \sqrt{3} \times 10^{-4} \text{ C}$$

$$\Rightarrow Q = 1.732 \times 10^{-4} \text{ C}$$

Illustrative Example 5.59

In figure-5.143 if S is closed at $t = 0$, find charge on capacitor as a function of time. The inductor is active with initial current I_0 in it.

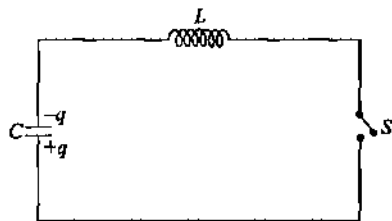


Figure 5.143

Solution

If q_0 is the maximum charge on capacitor then by energy conservation we have

$$\frac{1}{2} L I_0^2 = \frac{q_0^2}{2C}$$

$$\Rightarrow q_0 = \sqrt{LC} \cdot I_0$$

Charge on capacitor as a function of time is given as

$$q = q_0 \sin(\omega t + \alpha)$$

Where $\omega = \frac{1}{\sqrt{LC}}$ and at $t = 0$, $q = 0$ we get $\alpha = 0$.

Thus charge on capacitor is given as

$$q = \sqrt{LC} I_0 \sin\left(\frac{t}{\sqrt{LC}}\right)$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Self & Mutual Inductance

Module Number - 22 to 36

Practice Exercise 5.4

(i) Two thin concentric wires shaped as circle with radii a and b lie in the same plane. Allowing for $a \ll b$, find

(a) Their mutual inductance

(b) The magnetic flux through the surface enclosed by the outside wire, when the inside wire carries a current I .

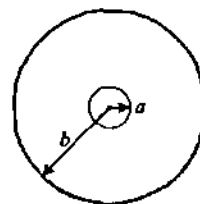


Figure 5.144

$$[(a) \frac{\mu_0 \pi a^2}{2b}; (b) \frac{\mu_0 \pi a^2 I}{2b}]$$

(ii) (a) Calculate the mutual inductance between two coils when a current of 4A in first coil changes to 12A in 0.5s and induces an EMF of 50mV in the other coil.

(b) Also calculate the induced EMF in the second coil if current in the first changes from 3A to 9A in 0.02s.

$$[(a) 3.125 \times 10^{-3} \text{ H}; (b) 0.9375 \text{ V}]$$

(iii) Find the mutual inductance of two thin coaxial loops of the same radius a if their centres are separated by a distance l , with $l \gg a$.

$$[\frac{\mu_0 \pi a^4}{2l^3}]$$

(iv) The equivalent inductance of two inductors is 2.4H when connected in parallel and 10H when connected in series. What is the value of inductance of the individual inductors. Neglect mutual inductance between the two inductors.

[4H , 6H]

(v) A circular coil P of 100 turns and radius 2cm is placed coaxially at the centre of another circular coil Q of 1000 turns and radius 20cm . Calculate

- The mutual inductance of the coils,
- The induced EMF in coil P when the current in the coil Q decreases from 5A to 3A in 0.04s
- The rate of change of flux through the coil P at this instant
- The charge passing through coil P if its resistance is 8Ω .

[(a) $3.94 \times 10^{-4}\text{H}$ (b) $19.72 \times 10^{-3}\text{V}$ (c) $19.72 \times 10^{-3}\text{Wb/s}$ (d) $9.86 \times 10^{-5}\text{C}$]

(vi) A long solenoid of length 1m , cross-sectional area 10cm^2 , having 1000 turns has wound about its centre a small coil of 20 turns. Compute the mutual inductance of the pair of solenoid and the coil. What is the induced EMF in the coil when the current in the solenoid changes at the rate of 10A/s ?

[$251\text{ }\mu\text{V}$]

(vii) An inductor with an inductance of 2.5H and a resistance of 8Ω is connected to the terminals of a battery with an EMF 6V and negligible internal resistance. Find

- The initial rate of increase of current in the circuit
- The rate of increase of current at the instant when the current is 0.50A
- The current 0.25s after the circuit is closed
- The final steady state current

[(a) 2.40A/s (b) 0.80A/s (c) 0.413A (d) 0.750A]

(viii) In the given circuit shown in figure-5.145, find the current through the 5mH inductor in steady state.

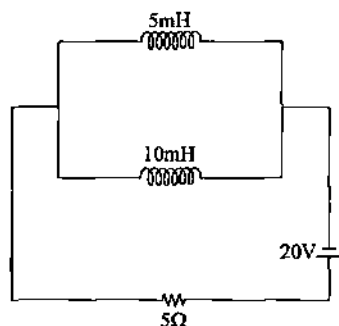


Figure 5.145

[$\frac{8}{3}\text{A}$]

(ix) In the LC circuit shown in figure-5.146, $C = 1\mu\text{F}$. With capacitor charged to 100V , switch S is suddenly closed at time $t = 0$. The circuit then oscillates at a frequency of 10^3Hz .

- Calculate ω and T
- Express q as a function of time
- Calculate L
- Calculate the average current during the first quarter-cycle.

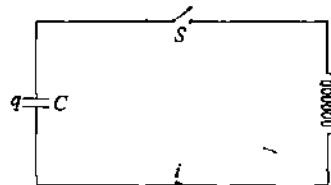


Figure 5.146

[(a) $6.28 \times 10^3\text{rad/s}$, 10^{-3}s (b) $10^{-4} \cos(6.28 \times 10^3 t)$ (c) 0.0253H (d) 0.4A]

(x) Two capacitors of capacitances $2C$ and C are connected in series with an inductor of inductance L . Initially capacitors have charge such that $V_B - V_A = 4V_0$ and $V_C - V_D = V_0$. Initial current in the circuit is zero. Find

- Maximum current that will flow in the circuit
- Potential difference across each capacitor at that instant
- Equation of current flowing towards left in the inductor

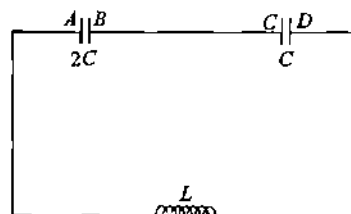


Figure 5.147

[(a) $2CV_0\omega$ (b) $3V_0, 3V_0$ (c) $i = q_0 \omega \sin \omega t$. Here, $q_0 = 2CV_0$ and $\omega = \sqrt{\frac{3}{2LC}}$]

(xi) A circuit containing capacitors C_1 and C_2 , shown in the figure-5.148 is in the steady state with key K_1 closed and K_2 opened. At the instant $t = 0$, K_1 is opened and K_2 is closed.

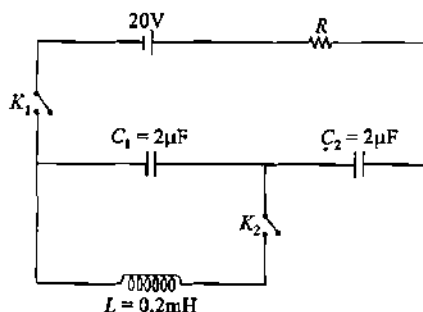


Figure 5.148

- (a) Find the angular frequency of oscillations of LC circuit.
 (b) Determine the first instant t , when energy in the inductor becomes one third of that in the capacitor.
 (c) Calculate the charge on the plates of the capacitor at that instant.

[(a) $5 \times 10^4 \text{ rad/s}$ (b) $1.05 \times 10^{-5} \text{ s}$ (c) $10\sqrt{3} \mu\text{C}$]

- (xii) In circuit shown in figure-5.149 if switches S_1 and S_2 are closed at $t = 0$, find the oscillation frequency of charge in this circuit. Neglect mutual induction between inductors.

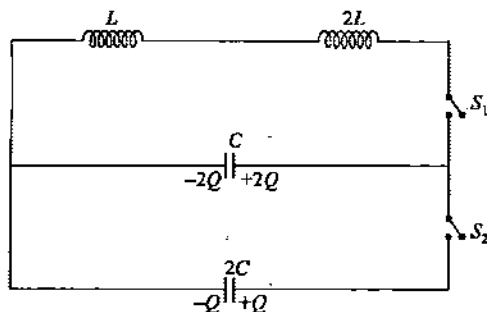


Figure 5.149

$$\left[\frac{1}{6\pi\sqrt{LC}} \right]$$

- (xiii) Figure-5.150 shows LC circuit with initial charge on capacitor $200 \mu\text{C}$. If at $t = 0$ switch is closed, find the first instant when energy stored in inductor becomes one third that of capacitor.

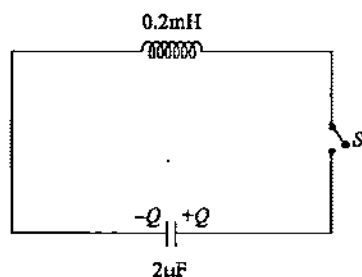


Figure 5.150

[10.5 μs]

5.9 Magnetic Properties of Matter

In previous chapter and articles we've discussed about the origin of magnetic field by currents whether in a coil or by motion of charge particles or orbital motion of electrons in atoms. It is observed when a material is placed in an external magnetic field then the magnetic field of atomic electrons in the atoms of material interact with the external field. The magnetic properties

of different materials depend upon how the atomic electrons interact with external magnetic field. The best way to discuss about the magnetic properties of materials is to analyze the behaviour and responses of different materials when placed in an external magnetic field. Based on the responses of materials in magnetic field, all materials are classified in three broad categories :

- (i) Diamagnetic Materials
- (ii) Paramagnetic Materials
- (iii) Ferromagnetic Materials

To understand the behaviour and classification of materials as discussed above first we need to understand the magnetic moment of atoms in different materials due to the orbital motion of electrons.

5.9.1 Magnetic Moment of Hydrogen Atom

In a hydrogen atom a single electron revolves around a proton and by Bohr's postulate of quantisation of angular momentum the angular momentum of electron in its ground state is given as

$$L = \frac{h}{2\pi} \quad \dots (5.114)$$

The magnetic moment μ of the atom due to electron motion can be directly given by equation-(4.168) as

$$\begin{aligned} \mu &= \frac{e}{2m} \times L \\ \Rightarrow \mu &= \frac{e}{2m} \times \frac{h}{2\pi} \\ \Rightarrow \mu &= \frac{eh}{4\pi m} \quad \dots (5.115) \end{aligned}$$

The value given in equation-(5.115) is termed as 'Bohr Magneton' and it is used as a basic unit for measurement of magnetic dipole moment due to electron spin or orbital motion in an atom.

5.9.2 Natural Magnetic Moment in Different Materials

In atoms of different materials there are many electrons in different orbits in their atoms. These electrons have orbital magnetic moment due to their orbital motion and in addition to that these electrons also have magnetic moment due to their spin angular momentum which may get added or subtracted in their resulting magnetic moment depending upon their direction.

Overall there can be two possibilities in a material. First is that magnetic moment due to an individual atom in a material is zero thus overall magnetic moment of the piece of material is also zero. Figure-5.151(a) shows magnified view of a small block of

material in which each dot represent an atom which has zero dipole moment. Another possibility is that magnetic moment due to individual atoms in the material is non zero but all such dipole moments are randomly scattered in the volume of material so overall magnetic moment of the piece of material is zero. Figure-5.151(b) shows magnified view of a small block of material in which each of its atom has non zero magnetic dipole moment which is represented by a small arrow but all these magnetic moments are distributed in random directions thus overall sum of all the dipole moments can be taken zero.

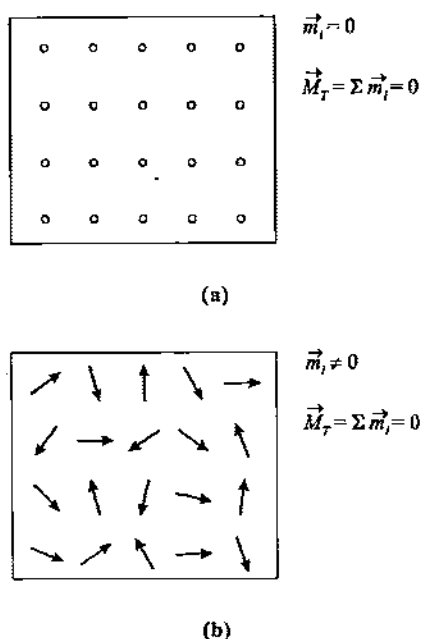


Figure 5.151

5.9.3 Diamagnetism

The property in a material because of which it has a tendency to repel external magnetic field is called '*Diamagnetism*'. To understand this see the figure-5.152 which shows a coil mounted on a rigid support and a bar magnet is moving toward the coil along its axis. We know that due to motion of magnet the magnetic flux through the coil changes and a current is induced in the coil in such a direction to oppose the motion of magnet. We can also say that a magnetic moment is induced in coil in opposition to the magnet.

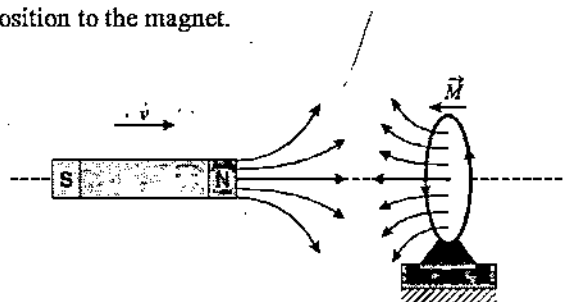


Figure 5.152

In every material there are atoms in which orbiting electrons behave like equivalent currents. Whenever an external magnetic field is applied on a material, all the orbits in atoms of the material have more or less tendency to oppose the external magnetic field and an opposite magnetic moment is induced in the coil which opposes the external field. Thus every material has some diamagnetic character in it.

5.9.4 Response of Diamagnetic Materials to External Magnetic Fields

Those materials in which individual magnetic moment of each atom or molecule is zero are called '*Diamagnetic Materials*' because when these materials are placed in external magnetic field, magnetic moment is induced in all these atoms or molecules of material due to which cumulatively all the atoms or molecules of material oppose the external magnetic field and repel it. Figure-5.153(a) shows a block of diamagnetic material of which initial magnetic moment is zero as discussed in article-5.9.2 and when a bar magnet is brought close to this block, due to its magnetic field, magnetic moments are induced in all the atoms in the block in opposite direction and because of all these dipoles block produces an induced magnetic field B_i in direction opposite to that of bar magnet as shown in figure-5.153(b) and this repels the bar magnet.

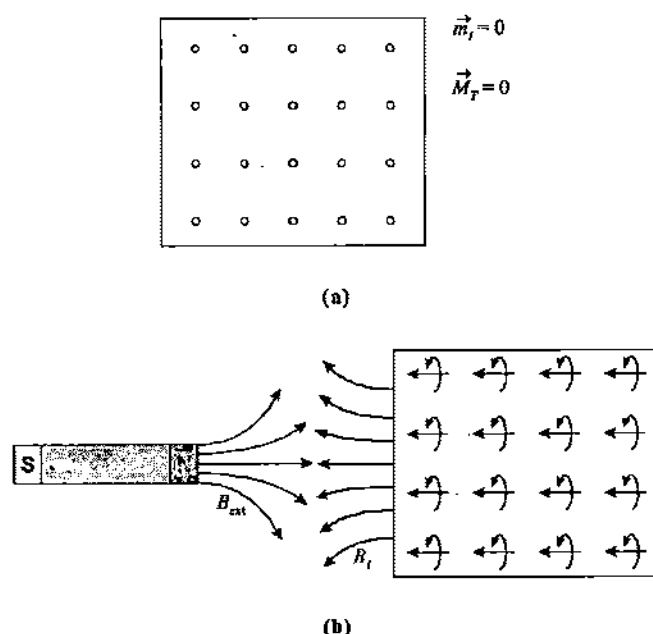


Figure 5.153

Always remember that diamagnetism is a weak phenomenon but in diamagnetic materials it can be seen in different ways like figure-5.154 shows a diamagnetic block placed between two opposite pole pieces repel the magnetic field and due to this magnetic lines bulge out as shown and almost every material has diamagnetic character upto some extent.

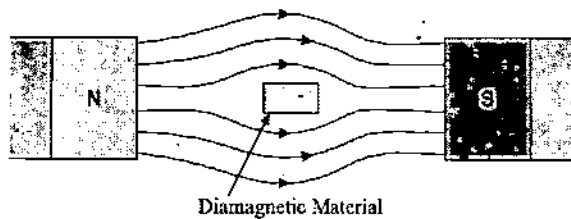
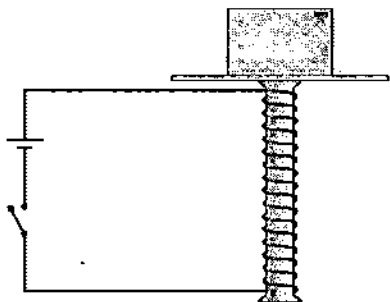


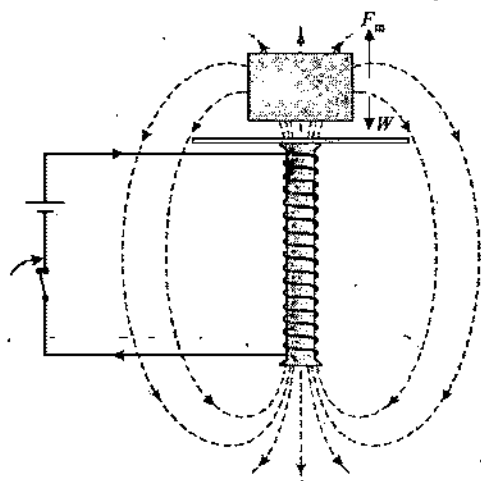
Figure 5.154

5.9.5 Magnetic Levitation in Diamagnetic Materials

As discussed in diamagnetic materials when subjected to external magnetic field then due to magnetization an opposite direction magnetic field is induced in the material and external magnetic field repels the magnetized block. This repulsive force can be used in suspension or propelling materials. This phenomenon is called 'Magnetic Levitation'. Figure-5.155(a) shows a diamagnetic block placed at the top of an electromagnet on an insulating stand and as soon as the circuit switch is closed a magnetic field by electromagnet passes through the diamagnetic block and it is repelled. If the repulsive force is more than the weight of block it gets suspended in space above the electromagnet as shown in figure-5.155(b).



(a)



(b)

Figure 5.155

5.9.6 Paramagnetism

Whenever a magnetic dipole is placed in an external magnetic field, it exerts a torque on the dipole to align it along the direction of magnetic field. Figure-5.156 shows a free magnetic dipole placed in external magnetic field due to which it experiences a torque $\vec{M} \times \vec{B}$ in clockwise sense and as the dipole is free to rotate it will align along the direction of magnetic field as shown.

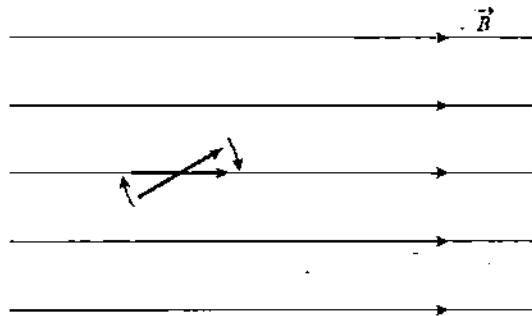
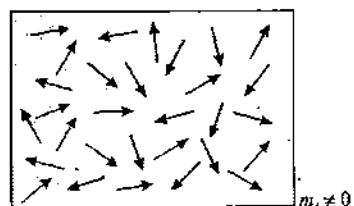


Figure 5.156

The materials in which their atoms or molecules have non zero magnetic moment randomly distributed in volume when placed in external magnetic field, these dipoles align along the direction of external field and then the total magnetic moment of the material will be non zero and along the direction of magnetic field. This phenomenon by which all the dipoles in a material align along the direction of external magnetic field and induces a magnetic field in the direction of external field is called 'Paramagnetism' and such materials are called 'Paramagnetic Materials'.

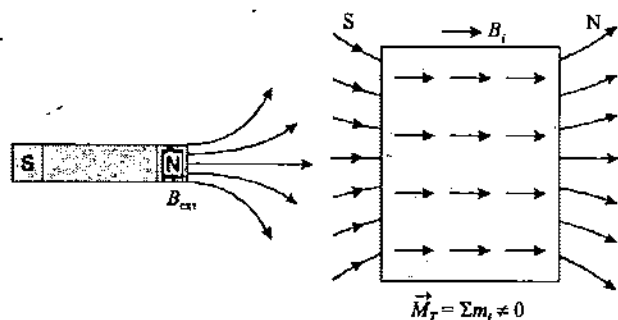
5.9.7 Response of Paramagnetic Materials to External Magnetic Fields

Figure-5.157(a) shows a block of paramagnetic material in which all its dipoles are randomly distributed with total dipole moment of the block zero as already discussed in article-5.9.2. When the block is subjected to an external magnetic field then all its dipoles align along the direction of external magnetic field as shown in figure-5.157(b). As all dipoles are in the direction of external magnetic field, thus the side of magnetically polarized block facing the magnet acts like a south pole which attracts the magnet or source of external magnetic field as shown in figure-5.157(b).



$$\vec{M}_r = \sum \vec{M}_i = 0$$

(a)



(b)
Figure 5.157

Paramagnetism is also a weak phenomenon and as already discussed that in every material upto some extent diamagnetism also exist which is in opposition to paramagnetism.

5.9.8 Magnetic Induction inside a Paramagnetic and Diamagnetic Materials

As already discussed in previous articles that in diamagnetism after polarization, material repel external magnetic field and the internal magnetic field produced by the material is in opposition to the external magnetic field. As shown in figure-5.158 if B_i is the internal magnetic induction produced by the dipoles of the material after it is placed in external magnetic induction B_0 then inside the material magnitude of net magnetic induction is given as

$$B_{\text{net}} = B_0 - B_i \quad \dots (5.116)$$

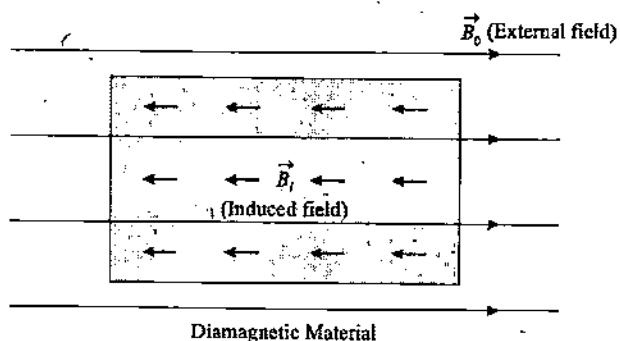


Figure 5.158

In paramagnetism after polarization, material attracts external magnetic field and the internal magnetic field produced by the material is in same direction as that of external magnetic field. As shown in figure-5.159 if B_i is the internal magnetic induction produced by the dipoles of the material after it is placed in external magnetic induction B_0 then inside the material magnitude of net magnetic induction is given as,

$$B_{\text{Net}} = B_0 + B_i \quad \dots (5.117)$$

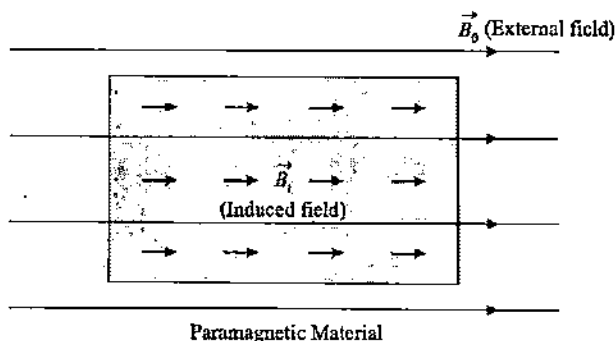


Figure 5.159

5.9.9 Intensity of Magnetization

As already discussed in article-5.9.2 due to random distribution of dipoles inside a material (if dipoles exist in the material) the total dipole moment of the whole block of material is zero but when it is subjected to an external magnetic field it gets magnetically polarized and net dipole moment exist. This net dipole moment direction is opposite to external magnetic field in diamagnetic materials and it is in same direction as that of external magnetic field in paramagnetic materials.

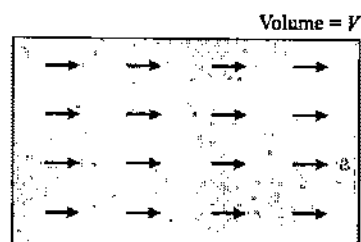


Figure 5.160

After magnetic polarization of a material the total dipole moment of the material per unit of its volume is called '*Intensity of Magnetization*'. Figure-5.160 shows a block of volume V which is magnetically polarized. If the total dipole moment of the block is $\vec{M}$ then the intensity of magnetization inside the block is given as

$$\vec{I} = \frac{\vec{M}}{V} \quad \dots (5.118)$$

Unit used for measurement of intensity of magnetization is '*ampere/meter*' denoted as '*A/m*'.

Inside a magnetically polarized material the magnetic induction due to its polarized dipoles is related to intensity of magnetization as

$$\vec{B} = \mu_0 \vec{I} \quad \dots (5.119)$$

Above relations given in equations-(5.119) is empirically defined relations which students can directly use in applications.

5.9.10 Magnetic Field Vectors $\vec{B}$ and $\vec{H}$

In previous chapter we've discussed that in a region of magnetic field the magnetic induction is measured in units of 'Tesla' and denoted by vector $\vec{B}$. In previous articles we've also discussed when external magnetic field passes through different materials then due to magnetization of materials net magnetic induction inside the magnetized (magnetically polarized) material changes so we need to define a physical quantity $\vec{H}$ for external magnetic field which does not change with medium. This $\vec{H}$ is called 'Magnetic Field Strength' or 'Magnetizing Field' of magnetic field. Magnetizing field in a given region is measured in units of 'ampere/meter' and denoted as 'A/m'.

In general both the vectors $\vec{B}$ and $\vec{H}$ are called commonly as magnetic field in practical situations. It is the unit of measurement which clarifies about which vector we are talking about. If unit used is 'tesla' then its $\vec{B}$ and if unit used is 'A/m' then its $\vec{H}$. In a given region of space both the vectors of magnetic field are related as

$$\vec{B} = \mu_0 \vec{H} \quad \dots (5.120)$$

Inside a material having magnetic permeability μ_r the relation is modified as

$$\vec{B} = \mu_0 \mu_r \vec{H} \quad \dots (5.121)$$

5.9.11 Relation in $\vec{B}$, $\vec{H}$ and $\vec{I}$ in Magnetization of Materials

When a material is placed in external magnetic field of magnetic induction $\vec{B}_0$ then as discussed in article-5.9.8 due to magnetization of material an induced magnetic induction inside the material $\vec{B}_i$ is developed and net magnetic induction inside the material vectorially is given as

$$\vec{B} = \vec{B}_0 + \vec{B}_i \quad \dots (5.122)$$

If magnetizing field of external magnetic induction is $\vec{H}$ and intensity of magnetization inside the material is $\vec{I}$ then from above equation-(5.122), we have

$$\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{I}$$

$$\Rightarrow \vec{B} = \mu_0 (\vec{H} + \vec{I}) \quad \dots (5.123)$$

5.9.12 Magnetic Susceptibility

This is a constant parameter for each material which gives an idea about the ease with which the material can be magnetized when subjected to external magnetic field. For any material magnetic susceptibility is defined as the ratio of intensity of magnetization inside the material to the magnetizing field applied for magnetization of the material so it is given as

$$\chi = \frac{I}{H} \quad \dots (5.124)$$

$$\Rightarrow H = \chi I \quad \dots (5.125)$$

Vectorially above relation in equation-(5.125) can be given as

$$\vec{H} = \chi \vec{I} \quad \dots (5.126)$$

For any material having magnetic permeability μ_r net magnetic induction inside is given by equation-(5.121) and (5.123) which is given as

$$\mu_0 \mu_r \vec{H} = \mu_0 (\vec{H} + \chi \vec{H})$$

$$\Rightarrow \mu_r = 1 + \chi \quad \dots (5.127)$$

As we know for diamagnetic materials $\vec{I}$ is in opposition to external field $\vec{H}$ thus χ is negative so magnetic permeability of diamagnetic materials is less than unity and for paramagnetic materials $\vec{I}$ is in same direction as that of external magnetic field $\vec{H}$ thus χ is positive so magnetic permeability of paramagnetic materials is more than unity.

5.9.13 Curie's Law

In previous article we've discussed that in paramagnetic materials intensity of magnetization is due to the alignment of all the dipoles of the material along external magnetic field after it gets magnetized. There are two main factors on which the intensity of magnetization depends. Intensity of magnetization is directly proportional to the external magnetizing field H . As H increases the dipoles of material will experience more torque and overall alignment of dipoles in the direction of external magnetic field increases. Thus we have

$$I \propto H \quad \dots (5.128)$$

Another factor on which intensity of magnetization depends is the temperature. As temperature increases then due to thermal agitation the alignment of dipoles gets disturbed and overall dipole moment of the material decreases. Intensity of magnetization is observed to be inversely proportional to the temperature of the material, thus we have

$$I \propto \frac{1}{T} \quad \dots (5.129)$$

From equations-(5.128) and (5.129) we have

$$I = C \left(\frac{H}{T} \right) \quad \dots (5.130)$$

In above equation C is called Curie's constant and at a specific temperature we can compare equation-(5.130) with equation-(5.131) which gives

$$\chi = \frac{C}{T} \quad \dots (5.131)$$

Thus the magnetic susceptibility of paramagnetic material is inversely proportional to its absolute temperature. This is called '*Curie's Law in Magnetism*'. In general diamagnetic materials do not obey Curie's law as in diamagnetic materials intensity of magnetization is weak and their susceptibility does not vary significantly with temperature whereas paramagnetic materials obey Curie law and on heating their susceptibility decreases to lower values and a paramagnetic material loses its magnetic character on heating.

5.9.14 Ferromagnetic Materials

There are some specific materials like Fe, Co, Ni and some more elements in which atoms have non zero magnetic moments and in their solid lattice due to a quantum phenomenon called '*Exchange Coupling*' atomic groups are formed in different pockets of material which have their magnetic moment aligned in same direction in that pocket. This phenomenon exchange coupling is due to the alignment of all dipoles by the magnetic field of neighbouring dipole in a short neighbourhood. Different pockets in the material may have atomic groups aligned with their magnetic moment in different directions as shown in figure-5.161. Each of these atomic groups aligned in same direction are called '*Domains*' and such materials are called '*Ferromagnetic Materials*'. As all the domains in a ferromagnetic material are having direction of their dipole moment randomly distributed the net magnetic moment of such materials is zero.

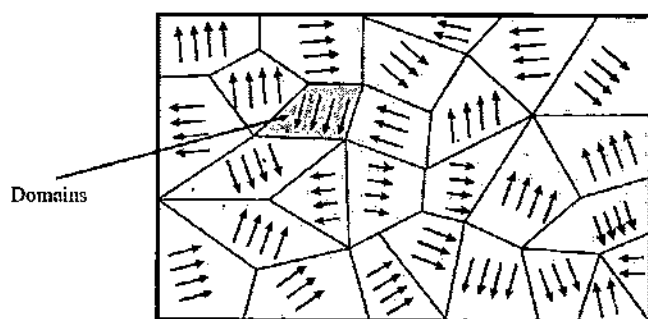


Figure 5.161

5.9.15 Magnetization of Ferromagnetic Materials

Due to exchange coupling all atoms in a domain have their individual dipole moment μ_i aligned in same direction. In this state even a very small magnitude external magnetic field can align all the domains along the direction of external magnetic field as shown in figure-5.162. This is because the internal magnetic field of already aligned atoms in a domain produces a torque on neighbouring domain atoms which supports the external field in this alignment process.

In ferromagnetic materials induced field B_i due to magnetization of material is much higher than the external field because of

exchange coupling. Thus we have

$$\begin{aligned} B_i &\gg B_0 \\ \Rightarrow I &\gg H \end{aligned} \quad \dots (5.132)$$

Thus for ferromagnetic materials we use $\chi \gg 1$.

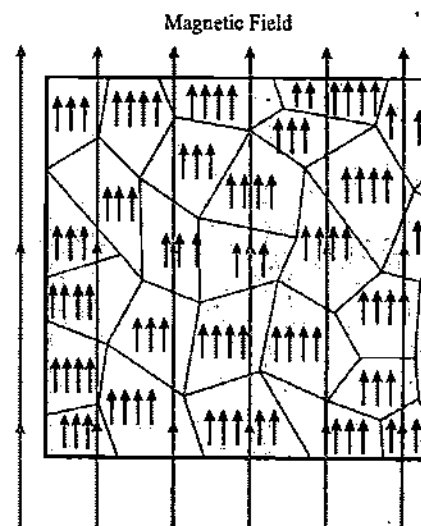


Figure 5.162

The susceptibility of ferromagnetic materials decreases with the rise in temperature in a different way. At a specific temperature called '*Curie Point*' or '*Curie Temperature*' a ferromagnetic material starts behaving like a paramagnetic material. Thus after Curie point, the susceptibility of a ferromagnetic material varies inversely with its absolute temperature or we can say that only above Curie point a ferromagnetic substance obeys Curie's law.

5.9.16 Superconductor as a Perfectly Diamagnetic Material

Superconducting materials are regarded as perfectly diamagnetic as on subjecting to external magnetic field, the free electrons in superconductors start developing an eddy current which grows until it balances the external magnetic field inside the material.

When inside a magnetic field a conductor is cooled to the superconducting state and at a specific temperature material loses all its electrical resistance and at the same instant the material expels all the magnetic field inside it. This is called '*Meissner's Effect*'. Even when the material is not acting as a superconductor, it has some eddy currents due to application of external magnetic field which grows as its temperature decreases and at the superconducting state the eddy currents will increase so as to balance the external magnetic field and resulting zero magnetic field inside the material.

Figure-5.163 shows a theoretical situation of coil made up of superconducting wire. When an external magnetic field is applied onto it then an induced current is developed in the coil which opposes the external magnetic field and due to zero resistance in the coil the current in it does not decay and grows to the limit till the net magnetic field at the plane of coil by its current balances the external magnetic field so theoretically we consider no magnetic flux passes through a superconducting coil when placed in an external magnetic field.

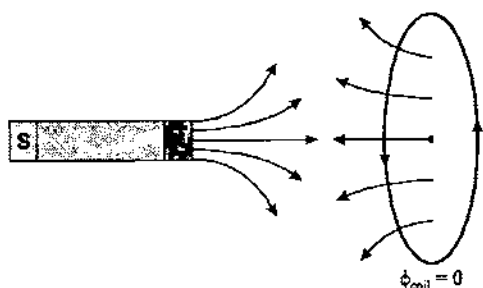


Figure 5.163

Another situation is shown in figure-5.164 in which a superconducting block is placed in an external magnetic field and in this situation inside the block eddy currents are induced to nullify the net magnetic field at all its interior points.

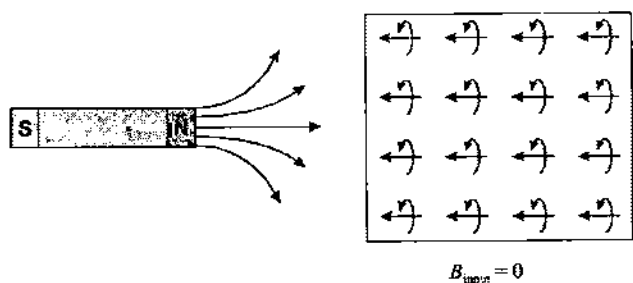


Figure 5.164

5.9.17 Magnetic Hysteresis

The phenomenon in which the value of a physical property lags behind the causes by which it is changing is called 'Hysteresis'. When a ferromagnetic material is magnetized by an external magnetizing field the intensity of magnetization in the material increases with the external field and when magnetizing field decreases then intensity of magnetization also decreases but it lags behind the change in magnetizing field. This is called 'Magnetic Hysteresis'. We will discuss magnetic hysteresis with the help of an illustration.

Consider a ferromagnetic material which is subjected to an external magnetic field which can be changed in magnitude as well as in direction. Figure-5.165 shows the variation of intensity

of magnetization in the material with change in magnetizing field. The curve starts from origin when initial value of magnetizing field was zero and at that instant polarization of material was also zero and hence no initial magnetization was there. When H increases, I also increases and curve follows path O to point A after which it no further increase in I is observed with increase in H . This state is called magnetic saturation material at which I approaches maximum value.

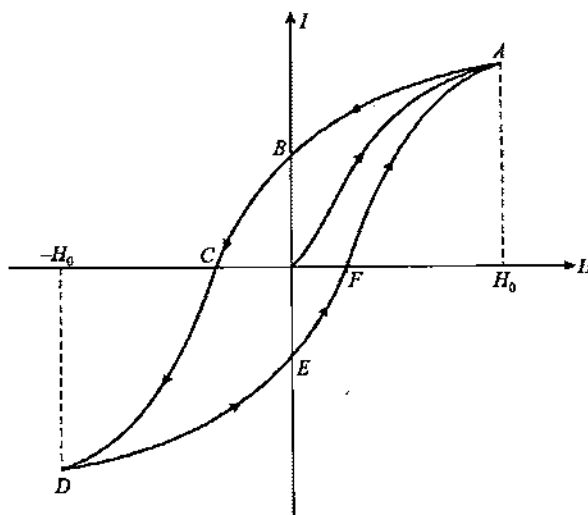


Figure 5.165

Now from the point of magnetic saturation H is decreases with which I also decreases along the path AB . When H becomes zero then at point B some magnetization is left in the material. This magnetization OB which is left in the material at the point when H is zero is called 'Retentivity' of the material. This intensity of magnetization left in the material when magnetizing field is reduced to zero from saturation is also called 'Remanence' or sometimes called 'Residual Magnetism'. This residual magnetism is also termed as 'Magnetic Inertia' due to which in absence of demagnetization the retention of magnetic flux in the material is there for a long term depending upon the retentivity of the material.

To reduce the intensity of magnetization to zero, the magnetising field is increased in reversed direction from B to C then at point C the intensity of magnetization becomes zero for which we had to apply a magnetizing field in reverse direction equal to OC . This magnetizing field OC is called 'Coercivity' of the material. Thus the value of reverse magnetizing field required to reduce the retentivity in material to zero is called coercivity of the material.

Further when the magnetizing field is increased in reverse direction, intensity of magnetization increases along the curve CD and at point D again it is the state of magnetic saturation in reverse direction. Now magnetizing field is increased in original

direction upto point A by passing through the point E and F which are corresponding to retentivity and coercivity in opposite directions as shown in figure-5.165. In this figure the closed curve ABCDEFA is traced between values of H and I is called 'Hysteresis Loop'. Based on several experiments and analysis it is found that the area of the hysteresis loop which is also called ' I - H Curve' is proportional to the net energy absorbed per unit volume by the material during one cycle of magnetization and demagnetization. This energy appears as heat in the material in the cycle.

Phenomenon of hysteresis discussed above is shown by ferromagnetic substances only and the shape of hysteresis curve gives the idea about some important magnetic properties of the material. Figure-5.166 shows two hysteresis curves plotted on same graph for steel and soft iron. Comparing the two curves we can make some inferences given below.

- (i) Area of hysteresis loop for soft iron is much smaller compared to that of steel which indicates that in case of soft iron energy loss per unit volume is very small as compared to steel when these are taken over cycle of magnetization and demagnetization.
- (ii) Soft iron gets magnetized to saturation value at less value of external magnetizing field compared to steel and on removing magnetizing field magnetic retention in case of soft iron is more than that of steel. That indicates that soft iron is strongly magnetized compared to steel by same magnetizing field.
- (iii) Coercivity of steel is much greater than soft iron that means the residual magnetism in steel sustains for a longer time and for soft iron residual magnetism can be destroyed by a smaller value of reverse magnetizing field easily.

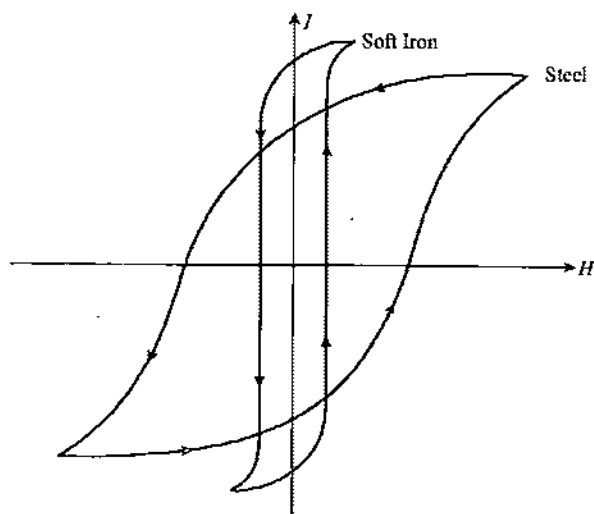


Figure 5.166

Illustrative Example 5.60

A bar magnet has its pole strength 4.5 Am and length 12 cm . The cross sectional area of the magnet is 0.9 cm^2 . Find the total dipole moment of the magnet and intensity of magnetisation of the magnetic material.

Solution

The intensity of magnetisation is the dipole moment per unit volume in the material. The effective total dipole moment of the magnet is given as

$$M = ml = 4.5 \times 0.12 = 0.54 \text{ Am}^2$$

Intensity of magnetisation of material is given as

$$I = \frac{M}{V} = \frac{ml}{Al} = \frac{m}{A} = \frac{4.5}{0.9 \times 10^{-4}} = 5 \times 10^4 \text{ A/m}$$

Illustrative Example 5.61

In a region at 280 K temperature, on an aluminium sample a magnetising field of intensity $2 \times 10^3 \text{ A/m}$ is applied due to which the sample acquires an intensity of magnetisation $4.8 \times 10^{-2} \text{ A/m}$. Calculate the susceptibility of aluminium at 280 K and if temperature of the sample is raised to 320 K , what will be the susceptibility and intensity of magnetisation.

Solution

Susceptibility of aluminium at 280 K is given as

$$\chi_{m1} = \frac{I}{H}$$

$$\Rightarrow \chi_{m1} = \frac{4.8 \times 10^{-2}}{2 \times 10^3} = 2.4 \times 10^{-5}$$

As aluminium is a paramagnetic material, it obeys Curie's law so if χ_{m2} is the susceptibility of aluminium at 320 K temperature we use

$$\frac{\chi_{m1}}{\chi_{m2}} = \frac{T_2}{T_1}$$

$$\Rightarrow \chi_{m2} = \chi_{m1} \times \frac{T_1}{T_2}$$

$$\Rightarrow \chi_{m2} = 2.4 \times 10^{-5} \times \frac{280}{320} = 2.1 \times 10^{-5}$$

If I' is the intensity of magnetisation of aluminium at 320 K temperature then we have

$$I' = \chi_{m2} H$$

$$\Rightarrow I' = 2.1 \times 10^{-5} \times 2 \times 10^3 \text{ A/m}$$

$$\Rightarrow I' = 4.2 \times 10^{-2} \text{ A/m}$$

Illustrative Example 5.62

A piece of iron of mass 8.4kg is repeatedly magnetised and demagnetised by an external periodic time varying magnetic field of frequency 50Hz. In the iron piece it is measured that rate of heat dissipation is 6.4×10^4 J/hr. If iron density is 7200 kg/m^3 find the energy dissipated in iron piece per cycle per unit volume of it.

Solution

The heat dissipation in the iron piece per unit volume per hour is given as

$$Q = \frac{\text{Heat}}{\text{Volume}} = \frac{6.4 \times 10^4}{8.4/7200} \text{ J/hr-m}^3$$

Period of one cycle of magnetisation is given as

$$T = \frac{1}{f} = \frac{1}{50 \times 3600} \text{ hr}$$

Thus heat dissipated per cycle per unit volume is given as

$$Q_v = \frac{6.4 \times 10^4}{8.4/7200} \times \frac{1}{50 \times 3600} = 305 \text{ J/m}^3$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Magnetic Properties of Materials

Module Number - 1 to 16

5.10 Alternating Current

A current which periodically changes direction is called an Alternating Current (AC). In an AC along with periodic change in direction of current its magnitude can also vary with time. Based on the ways in which magnitude and direction of an AC changes, several different types of ACs are analyzed.

Figure-5.167 shows some specific types of ACs in which current variation with time is shown. The most common type of AC which is widely used in AC applications is shown in figure-5.167(d) which varies with time according to sine function and it is called 'Sinusoidal AC'.

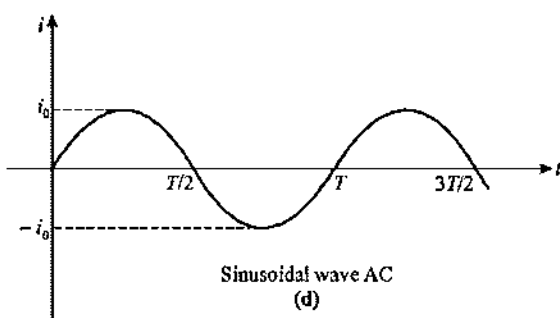
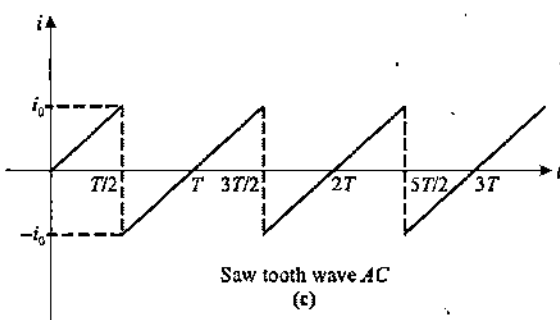
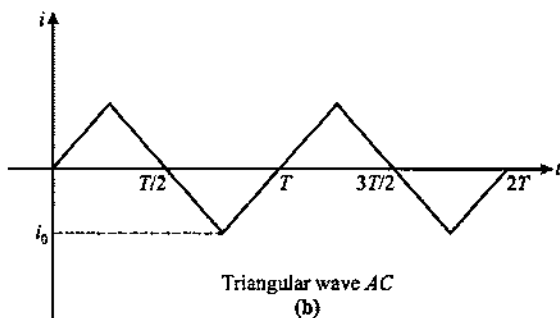
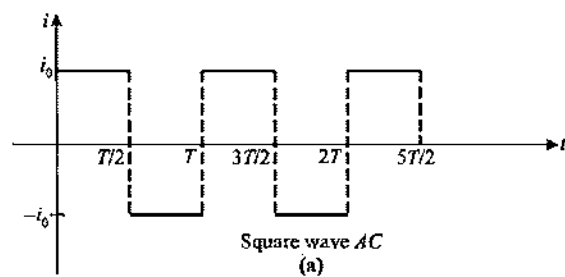


Figure 5.167

The alternating current which is used for domestic and industrial purposes is commonly the sinusoidal AC for which the time function is given as

$$i = i_0 \sin(\omega t) \quad \dots (5.133)$$

Here i_0 is called current amplitude or peak value of alternating current. ω is the angular frequency. The function given in equation-(5.133) gives the instantaneous value of alternating current.

Alternating current is produced due to an alternating EMF. An AC EMF also changes its direction periodically and its magnitude changes with time sinusoidally or in different ways. A sinusoidal EMF in general is expressed as

$$e = e_0 \sin \omega t \quad \dots (5.134)$$

The AC EMF shown in the above equation-(5.134) is considered as a reference for phase analysis of current and other circuit parameters in different types of AC circuits which we will discuss

in upcoming articles. Figure-5.168 shows the circuit symbol of an AC EMF used in drawing circuit diagrams.

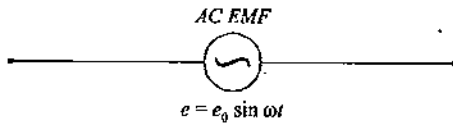


Figure 5.168

5.10.1 Mean or Average Value of an Alternating Current

Average value of an alternating current in a given time is equal to that steady or direct current due to which through a circuit branch same amount of charge is passed which passes due to the given alternating current in same time through the same circuit.

If in a situation time duration for which AC is to be calculated is not given then we calculate it for one time period of the AC.

If an alternating current is expressed as $i = f(t)$ then its average value I_{avg} for a time $t = 0$ to $t = T$ can be calculated by equating the amount of charge which flows due to these values of current in same time as

$$I_{avg} T = \int_0^T f(t) dt$$

$$\Rightarrow I_{avg} = \frac{1}{T} \int_0^T f(t) dt \quad \dots (5.135)$$

Expression in above equation-(5.135) is the average function formula used to evaluate the time average of any given function and here we will use this function for calculation of average or mean value of current for any AC of which the instantaneous variation of current with time is given.

If we carefully see the integral part of the equation-(5.135) it is the area of the curve $f(t)$ vs t between time $t = 0$ to $t = T$. For current this area is the total charge flown in the specified duration thus in case charge quantity is known which flows in a given time like if a charge Δq flows in time Δt then mean value of current can be directly given as

$$I_{avg} = \frac{\Delta q}{\Delta t} \quad \dots (5.136)$$

$$\Rightarrow I_{avg} = \frac{\text{Area under } i-t \text{ curve}}{\Delta t} \quad \dots (5.137)$$

5.10.2 Mean Value of a Sinusoidal Alternating Current

A general sinusoidal alternating current is given by the time function expressed in equation-(5.133). The curve of variation of current with time is shown in figure-5.169. For one complete cycle we can see in this figure the positive and negative area

are equal thus mean value of this sinusoidal current for one complete time period comes out to be zero.

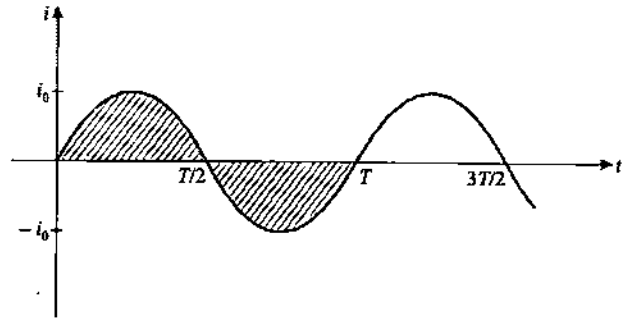


Figure 5.169

If we calculate the mean value of the sinusoidal current for half cycle then it can be calculated by using equation-(5.135) as

$$I_{avg} = \frac{1}{T/2} \int_0^{T/2} i_0 \sin(\omega t) dt$$

$$\Rightarrow I_{avg} = \frac{1}{T/2} \left(-\frac{i_0}{\omega} [\cos(\omega t)]_0^{T/2} \right)$$

$$\Rightarrow I_{avg} = \frac{1}{T/2} \left(\frac{i_0}{(2\pi/T)} [\cos(0^\circ) - \cos(\pi)] \right)$$

$$\Rightarrow I_{avg} = \frac{i_0}{\pi} [1 - (-1)]$$

$$\Rightarrow I_{avg} = \frac{2i_0}{\pi} = 0.636i_0 \quad \dots (5.138)$$

Above equation-(5.138) shows that average or mean value of a sinusoidal AC during half cycle is 63.6% of its peak value but the above equation is valid only if half cycle is considered from the duration $t = 0$ to $t = T/2$. For different half time periods the mean value may be different. Figure-5.170 shows the graph of same sinusoidal AC in which if we consider the time period from $t = T/4$ to $t = 3T/4$ then we can see that the total area of the graph is zero hence the mean current for this period which is also half of the total period comes out to be zero.

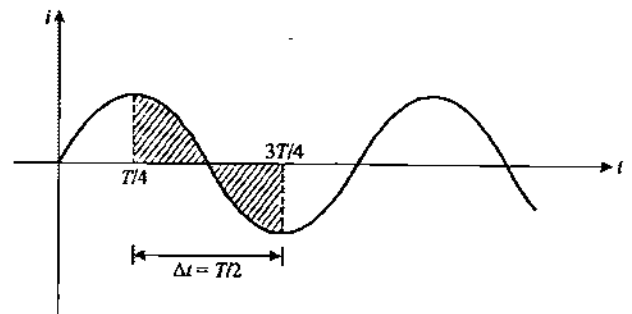


Figure 5.170

If we consider different time instants for the duration of half period then we can say that the mean value of a sinusoidal AC can range from 0 to $2i_0/\pi$.

5.10.3 Root Mean Square (RMS) value of an Alternating Current

RMS value of an alternating current for a given time duration is calculated by taking the square root of the average of square of the time function $i = f(t)$ of current as the name RMS implies. This can be mathematically calculated by using the average function formula for square of the time function of current which is given as

$$I_{RMS} = \sqrt{\frac{1}{T} \int_0^T [f(t)]^2 dt} \quad \dots (5.139)$$

Theoretically this RMS value is defined as a steady constant current when passes through a given resistor produces same amount of heat in a given duration which is produced by the given alternating current when passed through the same resistor in same time duration.

If the alternating current $i = f(t)$ is passed through a resistor of resistance R then for a given time 0 to T total heat produced in this resistor can be given as

$$H = \int_0^T i^2 R dt$$

$$\Rightarrow H = \int_0^T [f(t)]^2 R dt \quad \dots (5.140)$$

If I_{RMS} is the RMS value of the given alternating current then the total heat produced by this constant value in a resistance R in time 0 to T is given as

$$H = I_{RMS}^2 RT \quad \dots (5.141)$$

From equations-(5.140) and (5.141) we have

$$I_{RMS}^2 RT = \int_0^T [f(t)]^2 R dt$$

$$\Rightarrow I_{RMS}^2 = \frac{1}{T} \int_0^T [f(t)]^2 dt$$

$$\Rightarrow I_{RMS} = \sqrt{\frac{1}{T} \int_0^T [f(t)]^2 dt} \quad \dots (5.142)$$

We can see that equation-(5.142) is same as equation-(5.139).

5.10.4 RMS value of a Sinusoidal Alternating Current

A general sinusoidal alternating current is given by the time function expressed in equation-(5.133). For calculation of its RMS value as we will find the average of squares half cycle or full cycle does not make any difference as square of positive or negative both cycles will be positive. To determine the RMS value of a sinusoidal alternating current first we will calculate

the 'Mean Square' value of this current which is given as

$$I_{RMS}^2 = \frac{1}{T} \int_0^T [f(t)]^2 dt$$

$$\Rightarrow I_{RMS}^2 = \frac{1}{T/2} \int_0^{T/2} i_0^2 \sin^2 \omega t dt$$

$$\Rightarrow I_{RMS}^2 = \frac{\omega i_0^2}{\pi} \int_0^{\pi/\omega} \sin^2 \omega t dt$$

$$\Rightarrow I_{RMS}^2 = \frac{\omega i_0^2}{2\pi} \int_0^{\pi/\omega} [1 - \cos(2\omega t)] dt$$

$$\Rightarrow I_{RMS}^2 = \frac{\omega i_0^2}{2\pi} \left[t - \frac{1}{2\omega} \sin(2\omega t) \right]_0^{\pi/\omega}$$

$$\Rightarrow I_{RMS}^2 = \frac{\omega i_0^2}{2\pi} \left[\left(\frac{\pi}{\omega} - 0 \right) - (0 - 0) \right]$$

$$\Rightarrow I_{RMS}^2 = \frac{i_0^2}{2}$$

$$\Rightarrow I_{RMS} = \frac{i_0}{\sqrt{2}} = 0.707 i_0 \quad \dots (5.143)$$

The RMS value of sinusoidal alternating current as obtained in equation-(5.143) is a standard result for full or half cycle of it but for different periods the RMS value may come differently for this sinusoidal alternating current.

Illustrative Example 5.63

An electric bulb is designed to operate at 12V DC. If this bulb is connected to an AC source and it gives same brightness, what would be the peak voltage of AC.

Solution

As brightness is same, this indicates that heat produced in the two cases is same so the effective (RMS) current supplied by AC source is same as that of DC current through the bulb. Thus effective (RMS) voltage of AC must be 12V which is given in terms of peak value of voltage as

$$e_{rms} = \frac{e_0}{\sqrt{2}}$$

$$\Rightarrow e_0 = \sqrt{2} e_{rms} = 12\sqrt{2} \text{ V}$$

Illustrative Example 5.64

The current decaying in LR circuit has a function $i = I_0 e^{-t/\tau}$ where τ is the time constant of the circuit. Find RMS value of this current for the period $t = 0$ to $t = \tau$.

Solution

We know that root mean square value of current is given as

$$I_{RMS}^2 = \frac{1}{T} \int_0^T [f(t)]^2 dt$$

$$\Rightarrow I_{RMS}^2 = \frac{1}{\tau} \int_0^{\tau} I_0^2 e^{-2t/\tau} dt$$

$$\Rightarrow I_{RMS}^2 = \frac{I_0^2}{\tau} \left[\frac{e^{-2t/\tau}}{-2/\tau} \right]_0^{\tau}$$

$$\Rightarrow I_{RMS}^2 = -\frac{I_0^2}{2} [e^{-2} - 1]$$

$$\Rightarrow I_{RMS}^2 = \frac{I_0^2}{2} \left(1 - \frac{1}{e^2} \right)$$

$$\Rightarrow I_{RMS} = \frac{I_0}{\sqrt{2}} \sqrt{1 - \frac{1}{e^2}}$$

Illustrative Example 5.65

A direct current of 2A and an AC of peak value 2A flows through resistances 2Ω and 1Ω respectively. Find the ratio of heat produced in the two resistances in same interval.

Solution

In time t heat produced by a DC current I_1 is given as

$$H_1 = I_1^2 R t$$

$$\Rightarrow H_1 = (2)^2 (2) t = 8t$$

RMS value of AC is given as

$$i_{rms} = \frac{i_0}{\sqrt{2}} = \frac{2}{\sqrt{2}} = \sqrt{2} \text{ A}$$

Heat produced by the rms current in time t is given as

$$H_2 = i_{rms}^2 R t$$

$$\Rightarrow H_2 = (\sqrt{2})^2 (1) t = 2t$$

$$\Rightarrow \frac{H_1}{H_2} = \frac{8t}{2t} = 4$$

Illustrative Example 5.66

Find the RMS value of the sawtooth voltage of peak value V_0 per cycle for which the time function is shown in figure-5.171.

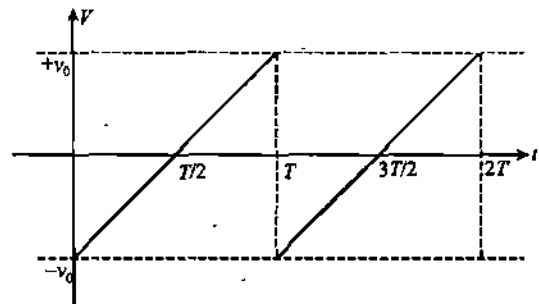


Figure 5.171

Solution

Time function of voltage for the above graph is given as

$$V = \frac{2V_0}{T} \cdot t - V_0$$

Mean square value of the voltage is given as

$$V_{ms} = \frac{1}{T} \int_0^T V^2 dt$$

$$\Rightarrow V_{ms} = \frac{1}{T} \int_0^T \left(\frac{2V_0}{T} t - V_0 \right)^2 dt$$

$$\Rightarrow V_{ms} = \frac{1}{T} \int_0^T \left(\frac{4V_0^2 t^2}{T} - \frac{4V_0^2 t}{T} + V_0^2 \right) dt$$

$$\Rightarrow V_{ms} = \frac{1}{T} \left(\frac{4}{3} V_0^2 T - \frac{4V_0^2}{2} T + V_0^2 T \right)$$

$$\Rightarrow V_{ms} = \left(\frac{V_0^2}{3} \right)$$

$$\Rightarrow V_{rms} = \sqrt{V_{ms}} = \frac{V_0}{\sqrt{3}}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Alternating Current

Module Number - 1 to 13

Practice Exercise 5.5

(i) An AC voltage is given as $e = e_1 \sin \omega t + e_2 \cos \omega t$. Find the RMS value of this voltage.

$$\left[\sqrt{\frac{e_1^2 + e_2^2}{2}} \right]$$

(ii) In a wire direct current i_1 and an AC current $i_2 = i_{20} \sin \omega t$ is superposed. Find the RMS value of current in wire.

$$\left[\sqrt{i_1^2 + \frac{i_{20}^2}{2}} \right]$$

(iii) Calculate the RMS value of the EMF for a cycle of which time variation is shown in figure-5.172

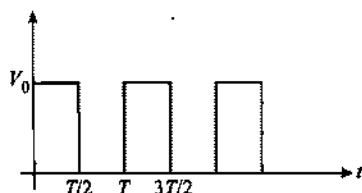


Figure 5.172

$$\left[\frac{V_0}{\sqrt{2}} \right]$$

(iv) In a given AC circuit through a specific branch the current varies as a function of time given as

$$i = i_0 \sin^2 \omega t \quad 0 \leq \omega t < \pi$$

$$\text{and } i = i_0 \sin \omega t \quad \pi \leq \omega t < 2\pi$$

Calculate the average current per cycle of this AC.

$$\left[i_0 \left(\frac{1}{4} - \frac{1}{\pi} \right) \right]$$

(v) Calculate the root mean square value of the voltage of the given variation of an alternating EMF with time in graph as shown in figure-5.173.

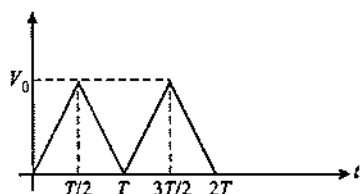


Figure 5.173

$$\left[\frac{V_0}{\sqrt{3}} \right]$$

(vi) In previous question calculate the average voltage per cycle of the alternating EMF.

$$\left[\frac{V_0}{\sqrt{2}} \right]$$

(vii) Calculate the average value of the AC per cycle for which time function of the current is shown in figure-5.174.

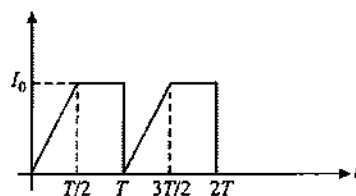


Figure 5.174

$$\left[\frac{3}{4} I_0 \right]$$

5.11 AC Circuit Components

In different types of AC circuits mainly three passive circuit components are used which are resistors, capacitors and inductors. These may be connected in different combinations across and AC EMF. In analysis of AC circuits, currents and potential difference across different circuit components are to be analyzed when circuit is closed. While analyzing AC circuits an important factor has to be taken care of along with magnitude of current and potential differences in different branches and for different circuit components which is the phase of current and potential difference being analyzed. Being a sinusoidal function when EMF is applied across a circuit, due to the behaviour of circuit components, currents in branches of circuit may lag or lead the applied EMF in phase so it is an essential parameter to be accounted in all calculations while solving an alternating current circuit. Later in upcoming articles we will see that power in AC circuit is also dependent upon the phase of current and potential difference for any circuit component. So we will build our understanding of solving AC circuits step by step.

5.11.1 A Resistor Across an AC EMF

Figure-5.175 shows a pure resistor of resistance R is connected across an AC EMF of which the time function is given as

$$e = e_0 \sin \omega t \quad \dots (5.144)$$

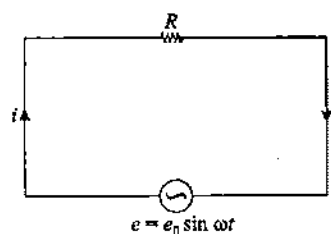


Figure 5.175

In above circuit due to applied EMF an alternating current i flows through the resistance. The direction of current shown in

figure is indicative and at an instant. By Ohm's Law the instantaneous current in the circuit is given as

$$i = \frac{e}{R}$$

$$\Rightarrow i = \frac{e_0 \sin \omega t}{R}$$

$$\Rightarrow i = i_0 \sin \omega t \quad \dots (5.145)$$

Where $i_0 = \frac{e_0}{R} \quad \dots (5.146)$

From equation-(5.144) and (5.145) we can see that alternating EMF and alternating current produced due to this EMF in a resistance are in same phase. Equation-(5.146) gives the current amplitude of the alternating current produced in the resistance.

Figure-5.176 shows the time variation of AC EMF and current through the resistor of resistance R when it is subjected to the AC EMF.

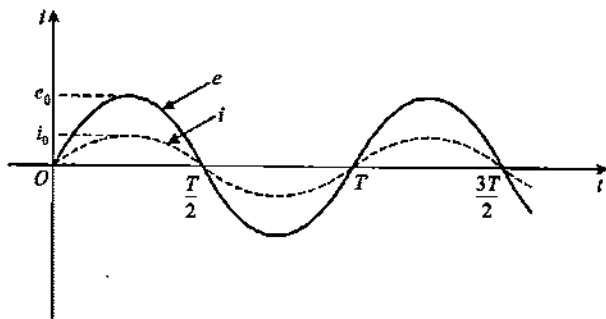


Figure 5.176

5.11.2 A Capacitor Across an AC EMF

Figure-5.177 shows a capacitor of capacitance C is connected across an AC EMF of which the time function is given as

$$e = e_0 \sin \omega t \quad \dots (5.147)$$

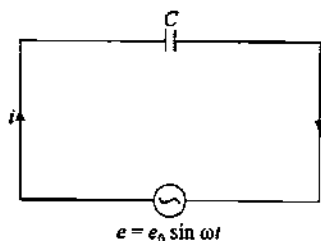


Figure 5.177

In above circuit due to applied EMF an alternating current i flows through the capacitor. The direction of current shown in above figure is indicative and at an instant. As capacitor is connected in parallel across the AC EMF, neglecting the resistance of connecting wires the instantaneous charge on

capacitor is given as

$$q = Ce$$

$$\Rightarrow q = Ce_0 \sin \omega t \quad \dots (5.148)$$

The instantaneous current in the circuit can be given as

$$i = \frac{dq}{dt}$$

$$\Rightarrow i = Ce_0 \omega \cos \omega t$$

$$\Rightarrow i = \frac{e_0}{(1/\omega C)} \cos \omega t$$

$$\Rightarrow i = i_0 \cos \omega t$$

$$\Rightarrow i = i_0 \sin(\omega t + \pi/2) \quad \dots (5.149)$$

Where $i_0 = \frac{e_0}{(1/\omega C)} \quad \dots (5.150)$

From equation-(5.147) and (5.149) we can see that alternating current produced due to the applied alternating EMF in a capacitor leads in phase over applied EMF by an angle $\pi/2$. Equation-(5.150) gives the current amplitude of the alternating current produced in the capacitor connected across AC EMF.

In equation-(5.150) the term $1/\omega C$ must have unit of 'ohm' for the equation-(5.150) to be dimensionally correct. This value $1/\omega C$ is the ohmic opposition offered by capacitor in path of alternating current. This is called '*Capacitive Reactance*' of the capacitor which is denoted by X_C and given as

$$X_C = \frac{1}{\omega C} \quad \dots (5.151)$$

Figure-5.178 shows the time variation of AC EMF and current through a capacitor of capacitance C when it is subjected to the AC EMF.

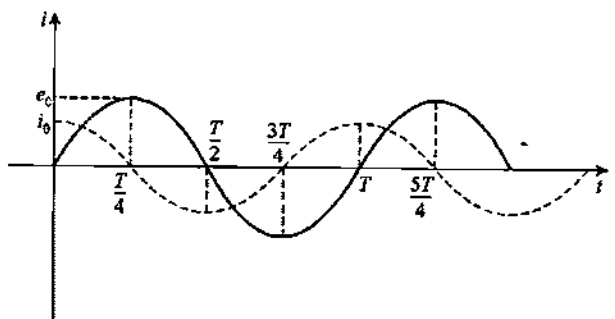


Figure 5.178

5.11.3 An Inductor Across an AC EMF

Figure-5.179 shows an inductor of self inductance L is connected across an AC EMF of which the time function is given as

$$e = e_0 \sin \omega t \quad \dots (5.152)$$

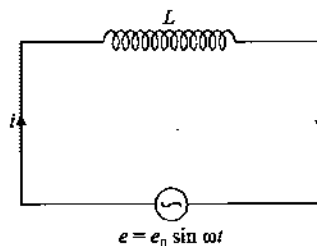


Figure 5.179

In above circuit due to applied EMF an alternating current i flows through the inductor. The direction of current shown in above figure is indicative and at an instant. As the inductor is connected in parallel across the AC EMF, neglecting the resistance of connecting wires we use

$$e = L \frac{di}{dt}$$

$$\Rightarrow e_0 \sin \omega t = L \frac{di}{dt}$$

$$\Rightarrow L di = e_0 \sin \omega t dt$$

Integrating the above expression we get

$$\int L di = \int e_0 \sin \omega t dt$$

$$Li = -\frac{1}{\omega} (e_0 \cos \omega t) + C$$

$$\Rightarrow i = -\frac{1}{L\omega} (e_0 \cos \omega t) + \frac{C}{L} \quad \dots (5.153)$$

In above equation first term on right hand side is the periodic current whereas second term is the direct current which does not change with time so alternating current produced in inductor due to AC EMF is given as

$$i = -\frac{1}{L\omega} (e_0 \cos \omega t)$$

$$\Rightarrow i = \frac{e_0}{L\omega} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\Rightarrow i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right) \quad \dots (5.154)$$

Where $i_0 = \frac{e_0}{\omega L} \quad \dots (5.155)$

From equation-(5.154) and (5.155) we can see that alternating current produced due to the applied alternating EMF in an inductor lags behind in phase with the applied EMF by an angle $\pi/2$. Equation-(5.155) gives the current amplitude of the alternating current produced in the inductor connected across AC EMF.

In equation-(5.155) the term ωL must have unit of 'ohm' for the equation-(5.155) to be dimensionally correct. This value ωL is

the ohmic opposition offered by an inductor in path of alternating current. This is called '*Inductive Reactance*' of the inductor which is denoted by X_L and given as

$$X_L = \omega L \quad \dots (5.156)$$

Figure-5.180 shows the time variation of AC EMF and current through an inductor of inductance L when it is subjected to the AC EMF.

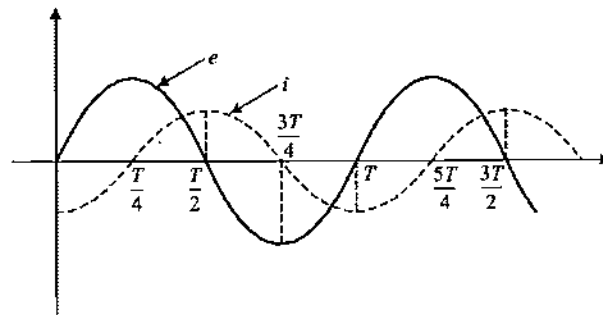


Figure 5.180

5.11.4 Comparison of AC Circuit Elements

In previous articles we've seen how a resistor, capacitor and an inductor behaves when connected across an AC EMF. In case of AC circuit a capacitor and an inductor also offers ohmic opposition which is called '*Reactance*' of the circuit measured in unit of ohm and denoted by ' Ω '. Unlike to the normal resistance of a resistor the reactance in AC circuit depends upon frequency as we've seen that both capacitive and inductive reactance are dependent upon frequency. Equation-(5.151) shows that capacitive reactance varies inversely with frequency of AC EMF whereas from equation-(5.156) shows that inductive reactance varies as directly proportional to the frequency.

Figure-5.181 shows the variation curve of the resistance and reactance with frequency of AC source.

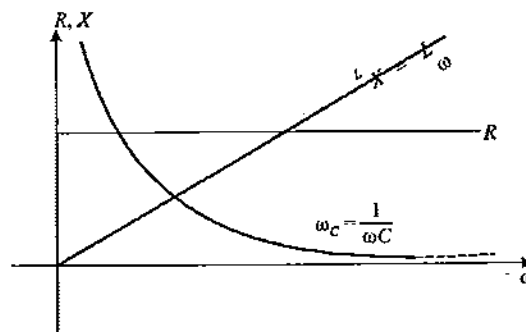


Figure 5.181

As discussed in previous articles we've also discussed the phase relationship between EMF and supplied current for different circuit components. Same phase relationships can be

used in different AC circuits between potential difference and current across AC components. Figure-5.182 shows a resistance through which an alternating current i flows. In this situation the potential difference across the resistance can be given as

$$V_{AB} = iR \quad \dots (5.157)$$

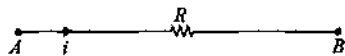


Figure 5.182

Students must keep in mind that in any type of AC circuit when an alternating current flows through a resistance then potential difference across the resistance and the current through it will always be in same phase. Figure-5.183 shows a capacitor through which an alternating current i flows. As we know the ohmic opposition or the reactance due to a capacitor in AC circuit is given as $X_C = 1/\omega C$ the potential difference across the capacitor is given by Ohm's law as

$$V_{AB} = iX_C \quad \dots (5.158)$$

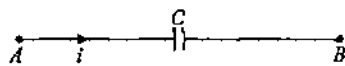


Figure 5.183

For all types of AC circuits when an alternating current flows through a capacitor then the potential difference across it lags behind in phase with the current flowing through it by a phase angle $\pi/2$. Figure-5.184 shows an inductor through which an alternating current i flows. As we know the ohmic opposition or the reactance due to an inductor in AC circuit is given as $X_L = \omega L$ the potential difference across the inductor is given by Ohm's law as

$$V_{AB} = iX_L \quad \dots (5.159)$$

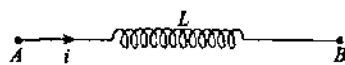


Figure 5.184

For all types of AC circuits when an alternating current flows through an inductor then the potential difference across it leads in phase over the current through it by a phase angle $\pi/2$. For any circuit component in AC circuit the potential difference can be calculated by the above equations-(5.157), (5.158) and (5.159).

5.11.5 Behaviour of L and C at High and Low Frequency AC

Equations-(5.151) and (5.156) gives the ohmic opposition or reactance offered by a capacitor and an inductor when AC is passes through these. For an inductor its reactance is given as $X_L = \omega L$ which is directly proportional to the frequency of applied AC thus as frequency of AC passing through an inductor increases its reactance also increases and at very high frequencies its X_L is also very high and an inductor behaves

like an open circuit at very high frequencies as shown in figure-5.185(a). At very low frequencies X_L may tend to negligible value and an inductor behaves like short circuit as shown in figure-5.185(b).

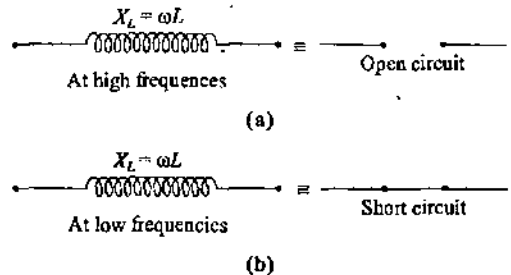


Figure 5.185

For a capacitor its reactance is given as $X_C = 1/\omega C$ which is inversely proportional to frequency of applied AC so as frequency of AC passing through the capacitor increases its reactance decreases and at very high frequencies X_C may tend to negligible value and a capacitor behaves like short circuit as shown in figure-5.186(a). At very low frequencies X_C will be very high and a capacitor behaves like open circuit as shown in figure-5.186(b).

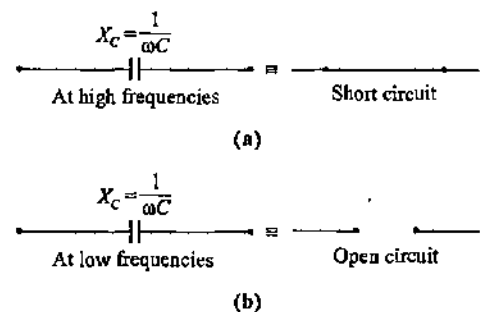


Figure 5.186

5.11.6 LR Series Circuit Across an AC EMF

Figure-5.187 shows a resistor of resistance R in series with an inductor of inductance L connected across an AC EMF. If at any instant current in circuit is i then KVL equation for the loop at that instant is written as

$$-iR - L \frac{di}{dt} + e_0 \sin \omega t = 0 \quad \Rightarrow \quad \frac{di}{dt} + \left(\frac{R}{L} \right) i = \frac{e_0}{L} \sin \omega t \quad \dots (5.160)$$

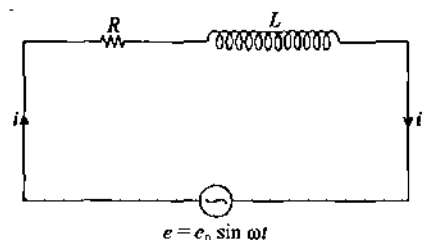


Figure 5.187

Equation-(5.160) is a first order differential equation of which the solution is given as

$$ie^{\int \frac{R}{L} dt} = \int \left(\frac{e_0 \sin \omega t}{L} \right) e^{\int \frac{R}{L} dt} dt + C$$

$$\Rightarrow ie^{\frac{Rt}{L}} = \int \left(\frac{e_0 \sin \omega t}{L} \right) e^{\frac{Rt}{L}} dt + C$$

$$\Rightarrow ie^{\frac{Rt}{L}} = \frac{e_0}{L} \int \sin \omega t \cdot e^{\frac{Rt}{L}} dt + C$$

In above expression we can drop the last term C as it is not a periodic term and it will not contribute in alternating current so considering only alternating current we have

$$ie^{\frac{Rt}{L}} = \frac{e_0}{L} \int \sin \omega t \cdot e^{\frac{Rt}{L}} dt = \frac{e_0}{L} (I_1) \quad \dots (5.161)$$

Where $I_1 = \int \sin \omega t \cdot e^{\frac{Rt}{L}} dt \quad \dots (5.162)$

Above integration I_1 can be solved by-parts as

$$I_1 = \frac{L}{R} \sin \omega t \cdot e^{\frac{Rt}{L}} - \int \omega \cos \omega t \cdot \left(\frac{L}{R} e^{\frac{Rt}{L}} \right) dt$$

$$\Rightarrow I_1 = \frac{L}{R} \sin \omega t \cdot e^{\frac{Rt}{L}} - \frac{\omega L}{R} \int \cos \omega t \cdot e^{\frac{Rt}{L}} dt$$

$$\Rightarrow I_1 = \frac{L}{R} \sin \omega t \cdot e^{\frac{Rt}{L}} - \frac{\omega L}{R} \left[\frac{L}{R} \cos \omega t \cdot e^{\frac{Rt}{L}} + \frac{\omega L}{R} \int \sin \omega t \cdot e^{\frac{Rt}{L}} dt \right]$$

$$\Rightarrow I_1 = \frac{L}{R} \sin \omega t \cdot e^{\frac{Rt}{L}} - \frac{\omega L^2}{R^2} \cos \omega t \cdot e^{\frac{Rt}{L}} - \frac{\omega^2 L^2}{R^2} I_1$$

$$\Rightarrow \left(\frac{R^2 + \omega^2 L^2}{R^2} \right) I_1 = \frac{L}{R} \sin \omega t \cdot e^{\frac{Rt}{L}} - \frac{\omega L^2}{R^2} \cos \omega t \cdot e^{\frac{Rt}{L}}$$

$$\Rightarrow I_1 = \frac{Le^{\frac{Rt}{L}}}{R^2 + \omega^2 L^2} [R \sin \omega t - \omega L \cos \omega t] \quad \dots (5.163)$$

Substituting the value of I_1 from above equation-(5.163) in equation-(5.161) gives

$$ie^{\frac{Rt}{L}} = \left(\frac{Le^{\frac{Rt}{L}}}{R^2 + \omega^2 L^2} [R \sin \omega t - \omega L \cos \omega t] \right)$$

$$\Rightarrow i = \frac{e_0}{R^2 + \omega^2 L^2} [R \sin \omega t - \omega L \cos \omega t] \quad \dots (5.164)$$

In above equation substituting

$$R = A \cos \phi$$

and $\omega L = A \sin \phi$

$$\Rightarrow A = \sqrt{R^2 + \omega^2 L^2} \quad \dots (5.165)$$

and $\phi = \tan^{-1} \left(\frac{\omega L}{R} \right) \quad \dots (5.166)$

Thus above equation-(5.164) can be reduced as

$$i = \frac{e_0}{R^2 + \omega^2 L^2} \left[\sqrt{R^2 + \omega^2 L^2} \sin \left(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \right]$$

$$\Rightarrow i = \frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \sin \left(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \quad \dots (5.167)$$

$$\Rightarrow i = i_0 \sin (\omega t - \phi) \quad \dots (5.168)$$

Where the peak current i_0 in circuit is given as

$$i_0 = \frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \quad \dots (5.169)$$

In above equation the denominator is the total ohmic opposition offered by LR circuit connected across the AC EMF which is called 'Impedance' of the AC circuit which is the resulting ohmic effect of the combination of resistance of resistor and inductive reactance of the inductor. So for a series RL circuit its impedance is given as

$$Z_{RL} = \sqrt{R^2 + \omega^2 L^2}$$

From equation-(5.168) we can see that the alternating current produced in a series LR circuit lags behind AC EMF by a phase angle ϕ which is given by equation-(5.166). As we know due to a pure resistance AC EMF and current remain in same phase whereas due to a pure inductor current lags behind EMF by a phase angle $\pi/2$. So when a pure resistance and an inductor are in series then due to the effect of both the current lags behind AC EMF but by an angle which is less than $\pi/2$.

Soon we will discuss a specific method of solving AC Circuits that is called 'Phasor Analysis' with which in a much simpler way we can determine the alternating current given in equation-(5.167) but students must understand the whole analysis we did in solving a series LR circuit to determine the alternating current in flowing in it.

Figure-5.188 shows the variation of AC EMF and alternating current flowing through a series LR circuit.

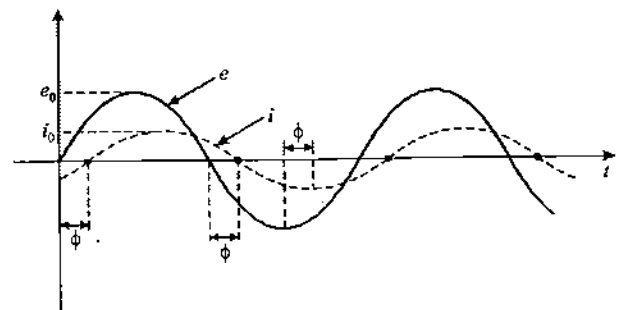


Figure 5.188

5.11.7 RC Series Circuit Across an AC EMF

Figure-5.189 shows a resistor of resistance R in series with a capacitor of capacitance C connected across an AC EMF. If at any instant current in circuit is i then KVL equation for the loop at that instant is written as

$$-iR - \frac{q}{C} + e_0 \sin \omega t = 0$$

Differentiating above equation, we get

$$\begin{aligned} \Rightarrow \quad \frac{di}{dt} R + \left(\frac{1}{C}\right)i &= e_0 \omega \cos \omega t \\ \Rightarrow \quad \frac{di}{dt} + \left(\frac{1}{RC}\right)i &= \frac{e_0 \omega}{R} \cos \omega t \quad \dots (5.170) \end{aligned}$$

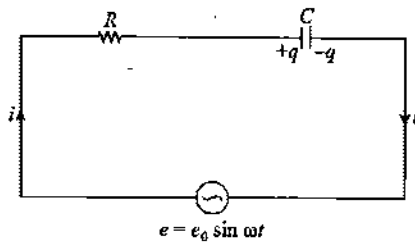


Figure 5.189

Equation-(5.170) is a first order differential equation of which the solution is given as

$$\begin{aligned} i e^{\int \frac{1}{RC} dt} &= \int \left(\frac{e_0 \omega \cos \omega t}{R} \right) e^{\int \frac{1}{RC} dt} dt + C_1 \\ \Rightarrow \quad i e^{\frac{t}{RC}} &= \int \left(\frac{e_0 \omega \cos \omega t}{R} \right) e^{\frac{t}{RC}} dt + C_1 \\ \Rightarrow \quad i e^{\frac{t}{RC}} &= \frac{e_0 \omega}{R} \int \cos \omega t \cdot e^{\frac{t}{RC}} dt + C_1 \end{aligned}$$

In above expression we can drop the last term C_1 as it is not a periodic term and it will not contribute in alternating current so considering only alternating current we have

$$i e^{\frac{t}{RC}} = \frac{e_0 \omega}{R} \int \cos \omega t \cdot e^{\frac{t}{RC}} dt = \frac{e_0 \omega}{R} (I_1) \quad \dots (5.171)$$

$$\text{Where } I_1 = \int \cos \omega t \cdot e^{\frac{t}{RC}} dt \quad \dots (5.172)$$

Above integration I_1 can be solved by-parts as

$$\begin{aligned} I_1 &= RC \cos \omega t \cdot e^{\frac{t}{RC}} + \int \omega \sin \omega t \cdot \left(RC e^{\frac{t}{RC}} \right) dt \\ \Rightarrow \quad I_1 &= RC \cos \omega t \cdot e^{\frac{t}{RC}} + \omega RC \int \sin \omega t \cdot e^{\frac{t}{RC}} dt \\ \Rightarrow \quad I_1 &= RC \cos \omega t \cdot e^{\frac{t}{RC}} + \omega RC \left[RC \sin \omega t \cdot e^{\frac{t}{RC}} - \omega RC \int \cos \omega t \cdot e^{\frac{t}{RC}} dt \right] \end{aligned}$$

$$\begin{aligned} \Rightarrow \quad I_1 &= RC \cos \omega t \cdot e^{\frac{t}{RC}} + \omega R^2 C^2 \sin \omega t \cdot e^{\frac{t}{RC}} - \omega^2 R^2 C^2 I_1 \\ \Rightarrow \quad (1 + \omega^2 R^2 C^2) I_1 &= RC \cos \omega t \cdot e^{\frac{t}{RC}} + \omega R^2 C^2 \sin \omega t \cdot e^{\frac{t}{RC}} \\ \Rightarrow \quad I_1 &= \frac{RC e^{\frac{t}{RC}}}{1 + \omega^2 R^2 C^2} [\cos \omega t + \omega RC \sin \omega t] \quad \dots (5.173) \end{aligned}$$

Substituting the value of I_1 from above equation-(5.173) in equation-(5.171) gives

$$i e^{\frac{t}{RC}} = \frac{e_0 \omega}{R} \left(\frac{RC e^{\frac{t}{RC}}}{1 + \omega^2 R^2 C^2} [\cos \omega t + \omega RC \sin \omega t] \right)$$

$$\Rightarrow \quad i = \frac{e_0 \omega C}{1 + \omega^2 R^2 C^2} [\cos \omega t + \omega RC \sin \omega t] \quad \dots (5.174)$$

In above equation substituting

$$I = A \cos \phi$$

$$\text{and } \omega RC = A \sin \phi$$

$$\Rightarrow \quad A = \sqrt{1 + \omega^2 R^2 C^2} \quad \dots (5.175)$$

$$\text{and } \phi = \tan^{-1} \left(\frac{1}{\omega RC} \right) \quad \dots (5.176)$$

Thus above equation-(5.174) can be reduced as

$$\begin{aligned} \Rightarrow \quad i &= \frac{e_0 \omega C}{1 + \omega^2 R^2 C^2} \left[\sqrt{1 + \omega^2 R^2 C^2} \sin \left(\omega t + \tan^{-1} \left(\frac{1}{\omega RC} \right) \right) \right] \\ \Rightarrow \quad i &= \frac{e_0}{\sqrt{R^2 + 1/\omega^2 C^2}} \sin \left(\omega t + \tan^{-1} \left(\frac{1}{\omega RC} \right) \right) \quad \dots (5.177) \\ \Rightarrow \quad i &= i_0 \sin (\omega t - \phi) \quad \dots (5.178) \end{aligned}$$

Where the peak current i_0 in circuit is given as

$$i_0 = \frac{e_0}{\sqrt{R^2 + 1/\omega^2 C^2}} \quad \dots (5.179)$$

In above equation the denominator is the total ohmic opposition offered by RC circuit connected across the AC EMF which is the resulting ohmic opposition of the combination of resistance of resistor and inductive reactance of the capacitor. So for a series RC circuit its impedance is given as

$$Z_{RL} = \sqrt{R^2 + 1/\omega^2 C^2}$$

From equation-(5.178) we can see that the alternating current produced in a series RC circuit leads AC EMF by a phase angle ϕ which is given by equation-(5.176). As we know due to a pure resistance AC EMF and current remain in same phase whereas

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due to a pure capacitor current leads EMF by a phase angle $\pi/2$. So when a pure resistance and a capacitor are in series then due to the effect of both the current leads AC EMF but by an angle which is less than $\pi/2$.

By using phasor analysis we can simplify the above calculations which we did for calculation of current obtained in equation-(5.177). Figure-5.190 shows the variation of AC EMF and current through a series RC circuit.

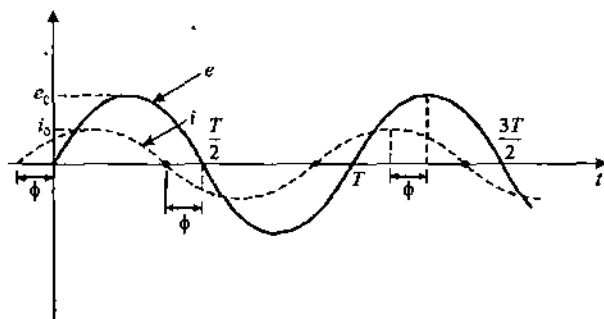


Figure 5.190

Illustrative Example 5.67

A $100 \mu\text{F}$ capacitor in series with a 40Ω resistor is connected to a 110V - 60Hz supply. What is the maximum current in the circuit? What is the time lag between current maxima and voltage maxima?

Solution

The peak current in a RC circuit is given as

$$i_0 = \frac{e_0}{\sqrt{[R^2 + (1/\omega C)^2]}}$$

Peak voltage is given as

$$e_0 = \sqrt{2}e_{\text{rms}} = 1.414 \times 110 = 155.5\text{V}$$

$$\Rightarrow i_0 = \frac{155.5}{\sqrt{[(40)^2 + \left(\frac{1}{376.8 \times 10^{-4}}\right)^2]}}$$

$$\Rightarrow i_0 = 3.24\text{A}$$

In RC circuit, the voltage lags behind the current by phase angle ϕ , where

$$\tan \phi = \frac{(1/\omega C)}{R} = \frac{1}{(376.8 \times 10^{-4})(40)}$$

$$\Rightarrow \tan \phi = 0.6635$$

$$\Rightarrow \phi = \tan^{-1}(0.6635) = 33.56^\circ$$

The time lag is given as

$$t = \frac{\phi}{360} \times T = \frac{\phi}{360} \left(\frac{1}{f} \right)$$

$$\Rightarrow t = \frac{33.56^\circ}{360} \left(\frac{1}{60} \right) = 1.55 \times 10^{-3}\text{s}$$

Illustrative Example 5.68

An electric lamp which runs at 40V and consumes 10A current is connected to AC mains at 100V , 50Hz supply. Calculate the inductance of the required choke for lamp to glow at full brightness.

Solution

Resistance of the lamp is given as

$$R = \frac{V}{I} = \frac{40}{10} = 4\Omega$$

Circuit current is given as

$$I = \frac{E}{\sqrt{[R^2 + (\omega L)^2]}}$$

$$\Rightarrow 10 = \frac{100}{\sqrt{[(4)^2 + (2 \times 3.14 \times 50 \times L)^2]}}$$

$$\Rightarrow L = 0.02916\text{H}$$

Illustrative Example 5.69

In the figure-5.191 shown the current in circuit is $i = 10e^{-4t}$ A. Find V_L and V_{ab} .



Figure 5.191

Solution

The voltage across the inductor is given as

$$V_L = L \frac{di}{dt}$$

$$\Rightarrow V_L = (2) \frac{d}{dt} (10e^{-4t})$$

$$\Rightarrow V_L = -80e^{-4t}$$

Writing equation of potential drop from point a to b gives

$$V_a - iR - L \frac{di}{dt} = V_b$$

$$\Rightarrow V_a - V_b = iR + L \frac{di}{dt}$$

$$\Rightarrow V_{ab} = (10e^{-4t})(4) - 80e^{-4t} = -40e^{-4t}$$

Illustrative Example 5.70

In the figure-5.192 shown, $i_1 = 10e^{-2t}$ A, $i_2 = 4$ A and $V_C = 3e^{-2t}$ V. Find

- (a) i_L and V_L , also plot the time functions of i_L and V_L
 (b) V_{ac} , V_{ab} and V_{cd} , also plot their time dependence curves.

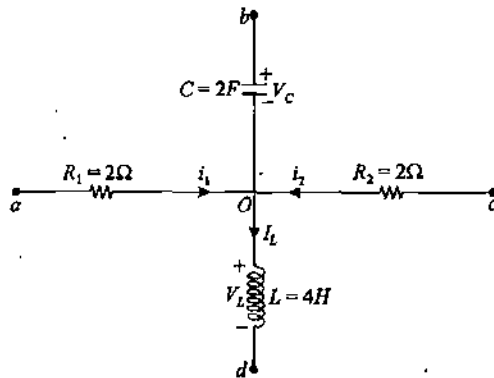


Figure 5.192

Solution

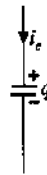
- (a) Charge stored in the capacitor at time t is given as

$$q = CV_C$$

$$q = (2)(3e^{-2t})$$

$$q = 6e^{-2t} \text{ C}$$

$$\Rightarrow i_c = \frac{dq}{dt} = -12e^{-2t} \text{ A}$$



Direction of current is from b to O . Applying KCL at O gives

$$i_L = i_1 + i_2 + i_c = 10e^{-2t} + 4 - 12e^{-2t} \text{ A}$$

$$\Rightarrow i_L = (4 - 2e^{-2t}) \text{ A}$$

$$\Rightarrow i_L = [2 + 2(1 - e^{-2t})] \text{ A}$$

i_L versus time graph is as shown in figure-5.193.

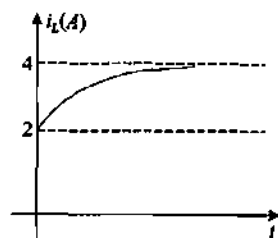


Figure 5.193

$$V_L = V_{od} = L \frac{di_L}{dt}$$

$$\Rightarrow V_L = (4) \frac{d}{dt} (4 - 2e^{-2t}) \text{ V}$$

$$\Rightarrow V_L = 16e^{-2t} \text{ V}$$

V_L versus time graph is as shown in figure-5.194.

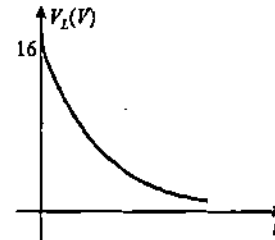


Figure 5.194

- (b) Writing equation of potential drop from point a to c gives

$$V_a - i_1 R_1 + i_2 R_2 = V_c$$

$$\Rightarrow V_a - V_c = V_{ac} = i_1 R_1 - i_2 R_2$$

Substituting the values we get

$$V_{ac} = (10e^{-2t})(2) - (4)(3)$$

$$V_{ac} = (20e^{-2t} - 12) \text{ V}$$

Time dependence curve of V_{ac} is shown in figure-5.195.

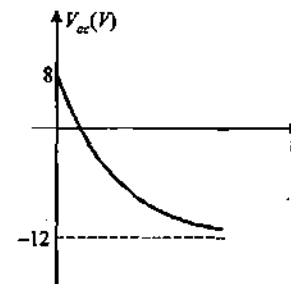


Figure 5.195

Writing equation of potential drop from point a to b gives

$$V_a - i_1 R_1 + V_C = V_b$$

$$\Rightarrow V_a - V_b = i_1 R_1 - V_C$$

Substituting the values we get

$$V_{ab} = (10e^{-2t})(2) - 3e^{-2t} \text{ V}$$

$$\Rightarrow V_{ab} = 17e^{-2t} \text{ V}$$

Time function curve of V_{ab} is shown in figure-5.196

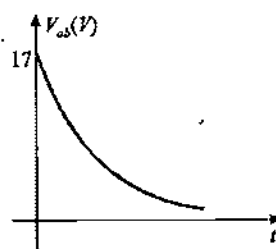


Figure 5.196

Writing equation of potential drop from point c to d gives

$$V_c - i_2 R_2 - V_L = V_d$$

$$\Rightarrow V_c - V_d = i_2 R_2 + V_L$$

Substituting the values we get

$$V_{cd} = (4)(3) + 16e^{-2t} \text{ V}$$

$$\Rightarrow V_{cd} = (12 + 16e^{-2t}) \text{ V}$$

Time function curve of V_{cd} is shown in figure-5.197.

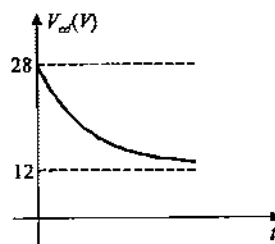


Figure 5.197

Illustrative Example 5.71

When 100V DC is applied across a solenoid, a steady current of 1A flows in it. When 100V AC is applied across the same solenoid, current drops to 0.5A. If the frequency of AC source is $\frac{150\sqrt{3}}{\pi}$ Hz, find the resistance and inductance of the solenoid.

Solution

Steady current through solenoid with DC source is given as

$$I = \frac{\epsilon}{R} = \frac{100}{R} = 1$$

$$\Rightarrow R = 100\Omega$$

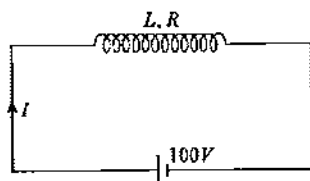


Figure 5.198

When AC is applied the current through the solenoid is given as

$$i = \frac{e}{z} = \frac{e}{\sqrt{R^2 + x_L^2}} = \frac{100}{\sqrt{R^2 + x_L^2}} = 0.5$$

$$\Rightarrow \sqrt{R^2 + x_L^2} = 200$$

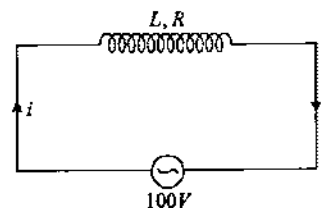


Figure 5.199

$$\Rightarrow x_L^2 = 40000 - 10000 = 30000$$

$$\Rightarrow x_L = 100\sqrt{3}\Omega$$

Inductive reactance is given as

$$x_L = \omega L$$

$$\Rightarrow L = \frac{x_L}{\omega} = \frac{100\sqrt{3}}{2\pi \cdot \frac{150\sqrt{3}}{\pi}} = \frac{1}{3} \text{ H}$$

Illustrative Example 5.72

In a circuit a resistance of 40Ω and capacitor of capacitance $\frac{1250}{9\pi} \mu\text{F}$ are connected in series across a 500V, 120Hz AC source. Find the effective current in circuit and phase difference between current and source EMF.

Solution

Circuit impedance of a RC Circuit is given as

$$\bar{Z} = \bar{R} + \bar{X}_C$$

$$\Rightarrow Z = \sqrt{R^2 + X_C^2}$$

Capacitive reactance is given as

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$\Rightarrow X_C = \frac{9\pi \times 10^6}{2\pi(120)(1250)} \Omega$$

$$\Rightarrow X_C = \frac{3 \times 10^6}{10^5} = 30\Omega$$

Circuit impedance is given as

$$Z = \sqrt{(40)^2 + (30)^2} = 50\Omega$$

Thus circuit current is given as

$$i_{rms} = \frac{e_{rms}}{Z} = \frac{500}{50} = 10A$$

Phase angle of Z phasor is given as

$$\phi = \tan^{-1}\left(\frac{X_C}{R}\right) = \tan^{-1}\left(\frac{30}{40}\right)$$

$$\Rightarrow \phi = \tan^{-1}\left(\frac{3}{4}\right) = 37^\circ$$

As in RC circuit current leads EMF so here current will lead EMF in phase by 37° .

5.12 Phasor Analysis

In the chapter of Simple Harmonic Motion (SHM) we've studied that instantaneous displacement of a particle in SHM can be represented by projection of tip of a rotating vector at uniform angular speed on the diameter of circle in which the tip is moving. The uniform circular motion of the tip can be used to analyze instantaneous displacement and phase of the particle as shown in figure-5.200. Here the displacement of P' which is in SHM can be given as

$$y = A \sin(\omega t + \alpha) \quad \dots (5.180)$$

Where α is the initial phase of the particle.

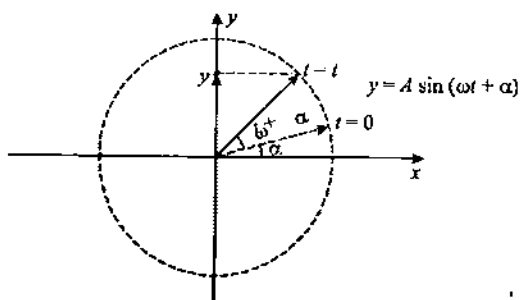


Figure 5.200

Above equation-(5.180) is similar function with which we represent AC EMF and current so this method can also be used to represent such alternating circuit parameters for sinusoidal functions. Such rotating vectors which are used to represent AC circuit parameters are called 'Phasors' and for time function and analysis AC circuits we used these phasors to represent the magnitude and phase of the circuit parameter. In general

phasors are drawn in static manner only. Rotation is considered only when we need to analyze phase at an instant, in general for AC circuits it is not needed. With illustrations taken here it will be more clear.

A phasor is denoted by placing a bar '-' over the symbol of physical quantity being represented along with its initial phase angle. If we consider y in equation-(5.180) represents a physical quantity of an AC circuit then its phasor is written as

$$\bar{y} = L \angle \alpha \quad \dots (5.181)$$

The diagrammatic representation of the static phasor as given by equation-(5.181) is shown in below figure-5.201. For diagrammatic representation the peak value of a phasor is drawn as length of phasor and its phase angle (also called as its 'argument') is the angle from the reference axis in anticlockwise sense at which it is drawn as shown in below figure.

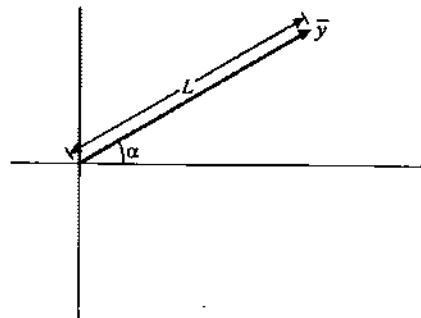


Figure 5.201

If in an AC circuit an AC EMF is used for which time function is given as $e = e_0 \sin \omega t$ then the phasor form of this AC EMF is written as

$$\bar{e} = e_0 \angle 0^\circ \quad \dots (5.182)$$

The diagrammatic representation of the EMF phasor as given by equation-(5.182) is shown in below figure-5.202.

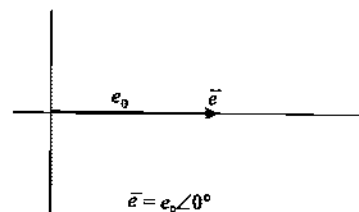


Figure 5.202

As discussed in previous articles generally we consider EMF as a reference and in different AC circuits we determine the phase lag or lead of current in the circuits so for an alternating

current $i = i_0 \sin(\omega t + \phi)$ its phasor form is written as

$$\vec{i} = i_0 \angle \phi \quad \dots (5.183)$$

The diagrammatic representation of the current phasor as given by equation-(5.183) is shown in below figure-5.203.

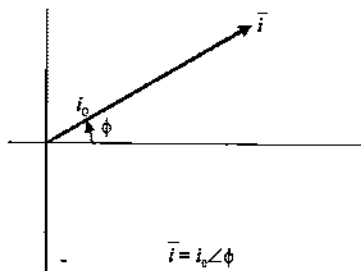


Figure 5.203

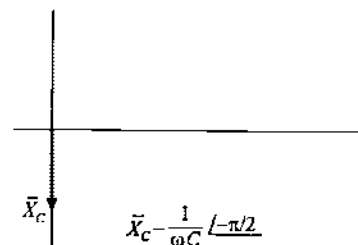


Figure 5.205

(iii) **Standard Phasor for Inductive Reactance :** For a pure inductor of inductance L its phasor for inductive reactance is written as

$$\vec{X}_L = \omega L \angle (+\pi/2) \quad \dots (5.186)$$

Diagrammatically it is represented as shown in figure-5.206.

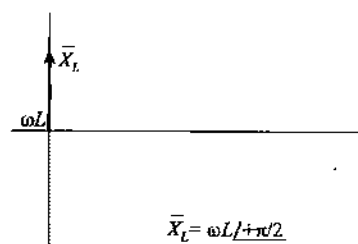


Figure 5.206

5.12.1 Standard Phasors for Resistances and Reactances in AC Circuits

In any AC circuit EMF and current are time varying functions for which phasors can be considered as rotating vectors like SHM if the time phase analysis of these is required. For mathematical analysis of AC circuits we also define standard phasors for resistances, capacitors and inductors with standard values for their magnitude and phasor angles. As resistances and reactances in a given AC circuit are not time varying so their vectors are never considered as rotating vectors. Standard phasors corresponding to these are given below.

(i) **Standard Phasor for a Resistance :** For a pure resistor of resistance R its phasor is written as

$$\vec{R} = R \angle 0^\circ \quad \dots (5.184)$$

Diagrammatically it is represented as shown in figure-5.204.

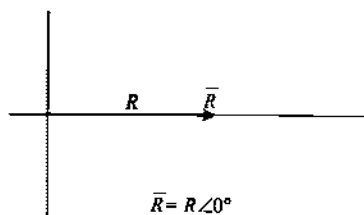


Figure 5.204

(ii) **Standard Phasor for Capacitive Reactance :** For a pure capacitor of capacitance C its phasor for capacitive reactance is written as

$$\vec{X}_C = \frac{1}{\omega C} \angle (-\pi/2) \quad \dots (5.185)$$

Diagrammatically it is represented as shown in figure-5.205.

In above equations-(5.184), (5.185) and (5.186) many times students get confuse about the phasor angles taken. Here the above values defined for phasor angles are standard values which are defined to analyze and solve AC circuits so one should not think about the derivation or approach of these values. With different illustrations of these standard phasor values applications can be understood at good level.

5.12.2 Phasor Algebra

For calculations in different AC circuits some rules are defined according to which different circuit parameters are used in mathematical operations, set of such rules for solving and analysis of AC circuit studied under 'Phasor Algebra'. We will discuss some basic rules and applications in solving simple AC circuits here. For basic mathematical operations these rules are listed below in pointwise manner.

(i) **Phasors as Complex Numbers :** For mathematical calculations phasors are treated like complex numbers with magnitude and argument and the diagrammatic representation of a phasor can be considered the representation of a complex number on argand plane with horizontal and vertical axes as real and imaginary axis of the argand plane. A specific phasor

$\bar{z}$ with its magnitude z_0 and argument α can be written as

$$\bar{z} = z_0 \angle \alpha \quad \dots (5.187) \Rightarrow$$

Above phasor if treated as complex number its diagrammatic representation on argand plane is shown in figure-5.207

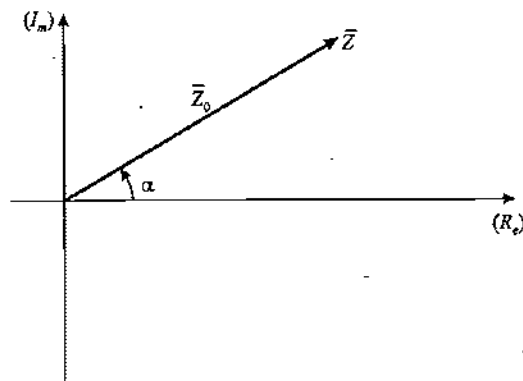


Figure 5.207

In complex form equation-(5.187) can be written as

$$\bar{z} = z_0 \cos \alpha + j z_0 \sin \alpha \quad \dots (5.188)$$

In above equation-(5.188) first term $z_0 \cos \alpha$ is the real part of the complex number and $z_0 \sin \alpha$ is the imaginary part of the complex number and $j = \sqrt{-1}$.

(ii) **Addition of Two Phasors :** When two phasors $\bar{z}_1$ and $\bar{z}_2$ are given as

$$\bar{z} = z_{10} \angle \alpha_1 \quad \dots (5.189)$$

$$\text{and} \quad \bar{z}_2 = z_{20} \angle \alpha_2 \quad \dots (5.190)$$

Sum of the above two phasors can be done diagrammatically as shown in figure-5.208.

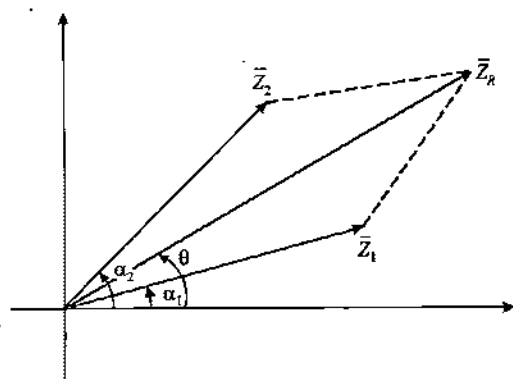


Figure 5.208

Thus the resultant phasor which is the sum of the two phasors is given as

$$\bar{z}_R = \bar{z}_1 + \bar{z}_2$$

$$\bar{z}_R = z_{10} \angle \alpha_1 + z_{20} \angle \alpha_2$$

$$\bar{z}_R = |\bar{z}_R| \angle \theta$$

Where $|\bar{z}_R|$ is the magnitude of resultant phasor and θ is its argument angle which can be calculated from figure-5.208 given as

$$|\bar{z}_R| = \sqrt{z_{10}^2 + z_{20}^2 + 2z_{10}z_{20} \cos(\alpha_2 - \alpha_1)} \quad \dots (5.191)$$

$$\text{and} \quad \theta = \tan^{-1} \left(\frac{z_{10} \cos \alpha_1 + z_{20} \cos \alpha_2}{z_{10} \sin \alpha_1 + z_{20} \sin \alpha_2} \right) \quad \dots (5.192)$$

In the same we can calculate if two phasors are subtracted by changing the direction of second phases and adding it to the first. It is done in the same way we add or subtract vectors or complex numbers diagrammatically.

(iii) **Product of Two Phasors :** When two phasors $\bar{z}_1$ and $\bar{z}_2$ as given by equations-(5.189) and (5.190) are multiplied then in the product phasor will have magnitude given by the product of their individual magnitudes and its argument angle will be given by the sum of the argument angles of the individual phasors as it is done in product of complex numbers.

$$\bar{z}_P = \bar{z}_1 \cdot \bar{z}_2$$

$$\Rightarrow \bar{z}_P = z_{10} \angle \alpha_1 \times z_{20} \angle \alpha_2$$

$$\Rightarrow \bar{z}_P = (z_{10} \times z_{20}) \angle (\alpha_1 + \alpha_2) \quad \dots (5.193)$$

(iv) **Division of Two Phasors :** When two phasors $\bar{z}_1$ and $\bar{z}_2$ as given by equations-(5.189) and (5.190) are divided then in the resultant phasor it will have magnitude given by the division of their individual magnitudes and its argument angle will be given by the difference of the argument angles of the individual phasors as it is done in case of division of complex numbers.

$$\bar{z}_P = \frac{\bar{z}_1}{\bar{z}_2}$$

$$\Rightarrow \bar{z}_P = \frac{z_{10} \angle \alpha_1}{z_{20} \angle \alpha_2}$$

$$\Rightarrow \bar{z}_P = \left(\frac{z_{10}}{z_{20}} \right) \angle (\alpha_1 - \alpha_2) \quad \dots (5.194)$$

With the above basic rules we can analyze all types of simple AC Circuits however the above calculations can also be done by considering phasors in complex notation form as given in equation-(5.188) and for calculations real and imaginary parts can be separated and finally given again in form of magnitude argument form which can be transformed as time function whenever needed.

In next upcoming articles we will re-analyze the cases of AC Circuits in which different circuit components are connected across an AC EMF by using phasor analysis.

5.12.3 Phasor Analysis of a Resistance Across an AC EMF

Figure-5.209 shows a resistor resistance R is connected across an AC EMF. Using Ohm's law the phasor current through the circuit is given as

$$\bar{i} = \frac{\bar{e}}{R}$$

$$\Rightarrow \bar{i} = \frac{e_0 \angle 0^\circ}{R \angle 0^\circ}$$

$$\Rightarrow \bar{i} = \frac{e_0}{R} \angle 0^\circ = i_0 \angle 0^\circ \quad \dots (5.195)$$

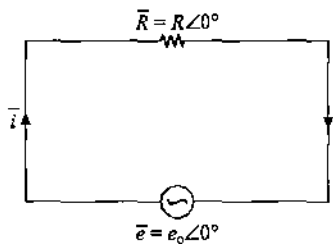


Figure 5.209

Using equation-(5.195) we can write the time function of the phasor current which is given as

$$i = i_0 \sin(\omega t + 0^\circ) \quad \dots (5.196)$$

Where $i_0 = \frac{e_0}{R} \quad \dots (5.197)$

Above equations are same as equations-(5.145) and (5.146) as discussed in article-5.11.1 but here these are obtained by using phasor analysis. Figure-5.210 shows the phasor diagram of AC EMF and current through a pure resistance.

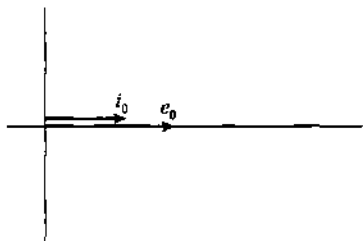


Figure 5.210

5.12.4 Phasor Analysis of a Capacitor Across an AC EMF

Figure-5.211 shows a capacitor of capacitance C connected across an AC EMF. Using Ohm's law in phasor form, phasor current in circuit is given as

$$\bar{i} = \frac{\bar{e}}{\bar{X}_C}$$

$$\bar{i} = \frac{e_0 \angle 0^\circ}{(1/\omega C) \angle (-\pi/2)}$$

$$\bar{i} = \frac{e_0}{1/\omega C} \angle (+\pi/2) = i_0 \angle (+\pi/2) \quad \dots (5.198)$$

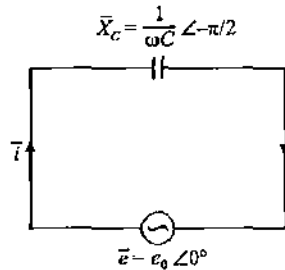


Figure 5.211

Using equation-(5.198) we can write the time function of the phasor current which is given as

$$i = i_0 \sin(\omega t + \pi/2) \quad \dots (5.199)$$

Where $i_0 = \frac{e_0}{1/\omega C} = \frac{e_0}{X_C} \quad \dots (5.200)$

Above equations are same as equations-(5.149) and (5.150) as discussed in article-5.11.2 but here these are obtained by using phasor analysis. Figure-5.212 shows the phasor diagram of AC EMF and current through a pure capacitance.

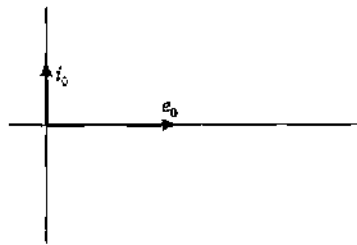


Figure 5.212

5.12.5 Phasor Analysis of an Inductor Across an AC EMF

Figure-5.213 shows an inductor of inductance L which is connected across an AC EMF. Using Ohm's law the phasor current through the circuit is given as

$$\bar{i} = \frac{\bar{e}}{\bar{X}_L}$$

$$\bar{i} = \frac{e_0 \angle 0^\circ}{\omega L \angle (+\pi/2)}$$

$$\bar{i} = \frac{e_0}{\omega L} \angle (-\pi/2) = i_0 \angle (-\pi/2) \quad \dots (5.201)$$

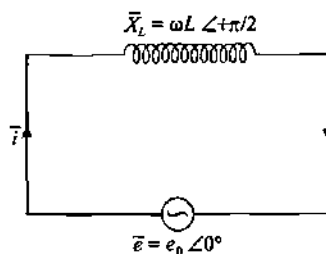


Figure 5.213

Using equation-(5.201) we can write the time function of the phasor current which is given as

$$i = i_0 \sin(\omega t - \pi/2) \quad \dots (5.202)$$

Where
$$i_0 = \frac{e_0}{\omega L} \quad \dots (5.203)$$

Above equations are same as equations-(5.154) and (5.155) as discussed in article-5:11.3 but here these are obtained by using phasor analysis. Figure-5.214 shows the phasor diagram of AC EMF and current through a pure inductor.

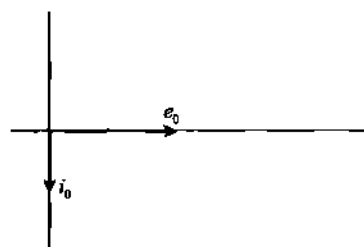


Figure 5.214

5.12.6 Impedance of an AC Circuit

We've already discussed in brief that in an AC circuit total ohmic opposition offered by the circuit including its all resistances and reactances is called 'Impedance' of the circuit and it is denoted by 'Z' and its phasor is denoted by $\bar{Z}$ and measured in unit of 'ohm'. In a circuit impedance is caused by either of its resistances or reactances. For purely resistive, capacitive and inductive circuits impedances are given as

Purely Resistive Impedance $\bar{Z} = \bar{R}$

Purely Capacitive Impedance $\bar{Z} = \bar{X}_C$

Purely Inductive Impedance $\bar{Z} = \bar{X}_L$

When different types of circuit components are connected in series or parallel then we calculate impedance of the circuit by using series and parallel combination of these components in the same way we used to calculate equivalent resistance for DC circuits. The only difference in case of AC circuits is that instead

of simple calculations we use rules of phasor algebra for calculations which will be clearly understood with the help of some illustrations. Figure-5.215 shows series combination of a resistor of resistance R with an inductor of inductance L. Here the impedance of the circuit across terminals A and B is given as

$$\bar{Z}_{AB} = \bar{R} + \bar{X}_L$$

$$\Rightarrow \bar{Z}_{AB} = R \angle 0^\circ + \omega L \angle (+\pi/2) \quad \dots (5.204)$$

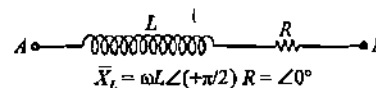


Figure 5.215

Above equation can be solve diagrammatically as shown in figure-5.216 where we add resistance phasor and inductive reactance phasor like vectors or complex numbers.

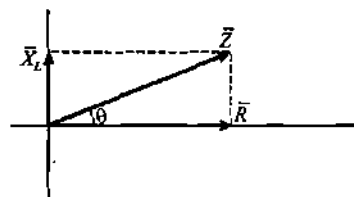


Figure 5.216

From above figure-5.216, the magnitude of impedance phasor is given as

$$Z = \sqrt{R^2 + X_L^2}$$

$$\Rightarrow Z = \sqrt{R^2 + \omega^2 L^2} \quad \dots (5.205)$$

The argument of the impedance phasor is given as

$$\theta = \tan^{-1} \left(\frac{\omega L}{R} \right) \quad \dots (5.206)$$

Thus the impedance of the series LR circuit is given as

$$\bar{Z}_{AB} = \sqrt{R^2 + \omega^2 L^2} \angle \left(\tan^{-1} \left(\frac{\omega L}{R} \right) \right) \quad \dots (5.207)$$

Consider another circuit shown in figure-5.217 in which a resistor or resistance R is connected in parallel with an inductor of inductance L. In this situation the impedance of the circuit is given as

$$\frac{1}{\bar{Z}_{AB}} = \frac{1}{\bar{R}} + \frac{1}{\bar{X}_L}$$

$$\Rightarrow \frac{1}{\bar{Z}_{AB}} = \frac{1}{R \angle 0^\circ} + \frac{1}{\omega L \angle (+\pi/2)}$$

$$\Rightarrow \frac{1}{\bar{Z}_{AB}} = \frac{1}{R} \angle 0^\circ + \frac{1}{\omega L} \angle (-\pi/2) \quad \dots (5.208)$$

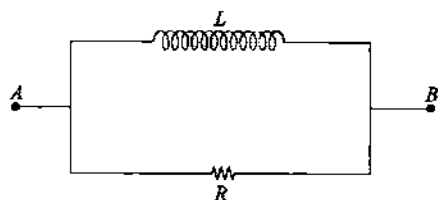


Figure 5.217

Above equation can be diagrammatically solved for adding the two terms on the RHS of the equation as shown in figure-5.218.

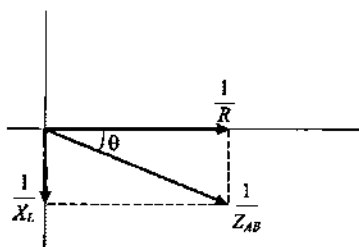


Figure 5.218

From the above figure the magnitude of the reciprocal of impedance phasor is given as

$$\left| \frac{1}{\bar{Z}_{AB}} \right| = \sqrt{\left(\frac{1}{R} \right)^2 + \left(\frac{1}{\omega L} \right)^2}$$

$$\Rightarrow \left| \frac{1}{\bar{Z}_{AB}} \right| = \frac{\sqrt{R^2 + \omega^2 L^2}}{R\omega L} \quad \dots (5.209)$$

The argument of the reciprocal of impedance phasor is given as

$$\theta = -\tan^{-1} \left(\frac{1/\omega L}{1/R} \right) = -\tan^{-1} \left(\frac{R}{\omega L} \right) \quad \dots (5.210)$$

Thus reciprocal of impedance phasor is given as

$$\frac{1}{\bar{Z}_{AB}} = \frac{\sqrt{R^2 + \omega^2 L^2}}{R\omega L} \angle \left(-\tan^{-1} \left(\frac{R}{\omega L} \right) \right)$$

$$\Rightarrow \bar{Z}_{AB} = \frac{R\omega L}{\sqrt{R^2 + \omega^2 L^2}} \angle \left(\tan^{-1} \left(\frac{R}{\omega L} \right) \right) \quad \dots (5.211)$$

Above equation-(5.211) gives the impedance of the parallel LR circuit shown in figure-5.217.

For any given AC circuit in which different circuit components are connected in some combination if it is connected across an AC EMF then alternating current supplied by the EMF can be given as

$$\bar{i} = \frac{\bar{e}}{\bar{Z}} \quad \dots (5.212)$$

5.12.7 Phasor Analysis of Series RL Circuit Across AC EMF

Figure-5.219 shows a series RL circuit connected across an AC EMF. As already discussed in previous article, circuit impedance in this case across AC EMF is given as

$$\bar{Z}_{AB} = \bar{R} + \bar{X}_L$$

$$\Rightarrow \bar{Z}_{AB} = R \angle 0^\circ + \omega L \angle (+\pi/2)$$

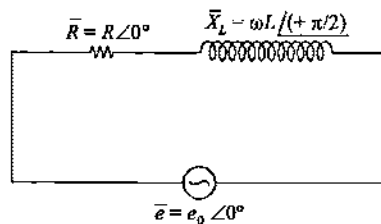


Figure 5.219

Above equation is solved graphically as shown in figure-5.216 which gives

$$\bar{Z}_{AB} = \sqrt{R^2 + \omega^2 L^2} \angle \left(\tan^{-1} \left(\frac{\omega L}{R} \right) \right) \quad \dots (5.213)$$

The current supplied by the AC EMF is given as

$$\bar{i} = \frac{\bar{e}}{\bar{Z}}$$

$$\Rightarrow \bar{i} = \frac{e_0 \angle 0^\circ}{\sqrt{R^2 + \omega^2 L^2} \angle \left(\tan^{-1} \left(\frac{\omega L}{R} \right) \right)}$$

$$\Rightarrow \bar{i} = \frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \angle \left(-\tan^{-1} \left(\frac{\omega L}{R} \right) \right) \quad \dots (5.214)$$

Current phasor as given in equation-(5.214) can be written in time function as

$$\bar{i} = \frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \sin \left(\omega t - \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \quad \dots (5.215)$$

$$\Rightarrow i = i_0 \sin(\omega t - \phi) \quad \dots (5.216)$$

$$\text{Where } i_0 = \frac{e_0}{\sqrt{R^2 + \omega^2 L^2}} \text{ and } \phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

Above equation-(5.215) is same as equation-(5.167) which was obtained by solving long calculations using Kirchhoff's law whereas in this case it is simply obtained by using phasor analysis without solving any differential equation and integration.

5.12.8 Phasor Analysis of Series RC Circuit Across AC EMF

Figure-5.220 shows a series LR circuit connected across an AC EMF. The circuit impedance in this case across AC EMF is given as

$$\bar{Z}_{AB} = \bar{R} + \bar{X}_C$$

$$\Rightarrow \bar{Z}_{AB} = R \angle 0^\circ + \frac{1}{\omega C} \angle (-\pi/2)$$

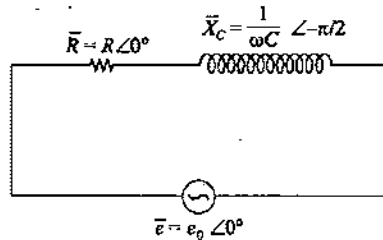


Figure 5.220

Above equation is solved graphically as shown in figure-5.221 which gives

$$\bar{Z}_{AB} = \sqrt{R^2 + \frac{1}{\omega^2 C^2}} \angle \left(-\tan^{-1} \left(\frac{1}{R\omega C} \right) \right) \quad \dots (5.217)$$

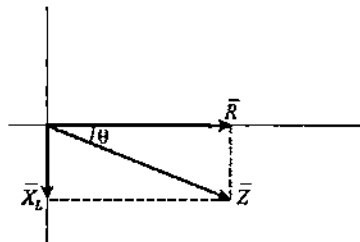


Figure 5.221

The current supplied by the AC EMF is given as

$$\bar{I} = \frac{\bar{e}}{\bar{Z}}$$

$$\Rightarrow \bar{I} = \frac{e_0 \angle 0^\circ}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}} \angle \left(-\tan^{-1} \left(\frac{1}{R\omega C} \right) \right)}$$

$$\Rightarrow \bar{I} = \frac{e_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \angle \left(\tan^{-1} \left(\frac{1}{R\omega C} \right) \right) \quad \dots (5.218)$$

Current phasor as given in equation-(5.218) can be written in time function as

$$i = \frac{e_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \sin \left(\omega t + \tan^{-1} \left(\frac{1}{R\omega C} \right) \right) \quad \dots (5.219)$$

$$\Rightarrow i = i_0 \sin(\omega t + \phi) \quad \dots (5.220)$$

$$\text{Where } i_0 = \frac{e_0}{\sqrt{R^2 + \frac{1}{\omega^2 C^2}}} \text{ and } \phi = \tan^{-1} \left(\frac{1}{R\omega C} \right)$$

Above equation-(5.219) is same as equation-(5.177) which was obtained by solving long calculations using Kirchhoff's law whereas in this case it is simply obtained by using phasor analysis without solving any differential equation and integration.

5.12.9 Phasor Analysis of Series LC Circuit Across AC EMF

Figure-5.222 shows a series LC circuit connected across an AC EMF. The circuit impedance in this case across AC EMF is given as

$$\bar{Z}_{AB} = \bar{X}_L + \bar{X}_C$$

$$\Rightarrow \bar{Z}_{AB} = \omega L \angle (+\pi/2) + \frac{1}{\omega C} \angle (-\pi/2)$$

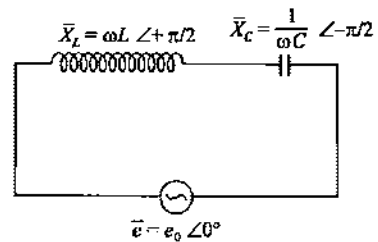


Figure 5.222

Above equation is solved graphically as shown in figure-5.223. Considering $\omega L > 1/\omega C$ gives

$$\bar{Z}_{AB} = \left(\omega L - \frac{1}{\omega C} \right) \angle (+\pi/2) \quad \dots (5.221)$$

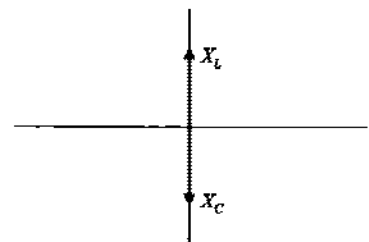


Figure 5.223

The current supplied by the AC EMF is given as

$$\bar{I} = \frac{\bar{e}}{\bar{Z}}$$

$$\Rightarrow \bar{I} = \frac{e_0 \angle 0^\circ}{\left(\omega L - \frac{1}{\omega C} \right) \angle (+\pi/2)}$$

$$\Rightarrow \bar{I} = \frac{e_0}{\left(\omega L - \frac{1}{\omega C} \right)} \angle (-\pi/2) \quad \dots (5.222)$$

Current phasor as given in equation-(5.222) can be written in time function as

$$i = \frac{e_0}{\left(\omega L - \frac{1}{\omega C}\right)} \sin\left(\omega t - \frac{\pi}{2}\right) \quad \dots (5.223)$$

$$\Rightarrow i = i_0 \sin(\omega t - \pi/2) \quad \dots (5.224)$$

Where $i_0 = \frac{e_0}{\left(\omega L - \frac{1}{\omega C}\right)}$

In case of an inductor in series with a capacitor across AC EMF always current is either lagging or leading in phase by an angle $\pi/2$ with EMF as phasors of capacitive or inductive reactance are in opposite directions. If capacitive reactance is more then current will lead EMF and if inductive reactance is more then current will lag behind EMF.

5.12.10 Series RLC Circuit Across AC EMF

Figure-5.224 shows a circuit in which R, L and C are connected in series across an AC EMF. In this case the circuit impedance is given as

$$\bar{Z}_{AB} = \bar{R} + \bar{X}_L + \bar{X}_C$$

$$\Rightarrow \bar{Z}_{AB} = R \angle 0^\circ + \omega L \angle (+\pi/2) + \frac{1}{\omega C} \angle (-\pi/2)$$

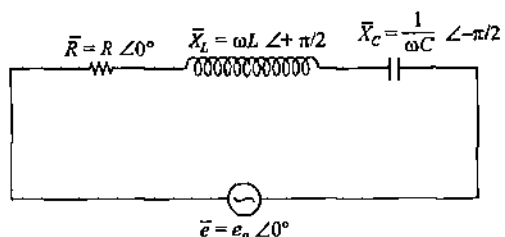


Figure 5.224

Above equation is solved graphically as shown in figure-5.225 in which we consider $\omega L > 1/\omega C$ which gives

$$\bar{Z}_{AB} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \quad \dots (5.225)$$

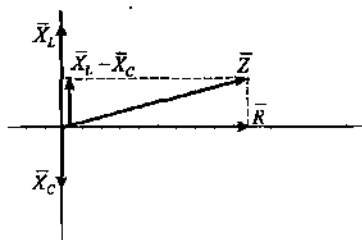


Figure 5.225

The current supplied by the AC EMF in above case is given as

$$\begin{aligned} \bar{i} &= \frac{\bar{e}}{\bar{Z}} \\ \Rightarrow \bar{i} &= \frac{e_0 \angle 0^\circ}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \left[\tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \right]} \\ \Rightarrow \bar{i} &= \frac{e_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \angle \left[-\tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \right] \quad \dots (5.226) \end{aligned}$$

Using above equation-(5.226) which gives the phasor current in series RLC circuit, we can write the time function of the current which is given as

$$i = \frac{e_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}} \sin \left[\omega t - \tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \right] \quad \dots (5.227)$$

5.12.11 Phase Relations in Series RLC Circuit

Figure-5.226 shows a series RLC components in a branch of an AC circuit in which a current i flows. The impedance of the branch across terminals A and B is given as

$$\bar{Z}_{AB} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} \angle \tan^{-1} \left(\frac{\omega L - \frac{1}{\omega C}}{R} \right) \quad \dots (5.228)$$

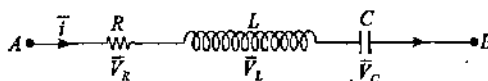


Figure 5.226

The potential difference across terminals A and B is given by Ohm's law as

$$\bar{V}_{AB} = \bar{i} \cdot \bar{Z}$$

As in above equation-(5.228) we considered $\omega L > 1/\omega C$ so in the circuit current will be lagging behind the potential difference. Actually the phase relation between the current and potential difference depends upon the frequency of AC EMF in above circuit. There are three cases in different situations.

Case-I : When $\omega L = \frac{1}{\omega C} \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

In this case $\theta = 0$ where θ is the phase with which current will lag behind the potential difference across terminals A and B or

at the above frequency current and potential difference will be in same phase.

Case-II : When $\omega L > \frac{1}{\omega C} \Rightarrow \omega > \frac{1}{\sqrt{LC}}$

In this case θ is positive and current will lag behind the potential difference across terminals A and B .

Case-III : When $\omega L < \frac{1}{\omega C} \Rightarrow \omega < \frac{1}{\sqrt{LC}}$

In this case θ is negative thus current will lead the potential difference across terminals A and B .

If we analyze the potential differences across each of R , L and C and these are given as $\bar{V}_R$, $\bar{V}_L$ and $\bar{V}_C$ then we use

$$\bar{V}_{AB} = \bar{V}_R + \bar{V}_L + \bar{V}_C \quad \dots (5.229)$$

As current $\bar{i}$ through all the three components is same the potential difference across resistance $\bar{V}_R$ is in same phase with this current, potential difference across inductor $\bar{V}_L$ will lead this current by an angle $\pi/2$ and the potential difference across the resistance $\bar{V}_C$ will lag behind this current by an angle $\pi/2$.

Figure-5.227 shows the phasors of current and potential differences in this circuit branch and by this we can write the relation in magnitudes of all the potential difference mentioned in equation-(5.229) which is in phasor form.

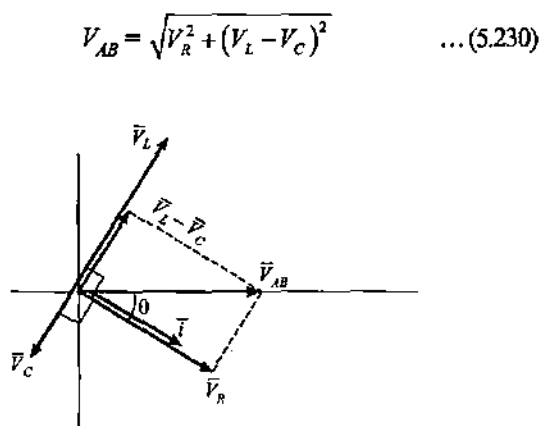


Figure 5.227

Illustrative Example 5.73

A 12Ω resistance and an inductance of $0.05/\pi$ H with negligible resistance are connected in series. Across the end of this circuit is connected a 130V alternating voltage of frequency 50Hz. Calculate the alternating current in the circuit and potential difference across the resistance and that across the inductance.

Solution

The impedance of the circuit is given as

$$Z = \sqrt{R^2 + \omega^2 L^2}$$

$$\Rightarrow Z = \sqrt{R^2 + (2\pi fL)^2}$$

$$\Rightarrow Z = \sqrt{[(12)^2 + \{2 \times 3.14 \times 50 \times (0.05/3.14)\}^2]}$$

$$\Rightarrow Z = \sqrt{(144 + 25)} = 13\Omega$$

Current in the circuit is given as

$$i = \frac{E}{Z} = \frac{130}{13} = 10A$$

Potential difference across resistance is given as

$$V_R = iR = 10 \times 12 = 120V$$

Inductive reactance of coil is given as

$$X_L = \omega L = 2\pi fL$$

$$\Rightarrow X_L = 2\pi \times 50 \times \left(\frac{0.05}{\pi}\right) = 5\Omega$$

Potential difference across inductance is given as

$$V_L = i \times X_L = 10 \times 5 = 50V$$

Illustrative Example 5.74

An AC source of angular frequency ω is fed across resistor R and a capacitor C in series. The current registered is i . If now the frequency of the source is changed to $\omega/3$ but maintaining the same voltage, the current in the circuit is found to be halved. Calculate the ratio of reactance to resistance at the original frequency.

Solution

At angular frequency ω , the current in RC circuit is given as

$$i_{rms} = \frac{E_{rms}}{\sqrt{R^2 + (1/\omega^2 C^2)}} \quad \dots (5.231)$$

When frequency is changed to $\omega/3$, the current is halved so we have

$$\frac{i_{rms}}{2} = \frac{E_{rms}}{\sqrt{R^2 + 1/(\omega/3)^2 C^2}}$$

$$\Rightarrow \frac{i_{rms}}{2} = \frac{E_{rms}}{\sqrt{R^2 + (9/\omega^2 C^2)}} \quad \dots (5.232)$$

From equation-(5.231) and (5.232), we have

$$\frac{1}{\sqrt{R^2 + (1/\omega^2 C^2)}} = \frac{2}{\sqrt{R^2 + (9/\omega^2 C^2)}}$$

$$\Rightarrow 3R^2 = \frac{5}{\omega^2 C^2}$$

Thus the ratio of reactance to resistance is given as

$$\frac{X_C}{R} = \frac{(1/\omega C)}{R} = \sqrt{\left(\frac{3}{5}\right)}$$

Illustrative Example 5.75

In a series RL circuit with $L = \frac{175}{11}$ mH and $R = 12\Omega$ an AC source of emf $e = 130\sqrt{2}$ V, 50Hz is applied. Find the circuit impedance and phase difference of EMF and current in circuit.

Solution

For a series RL circuit, impedance is given as

$$\bar{Z} = \bar{R} + \bar{X}_L$$

$$\Rightarrow Z = \sqrt{R^2 + X_L^2}$$

Inductive reactance is given as

$$X_L = \omega L = 2\pi fL$$

$$\Rightarrow X_L = 2 \cdot \left(\frac{2L}{7}\right) \cdot 50 \frac{175}{11} m$$

$$\Rightarrow X_L = 5000 \times 10^{-5} = 5\Omega$$

Circuit impedance is calculated as

$$Z = \sqrt{(12)^2 + (5)^2}$$

$$\Rightarrow Z = \sqrt{144 + 25} = \sqrt{169} = 13\Omega$$

With phase angle of impedance calculated from phasor diagram, given as

$$\phi = \tan^{-1} \left(\frac{X_L}{R} \right)$$

$$\Rightarrow \phi = \tan^{-1} \left(\frac{5}{12} \right)$$

Illustrative Example 5.76

A coil has inductance $2.2/\pi$ H and is joined in series with a resistance of 220Ω . An AC EMF 220V, 50 Hz is applied to it, find the wattless current in circuit.

Solution

In series RL circuit, impedance is given as

$$Z = \sqrt{R^2 + X_L^2}$$

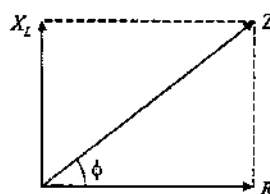


Figure 5.228

Inductive reactance is given as

$$X_L = \omega L$$

$$\Rightarrow X_L = 2\pi(50) \times \frac{2.2}{\pi}$$

$$\Rightarrow X_L = 220\Omega$$

As $X_L = R$ we have $\phi = 45^\circ$

Circuit current is given as

$$i = \frac{e}{Z} = \frac{220}{220\sqrt{2}} = \frac{1}{\sqrt{2}} A$$

Wattless current in AC circuit is given as

$$i_{WL} = i \sin \phi$$

$$\Rightarrow i_{WL} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2} = 0.5A$$

Illustrative Example 5.77

A 200 km long telegraph wire has a capacitance of $0.014 \mu\text{F}/\text{km}$. If it carries an alternating current of 50kHz, what should be the value of an inductance required to be connected in series so that impedance is minimum.

Solution

The total capacitance of telegraph line is given as

$$C = 0.014 \times 200 = 2.8 \mu\text{F} = 2.8 \times 10^{-6}\text{F}$$

For minimum impedance, we have

$$\omega L = \frac{1}{\omega C}$$

$$\Rightarrow L = \frac{1}{\omega^2 C} = \frac{1}{(2\pi f)^2 C} = \frac{1}{4\pi^2 f^2 C}$$

$$\Rightarrow L = \frac{1}{4 \times (3.14)^2 \times (50 \times 10^3)^2 \times (2.8 \times 10^{-6})} \text{H}$$

$$\Rightarrow L = 0.36 \times 10^{-3} = 0.36\text{mH}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Alternating Current

Module Number - 14 to 33

Practice Exercise 5.6

(i) A series circuit consists of a resistance of 15Ω , an inductance of 0.08H and a capacitor of capacitance $30\mu\text{F}$. The applied voltage has a frequency of 500 rad/s . Does the current lead or lag the applied voltage and by what angle.

[60.65° lead]

(ii) An 220V AC voltage at a frequency of 40 cycles/s is applied to a circuit containing a pure inductance of 0.01H and a pure resistance of 6Ω in series. Calculate

- The current supplied by source
- The potential difference across the resistance
- The potential difference across the inductance
- The time lag between maxima of current and EMF in circuit

[(a) 33.83mA (b) 202.98V (c) 96.83V (d) 0.01579s]

(iii) A box P and a coil Q are connected in series with an AC source of variable frequency. The EMF of source is constant at 10V . Box P contains a capacitance of $1\mu\text{F}$ in series with a resistance of 32Ω . Coil Q has a self inductance 4.9mH and a resistance of 68Ω . The frequency is adjusted so that the maximum current flows in P and Q . Find the impedance of P and Q at this frequency. Also find the voltage across P and Q respectively.

[76Ω , 98Ω , 7.6V , 9.8V]

(iv) An AC source is connected to two circuits as shown in figure-5.229. Obtain the current through resistance R at resonance in both the circuits.

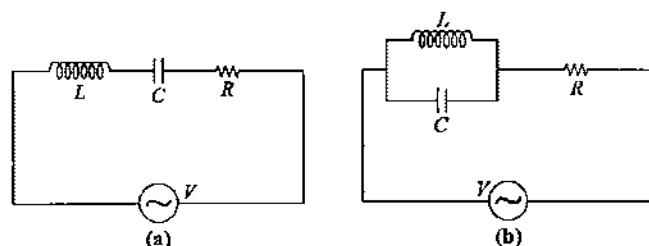


Figure 5.229

[$\frac{V}{R}$, 0]

(v) A resistance R , and inductance L and a capacitor C are all connected in series with an AC supply. The resistance is of 16Ω and for a given frequency, the inductive reactance of L is 24Ω and capacitive reactance of C is 12Ω . If the current in the circuit is 5A , find

- The potential difference across R , L and C
- The impedance of the circuit
- The voltage of AC supply
- The phase angle between current and voltage

[(a) 80V , 120V , 60V (b) 20Ω (c) 100V (d) 37°]

(vi) A series circuit consists of a resistance, inductance and capacitance. The applied voltage and the current at any instant are given as

$$e = 141.4 \cos(300t - 10^\circ)$$

and $i = 5 \cos(3000t - 55^\circ)$

The inductance is 0.01H . Calculate the values of the resistance and capacitance.

[$33.33\mu\text{F}$]

(vii) A circuit with $R = 70\Omega$ in series with a parallel combination of $L = 1.5\text{H}$ and $C = 30\mu\text{F}$ is driven by a 230V supply of angular frequency 300 rad/s .

- Find the impedance of the circuit.
- What is the RMS value of total current?
- What are the current amplitudes in the L and C arms of the circuit?
- How will the circuit behave at $\omega = 1/\sqrt{LC}$?

[(a) 163.3Ω (b) 1.414A (c) 0.653A , 2.64A (d) Circuit current becomes zero]

(viii) A resistance of 10Ω is joined in series with an inductance of 0.5H . What capacitance should be put in series with the combination to obtain the maximum current? What will be this maximum current? What will be the potential difference across the resistance, inductance and capacitance? The current is being supplied by 200V , 50Hz AC mains.

[$20.24\mu\text{F}$, 20A , 200V , 3142V , 3142V]

(ix) In the circuit shown in figure-5.230 AC EMF is $e = 10 \cos 100\pi t$. Find circuit impedance and the potential difference across points A and B when it is half of the voltage of source at that instant.

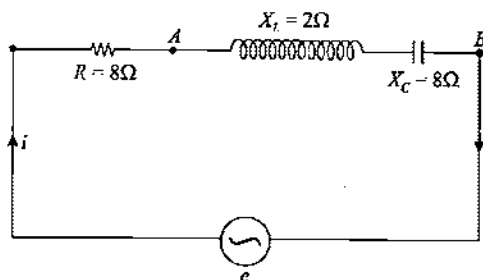


Figure 5.230

5.13 Power in AC Circuits

In the chapter of current electricity we've studied that the power supplied by a source to the circuit is given by the product of the EMF and the current supplied by the source. In DC circuits current and EMF remain constant in steady state but in AC circuits with time EMF and current continuously vary and depending upon the phase difference between EMF and the current their instantaneous values will be different for different circuits as phase relationship between EMF and current depend upon the type of circuit used so instantaneous power supplied by an AC source continuously vary with time that's why in AC circuits we analyze average power supplied by the AC source.

5.13.1 Average Power in AC Circuits

Figure-5.231 shows an AC Circuit in which an AC EMF is connected across an impedance $\bar{Z}$ which may be a combination of two or more components or a single component. The current in this circuit is given by Ohm's law as

$$\bar{i} = \frac{\bar{e}}{\bar{Z}}$$

$$\Rightarrow \bar{i} = \frac{e_0 \angle 0^\circ}{Z_0 \angle \phi} = \frac{e_0}{Z_0} \angle -\phi = i_0 \angle -\phi$$

$$\Rightarrow i = i_0 \sin(\omega t - \phi) \quad \dots (5.233)$$

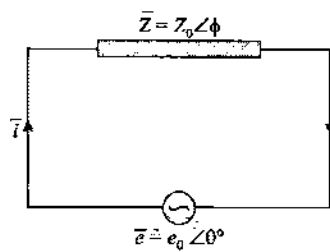


Figure 5.231

In this circuit instantaneous power supplied by the AC source can be given as

$$P = ei$$

$$\Rightarrow P = e_0 \sin \omega t \times i_0 \sin(\omega t - \phi)$$

$$\Rightarrow P = e_0 i_0 \sin \omega t \times (\sin \omega t \cos \phi - \cos \omega t \sin \phi)$$

$$\Rightarrow P = e_0 i_0 \sin^2 \omega t \cos \phi - e_0 i_0 \sin \omega t \cos \omega t \sin \phi \quad \dots (5.234)$$

Above equation-(5.234) gives the instantaneous power supplied by the AC source of which average value can be given by replacing the sinusoidal time functions on RHS of this equation by their average values which gives

$$P_{avg} = e_0 i_0 \langle \sin^2 \omega t \rangle \cos \phi - e_0 i_0 \langle \sin \omega t \cos \omega t \rangle \sin \phi$$

$$\Rightarrow P_{avg} = e_0 i_0 \left(\frac{1}{2} \right) \cos \phi - e_0 i_0 (0) \sin \phi$$

$$\Rightarrow P_{avg} = \frac{1}{2} e_0 i_0 \cos \phi$$

$$\Rightarrow P_{avg} = \frac{e_0}{\sqrt{2}} \frac{i_0}{\sqrt{2}} \cos \phi$$

$$\Rightarrow P_{avg} = e_{rms} i_{rms} \cos \phi \quad \dots (5.235)$$

Above equation-(5.235) gives the average power supplied by an AC source to the connected circuit. In general average is defined for one cycle or for large number of cycles or over a significant duration. The term of ' $\cos \phi$ ' on RHS of equation-(5.235) is called '*Power Factor*' of the AC circuit. Thus average power supplied to an AC circuit depends upon the phase difference between the AC EMF and the current supplied by the source.

5.13.2 Wattless Current in AC Circuits

If in a given AC circuit ϕ is the phase difference between the applied EMF and the current then phasor diagram for EMF and current drawn is shown in figure-5.232.

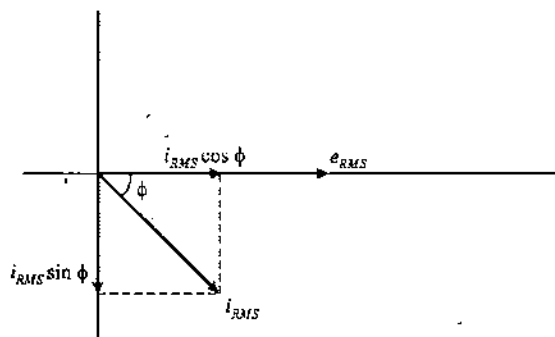


Figure 5.232

In above figure we've resolved the current phasor in two components, one along the EMF and other normal to it. For this circuit the average power delivered by the AC source can be given as

$$P_{avg} = e_{rms} i_{rms} \cos\phi$$

We can see in above expression of average power that it is due to the current component $i_{rms} \cos\phi$ and there is no contribution of the current component $i_{rms} \sin\phi$ in average power supplied by the AC source so this current component ' $i_{rms} \cos\phi$ ' is called '*Wattless Current*' of the AC circuit.

This wattless current does not have significance for calculation of average power in an AC circuit but for current calculations it is useful if mentioned specifically in some situation.

5.13.3 Maximum Power in an AC Circuit

As discussed in article-5.13.1 that average power supplied by an AC source to the circuit is given as

$$P_{avg} = e_{rms} i_{rms} \cos\phi$$

In a given AC circuit RMS values of current and EMF can be related as

$$i_{rms} = \frac{e_{rms}}{Z}$$

Where Z is the circuit impedance across the AC source. Using this average power can be written as

$$P_{avg} = \frac{e_{rms}^2}{Z} \cos\phi$$

With the change in phase difference between AC EMF and current in the circuit average power supplied changes. If in a circuit $\phi = 0$ then in that circuit, impedance will be purely resistive ($Z = R$) and power supplied will be maximum which is given as

$$P_{avg} = i_{rms} e_{rms} = i_{rms}^2 R = \frac{e_{rms}^2}{R} \quad \dots (5.236)$$

Thus when $\phi = 0$ then in the circuit capacitive reactance and inductive reactance nullify the effect of each other and circuit behave as a purely resistive circuit across AC EMF.

If in a circuit $\phi = \pm \pi/2$ then the circuit impedance will be either purely capacitive or purely inductive and in this case average power supplied by the AC source will be zero. It is to be noted that in AC circuits inductors and capacitors never consume any average power. This is because in one half cycle for reactive components of AC circuits power is positive and for other half

cycle power is negative so these components do not contribute in power consumption per cycle independently but due to reactive components of circuit, the circuit impedance and phase relationship between applied EMF and current changes thus overall power supplied by the source is affected.

5.13.4 Resonance in Series RLC Circuit

For a series RLC circuit we've studied the phasor analysis in article-5.12.10 and 5.12.11. The current amplitude supplied by the AC source in series RLC circuit is given as

$$i_0 = \frac{e_0}{Z} = \frac{e_0}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

In above equation we can see that i_0 will be maximum when in denominator $X_L = X_C$ which occurs at a specific frequency for a specific circuit which is given as

$$\omega = \frac{1}{\sqrt{LC}} \quad \dots (5.237)$$

At above frequency $Z = R$, $\phi = 0$ and circuit impedance becomes minimum and behave as a pure resistive impedance at which circuit current becomes maximum which is given as

$$i_0 = \frac{e_0}{R} \quad \dots (5.238)$$

At above frequency given by equation-(5.237) the current as well as power supplied by the AC source to the circuit becomes maximum and circuit impedance is purely resistive. This state is called '*Resonance*' or '*Resonating State*' of AC circuit and the frequency given by equation-(5.237) at which it occurs is called '*Resonating Frequency*'.

5.13.5 Variation of Current with Frequency in Series RLC Circuit

In series RLC circuit the RMS current is given as

$$i_{rms} = \frac{e_{rms}}{Z} = \frac{e_{rms}}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$$

In previous article we've discussed that at resonating frequency the current and power supplied in circuit becomes maximum thus at frequencies below and above the resonating frequency current will be less than maximum of which the variation is shown in figure-5.233.

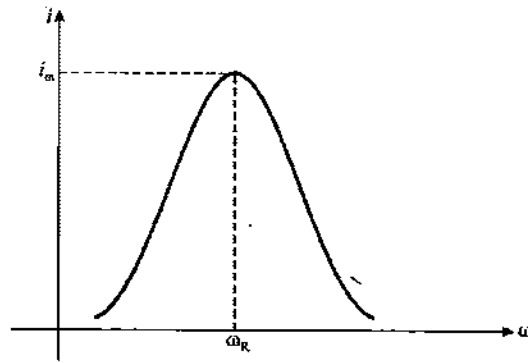


Figure 5.233

The resonating frequency at which current and power in circuit becomes maximum as shown above is given as

$$\omega_R = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow f_R = \frac{1}{2\pi\sqrt{LC}}$$

It is already discussed when frequency of AC source is $f = f_R$ then $\phi = 0$ and circuit impedance is purely resistive with current approaches to maximum value. When $f < f_R$ circuit impedance behave as combination of resistance and capacitance and ϕ is positive in this case and current leads EMF and when $f > f_R$ circuit impedance behave as combination of resistance and inductance and ϕ is negative so in this case current lags behind EMF.

5.13.6 Half Power Frequencies in Series RLC Circuit

As discussed in previous article that current and power in series RLC circuit are maximum at resonant frequency given by equation-(5.237). In the graph shown in figure-5.233 on the two sides of resonant frequency there are two side frequencies ω_1 and ω_2 at which power supplied is half of the maximum power. These frequencies are shown in figure-5.234.

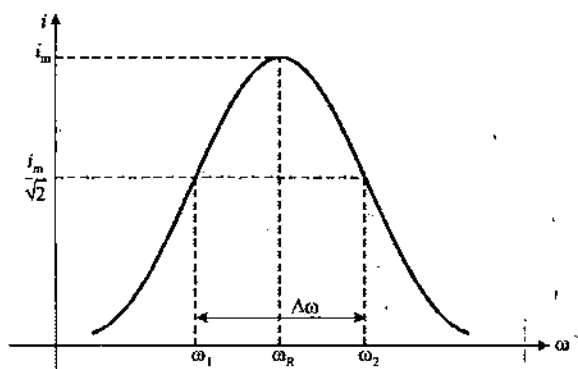


Figure 5.234

In AC circuits power consumed is only by the resistance of circuit as average power consumed by inductor and capacitor is always zero. Thus power of an AC Circuit is given as

$$P_{avg} = i_{rms} e_{rms} \cos \phi$$

$$\Rightarrow P_{avg} = i_{rms}^2 Z \cos \phi$$

$$\Rightarrow P_{avg} = i_{rms}^2 R \quad \dots (5.239)$$

Power is maximum when current is maximum at resonant frequency which is given as

$$P_m = i_m^2 R$$

When frequency is deviated from the resonant frequency power in circuit decreases and when power is reduced to half then current in circuit is called half power current which is given as

$$i_{hp} = \sqrt{\frac{(P_m/2)}{R}} = \frac{i_m}{\sqrt{2}}$$

The range of frequencies ω_1 to ω_2 at which the power in circuit is more than half of maximum is called 'Circuit Bandwidth' which is given as

$$\Delta\omega = \omega_2 - \omega_1$$

5.13.7 Bandwidth for a Series RLC Circuit

The effective current in a series RLC circuit is given as

$$i_{rms} = \frac{e_{rms}}{Z} \quad \dots (5.240)$$

At maximum power in circuit we've discussed that $Z = R$ thus we use

$$i_m = \frac{e_{rms}}{R} \quad \dots (5.241)$$

Half power current is given as

$$i_{hp} = \frac{i_m}{\sqrt{2}}$$

From equations-(197) and (198) we can see that at half power frequency we have

$$Z = \sqrt{2}R \quad \dots (5.242)$$

For Series RLC circuit impedance is given as

$$Z = \sqrt{R^2 + X^2} = \sqrt{2}R$$

$$\Rightarrow X = \pm R \quad \dots (5.243)$$

$$\Rightarrow \left(\omega L - \frac{1}{\omega C} \right) = \pm R \quad \dots (5.244)$$

There are two quadratic equations in equation-(5.244) for the '+' and '-' signs on the RHS of the equation. Positive sign will correspond to positive reactance of series RLC circuit in which current will lag behind the EMF so for the right side half power frequency ω_2 we use positive sign and similarly negative sign correspond to negative reactance of series RLC circuit in which current will lead the EMF so for the left side half power frequency ω_1 we use negative sign.

For ω_1 we use

$$\omega L - \frac{1}{\omega C} = -R \quad \dots (5.245)$$

Solving the above equation gives

$$\omega_1 = -\frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}} \quad \dots (5.246)$$

For ω_2 we use

$$\omega L - \frac{1}{\omega C} = +R \quad \dots (5.247)$$

Solving the above equation gives

$$\omega_2 = \frac{R}{2L} + \sqrt{\frac{R^2}{4L^2} + \frac{1}{LC}} \quad \dots (5.248)$$

From equations-(5.246) and (5.248) bandwidth of series RLC circuit is given as

$$\Delta\omega = \omega_2 - \omega_1 = \frac{R}{L} \quad \dots (5.249)$$

5.13.8 Quality Factor of Series RLC Circuit

The ratio of energy stored in circuit to the energy loss per cycle at resonance in an AC circuit is defined as '*Quality Factor*' or '*Q-Factor*' of the circuit. This is given as

$$Q = \frac{\text{Maximum Energy Stored per Cycle}}{\text{Energy Loss per Cycle}}$$

It is numerically calculated as the ratio of resonating frequency to the circuit bandwidth which is given as

$$Q = \frac{\omega_R}{\Delta\omega} \quad \dots (5.250)$$

For series RLC circuit we substitute the values of resonating frequency and bandwidth which gives

$$Q = \frac{\omega_R L}{R}$$

$$\Rightarrow Q = \frac{1}{R} \sqrt{\frac{L}{C}} \quad \dots (5.251)$$

Above equation-(5.251) gives the *Q*-factor for a series RLC circuit whereas equation-(5.250) gives the general function for *Q*-factor for any AC circuit.

5.13.9 Selectivity of a Resonance Circuit

Resonance circuits are used to respond selective signals of some specific frequency in communication systems. If in a circuit resonance is having a narrow peak at the chosen frequency then the circuit is called '*Highly Selective*'. Figure-5.235 shows the current variation with frequency for three different circuits named 1, 2 and 3. For circuit-1 at the same resonant frequency bandwidth is less and the curve is narrow compared to circuit-2 and 3 so in very short neighbourhood of the circuit, the power drops to a level below half power level and because of this, its *Q*-factor is high and this circuit is said to have '*Selectivity*' more than circuit-2 and 3. Circuit-3 has higher bandwidth at same resonance frequency in which to a far variation of frequency away from the resonance frequency, circuit power is not going below the half power level so the frequency band upto its half power frequencies is blocked by this channel only so this is said to be a '*Poorly Selective*' circuit. Indirectly we can say that selectivity is a qualitative term and *Q*-factor is a quantitative term used to analyze the suitability of circuit to be used for a communication system.

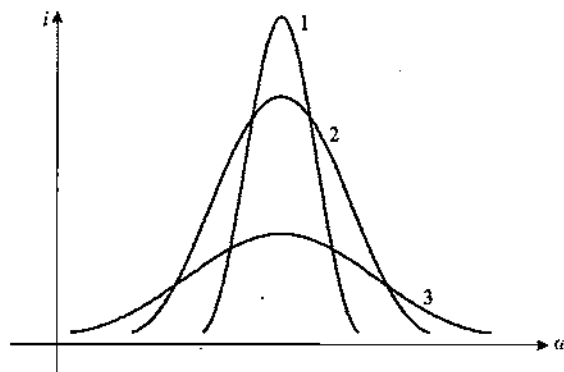


Figure 5.235

5.13.10 Parallel RLC Circuit

Figure-5.236 shows combination of a resistance, capacitance and an inductor connected in parallel across an AC source. In series RLC circuit we've studied that for all components current remains same and applied EMF is divided across all components in phasor form. In this case the applied potential difference is same across all components and current is divided in the three components which can be given as

$$\vec{i} = \vec{i}_R + \vec{i}_C + \vec{i}_L \quad \dots (5.252)$$

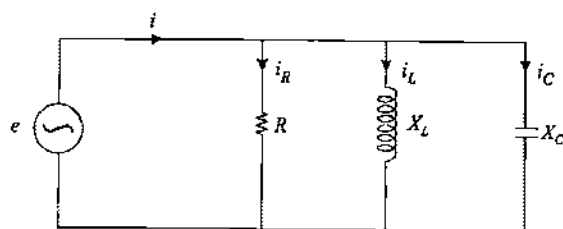


Figure 5.236

In above case the total current supplied by the AC source will not be the mathematical sum of the three currents through the three components as the phase of the three currents will differ. As the applied voltage is same to all, the current through resistance will be in same phase as that of the applied voltage, the current in inductor will lag behind the applied voltage by 90° and current in capacitor will lead the applied voltage by 90° . If the reference phase is taken from EMF then equation-(5.252) can be rewritten in phasor forms using the phasor diagram as shown in figure-5.237 of currents as

$$\vec{i} = i_R \angle 0^\circ + i_C \angle 90^\circ + i_L \angle -90^\circ \quad \dots (5.253)$$

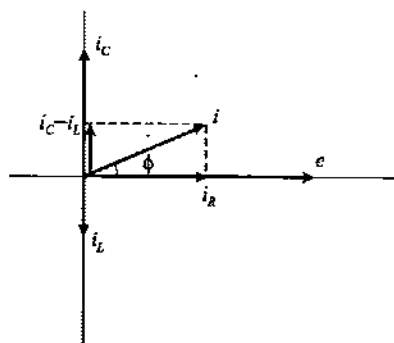


Figure 5.237

Thus phasor sum of the three currents in parallel RLC circuit can be given for their magnitudes as

$$i = \sqrt{i_R^2 + (i_C - i_L)^2} \quad \dots (5.254)$$

Impedance in parallel RLC Circuit can be given as

$$\frac{1}{Z} = \frac{1}{R} + \frac{1}{X_L} + \frac{1}{X_C}$$

$$\Rightarrow \frac{1}{Z} = \frac{1}{R \angle 0^\circ} + \frac{1}{\omega L \angle 90^\circ} + \frac{1}{1/\omega C \angle -90^\circ}$$

$$\Rightarrow \frac{1}{Z} = \frac{1}{R} \angle 0^\circ + \frac{1}{\omega L} \angle -90^\circ + \frac{1}{1/\omega C} \angle 90^\circ$$

$$\Rightarrow \frac{1}{Z} = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

$$\Rightarrow Z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

Thus current in parallel RLC Circuit can be given as

$$i = \frac{e}{Z} = e \sqrt{\left(\frac{1}{R}\right)^2 + \left(\omega C - \frac{1}{\omega L}\right)^2}$$

For the above current the frequency response curve is shown in figure-5.238. Here we can see that the frequency at which capacitive and inductive reactance are equal and impedance is purely resistive, the current in circuit is minimum. Such a circuit is called '*Rejactor Circuit*' in which at a specific frequency the power supplied or current from the source is minimum. This state is called resonance of parallel RLC circuit and the resonant frequency ω_R is given as

$$\omega_R C - \frac{1}{\omega_R L} = 0$$

$$\Rightarrow$$

$$\omega_R = \frac{1}{\sqrt{LC}}$$

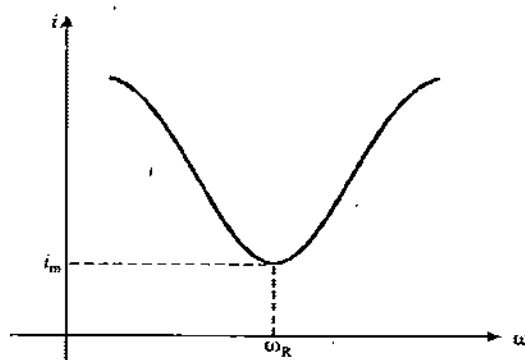


Figure 5.238

Illustrative Example 5.78

A 750Hz, 20V source is connected to a resistance of 100Ω , an inductance of 0.1803H and a capacitance of $10\mu\text{F}$ in series. Calculate the time in which the resistance which has thermal capacity $2\text{ J/}^\circ\text{C}$ will get heated by 10°C .

Solution

The inductive and capacitive reactance are given as

$$X_L = \omega L = (2 \times 3.14 \times 750) \times 0.1803 = 850\Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(2 \times 3.14 \times 750) \times (10^{-5})} = 21.2\Omega$$

Impedance of series RLC circuit is given as

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\Rightarrow Z = \sqrt{[(100)^2 + (850 - 21.2)^2]} = 835\Omega$$

Power dissipated in circuit is given as

$$P = E_{rms} \times I_{rms} \times \cos \phi$$

Where power factor is given as

$$\cos \phi = \frac{R}{Z}$$

$$\Rightarrow P = E_{rms} \times I_{rms} \times \frac{R}{Z}$$

$$\Rightarrow P = E_{rms} \times \left(\frac{E_{rms}}{Z}\right) \times \frac{R}{Z}$$

$$\Rightarrow P = \frac{E_{rms}^2 \times R}{Z^2}$$

$$\Rightarrow P = \frac{(20)^2 \times 100}{(835)^2} = 0.0574W$$

Heat produced in resistance to raise its temperature by 10°C is given as

$$H = 2 \text{ J/}^\circ\text{C} \times 10^\circ\text{C} = 20\text{J}$$

If this heat is produced in t s then we use

$$Pt = 20 \text{ J}$$

$$\Rightarrow t = \frac{20}{0.0574} = 384\text{s}$$

Illustrative Example 5.79

A choke coil is needed to operate an arc lamp at 160V, 50Hz AC supply. The arc lamp has an effective resistance of 5Ω when running at 10A. Calculate the inductance of the choke coil if the same arc lamp is to be operated on 160V DC supply, what additional resistance is required. Compare the power losses in both cases.

Solution

The effective current through the circuit with lamp and choke coil in series is given as

$$I_{rms} = \frac{E_{rms}}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\Rightarrow 10 = \frac{160}{\sqrt{25 + (2\pi \times 50L)^2}}$$

$$\Rightarrow 10 = \frac{160}{\sqrt{25 + (100\pi L)^2}}$$

$$\Rightarrow \sqrt{25 + (100\pi L)^2} = 16$$

$$\Rightarrow 25 + (100\pi)^2 L^2 = 256$$

$$\Rightarrow 25 + 10^5 L^2 = 256$$

$$\Rightarrow L^2 = \frac{231}{10^5} = 231 \times 10^{-5}$$

$$\Rightarrow L = 0.048\text{H}$$

For 160V DC if R_A be the additional resistance, then we have

$$10 = \frac{160}{(R_A + 5)}$$

$$\Rightarrow R_A + 5 = 16$$

Additional resistance required is given as

$$R_A = 16 - 5 = 11\Omega$$

AC power consumed by the circuit is given as

$$P = E_{rms} I_{rms} \cos \phi$$

The power factor of circuit is given as

$$\cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$

$$\Rightarrow \cos \phi = \frac{5}{\sqrt{25 + 231}} = \frac{5}{16}$$

$$\Rightarrow P_{ac} = 160 \times 10 \times (5/16) = 500\text{W}$$

Across DC supply the power loss is given as

$$P_{dc} = 160 \times 10 = 1600\text{W}$$

Illustrative Example 5.80

An inductor of 2H , capacitance $18\mu\text{F}$ and a resistance of $10\text{k}\Omega$ are connected in series to an AC source of 20V with adjustable frequency. At what frequency the current in circuit will be maximum and what will be this maximum current.

Solution

If i is maximum current at Resonance in series RLC circuit, the

resonant frequency is given as

$$f_R = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{2 \times 18 \times 10^{-6}}}$$

$$\Rightarrow f_R = \frac{1}{12\pi} \times 10^3 \text{ Hz} = \frac{250}{3\pi} \text{ Hz}$$

Maximum current in circuit is given as

$$i_{\max} = \frac{e}{R} = \frac{20}{10 \times 10^3} = 2 \text{ mA}$$

Illustrative Example 5.81

A series RLC circuit with 100Ω resistance is connected to an AC source of 200V and angular frequency 300rad/s . When only the capacitance is removed, the current lags behind the voltage by 60° . When only the inductance is removed, the current leads the voltage by 60° . Calculate the current and power dissipated in RLC circuit.

Solution

As per given condition in RLC circuit we have

$$\tan 60 = \frac{\omega L}{R}$$

$$\text{and } \tan 60 = \frac{1/\omega C}{R}$$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

Impedance of circuit is given as

$$Z = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = R$$

Current in the circuit is given as

$$I_0 = \frac{V_0}{Z} = \frac{V_0}{R} = \frac{200}{100} = 2 \text{ A}$$

Average power dissipated in circuit is given as

$$\bar{P} = \frac{1}{2} V_0 I_0 \cos \phi$$

$$\Rightarrow \bar{P} = \frac{1}{2} \times 200 \times 2 \times 1 = 200 \text{ W}$$

Illustrative Example 5.82

A RLC series circuit has $L = 10\text{mH}$, $R = 3\Omega$ and $C = 1\mu\text{F}$ connected in series to a source of $15\cos\omega t$ volt. Calculate the current amplitude and the average power

dissipated per cycle at a frequency that is 10% lower than the resonant frequency.

Solution

Resonant frequency of circuit is given as

$$\omega_R = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_R = \sqrt{\frac{1}{(10 \times 10^{-3})(1 \times 10^{-6})}} = 10^4 \text{ rad/s}$$

10% less frequency of above resonant frequency is given as

$$\omega = 10^4 - 10^4 \times \frac{10}{100} = 9 \times 10^3 \text{ rad/s}$$

At this frequency, the reactances are given as

$$X_L = \omega L = 9 \times 10^3 \times (10 \times 10^{-3}) = 90\Omega$$

$$\text{and } X_C = \frac{1}{\omega C} = \frac{1}{(9 \times 10^3)(1 \times 10^{-6})} = 11.11\Omega$$

$$\Rightarrow Z = \sqrt{R^2 + (90 - 11.11)^2} = 21.32\Omega$$

Current amplitude is given as

$$I_0 = \frac{E_0}{Z} = \frac{15}{21.32} = 0.704 \text{ A}$$

Average power consumed by circuit is given as

$$P = \frac{1}{2} E_0 I_0 \cos \phi$$

Power factor of the circuit is given as

$$\cos \phi = \frac{R}{Z} = \frac{3}{21.32} = 0.141$$

$$\Rightarrow P = \frac{1}{2} \times 15 \times 0.704 \times 0.141 = 0.744 \text{ W}$$

Illustrative Example 5.83

In an AC circuit EMF applied is $e = 5\sin\omega t$ due to which a current $i = 3\cos\omega t$ flows in circuit. Find the average power dissipated in the circuit.

Solution

From the given information we can see that the phase difference between EMF and the current supplied is $\pi/2$ so the circuit impedance is reactive for which the average power dissipated in circuit will be zero.

Illustrative Example 5.84

A current of 4A flows in a coil when connected to a 12V DC source. If the same coil is connected to a 132V, 50 rad/s, AC source, a current of 2.4A flows in the circuit. Determine the inductance of the coil. Also find the power developed in the circuit if a 2500 μ F capacitor is connected in series with the coil.

Solution

When the coil is connected to a DC source, its resistance R is given as

$$R = \frac{V}{I} = \frac{12}{4} = 3\Omega$$

When it is connected to AC source, the impedance Z of the coil is given as

$$Z = \frac{V_{rms}}{I_{rms}} = \frac{12}{2.4} = 5\Omega$$

For a coil its impedance is given as

$$Z = \sqrt{[R^2 + (\omega L)^2]}$$

$$\Rightarrow 5 = \sqrt{[(3)^2 + (50L)^2]}$$

$$\Rightarrow 25 = [(3)^2 + (50L)^2]$$

$$\Rightarrow L = 0.08\text{H}$$

When the coil is connected with a condenser in series, its impedance Z' is given by

$$Z' = \sqrt{\left[R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2 \right]}$$

$$\Rightarrow Z' = \left[(3)^2 + \left(50 + 0.8 - \frac{1}{50 \times 2500 \times 10^{-6}} \right)^2 \right]^{1/2}$$

$$\Rightarrow Z' = 5\Omega$$

Average power developed in circuit is given as

$$P = V_{rms} \times I_{rms} \times \cos \phi$$

Where power factor of circuit is given as

$$\cos \phi = \frac{R}{Z'} = \frac{3}{5} = 0.6$$

$$\Rightarrow P = 12 \times 2.4 \times 0.6 = 17.28\text{W}$$

5.14 Transformer

A transformer is a device used to increase or decrease AC voltage by using the concept of mutual induction between two coils. Figure-5.239 shows a transformer in which on a soft iron core two coils are wound named 'Primary Coil' and 'Secondary Coil' of the transformer. These are also called 'P-Coil' and 'S-Coil' of transformer. Primary coil of transformer is considered as input coil at which an AC source is connected of which the voltage is to be increased or decreased and secondary coil is considered as output coil from which desired output voltage is taken out. Figure-5.239(b) shows a three dimensional view of transformer for understanding its structure.

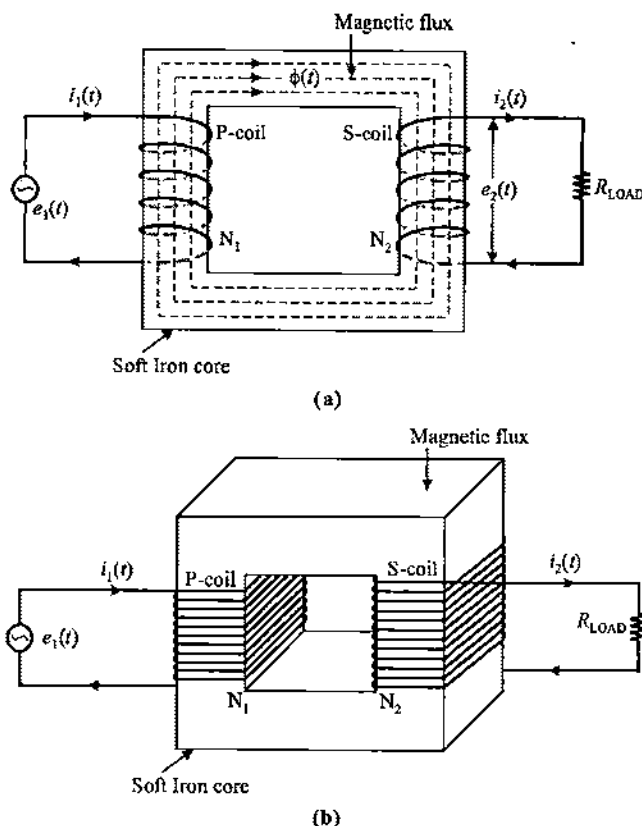


Figure 5.239

When an AC source of EMF $e(t)$ is connected across the primary coil of the transformer which is considered to be supplying a current $i_1(t)$ in primary coil as shown in figure. This current produces a total magnetic flux $\phi(t)$ in the core of transformer. If N_1 and N_2 are the total number of turns in primary and secondary coils then the magnetic flux linked with primary coil ϕ_1 and ϕ_2 are given as

$$\phi_1 = N_1 \phi(t)$$

and

$$\phi_2 = N_2 \phi(t)$$

Due to above time varying magnetic flux linked with the two coils of transformer the EMF induced in the two coils are given as

$$e_1 = \frac{d\phi_1}{dt} = N_1 \frac{d\phi(t)}{dt} \quad \dots (5.255)$$

and

$$e_2 = \frac{d\phi_2}{dt} = N_2 \frac{d\phi(t)}{dt} \quad \dots (5.256)$$

Dividing equations-(5.255) and (5.256) gives

$$\frac{e_1}{e_2} = \frac{N_1}{N_2} \quad \dots (5.257)$$

Above equation-(5.257) which relates the EMFs in primary and secondary coils with their number of turns is called '*Transformer Equation*'.

5.14.1 Power Relations in a Transformer

Figure-5.240 shows a transformer with AC source of EMF e_1 connected at primary coil and supplying a current i_1 into it. From secondary coil a load resistance R is connected across which the EMF induced is e_2 and current through it is i_2 .

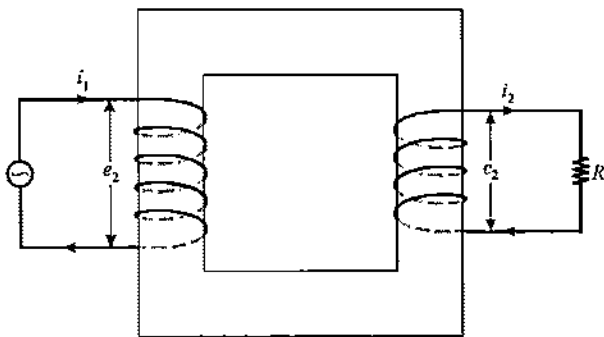


Figure 5.240

In above situation the power supplied at primary coil of transformer by AC source is given as

$$P_1 = e_1 i_1 \quad \dots (5.258)$$

At the secondary coil power delivered to the load resistance is given as

$$P_2 = e_2 i_2 \quad \dots (5.259)$$

For an ideal transformer energy losses are neglected so we have by conservation of energy

$$P_1 = P_2$$

$$\Rightarrow e_1 i_1 = e_2 i_2 \quad \dots (5.260)$$

Using transformer equation, above equation-(5.260) can be rewritten as

$$\frac{e_1}{e_2} = \frac{i_2}{i_1} = \frac{N_1}{N_2} \quad \dots (5.261)$$

Above equation-(5.261) is called '*Ideal Power Equation*' of a transformer. In case of power losses in a transformer if total

power losses is P_L then power equation is written as

$$e_1 i_1 - P_L = e_2 i_2 \quad \dots (5.262)$$

5.14.2 Types of Transformers

Depending upon the output voltage of a transformer it is classified in two categories - '*Step-up Transformers*' and '*Step-down Transformers*'.

Step-up Transformer : In step-up transformer the output voltage at secondary coil is more than voltage applied at primary coil. As $e_2 > e_1$ in a step-up transformer then from transformer equation we get $N_2 > N_1$ and from ideal power equation we get $i_2 < i_1$.

Step-down Transformer : In step-down transformer the output voltage at secondary coil is less than voltage applied at primary coil. As $e_2 < e_1$ in a step-down transformer then from transformer equation we get $N_2 < N_1$ and from ideal power equation we get $i_2 > i_1$.

5.14.3 Losses in Transformers

There are two types of power losses occur in transformer when its operating. These are - '*Copper Losses*' and '*Iron Losses*'. We will discuss both of these in some details.

Copper Losses in Transformer : These are dissipation losses in primary and secondary coils of transformer due to the resistance of coil windings. If resistance of the primary and secondary coil of a transformer are R_p and R_s and current in the two coils are i_p and i_s then total copper losses in a transformer circuit are given as

$$P_{CL} = i_p^2 R_p + i_s^2 R_s \quad \dots (5.263)$$

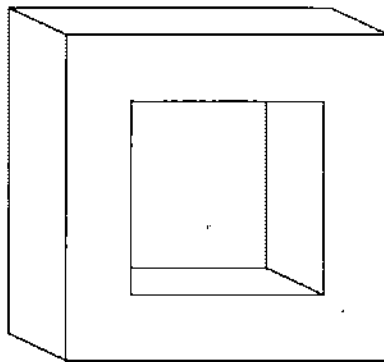
Copper losses in a transformer can only be reduced by proper selection of low resistance windings.

Iron Losses in Transformer : These are the dissipation losses in the core of transformer due to eddy currents. As time varying magnetic field continuously exist in the core of transformer due to which in all the cross sectional planes of the core eddy currents develop and produces heat. There is no direct mathematical way to analyze iron losses in a transformer. If P_T is the total power losses in a transformer then iron losses can be calculated by subtracting copper losses from it which is given as

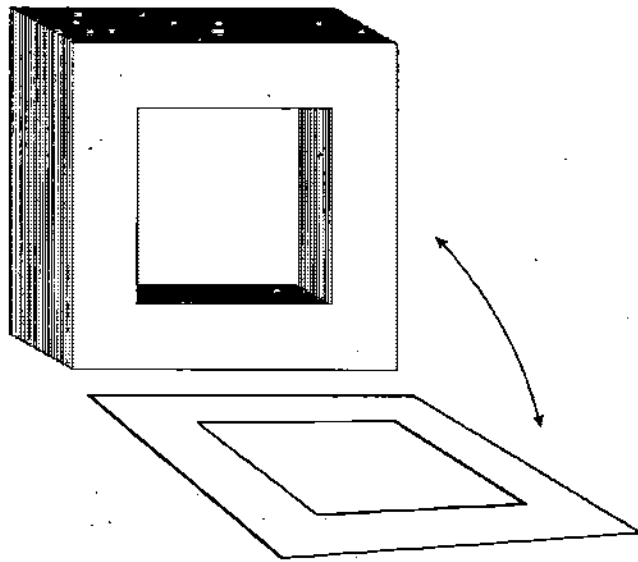
$$P_{IL} = P_T - P_{CL} \quad \dots (5.264)$$

To reduce iron losses we can break eddy currents by making the transformer core by joining laminated soft iron thin sheets

instead of using a solid soft iron core. Figure-5.241(a) shows a solid transformer core and figure-5.241(b) shows a transformer core which is made up of joining thin laminated sheets. In solid core cross section eddy currents flow in larger paths as shown in figure-5.242(a) which causes severe losses whereas figure-5.242(b) shows the cross section of a core made up of laminated sheet in which eddy currents are restricted within the cross sections of thin sheets as lamination does not allow conduction between the sheets due to which eddy current losses are reduced.

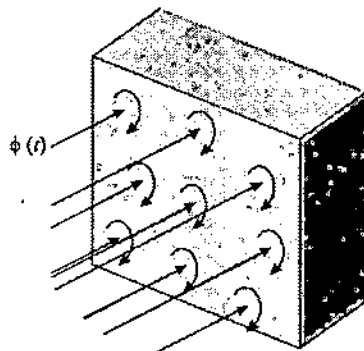


(a)

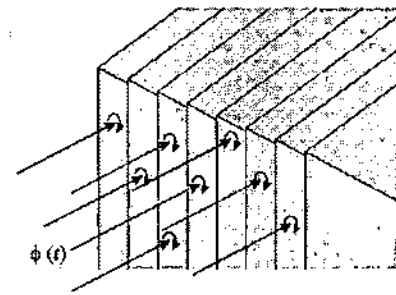


(b)

Figure 5.241



(a)



(b)

Figure 5.242

5.14.4 Efficiency of a Transformer

Due to power losses in a transformer its efficiency decreases and due to losses the power output taken from the secondary coil of the transformer is less than the power input at the primary coil. The efficiency of a transformer is given as

$$\eta = \frac{\text{Power Delivered at S-Coil}}{\text{Power Supplied at P-Coil}} \times 100\%$$

$$\Rightarrow \eta = \frac{e_s i_s}{e_p i_p} \times 100\% \quad \dots (5.265)$$

$$\Rightarrow \eta = \frac{e_p i_p - P_L}{e_p i_p} \times 100\% \quad \dots (5.266)$$

In case of an ideal transformer power losses are considered zero so for $P_L = 0$ we have $\eta = 100\%$.

Illustrative Example 5.85

An ideal transformer has 50 turns in its primary winding and 25 turns in its secondary winding. If the current in the secondary winding is 4A what is the current in primary winding if a 200V AC is applied across it?

Solution

With the ratio of turns in primary to secondary winding it is a 2:1 step down transformer for which by transformer equation we have

$$\frac{e_s}{e_p} = \frac{N_s}{N_p}$$

$$\Rightarrow e_s = \frac{N_s}{N_p} \times e_p$$

$$\Rightarrow e_s = \frac{25}{50} \times 200 = 100V$$

Power equation of ideal transformer gives

$$e_s i_s = e_p i_p$$

$$\Rightarrow i_p = \frac{e_s i_s}{e_p} = \frac{100 \times 4}{200} = 2\text{A}$$

Illustrative Example 5.86

An ideal power transformer is used to step up an alternating EMF of 220V to 4.4kV to transmit 0.6kW of power. If the primary coil has 1000 turns, find

- The number of turns in the secondary coil
- The current rating of the secondary

Solution

- By transformer equation we have

$$N_s = \frac{e_s}{e_p} \times N_p$$

$$\Rightarrow N_s = \left(\frac{4.4 \times 1000}{220} \right) \times 1000 = 20000 \text{ turns}$$

- Power supplied at primary coil is given as

$$i_p e_p = 6.6 \times 10^3 \text{W}$$

$$\Rightarrow i_p = \frac{6.6 \times 10^3}{220} = 30\text{A}$$

For an ideal transformer we use

$$\frac{i_s}{i_p} = \frac{N_p}{N_s} = \left(\frac{1000}{20,000} \right) = \frac{1}{20}$$

$$\Rightarrow i_s = \left(\frac{1}{20} \times i_p \right) = 1.5\text{A}$$

Illustrative Example 5.87

A transformer has turns ratio $N_s/N_p = 4$. If a 200V AC voltage is applied across its primary and it carries 1A current, find current in circuit connected to secondary coil if transformer is 80% efficient.

Solution

Power supplied at primary coil is

$$P_1 = e_1 i_1 = 200 \times 1 = 200\text{W}$$

Voltage across secondary coil is

$$e_2 = e_1 \times \frac{N_2}{N_1} = 200 \times 4 = 800\text{V}$$

In the given conditions power losses are 20%

$$\Rightarrow P_L = 200 \times \frac{20}{100} = 40\text{W}$$

Power available at secondary coil is

$$P_2 = 200 - 40 = 160\text{W}$$

$$\Rightarrow P_2 = e_2 i_2 = 160\text{W}$$

$$\Rightarrow i_2 = \frac{160}{800} = 0.2\text{A}$$

Illustrative Example 5.88

In a transformer there are 10000 turns in primary coil and 25000 turns in secondary coil. An AC EMF $e = 50 \sin \pi t$ is applied across primary coil, find the peak EMF across secondary coil in ideal conditions.

Solution

As per given turns ratio, this is a 1:2.5 step up transformer. Using transformer equation we have

$$\frac{e_1}{e_2} = \frac{N_1}{N_2}$$

$$\Rightarrow e_2 = \frac{N_2}{N_1} \times e_1 = \frac{25000}{10000} \times 50\text{V}$$

$$\Rightarrow e_2 = 125\text{V}$$

Web Reference at www.physicsgalaxy.com

Age Group - Grade 11 & 12 | Age 17-19 Years

Section - Magnetic Effects

Topic - Alternating Current

Module Number - 34 to 60

Practice Exercise 5.7

- A series LCR circuit containing a resistance of 120Ω has resonant frequency $4 \times 10^5 \text{ rad/s}$. At resonance the voltage across resistance and inductance are 60V and 40V respectively. Find the values of L and C and at what frequency the current in the circuit lags the voltage by 45°

$$[2 \times 10^{-4} \text{H}, \frac{1}{32} \mu\text{F}, 8 \times 10^5 \text{ rad/s}]$$

- 2000V – 200V, 20kVA transformer has 66 turns in the secondary. Calculate the primary and secondary full-load current, neglect power losses in transformer.

$$[10\text{A}, 100\text{A}]$$

- A circuit draws a power of 550W from a source of 220 V, 50Hz. The power factor of the circuit is 0.8 and the current lags in phase behind the potential difference. To make the power